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SCIENCE & TECHNOLOGY

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1 CHAPTER BIOTECHNOLOGY

➔ BASICS: DNA, RNA, CHROMOSOME, GENOME

1. WHAT IS DNA?

DNA (deoxyribonucleic acid) can be looked at in two ways:

- As code of life
- As vehicle of heredity

DNA AS CODE OF LIFE

- DNA carries the genetic information in the form of genes, which encode a complete set of instructions for building and maintaining an organism.
- The code is present in the nucleus of every cell of every living organism (with few exceptions like RBC, certain viruses etc.)
- The code is read by a host of machineries in the cell to build proteins which are the building blocks of life. (Proteins constitute 55% of the dry mass of the cell. Everything from the cell's structure to enzymes are made of proteins.)

2. WHAT IS CHROMOSOME?

- Chromosomes are thread-like structures found in the nucleus of cells that carry genetic information. They are composed of DNA (deoxyribonucleic acid) and proteins.
- Normally, each cell in the human body has 23 pairs of chromosomes (46 total chromosomes). This includes 22 pairs of autosomes plus one pair of sex chromosomes. Females have two X chromosomes (XX), while males have one X and one Y chromosome (XY).
- The 23 pairs of chromosomes together have **3.2 billion base-pairs of DNA.**

3. WHAT IS GENOME?

A genome is an organism's complete set of DNA or genetic material. It is a collection of all the genes and the regions between the genes contained in our 23 pairs of chromosomes.

BUILDING BLOCKS OF DNA

- The building blocks of DNA are called nucleotides. Each nucleotide is made up of three main components:

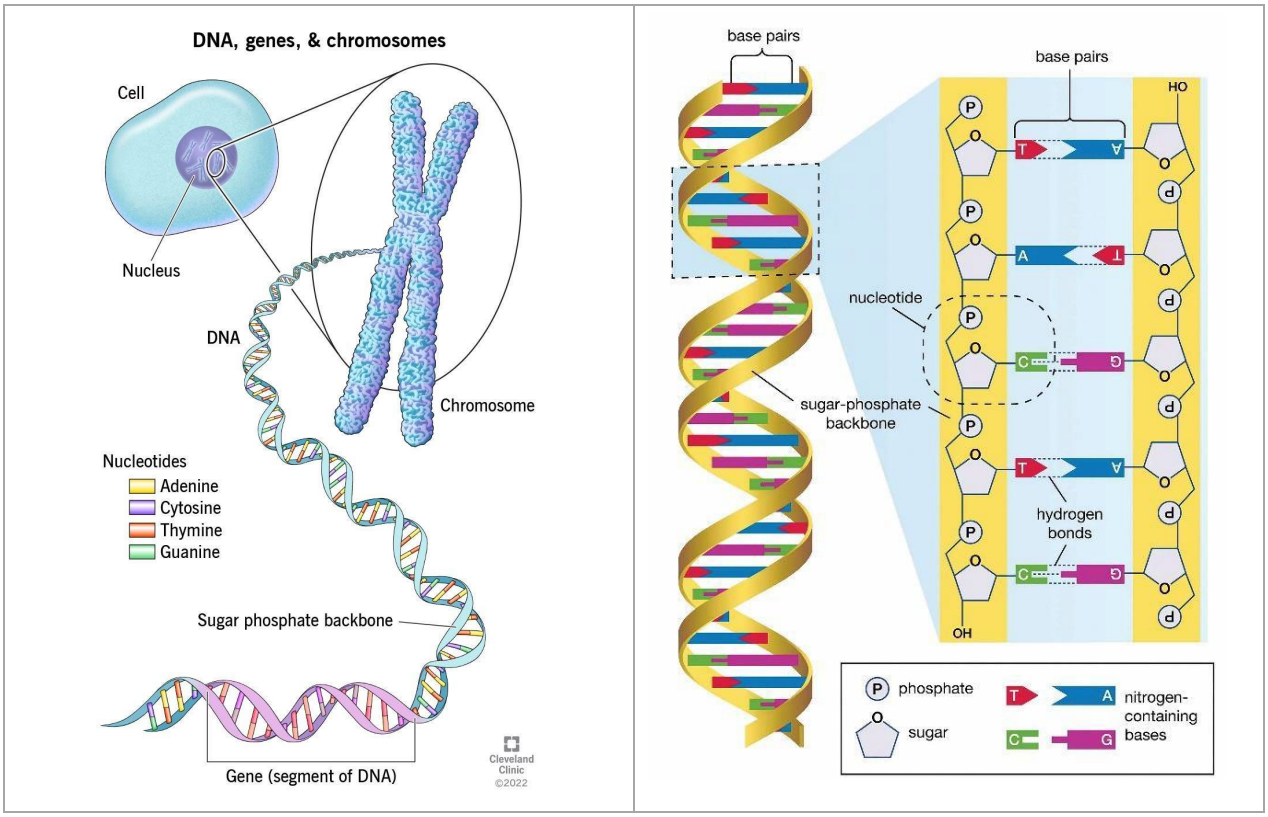
(i) **Sugar molecule**: A five-carbon sugar molecule, called deoxyribose, provides structural support to the DNA molecule.

(ii) **Phosphate group** attached to the deoxyribose sugar molecule. The phosphate groups contribute to the overall negative charge of the DNA molecule and play a role in linking nucleotides together to form the sugar-phosphate backbone of the DNA strand.

(iii) **Nitrogenous base (4 bases):** Adenine (A), Guanine (G), Cytosine (C) and Thymine (T).

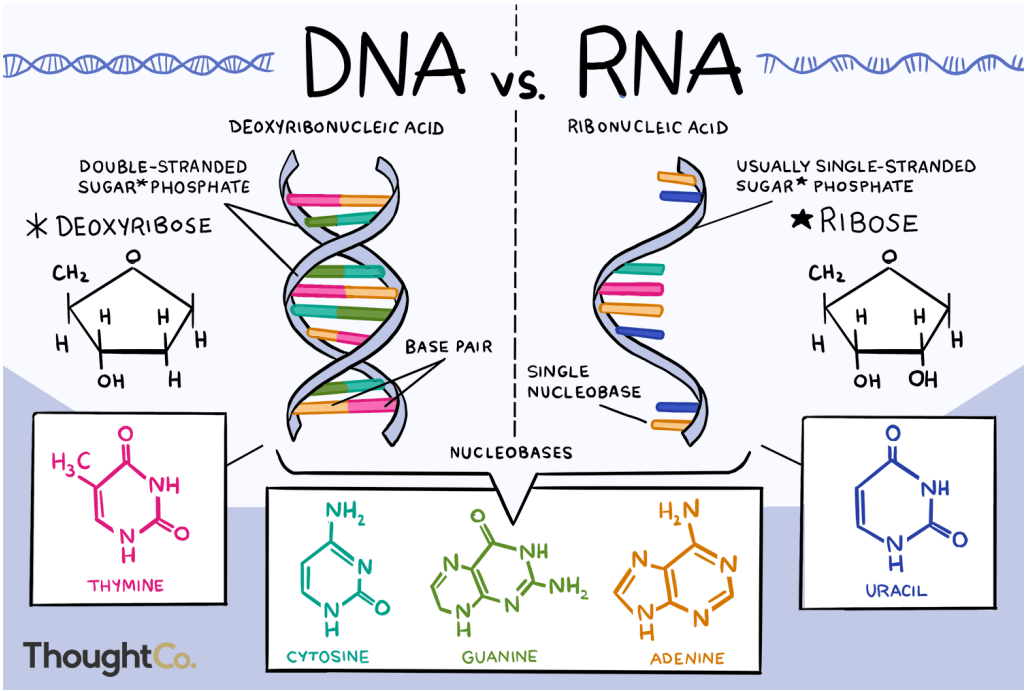
The specific sequence of these four nitrogenous bases determines the genetic code. These bases pair up specifically (A-T and C-G) forming the "rungs" of the double helical structure of DNA. (So if you know one strand you know what is on the other)

Note: While in nature only 4 lettered-DNA is found, scientists have recently created a synthetic DNA with 8 letters. The human genome is made of 3 billion base pairs. (So total 6 billion bases)



4. WHAT IS MITOCHONDRIAL DNA?

- In addition to the nucleus, some DNA is also present in the mitochondria (mtDNA).
- Mitochondria are cellular organelles found in the **cytoplasm.** They are called the powerhouse of the cell as they generate most of the cell's energy currency- **ATP** (adenosine triphosphate).
- mtDNA codes for a small number of genes, essential for the proper functioning of mitochondria. In humans, mtDNA is circular and much smaller (consists of about 16,500 base pairs), compared to nuclear DNA.
- **Inheritance:** Unlike nuclear DNA which come from both parents, mtDNA is inherited almost exclusively from the mother. This is because mitochondria are primarily contributed by the egg cell during fertilisation. While sperm cells do contain mitochondria, these mitochondria are typically not incorporated into the developing embryo.



5. WHAT IS RNA?

- RNA (Ribonucleic acid) in a single-stranded molecule/ nucleic acid found in most living organisms and viruses. It is made up of nucleotides, which are ribose sugars attached to nitrogenous bases and phosphate groups.
- The nitrogenous bases include adenine (A), guanine (G), uracil (U) and cytosine (C).
- **Major types of RNA:**
  - **mRNA:** messenger RNA molecules copies the genetic information/ instruction from DNA and carries it to ribosomes, where protein synthesis occurs. In order to transcribe, mRNA uses an enzyme called RNA polymerase.
  - **tRNA:** transfer RNA assists in translation of mRNA into proteins by bringing amino acids to the ribosome. t-RNA brings amino acids that are lying in the cell after reading 3 letters of mRNA at a time (3 letters or codon translate into one amino acid).
  - **rRNA:** ribosomal RNA molecules are structural components of ribosomes (cellular structures where protein synthesis takes place).
- RNA is generally less stable than DNA due to the presence of an extra hydroxyl (OH) group on the ribose sugar in RNA. This makes RNA more prone to degradation by enzymes called ribonucleases. DNA, with its deoxyribose sugar lacking the extra OH group, is more stable and less susceptible to enzymatic degradation.

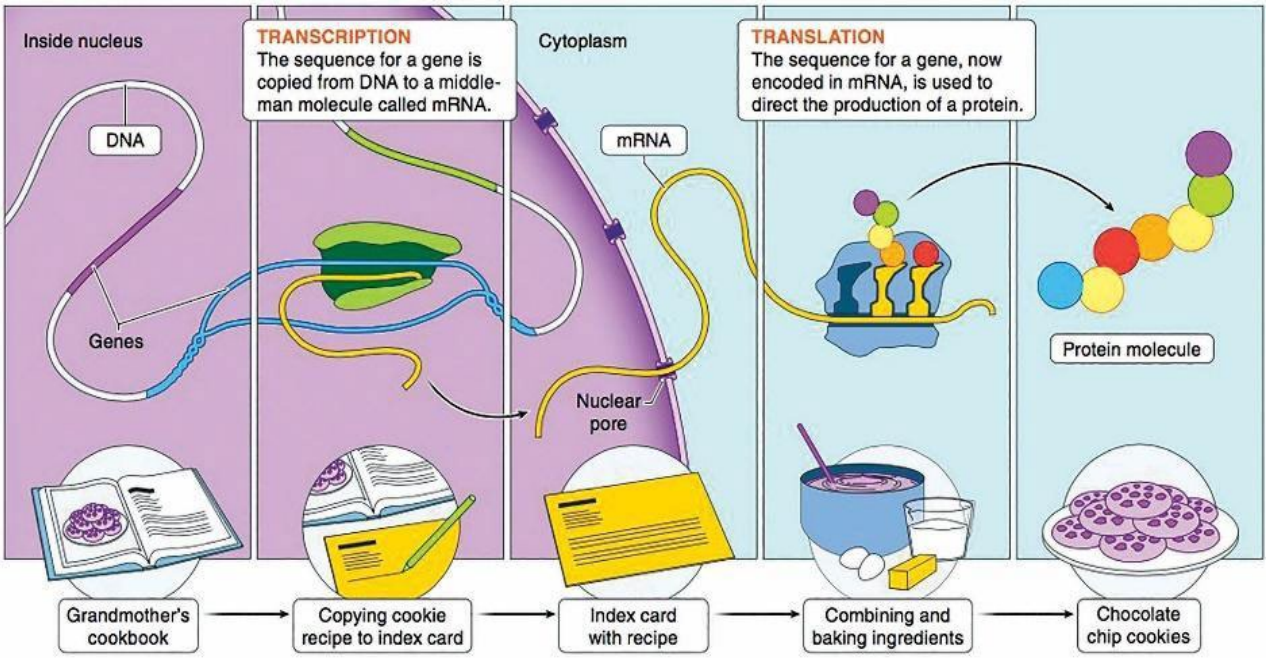
6. CENTRAL DOGMA



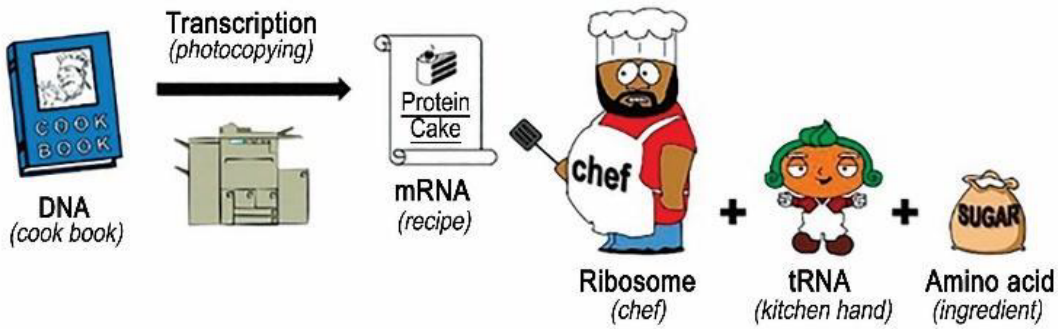
Central dogma of molecular biology explains how genetic information flows from DNA to RNA to proteins.

- **Gene expression** is the process our cells use to convert instructions in DNA into proteins. This happens in two-stages called transcription (occurs in the nucleus, in eukaryotic organisms) and translation (occurs in cytoplasm).
  - **Transcription**: mRNA copies the sequence of a gene from DNA, and then leaves the nucleus and travels to cytoplasm.

- **Translation**: Using the template encoded by mRNA, translation is carried out by ribosomes, i.e., the information in the mRNA is used to synthesise proteins (building blocks of life).



CENTRAL DOGMA (COOKBOOK ANALOGY)



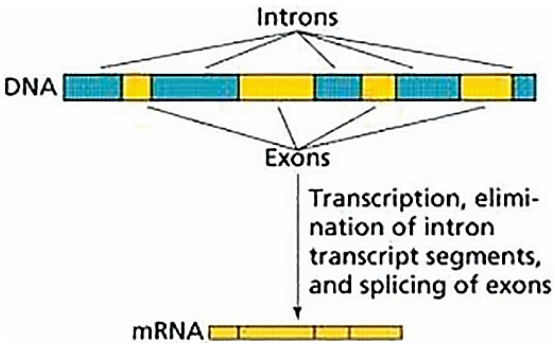
- Consider DNA to be a cookbook of recipes for making proteins. Transcription is the process of photocopying the recipe (done by mRNA inside the nucleus).
- Once copied the recipe mRNA goes outside the nucleus to the chef (ribosome). There is also a translator, tRNA who translates mRNA recipe to rRNA of the ribosome which then makes Amino acids (the building blocks of proteins).
- A block of 3 letters in the mRNA corresponds to 'cooking' of 1 Amino acid. Each 3-letter base in mRNA that is read by tRNA that corresponds to one amino acid is called 'codon'.
- Many Amino acids come together to form the primary structure of protein. The primary structure is twisted and folded to make a 3-D structure of protein.

7. GENES & EPIGENETICS

- The **Human Genome Project** (completed in 2003) found that only 1.5% of the total human genome (3 billion base pairs) codes for proteins. (The coding parts of protein are called **genes**)
- It also found that humans have about 20,000–25,000 genes. The remaining 98.5% of the genome is non-coding, meaning it does not provide instructions for making proteins.

EPIGENETICS: IT'S ALL ABOUT GENE EXPRESSION

- For a long time, non-coding regions of the genome (98.5%) were thought to be 'dark area' or 'Junk DNA', because they did not appear to have any specific purpose.
- However, new studies suggest that they play a huge role in gene expression which has become an important area of study in recent times called epigenetics.
- **Epigenetics** deals with the processes that control how the genes are expressed.
  - We know that all the cells in our body have the same genome. Average human body contains approximately 37.2 trillion cells and about 220 different cell types.
  - If the same code (genome) exists in the nucleus of all these cells, how 220 odd cell types make up for 4 different tissue types and 78 different organ types, all working in unison to make human life possible?
- The answer lies in **gene expression**. Different genes are expressed in different cells that perform different functions that look differently. (like your heart cell and your kidney cell)



INTRONS, EXONS AND RNA SPLICING

- As we have seen, mRNA is a copy of only the coding part of DNA (gene).
- The coding part does not occur on chromosomes in one single sequence as one whole. It is spread out on a chromosome in parts. Each part is separated by a non-coding part of the genome.
- In fact, 25% of all the non-coding parts occur in between genes.
- The non-coding parts between genes are called **introns** and the coding parts that mRNA is interested in are called **exons.**
- So, mRNA must copy only exons and cut out all the introns. This cutting of introns to join only exons is called **RNA splicing.**
- The final mRNA after splicing of introns is called **exome** (which represents only 1.5% of the genome). It is this final mRNA after splicing that is important for coding for protein.

MUSCULAR DYSTROPHY

- One type of muscular dystrophy, a genetic disease, is a result of defect in RNA splicing while copying X-chromosome.
- Since it is associated with the X-chromosome it is more prevalent in males as they have only one X-chromosome.

8. GENOME SEQUENCING: READING THE CODE OF LIFE

Genome sequencing in simple words is reading the entire book of genome letter by letter (base by base). It is a procedure for determining the precise order of the four nucleotide bases (A,T,C,G) in DNA.

SEQUENCING TECHNOLOGIES:

- **Short-read**: Earlier approach to sequencing, where the genome was chopped into small fragments which could be reassembled like a jig-saw puzzle. In this approach we read 150 bases at a time. Reading short segments of the genome was time-consuming and labour intensive.
- **Next-Generation sequencing:** Use of computers to read multiple fragments at the same time in an automated process which makes whole-genome sequencing relatively faster, accurate, automated and cheap. (dominant method)
- **Nanopore sequencing:** Use of nanotechnology enabled reading longer sequences of the genome (2.3 million base long sequences at once). The technique is also suitable for identifying pathogens in food, disease control and diagnosis of infection.

APPLICATIONS OF GENOME SEQUENCING

- **Disease diagnosis and Personalised drug development**: Identify genetic mutations and variations, evaluate rare disorders and even cancer from a genetic viewpoint. **E.g.,** Mutation in MYBPC3 (Cardiac Myosin binding protein) leads to cardiac arrest at young age (found in 4.5% of Indian population but is rare globally).
- **Agriculture:** Help identify genes that contribute to desirable traits in plants and animals, allowing for selective breeding of crops and livestock.
- **Tool for prenatal screening** to investigate whether a foetus has genetic disorders/anomalies.
- **Identify suspects in criminal investigations** and establish paternity in cases of disputed parentage.
- **Help trace the evolutionary history** of species and understand the mechanisms underlying evolution.

IMPORTANT GENOME SEQUENCING INITIATIVES

NAME	SIGNIFICANCE
Human Genome Project (1990-2003)	Led to the decoding of the entire human genome for the first time, giving the ability to read nature's complete genetic blueprint.
IndiGen (completed 2019)	Initiated by Council of Scientific and Industrial Research (CSIR) from April 2019 to October 2019. **Aim:** To sequence whole genomes of 1029 individuals from diverse ethnic groups across India.
Genome India Project (2020-2023)	To create a comprehensive reference database of genetic variations and subsequent data analysis of 10,000 individuals representing India's diverse population. Entire dataset will be stored at **Indian Biological Data Centre** (India's only data bank for life-sciences) and made available as a digital public good or research.
Genomics in Indian Agriculture	Wheat Genome Sequencing Programme Rice Functional Genomics National Plant Gene Repository programme Next Generation Challenge Programme on Chickpea Genomics **INDIGAU**: Genomics of cow
National Genomic Grid	**Aim:** To collect cancer cells and tissues to facilitate cancer research in India. Uses genome sequencing of cancer cells to study genomic factors influencing cancer in the Indian population.
Earth Bio-genome Project	International effort to sequence and digitise the genomes of every eukaryotic biodiversity on Earth over a period of 10 years. It is an open-source DNA database. **Application:** Planning environmental conservation initiatives.
Human Microbiome Project	To study genes of microbes in the human body including gut, skin, oral cavity and vagina to study their role in human health and diseases. **Note:** The human body contains 10 times as many microbes as human cells.
DeepVariant: AI in genomics	Google's AI system that converts sequencing data from high-throughput sequencing into an accurate picture of the entire genome.
AlphaFold (Nobel Prize in Chemistry 2024)	Google's AI system that is capable of predicting protein modelling. It is important to understand diseases and corresponding drug development.

Data-Driven Research to Eradicate TB Initiative	Genome sequencing initiative launched in 2022. **Aim:** To improve the understanding of drug-resistant TB and capture unique genomic features of the bacterium in India. **Target:** Genome sequencing of 32,500 samples of Mycobacterium tuberculosis. (10,000 samples have been sequenced till March 2025) Genomic dataset developed has potential to revolutionise TB diagnostics and drug resistance prediction: i) Improve diagnostic accuracy ii) Enable faster resistance profiling, iii) Develop advanced tests and reduce the time taken to confirm TB to from weeks to mere hours or days. iv) Tailor treatment regimens to individual patient needs.

9. CELL-FREE DNA

- Cell-free DNA (cfDNA) refers to fragments of DNA found in bodily fluids such as blood, urine, saliva, and cerebrospinal fluid.
- They are released into the bloodstream by cells that underwent programmed cell death or an unplanned cell death due to any injury or disease.
- These degraded (non-functional) fragments of nucleic acids do not possess the ability to replicate or carry out cellular processes.
- However, cfDNA can carry genetic information from the cells they originated from, making them a valuable source for various applications.

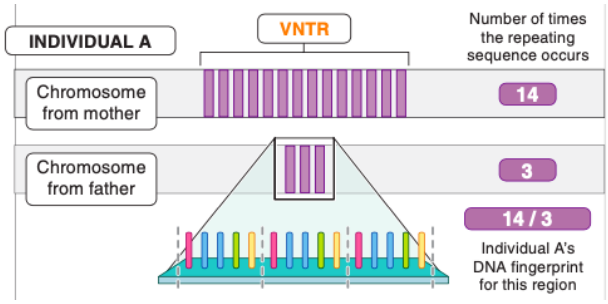
IMPORTANT APPLICATIONS OF CFDNA

- Non-invasive prenatal testing to screen for chromosomal abnormalities in a developing foetus like Down syndrome, Edwards syndrome etc.
- Can provide information about genetic mutations and alterations present in tumour cells and early cancer detection.
- Aid in diagnosis of infectious diseases and in forensics.
- Used as a biomarker for neurological disorders like Alzheimer's disease etc.
- Provide an early indication of graft rejection in patients who underwent organ transplants.

10. VARIATIONS IN THE GENOME: IDENTITY MARKERS

- Humans are 99.9% identical in genome to one another. But given the size of the human genome (3-billion base pairs), even a small (0.1%) proportion of variation is huge.
- The variation in the base pairs is called polymorphism. They are very important to decide the complete make-up of an individual including one's eye colour, susceptibility to disease, one's parentage and even ancestry etc.
- Further variations can be of many types, like single base variation, large sequence variation, variations in the way sequences are structured etc.
- Study of variations has interesting applications. (see table)

DNA Fingerprinting	• There are some sequences in our genome (15 to 100 bases) that keep repeating over and over again. Say, like a word 'green' keeps on repeating in a book. • It so happens that the number of times this repetitive sequence occurs on a chromosome differs. These are called VNTR (variable number of tandem repeats)

	• By counting the number of times the repetitive sequence occurs on chromosomes (both mother's and father's versions), we can **establish the identity of an individual.** This is the DNA fingerprint of that individual. **Applications:** • Forensics • Establishing parentage 
Single Nucleotide Polymorphism and Population genetics (related Nobel Prize 2022)	• Single letter changes in DNA is called Single Nucleotide Polymorphism. • They occur throughout a person's DNA, one in every 300 letters on an average. • It could occur due to mutation during DNA replication (cell division) or may be inherited. **Applications:** • Can act as biological markers to locate a gene associated with disease. Could help in dealing with future pandemics. • Some SNPs which are inherited act as markers to indicate ancestry. (very important in studying population genetics). SNPs keep changing as they flow from one generation to another. Tracing gene flow by observing SNPs is one way of establishing ancestral links. **Nobel Prize for 2022 (Paleogenetics):** • The scientist who won the Nobel prize in 2022 had studied the gene flow from hominins to homo sapiens after the migration out of Africa, 70000 years ago. • He studied the gene flow of Neanderthals and Denisovan (both extinct hominin species) to Homo sapiens. • Some genes inherited from these ancient relatives might influence how our immune system reacts to infections.
1000 Genome Project	It is an effort to study different variations in DNA including SNPs and also larger organisational variations in the genome.

11. ONE DAY ONE GENOME INITIATIVE

- **Initiative of:** Department of Biotechnology and Biotechnology Research and Innovation Council
- It will highlight the unique bacterial species found in India, and emphasise their critical roles in environment, agriculture and human health.

BACTERIA

- Microscopic, single-celled organisms (prokaryotic microorganisms).
- Bacteria inhabit the air, soil, water, acidic hot springs, radioactive waste, and the deep biosphere of Earth's crust.

BENEFICIAL USE CASES OF BACTERIA

- **Soil fertility**: Convert atmospheric nitrogen into plant-usable form; aid in nutrient uptake, especially phosphorus.
- **Food Industry**: Fermentation of various food products (curd, cheese), preservation of food; development of distinct flavours and textures; probiotics.

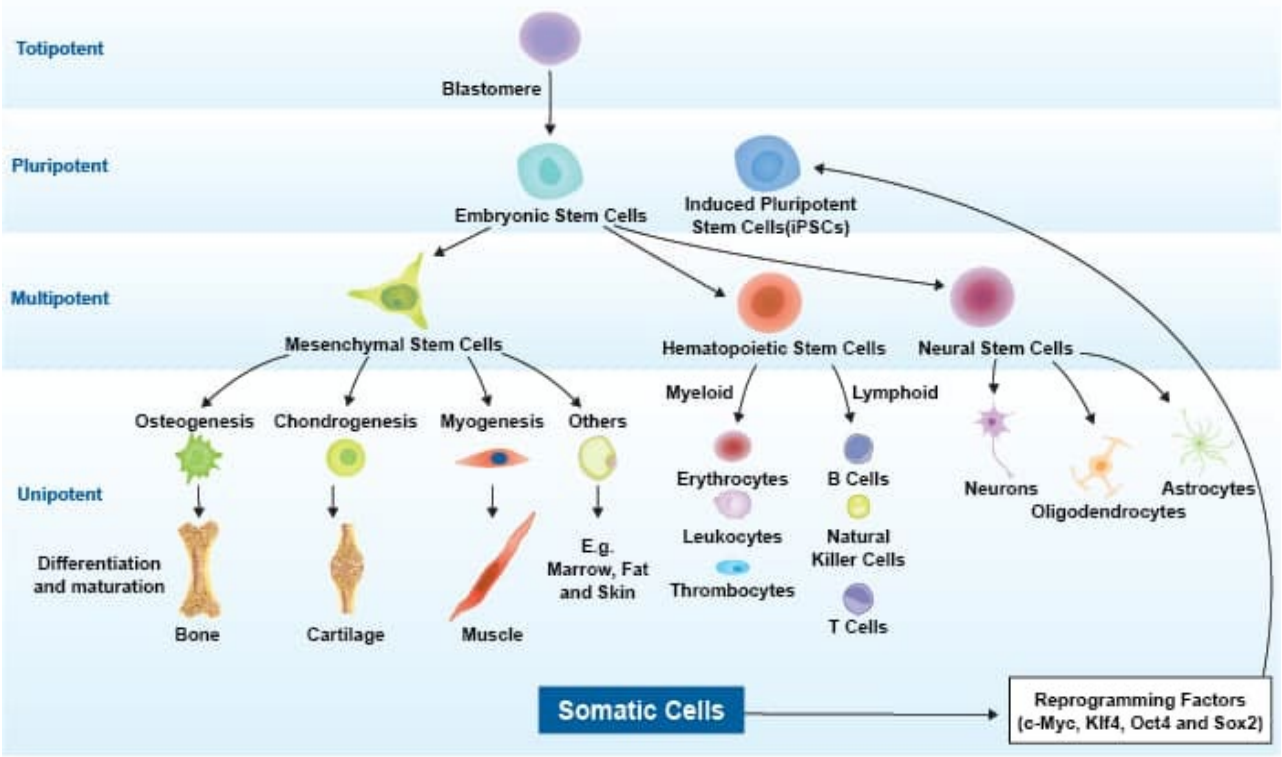
- **Biomedical use**: Used as a host organism in genetic engineering to produce various proteins, enzymes, and other products (like insulin, growth hormones etc). Used as vectors (vehicles) to carry therapeutic genes into human tissues.
- **Biofuel production**: GM bacteria and cyanobacteria can be used for biofuel production.
- **Biogas production**: Break down organic matter in the absence of oxygen, producing biogas (methane and carbon dioxide) as a byproduct.
- **Bioremediation:** Can break down a range of pollutants, including hydrocarbons and toxic chemicals to clean up contaminated environments.

12. STEM CELLS

Stem cells: Undifferentiated cells with the potential to develop into specialised cell types in the body.

Types of Stem cells:

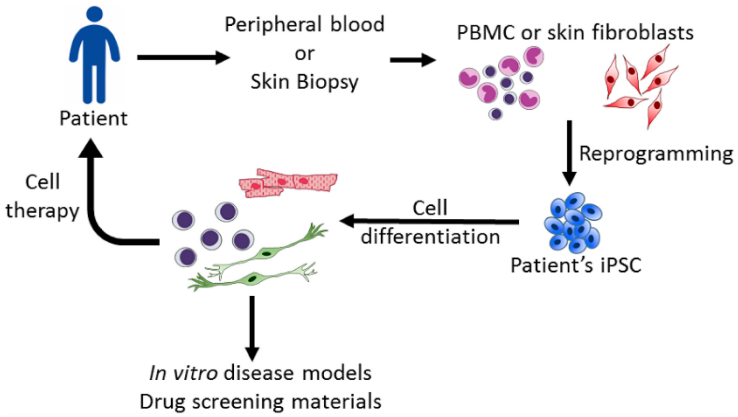
- **Totipotent**: Most versatile with potential to differentiate into all different cell types of an organism and extraembryonic cells (including placenta and umbilical cord). Only found in fertilised egg (zygote).
- **Pluripotent**: Derived from inner cell mass of a blastocyst (an early-stage embryo). Can differentiate into all three primary germ layers (endoderm, mesoderm, and ectoderm), which give rise to all different cell types in the body (except extraembryonic cells). **E.g.,** Embryonic stem cells; and induced pluripotent stem cells (reprogrammed from adult cells).
- **Multipotent**: More specialised than pluripotent stem cells and can differentiate into limited range of cell types within a specific tissue or organ. **E.g.,** Hematopoietic stem cells (give rise to various blood cell types), mesenchymal stem cells (can differentiate into bone, cartilage, fat cell etc.)
- **Unipotent**: Most specialised type of stem cells, can only renew themselves and differentiate into one specific cell type. **E.g.,** Stem cells in skin that can only differentiate into another skin cell.



13. INDUCED-PLURIPOTENT STEM CELLS: IPSCs

- iPSCs are a type of pluripotent stem cell that can be generated directly from a somatic cell.

- If one can reprogram specialised cells (somatic cells) to tweak the transcriptional factors, it can reverse into embryonic stem cells.
- Reprogramming adult cells to become pluripotent is done by introduction of four specific genes, collectively known as Yamanaka factors. It was awarded the Nobel prize in 2012.

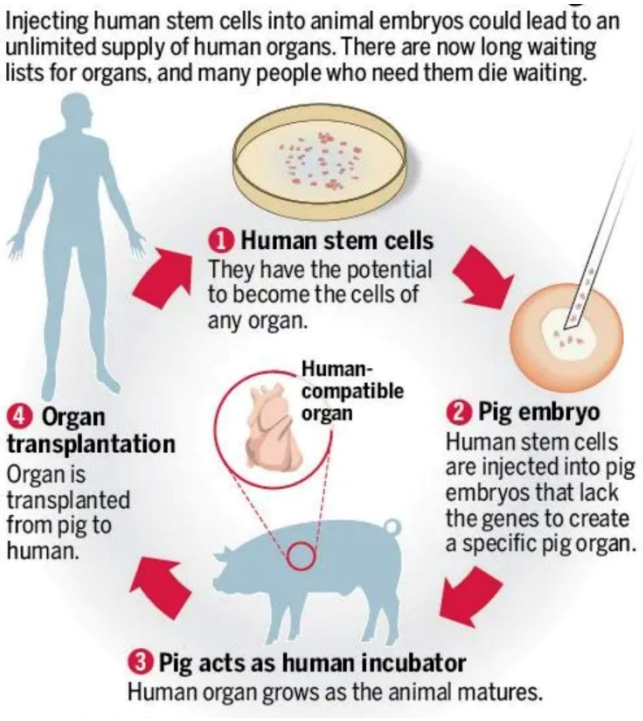


- **Advantage:** Since iPSCs can be derived directly from a patient's own cells, it eliminates the need for embryonic cells and avoids issues related to immune rejection.
- iPS cells hold promise in both regenerative medicine (tissue engineering/organ transplantation) and drug development, and allow researchers to study diseases at cellular level.

14. STEM CELL THERAPY

Stem cell therapy: Use of stem cells to grow healthy adult cells in the lab to replace damaged, defective, or degraded adult cells in the body.

- It is particularly useful when:
  - cells are degraded as in case of neurological disorder with degeneration of neurons (Alzheimer's, Parkinson's etc.) as neurons do not multiply.
  - damage of organs due to accidents, old age etc.
  - body produces defective cells as in case of blood related disorders like sickle cell anaemia, beta thalassemia where regular blood transfusion is needed. In this case there are 3 ways:
    i. Stem cell transplantation of a healthy donor
    ii. Make iPS cells and grow healthy adult cells
    iii. Gene-edited stem cells from one's own body



- As per the **National Guidelines for Stem Cell Research** (NGSCR) 2017, only bone marrow transplant or hematopoietic stem cell transplantation for blood disorders (including blood cancers and thalassemia) is permitted in India.
- **Note:** Stem cells from humans can be introduced in pigs to grow organs in them as they have a faster life cycle.

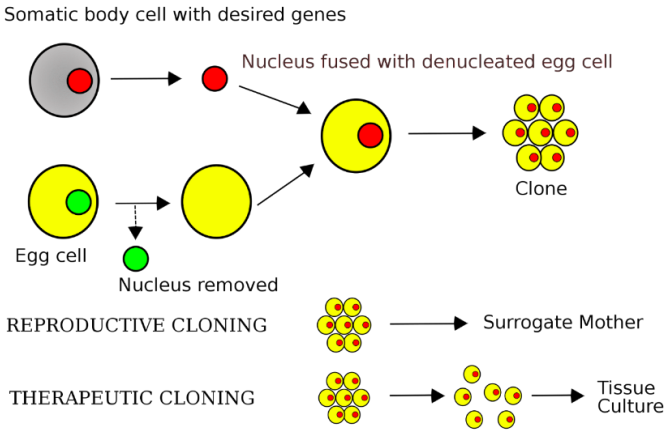
STEM CELL THERAPY IN TYPE 1 DIABETES

- A woman with type 1 diabetes (T1D) in China has regained her insulin production **through stem cell therapy.** She received a transplantation of reprogrammed stem cells that reportedly "reversed" the disease.
- **Pluripotent stem cells** are **pre-programmed into insulin-producing cells** (similar to the ones found in a healthy pancreas) and **transplanted** into the human body.

- After transplantation, these pre-programmed cells will **start producing insulin** and help regulate glucose. This is a potential game-changer in the treatment of T1D.

15. SOMATIC CELL NUCLEAR TRANSFER (SCNT) OR REPRODUCTIVE CLONING

- SCNT technology, also known as "cloning by nuclear transfer," is a technique used in the field of reproductive cloning.
- **Cloning involves 3 steps:**
  1. Take any adult cell (skin, hair, nail etc) and suck out its nucleus.
  2. Take an egg cell from a donor and remove the nucleus from the egg/oocyte. (enucleated)
  3. Inject the nucleus of the adult cell in the enucleated egg. You have an oocyte with somatic cell nucleus.



- Oocyte cytoplasm reprograms the nucleus of the somatic cell and it makes an embryonic stem cell.
- The embryonic stem cell can then be grown into an organism identical to the person from whom the adult cell was taken. (since it is the nuclear DNA that codes for proteins) This is how Dolly the sheep was cloned.
- **Applications:** Help preserve endangered species, replicate highly valuable animals, and advance scientific research in areas such as regenerative medicine and genetic engineering.
- **Disadvantage:** Reproductive cloning is very inefficient as the nucleus from somatic cells resists reprogramming.

16. TISSUE ENGINEERING/ REGENERATIVE MEDICINE/ ORGAN TRANSPLANTATION

- One major challenge of organ transplantation or stem cell transplantation is finding suitable donors, as matching of HLA (human leukocyte antigen) plays a crucial role in the success of these procedures.
- HLA genes encode proteins essential for the immune system to distinguish between self and non-self cells and respond to foreign substances (pathogens or transplanted tissues).
- When the HLA match is absent the immune system rejects organ transplants. HLA matches are best among same-sex siblings. Even between parents and offspring, there is no suitable HLA match.

Note: To find healthy donors for stem cell transplantation, a database called **National Stem Cell Registry** is maintained in India.

17. AUTOLOGOUS, ALLOGENIC AND XENOGENIC TRANSPLANTS

- In the context of organ or stem cell transplantation, depending on its source, donor cells plantation can be classified as:
  1. **Autologous:** Cells from one's own body (including stem cell therapy, iPS cells etc).
  2. **Allogenic:** Cells from another's body but of the same species.
  3. **Xenogenic:** Cells from another species.

- With advancements in gene editing techniques, gene therapies and Xenotransplantation are seen as the potential alternatives to supply organs for transplantation. However, the major challenge is immune reaction from the human recipient.

18. XENOTRANSPLANTATION



Scientists Successfully Transplant Two Kidneys From a Genetically Modified Pig Into Human Recipient

TOPICS: Kidney Organ Transplant Popular Surgery University Of Alabama

By UNIVERSITY OF ALABAMA AT BIRMINGHAM JANUARY 20, 2022

SCIENCE

U.S. man who got first pig heart transplant dies two months after surgery

AP

MARCH 09, 2022 21:00 IST

UPDATED: MARCH 10, 2022 08:45 IST

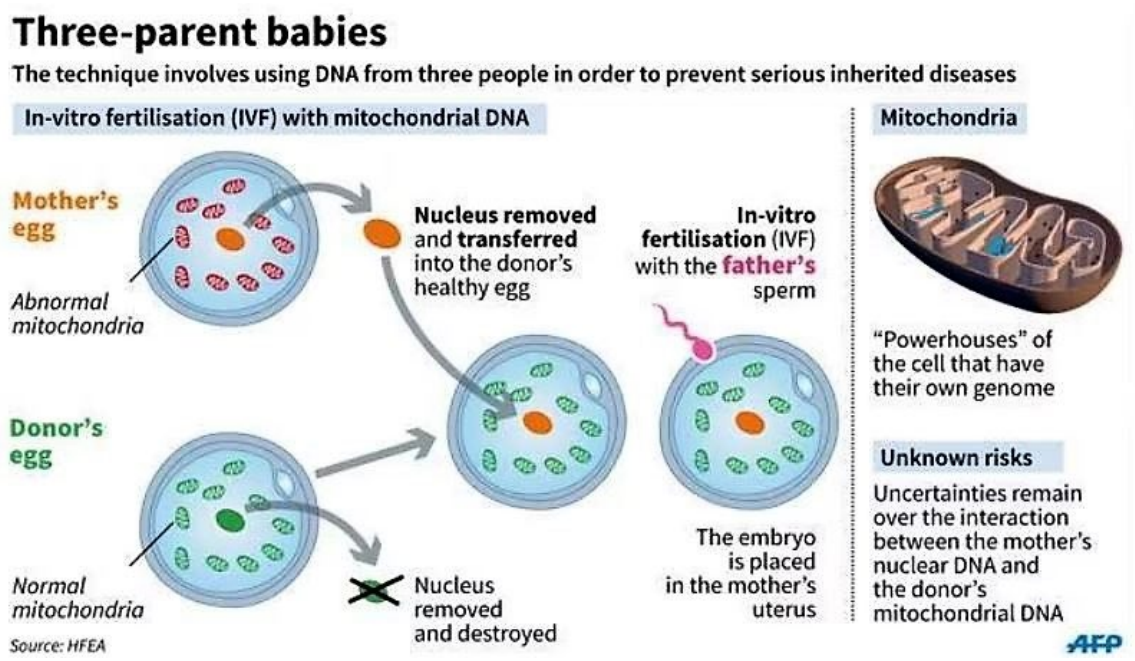
- Organ transplantation from one species to another is called xenotransplantation.

NEED FOR XENOTRANSPLANTATION

- While Autologous is the best method (as there would not be immune rejection), making iPS cells takes a lot of time. Allogenic is good when there are suitable donors.
- Given the shortage of donors with HLA match, scientists have been increasingly working on Xenotransplantation, as animals like pigs with lower lifespan can be bred to grow human organs.
- Gene editing in pigs to reduce immune rejection has made organ transplants from pigs to humans possible, which could offer help to thousands of people who face organ failure, disease, or injury.
- Natural lifespan of a pig is 30 years, they are easily bred and can have organs of similar size to humans.

19. MITOCHONDRIAL DNA TREATMENT

- Mitochondrial DNA is more prone to mutations compared to nuclear DNA. This is because mitochondria are exposed to free radicals generated during energy production, which can damage DNA.
- When mitochondria are impaired they do not produce sufficient energy, which affects how organs function and cause inherited conditions like heart problems, liver failure, brain disorders, blindness and muscular dystrophy. There is no cure for mitochondrial DNA diseases at present.



MITOCHONDRIAL DONATION TREATMENT (THREE PARENT BABY)

- Mitochondrial donation involves replacing unhealthy mitochondria in the mother with healthy mitochondria from a donor through in-vitro fertilisation (IVF). This would avoid passing faulty mitochondria to the child.

- **The Process:**
  - The mother's egg with faulty mitochondria is collected in the laboratory.
  - Eggs from the donor (female) with healthy mitochondria are collected, and its nucleus is removed.
  - The nucleus carrying the mother's DNA is inserted into the donor's enucleated egg.
  - The 'reconstructed egg' is fertilised with the father's sperm via IVF.
  - The final product — 'resulting embryo' — having genetic material (DNA) from parents, and healthy mitochondria from the donor, is implanted into the mother's uterus. The baby thus produced has three genetic parents.

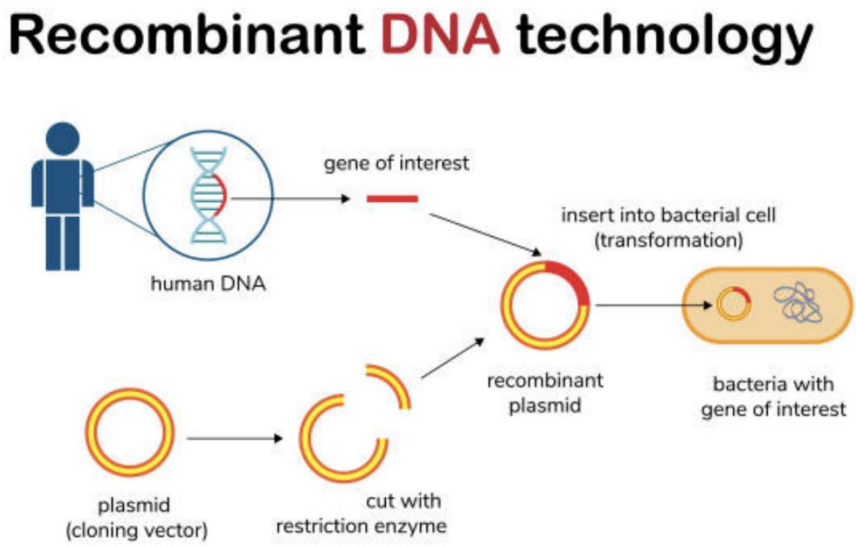
➔ RECOMBINANT DNA TECHNOLOGY: BIRTH OF GENETIC ENGINEERING

20. R-DNA TECHNOLOGY

- Recombinant DNA technology or genetic engineering involves manipulation of the genetic code or DNA of living organisms.
- **Basic principle:** Isolating a specific gene or DNA sequence of interest from one organism and inserting it into the genome of another organism. The inserted gene can be from the same species or from a different species.

Key steps in R-DNA technology:

- **Isolation** of the gene or DNA sequence of interest from the source organism.
- **Fragmenting** this DNA using 'molecular scissors' (Restriction endonuclease Enzymes).
- **Screening** the fragments for a 'desired gene'.
- **Inserting** the fragments with the desired gene into a 'vector' (plasmids, bacteriophage, cosmid) to develop a recombinant DNA. (done using an enzyme called DNA ligase which acts like molecular glue)
- **Introducing** the recombinant vector into the target organism or host cell. The vector integrates into the host's genome and the gene of interest is expressed.
- **Expression:** The target organism produces the protein encoded by the inserted gene.



APPLICATIONS OF R-DNA TECHNOLOGY

- Creation of Genetically modified (GM) crops with desirable traits (resistance to pests, diseases, or herbicides).
- Production of therapeutic proteins such as insulin, interferon and human growth hormone (Human insulin was the 1st therapeutic protein to be genetically cloned in E.coli using R-DNA technology).
- Creation of Mono-clonal antibodies.

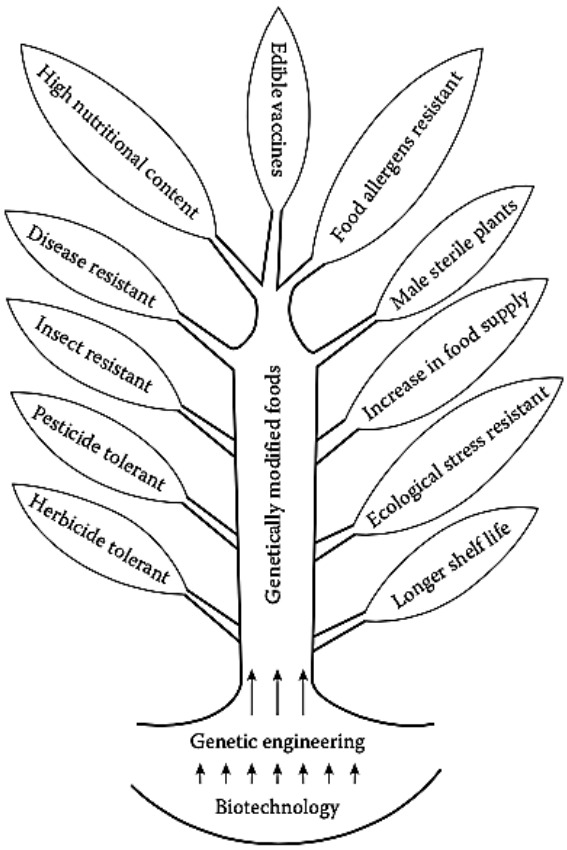
- Production of vaccines. E.g., Hepatitis-B vaccine
- Backbone of diagnostic tests for diseases like HIV and Hepatitis.
- Produce clotting factors for treating Haemophilia.
- Development of synthetic anti-venom, free from animal-derived proteins.
- Create genetically engineered microorganisms for bioremediation and cleaning up environmental pollutants.

CONCERNS OF R-DNA TECHNOLOGY

- Environmental release of GMOs may have unintended consequences. **E.g.,** Herbicide-resistant crops could lead to evolution of superweeds resistant to herbicides.
- Horizontal gene transfer from GMOs to other organisms could lead to spread of unwanted traits, such as antibiotic resistance.
- Allows germline editing or genetic changes to human germline which would be passed onto future generations.
- Accidental release of GMOs/superbugs from laboratories may cause future pandemic and environmental damage.

21. GM CROPS

- Transgenic plants are those that have been genetically modified using recombinant DNA technology.
- Genetic modification is done to confer a particular trait to the plant with one of the following properties:
  - Increased yield of a crop
  - Increased nutritional content of a crop
  - Developing resistance to:
    - Abiotic stresses like temperature, salinity or herbicide-resistant
    - Biotic stresses like insect-resistant crops.
- BT cotton is the only genetically modified crop that is commercially allowed in India from 2002.



REGULATION OF GM CROPS IN INDIA

- Genetically Modified Organisms (GMOs) and the products thereof are regulated under the "Rules for the Manufacture, Use, Import, Export and Storage of Hazardous Microorganisms/Genetically Engineered Organisms or Cells, 1989" under the Environment (Protection) Act, 1986.
- The rules include within its purview all molecular genetics techniques including R-DNA, gene editing techniques, gene drive etc. The rules are enforced by the MOEFCC, DBT and State Governments.
- Genetic Engineering Appraisal Committee (GEAC), a statutory body under MoEFCC, is the final approval authority for allowing field trials and commercial cultivation of GM crops. GEAC is also responsible for certification of GMOs.
- Food has been moved out of Regulation of Genome Engineering Technologies recently. Accordingly, the Food Safety and Standards Authority of India (FSSAI) regulates manufacture, storage, distribution, sale and import of GM food in India.

22. GM CROPS: RECENT DEVELOPMENTS

Bt Crops: Importance of Bacillus thuringiensis	• Soil-dwelling bacteria Bacillus thuringiensis produce spores containing crystals that are poisonous to the insect's digestive system (its gut) but harmless to crop plants and to people. • Thus, all Bt crops involve inserting a gene from soil bacteria to the plant to make it express a crystal that kills the insect by attacking its digestive tract.
Bt cotton	• Resistance to bollworm pest • Only GM crop allowed in India
Bt brinjal	• Bt brinjal is genetically engineered by inserting a gene from the soil bacterium Bacillus thuringiensis for its insecticidal property. • Since 2010 there has been an indefinite moratorium on commercial cultivation of Bt Brinjal in India.
HTBT Cotton	• Short for Herbicide Resistant Bt Cotton • The cotton seed is inserted with a gene from a soil bacterium. • This produces a modified protein glyphosate which makes it herbicide resistant. **Problem with Glyphosate:** • Glyphosate is a herbicide used in cotton fields which is known to be harmful to soil and also classified as 'probable human carcinogen' by WHO. • Besides glyphosate also impacts honeybee gut microbiome thereby affecting pollination.
GM Mustard (DMH-11) **Dhara Mustard Hybrid**	Recently, GEAC approved GM mustard for commercial use, the first genetically modified crop in the country to get approval in two decades. **About GM Mustard** • Transgenic hybrid variety. • Under normal conditions mustard plants do not cross across varieties as they are self-pollinating in nature. • As a result we do not have hybrid varieties of mustard due to which plant-breeders are not able to induce desired traits in mustard plants. • Genetic modification is done to alter the genes of 2 mustard varieties including Varuna and Heera. • Genetic modification is done with the help of soil bacteria barnase and barstar which help create hybrid varieties of mustard. **Benefits:** • Increase in yield up to 20-30% • Seeks to attain self-sufficiency in edible oil seeds production **Issues:** • The transgenic hybrid variety becomes tolerant to a herbicide called glufosinate-ammonium upon genetic modification. • Herbicides are extensively used in field trials including glyphosate, endosulfan. • These may harm honey bees (the biggest pollinators of mustard fields).

Golden Rice	Vitamin A (2 genes from daffodil and 1 from bacteria)
Soybean Sugarbeet Sugarcane	Herbicide tolerant (glyphosate, glufosinate)
Edible vaccines	A new approach for delivering vaccine is transgenic food crops **E.g.,** A rice-based mucosal vaccine for cholera. Plant-based mRNA vaccines using transgenic lettuce are developed.
GM Rubber	World's 1st planted in Assam. To make it more tolerant to colder climates as rubber is native to warm humid Amazon forests.
GM Soymeal	India will import GM soymeal for the 1st time to be used as livestock-feed particularly in poultry.
Protato	Protein packed GM potato that contains 60% more protein than a wild-type potato.
GM bacteria for nitrogen fixation	Genetically modified bacteria are made that copy the action of Rhizobium to induce the ability of nitrogen fixation in the roots of crops other than legumes.

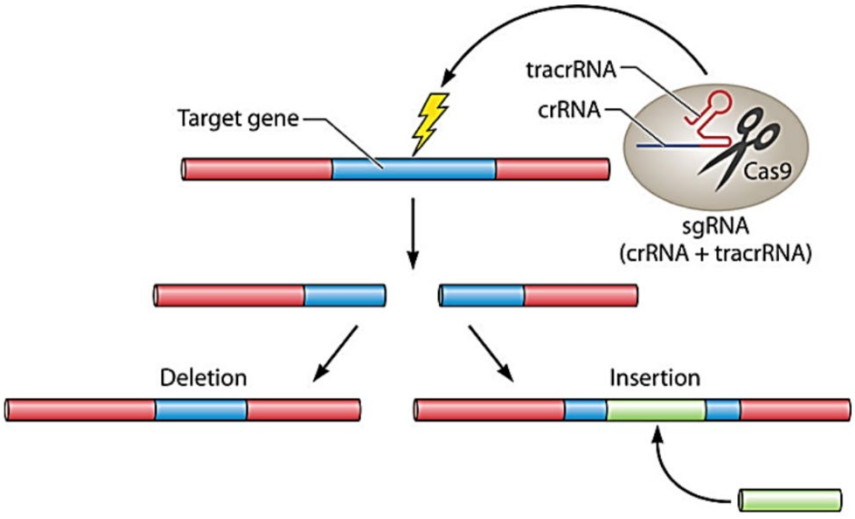
➔ GENE EDITING: CRISPR-CAS9: REVOLUTION IN GENETIC ENGINEERING

23. CRISPR-BASED GENE EDITING

- CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) and Cas9 (CRISPR-associated protein 9) are components of the bacterial immune system that have been repurposed for gene editing.

WORKING OF CRISPR

- In CRISPR-based genome editing, the first step is identification of the targeted DNA sequence.
- Once identified, a **guide RNA** molecule brings **Cas9 protein** to the specific spot in the DNA. Cas9 protein acts like a pair of molecular scissors and cuts the DNA at that specific spot.
- Once the RNA-guided Cas9 enzyme cuts the DNA at a specific site, we could make the following changes to the DNA:
  - Inserting a new sequence (gene insertion)
  - Deleting the sequence (gene removal)



- Modifying the sequence (gene editing)

- If the existing (native) genome is altered, the **end product would be genetically modified** (GM), and if a foreign genome is inserted then the end product would be **transgenic.**

ADVANTAGES OF CRISPR

- Greater precision in modifying DNA, and targets specific locations with much greater accuracy compared to R-DNA technology.
- Used for a wider range of applications than R-DNA. This is because CRISPR is not just limited to introducing new genes, but can also be used to make precise edits within existing genes.

CONCERNS

- Potential to make irreversible changes to the human genome, and Risk of passing erroneous CRISPR-induced changes to successive generations.
- Special traits in offspring/designer-babies can be created through CRISPR posing ethical concerns.
- Potential weapon of mass destruction by creating harmful biological agents or products.
- Eliminating species by gene-editing, or recreating extinct species may disrupt the food web/ecological cycle.

LIMITATIONS OF CRISPR-BASED GENE EDITING

- In CRISPR-based editing, mutations are corrected by cutting the double strand of the DNA. As a result, it can cut only large portions of the genome and not single letter changes.
- Additionally, it can induce unintended off-the-target mutations or incomplete edits in some places.
- CRISPR uses one of two proteins, Cas9 or Cas12, to target specific parts of the DNA. But they are too bulky for plant cells to accommodate. (**#Recent development**: Alternative to Cas9 and Cas12 proteins- ISDra2TnpB)

24. ADVANCEMENTS IN GENE EDITING SYSTEMS

BASE EDITING

- Advanced form of CRISPR, which is suitable for single letter editing.
- While Cas9 cuts double stranded DNA targets, base editing uses an enzyme to rearrange some atoms in the base molecule of the DNA, thereby altering it.
- It involves rewriting the DNA instead of cutting the target sequence and adding new base molecules. This makes it suitable for single-letter mutations.

PRIME EDITING

- Another advanced form of CRISPR, but unlike CRISPR which chops DNA in half, prime editing nicks it (small nick instead of complete cut) and writes a new section of DNA in the specified region similar to base editing.
- Though it is similar to base editing, it includes an additional enzyme, a reverse transcriptase, to copy and paste new DNA sequences into the genome.
- Thus, it is suitable for precise insertions, deletions, and augmentation of the genome. While base-editing is suitable for single-letter modifications, prime editing is suitable for more extensive large edits.

ISDRA2TNPB PROTEIN (MINIATURE PLANT GENOME-EDITING TOOL)

- Researchers have developed a **plant genome editor** consisting of a protein called **ISDra2TnpB,** derived from a bacteria, *Deinococcus radiodurans* (capable of surviving extreme environmental conditions). ISDra2TnpB is less than half the size of Cas9 and Cas12.

- **TnpB proteins** are found in a variety of bacteria and archaea. TnpB protein is made up of around 400 amino acid units. It belongs to a family of transposons or jumping genes. (Transposons are parts of a genome that can move from one location to another).

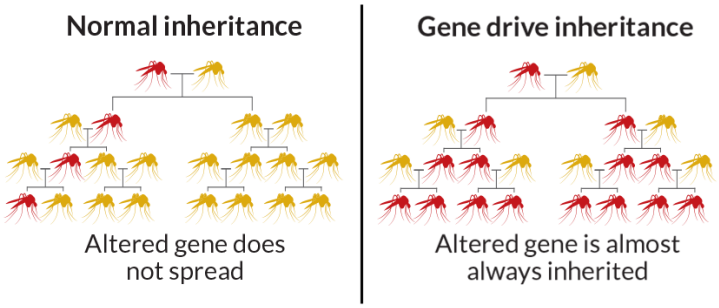
REGULATION OF GENE EDITING IN AGRICULTURE IN INDIA: (SDN1 AND SDN2)

- "Rules for the Manufacture, Use, Import, Export and Storage of Hazardous Microorganisms/Genetically Engineered Organisms or Cells, 1989" notified under the **Environment Protection Act, 1986,** regulate genetically modified organisms.
- There is no explicit mention of the term gene editing. Recently an amendment was introduced to the aforesaid regulation in order to encourage gene editing in agriculture.
- Gene editing techniques using native genome are allowed in India. However, gene editing techniques using foreign genomes are not allowed.
- The gene editing techniques called **SDN 1 and 2** (Site-Directed Nuclease) use native genome and hence, are allowed under the Rules under EPA, 1986.
- However, gene editing techniques using foreign genomes like **SDN 3, 4 and 6** are placed under GMO regulation and therefore not allowed in India.

25. ACHIEVEMENTS OF THE CRISPR SYSTEM

<table>
<tr><th colspan="2">GENETIC ENGINEERING: RECENT ADVANCEMENTS</th></tr>
<tr><td>What can all CRISPR systems do?</td><td>GENE EDITING: DNA AND RNA BOTH
• Both coding and non-coding parts including RNAs (gene expression)
• Initially only DNA was the target, recently RNA is also being modified using the CRISPR system.
• Conventionally could change a section of DNA (large cuts)
• Now with base editing suitable for single-letter mutations.
APPLICATION
Gene therapies can not only repair faulty genes but also regulate healthy genes, either by silencing certain genes or activating silent/non-expressive genes.
MOLECULAR DIAGNOSIS
In addition to editing, CRISPR can be used to detect single target DNA or RNA molecule (CRISPR-Cas13). This makes it a sensitive diagnostic tool to detect mutations.
SHERLOCK
A biological detective that uses CRISPR-Cas13 to detect RNA sequences associated with diseases like Zika virus and Dengue virus.</td></tr>
<tr><td>CRISPR in Gene therapy</td><td>• Scientists have demonstrated administration of CRISPR-based gene therapy both in-vivo (directly in the human body) and ex-vivo (from lab culture to human body).
• CRISPR-based gene therapy is most suitable in mono-genetic diseases in which we can identify a single gene responsible for the disease trait. Examples of such diseases:
○ Blood-related disorders: Sickle-cell anaemia, beta thalassemia, Haemophilia etc.
○ Corneal diseases and skin diseases
○ Degenerative neurological diseases
○ Immunological diseases like HIV, bubble boy syndrome etc.</td></tr>
</table>

	○ Immune-therapy for cancer (CAR T-cell therapy) ● Scientists have demonstrated how CRISPR-CAS9 can be used to eliminate HIV in infected mice. ● Gene editing is tried in mice to correct genes involved in muscular dystrophy. ● Gene editing has been carried out inside the human body for the 1st time to treat Hunter's syndrome. ● The US has approved CAR-T cell therapy which involves modifying immune cells to attack cancer cells in case Leukaemia. In 2021 Department of Biotechnology supported 1st CAR-T cell therapy was conducted in India. ● 1st in-vivo administration of gene therapy to modify photoreceptor cells in the eye to treat blindness was done in 2020.
CRISPR in tissue engineering and organ transplantation	**CRISPR AND CELL THERAPY** CRISPR is increasingly being used in tissue engineering to genetically modify pig cells to make them suitable for growing human organs in them. Stem cells from humans can be introduced in pigs to grow organs in them as they have a faster life cycle. The challenge was that a section of the pig genome was known to cause cancer which acted as a major hurdle in organ transplantation from pigs to humans. These cancer-causing genes are edited using CRISPR to silence them.
CRISPR in Agriculture: GM Crops	● Engineer crops to increase their nutritional value, pest-resistance, drought-tolerance. ● ICAR is using SDN 1 and 2 to produce rice varieties which are drought-resistant, salinity-resistant and high-yielding. **CRISPR IN CLIMATE-SMART AGRICULTURE** CRISPR is used in plants to increase their photosynthetic efficiency by up to 25%. E.g., tobacco plants. **CRISPR AND CCUS TECHNOLOGIES** The CRISPR system is being used to increase the carbon fixing property in algae by increasing its photosynthetic efficiency. **CRISPR AND BIOFUELS** ● CRISPR is used to genetically modify microbes like yeast to improve its efficiency of fermentation and producing ethanol at a faster rate. ● CRISPR is used to genetically modify methanogens (microbes) to improve their performance in biogas production.
GM Food	**LAB-GROWN MEAT** ● Cell-based meat produced by culturing cells in a lab does not have a suitable texture and flavour. CRISPR systems are used to genetically modify cells to produce proteins responsible for texture and flavour. **CLEAN MEAT PROJECT: INDIA** ● The Clean Meat project will be taken up by CCMB and the National Research Centre on Meat of ICAR. Gene edited lab-cultured meat to augment its nutritional content.

	INDUSTRIAL FERMENTATION CRISPR is used to genetically modify microbes like yeast used in wine making, baking and brewing to improve its efficiency of fermentation.
Gene Editing and embryonic cells	**MORATORIUM ON GENE EDITING OF EMBRYONIC CELLS** • In 2018, a Chinese scientist announced the birth of the 1st gene-edited babies. After this WHO urged countries to ban experiments that would lead to more gene-edited babies. • Gene editing of embryonic cells is banned across the world. In 2019, ICMR issued the National Guidelines for Gene Therapy which also **bans gene-editing** of embryonic cells. **Note**: The Chinese scientist modified the CCR5 gene on the embryonic cells of the couple to make them resistant to the HIV virus. (**CCR5** is a gene that codes for receptors in our immune cells which HIV uses like a gateway to get inside the cell). **RESEARCH ON EMBRYONIC CELLS** • In spite of the world-wide moratorium on gene editing of embryonic cells, research is allowed on embryonic cells less than 14-days old to understand the nature of many inherited diseases. • Accordingly following research is going on: ○ Pre-implanted human embryos are being tested for understanding inherited heart disease. ○ Genome of embryonic cells is edited using CRISPR-Cas9 to study the cause of infertility. ○ Research on genes linked to beta thalassaemia, inherited blood disorder, in human embryos using base editing techniques is being carried out.
CRISPR and Gene Drive: Global fight for Malaria	**WHAT IS GENE DRIVE?** • **Under normal law of inheritance,** a specific trait from an organism has a 50/50 chance to be passed. (Offspring inherit one copy of each gene from each parent, resulting in a random chance of expressing either mother's version or father's version) • **Gene Drive** is the use of gene-editing techniques to alter the law of inheritance to pass on a particular genetic trait from one generation to next, faster than normal. • This is achieved by editing a particular gene in a way that it can copy and paste itself into its corresponding location on the other chromosome, instead of the 50/50 inheritance pattern that occurs naturally. 

	• This copying and pasting occurs during the production of reproductive cells (sperm or egg), resulting in a higher probability that the gene will be passed down to the next generation. • CRISPR is being used to edit a gene called 'doublesex' in female mosquitoes which are the main transmitters of malaria. • When the female mosquitoes inherit two copies of the disrupted gene, they develop like males and are unable to bite or lay eggs. • This genetic tweak of double-sex gene follows gene drive inheritance. With this, in 8 generations female mosquitoes were completely eliminated.
GM Mosquitoes	• Genetically modified (GMO) mosquitoes have been released in Djibouti in East Africa to fight malaria. • The lab-produced male mosquitoes carry a "self-limiting" gene. This gene is passed on during mating, and prevents female offspring from surviving to adulthood. Only the male offspring survives. • Male mosquitoes do not bite, and therefore cannot transmit Malaria.

26. CAR-T CELL THERAPY: MAINSTAY OF CANCER TREATMENT

- **Cancer** is a condition of uncontrolled cell growth and division due to absence of apoptosis. Cancer occurs when some disruption of the DNA in a normal cell interferes with the cell's ability to regulate cell division.
- DNA disruption can be caused by chemicals that mutate DNA or by sources of high energy (Sun, X-rays, Nuclear radiation), even infection by viruses and also due to environmental, and lifestyle factors.

TREATMENT OF CANCER

- To treat cancer, the rapidly dividing cells must be removed surgically or killed, or their division slowed down. This is done in two ways: (i) Chemotherapy and (ii) Radiation therapy.

CHEMOTHERAPY

- Drugs that interfere with cell division are administered, slowing down the growth of tumours. But these drugs disrupt the cell division of normal cells too, causing complications. **E.g.**,
  - Extreme fatigue due to reduction in production rate of red blood cells (RBCs).
  - Increases bruising and bleeding, as well as a susceptibility to infection as it reduces production of platelets and white blood cells (WBCs).

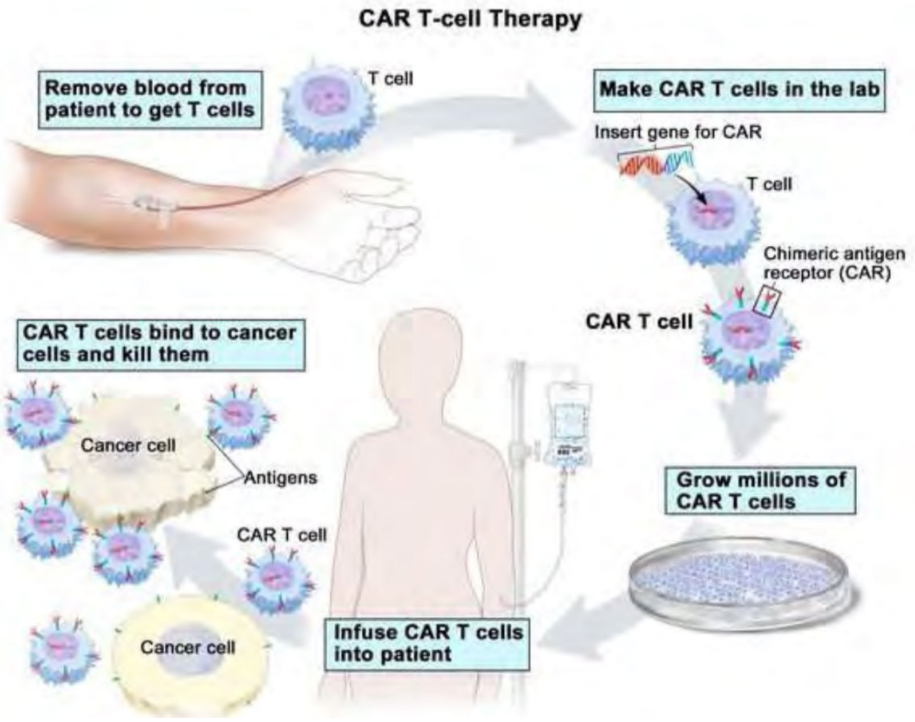
RADIATION

Also works by disrupting cell division, but it is more targeted. It directs high-energy radiation only at the part of the body where a tumour is located. However, the radiation process is not perfect, and the nearby tissues are often harmed.

CAR T-CELL THERAPY

- Chimeric antigen receptor (CAR) T-cell therapy is a cell-based gene therapy which involves genetically modifying our immune cells called T-cells to help them attack cancer cells.
  - T-cells are special cells (a type of WBCs) whose primary function is cytotoxic (attack and kill pathogens and foreign cells).
  - It is the job of B-cells (another type of WBCs) to identify the pathogen and signal other immune cells to do their job. Now, as cancer cells are not foreign cells, B-cells cannot recognise them as foreign bodies and hence, cannot present it to the immune system (T-cells) as enemy cells.

- Alternatively, if we can genetically alter T-cells to recognise the cancer cells, it will kill the cells. This is what CAR T-cell therapy does.



Mechanism:

- In the therapy, T-cells are harvested from the patient's blood.
  - Researchers genetically modify these cells, using the CRISPR system, so that they express specific proteins on their surface known as chimeric antigen receptors (CAR).
  - These cells are then multiplied in the laboratory and inserted back into the patient.
  - This genetic modification allows CAR T-cells to effectively bind to the cancer cells (identify them) and destroy them.
- As CAR T-cells directly activate the patient's immune system against cancer, it makes the treatment more clinically effective than Radiotherapy and Chemotherapy.
- Presently, CAR T-cell therapy has been approved for leukaemia and lymphoma.

Limitations:

- CAR T-cell therapy could induce an immune response like increased cytokine release which can lead to fever, low blood pressure, neurological symptoms and organ damage.
- Further, CAR T-cell therapy is best suited for certain blood cancers and may not be effective for all types of cancer.

NexCAR19: INDIA'S OWN CAR-T CELL THERAPY

- Central Drugs Standard Control Organisation (CDSCO) has granted market authorisation/approval for NexCAR19. With this India has its indigenous CAR-T and gene therapy platform.
- NexCar19 is developed indigenously by ImmunoACT (a company incubated at IIT Bombay).
- The therapy is designed to target cancer cells that carry CD19 protein. This protein acts like a flag on cancer cells, which allows CAR-T cells to recognise and attach themselves to the cancer cells and start the process of elimination.
- The therapy is for people with B-cell lymphomas who did not respond to standard treatments like chemotherapy, leading to relapse or recurrence of the cancer.

IMPORTANT CUTTING-EDGE TECHNOLOGIES FOR CANCER CARE

S. No.	Technology	Details
1.	Image-guided Radiation Therapy	Form of radiation therapy that uses imaging techniques to precisely locate and treat cancerous tissue. Allows for higher doses of radiation to be delivered to tumour while minimising exposure to surrounding healthy tissues.
2.	Proton Therapy	Type of radiation therapy that uses high-energy proton beams to destroy cancerous cells. Particularly useful to treat tumours located in sensitive areas (brain, eyes and spinal cord) where traditional radiation therapy can cause severe side effects.
3.	Precision Medicine	Uses genomic information (genetic profile of their tumour) to personalise treatment for cancer patients.
4.	Radiomics	Uses advanced computer algorithms (AI-powered healthcare) to analyse medical images, such as X-rays, CT scans, and MRIs to improve diagnosis, treatment planning and prognosis.
5.	Immunotherapy	i. CAR-T Cell Therapy ii. Utilising Monoclonal Antibodies iii. Injecting Cancer Vaccines

27. VACCINES AND THEIR TYPES

Type of vaccines	Description	Diseases covered
Live attenuated vaccines	Contains a weakened version of the living microbe so that it can not cause disease.	Measles, mumps, rubella (MMR combined vaccine) Varicella (chickenpox) Influenza (nasal spray) Rotavirus
Inactivated vaccines	Virus is first killed with chemicals, heat, or radiation and then is used to make the vaccine. Inactivated vaccines usually do not require refrigeration, and they can be easily stored and transported in a freeze-dried form.	Hepatitis A, Influenza, polio, rabies
Sub-unit vaccine	A piece of the virus that is important for immunity, like the spike protein of COVID-19, is used to make the vaccine.	Hepatitis B Human papillomavirus vaccines
Toxoid vaccines	Contains a toxin or chemical made by the bacteria or virus. They make a person immune to the harmful effects of the infection, instead of to the infection itself.	Diphtheria and tetanus

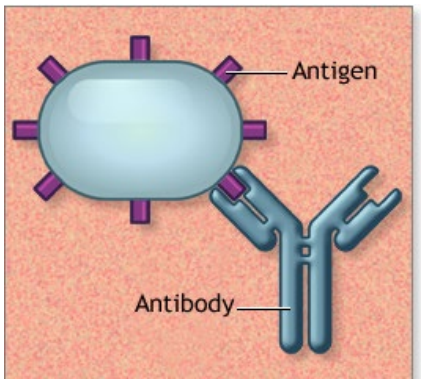
Polysaccharide Vaccines	Unique type of inactivated subunit vaccine composed of long chains of sugar molecules that make up the surface capsule of certain bacteria.	Pneumococcal disease, meningococcal disease, and Salmonella Typhi
Biosynthetic vaccines	Contains man-made substances that are very similar to pieces of the virus or bacteria.	HIV
DNA vaccine	The gene that codes for the COVID-19 spike protein is inserted into a small, circular piece of DNA, called a plasmid. The plasmids are then injected as the vaccine.	ZyCoV-D
m-RNA vaccine	The vaccine contains mRNA molecules instead of an actual bacteria or virus. When mRNA enters the human body through a vaccine, our cells read the mRNA as a set of instructions to build proteins (translation) that match up with parts of the pathogen called antigens. **E.g.,** Cells produced the spike proteins found on the surface of COVID-19 virus. Our immune system recognises the foreign protein as an invader and triggers an immune response by creation of antibodies, and T-cells target that protein.	Covishield Covaxin

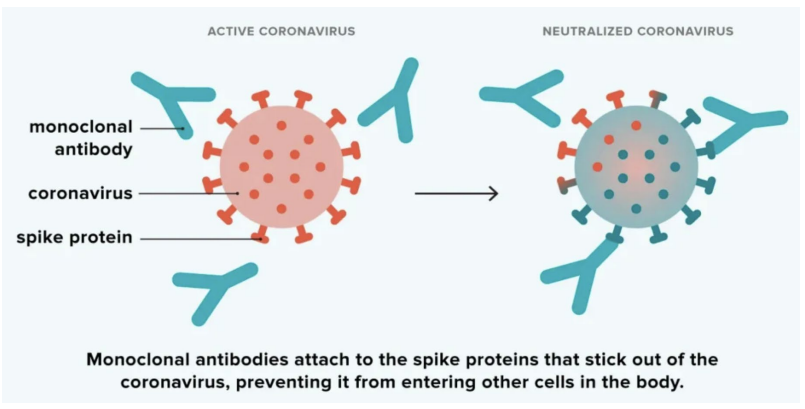
28. MONOCLONAL ANTIBODIES

- Recently, India reached out to Australia to procure monoclonal antibody doses to combat the Nipah virus outbreak in Kerala. The dosage has to be administered at an early stage of infection.

ANTIBODY

- Proteins produced by the immune system that neutralise any foreign substance (bacteria, virus) entering the human body.





- Antibody attaches itself to an antigen (a foreign substance, usually a disease-causing molecule) and helps the immune system eliminate it from the body.

MONOCLONAL ANTIBODY

- Laboratory-made proteins to serve as substitute antibodies and mimic their behaviour to protect against diseases and foreign substances.
- They are specifically engineered and generated to target certain diseases and are meant to attach themselves to the specific disease-causing antigen.

- **E.g.,** Monoclonal antibodies during COVID-19 pandemic were engineered to bind to the spike protein of SARS-CoV-2 virus. The binding prevents the protein from exercising its regular functions, including its ability to infect other cells.

29. ANTIMICROBIAL RESISTANCE (AMR)

- AMR is the resistance acquired by any microorganism (bacteria, viruses, fungi, parasite) against antimicrobial drugs or treatments to which they were previously susceptible.

SOURCES OF DEVELOPMENT OF AMR

- Microbes may develop resistance to antibiotics over time through natural mutation.
- Can be acquired from vertical gene transfer i.e., during the process of bacterial division, the drug resistance gene is transmitted from parent to offspring.
- Can be acquired by a microbe from other microbes through swapping genes via horizontal gene transfer i.e., genetic sequences associated with antimicrobial resistance may pass on from one microbe to another in the community.
- Incomplete doses of medication and self-medication.
- Inappropriate disposal of unused or expired medication can expose microbes in the environment.
- Using antibiotics in farm animals and herbicide use to control weeds may enrich Antimicrobial resistance gene (ARGs) and Mobile genetic element (MGEs) by altering soil microbiomes.
- Microplastics and untreated solid/liquid waste and **biofilms** (communities of microorganisms) can act as a reservoir of AMR microbes.

Note: In India, Antibiotics are included in Schedule H and H1 of the Drugs Rules, 1945. These drugs have specific caution labelling requirements and are sold by retail only under the prescription of a Registered Medical Practitioner.

BREAKING DOWN BIOFILM BARRIER

- Disease-causing bacteria secrete matrix-like layers around themselves called **biofilms**. These biofilms act as thick barriers to antibiotics. They **limit the entry of drugs,** and help the bacteria become resistant to antibiotics.
- To break down this biofilm barrier, scientists at IISc have used an enzyme from the cow's digestive tract. The enzyme is able to break down the 'biofilm barriers' and thus, can aid the entry of drugs.

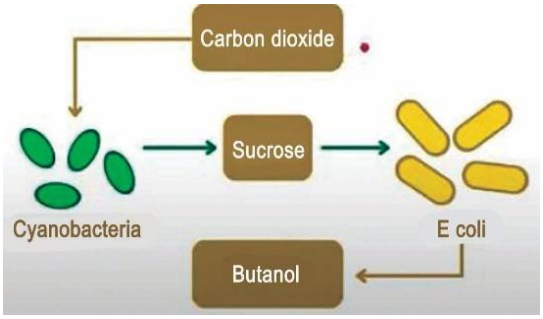
➔ SYNTHETIC BIOLOGY: FUTURE OF BIOENGINEERING

- In February 2022, DBT mooted the need for a National Policy on Synthetic Biology aimed at striking a balance between harnessing the potential benefits and mitigating potential risks of synthetic biology.

30. WHAT IS SYNTHETIC BIOLOGY?

- While biotechnology is the use and modification of biological organisms to produce useful products, synthetic biology is the construction of novel biological systems to produce useful products.
- It is a novel field in biology that works bottom-up as opposed to the top-down approach of biotechnology.
- This has been made possible as a result of the development of bioinformatics which has opened the gate for producing novel products by mimicking nature.

- As we understand the design of plants at the molecular level, we will be able to design similar systems to do the same artificially. This is what we do in an artificial leaf.
- Another way is to engineer two or more biological systems to produce useful products. Under the following example we are merging photosynthesis of cyanobacteria to produce a chain of carbohydrates (starch) and anaerobic respiration of so-produced starch by E.coli to produce alcohol.



1ST ARTIFICIAL CELL: SYNTHIA

- In 2010, a US scientist created of the world's first artificial cell with a synthetic chromosome. (Synthia)
- In 2016, the same scientist created an artificial cell with the smallest known genome ever, that he created completely from scratch (473 genes) called syn 3.0.(Synthia 3.0).

SYNTHETIC BASES: A 8-LETTER DNA: HACHIMOJO

- Japanese scientists have produced an 8-letter DNA instead of the 4-lettered one that nature has produced.
- 4 out of 8 were natural (AGCT) and 4 artificial ones (SBPZ). This could have potential benefits in DNA Data storage.

SYNTHETIC E.COLI: THIS IS THE LATEST SYNTHETIC GENOME WE HAVE CREATED TILL DATE.

POTENTIAL APPLICATIONS:

- With the advancement in our understanding of genes and gene expression we could make synthetic genomes to express very specific traits with specific applications.
- Some potential areas could be:
  - Producing synthetic algae for high-efficiency photosynthesis that can be used in CCUS technology.
  - Synthetic microbes as agents of bioremediation: **E.g.,** Plastic-eating bacteria are suitable only for certain kinds of plastics. If we can mimic the process with a synthetic genome, we can have novel organisms that can be used for bioremediation.
  - Antibiotics are now being made by engineering a completely artificial gene sequence to code for a protein (polypeptides). These are known to kill E.coli, Staphylococcus aureus etc.
  - Can replace Genetically Modified biologics like Insulin.
  - Anti-malarial polypeptide is made using an artificial sequence.

31. BIOE3 POLICY

- BioE3 (Biotechnology for Economy, Environment and Employment) is a policy of the Department of Biotechnology.
- **Aim:** Fostering high-performance biomanufacturing which involves the production of bio-based products across various sectors.
- India's bio economy has skyrocketed from $10 billion in 2014 to over $130 billion in 2024, with projections to reach $300 billion by 2030.
- BioE3 Policy would focus on the following strategic and thematic sectors:
  - Smart proteins and functional foods
  - High value bio-based chemicals, biopolymers, and enzymes
  - Precision biotherapeutics
  - Climate resilient agriculture

- Carbon capture & its utilisation
- Marine and space research

Implementation Strategies:

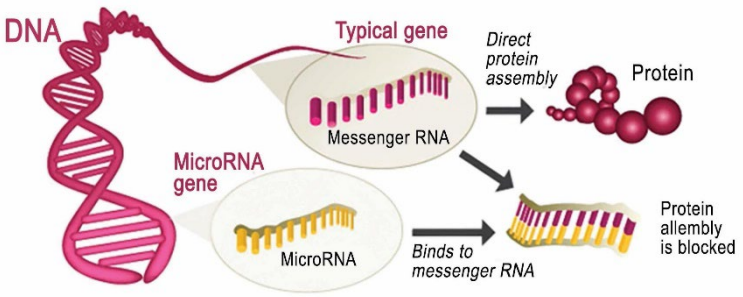
- Support innovation-driven research and development (R&D) and entrepreneurship.
- Establish biomanufacturing hubs, Bio-AI centers, and biofoundries.
- Expand India's skilled biotechnology workforce, especially in tier-II and tier-III cities.
- Align with initiatives like 'Net Zero' carbon economy and 'Lifestyle for Environment' (LiFE) to promote a circular bioeconomy.

32. NOBEL PRIZE IN MEDICINE 2024

The Nobel Prize for Medicine for 2024 has been awarded to **Victor Ambros** and **Gary Ruvkun**. The American scientists were honoured for the **discovery of microRNA** and its **role in post-transcriptional gene regulation**.

WHAT IS MICRORNA?

- MicroRNA (miRNA) is a **small RNA molecule** that consists of 22 nucleotides.
- It is a **non-coding RNA** (does not code for proteins), rather it helps **cells control their protein production.**
- It is found in all multi-celled organisms - plants, animals, mammals, and fungi. As per current knowledge, the human genome codes for **over 1,000 microRNAs.**
- **MicroRNA controls gene expression** by binding with another type of molecule called **messenger RNA (mRNA)** in the cell cytoplasm. Instead of being translated quickly into a protein, the **marked mRNA will be either destroyed and its components recycled**, or it will be preserved and translated later.



- **Functions:**
  - Inhibit protein formation by either degrading the mRNA or by blocking the protein assembly.
  - Post-transcriptional regulation of gene expression.

ROLE AND SIGNIFICANCE OF MICRORNA

- MicroRNA (miRNA) plays a crucial role in **regulating gene expression**, and thus impacts cellular processes.

1. **microRNA and Diseases:**

- **Dysregulation of miRNA expression** can lead to severe consequences for human health. Conditions such as cancer, diabetes, and autoimmune diseases are strongly associated with altered expression of miRNA. **E.g.,**
  - **Cancer:** miRNA dysregulation (from the amplification or deletion of miRNA genes), or abnormal transcriptional control can cause cancer.
  - **Autoimmune Diseases:** Dysfunction of miRNA can disrupt immune responses and trigger autoimmune diseases, such as rheumatoid arthritis and multiple sclerosis.
  - **Inherited diseases:** Dysregulated miRNAs can cause skeletal growth defects, cataract, progressive hearing loss, kidney diseases, neurological disorders, even obesity and diabetes.

2. **Role in Diagnostics and Therapeutics:**

- **Diagnostic Biomarkers:** Certain miRNAs can **serve as biomarkers** for diagnosing human cancers. Specific miRNAs have been linked to prognosis and can help predict the course of the disease.

- **Therapeutic Targets:** Drugs are being designed to **modulate miRNA activity** (to either inhibit harmful miRNAs or enhance beneficial ones) to treat diseases like cancer and autoimmune disorders.

33. NOBEL PRIZE IN CHEMISTRY 2024

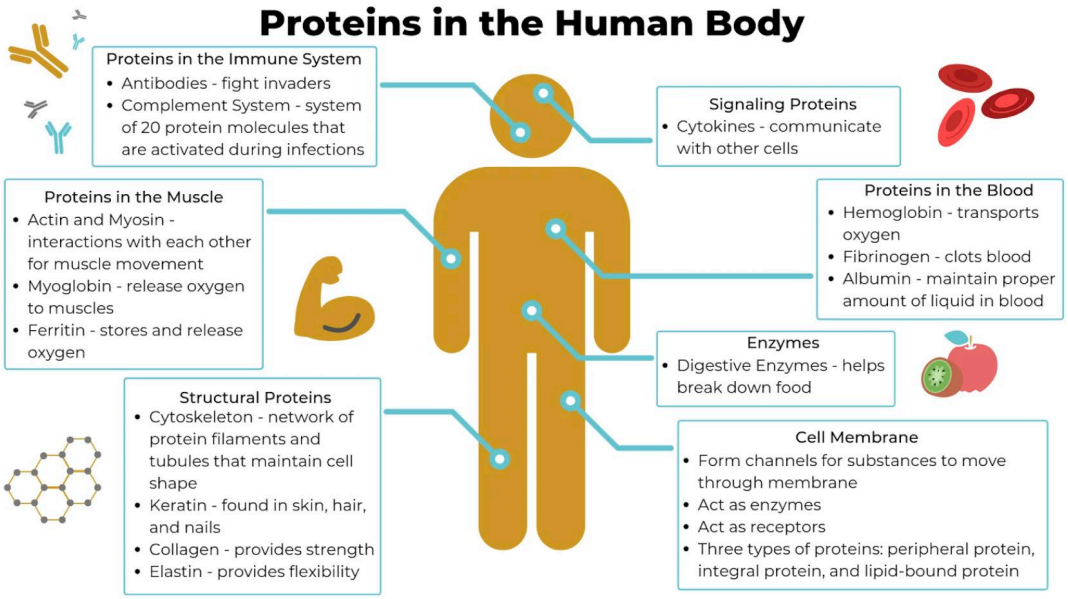
The Nobel Prize in Chemistry for 2024 has been jointly awarded to **David Baker** for his work on computational protein design **(building new proteins)** and to **Demis Hassabis** and **John M. Jumper** for **predicting proteins' complex structures** using an artificial intelligence (AI) model called AlphaFold.

PROTEINS: STRUCTURE & FUNCTION

- Proteins are the building blocks of life. Proteins are **biomolecules or polymers** formed from the **sequences of amino acids** (monomers).
- While there are many types of amino acids in nature, **only 20 of them in different combinations make up all the proteins** in the human body and in most life-forms.

ALPHAFOLD AI MODEL

- **AlphaFold is a deep-learning model** able to predict the 3D structures of almost all proteins after training on the set of known structures.
- **AlphaFold 3 (2024)** is **able to predict the structures of various proteins** as well as how two proteins and/or a protein and another molecule might interact.
- These machine-learning models are **capable of deducing the 3D shapes of most proteins** in a matter of hours — a task that once required several months/years.



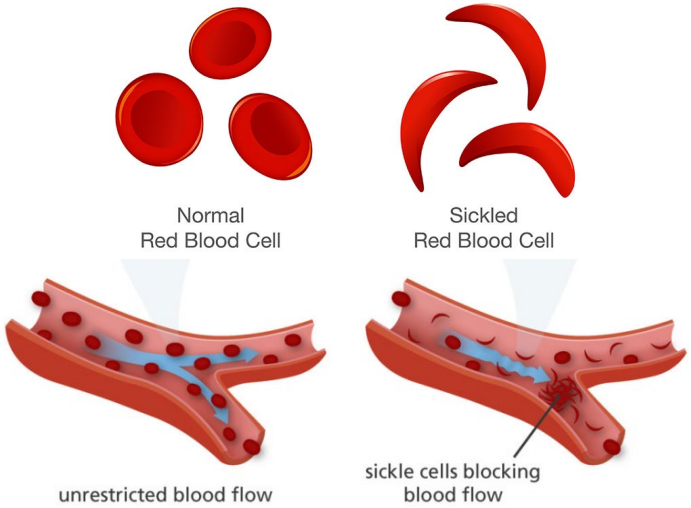
➔ DISEASES

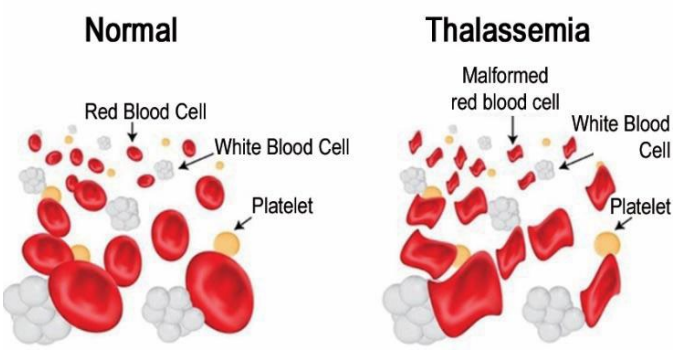
34. DISEASES IN NEWS

Nipah Virus (NiV)	• Zoonotic disease that can be transmitted to humans through direct contact with infected animals, especially bats and pigs. Can also be transmitted through contaminated food or directly from person to person.

	• Fruit Bats (flying foxes) are natural reservoir/primary carriers of Nipah virus. • **Symptoms:** Fever, muscle pain, and respiratory problems (similar to that of influenza). Inflammation of the brain and late onset of Encephalitis can also occur. • **Treatment:** Currently, there are no approved vaccines available.
Human papillomavirus (HPV)	• Small, double-stranded DNA virus capable of infecting women and men. • Can cause cancer in the cervix, vulva, vagina, penis, anus, and throat. • **Transmission:** Any intimate skin-to-skin contact and sexually transmitted via vaginal, anal, or oral sex. • **Cervavac,** India's first Quadrivalent HPV Vaccine, developed by the Serum Institute of India, prevents entry of four of the most common types of HPV 16, 18, 6 and 11. • The vaccine has been approved for girls and boys aged between 9 and 26 years.
Human metapneumovirus (HMPV)	• Respiratory virus that typically causes upper and lower respiratory infections. • Symptoms similar to common cold or flu. • Can affect individuals across all age groups. • Transmission through droplets from coughing, sneezing, close contact with contaminated surfaces.
Monkey Pox	• Zoonotic disease caused by the virus belonging to the Orthopoxvirus genus. • Animal-to-human transmission can occur from direct contact with the blood, bodily fluids, or cutaneous or mucosal lesions of infected animals. • **Symptoms:** Characterised by rash or skin lesions that are usually concentrated on the face, palms of the hands, and soles of the feet.
Chandipura virus	• Single-stranded RNA virus. Largely endemic to central India. • **Vectors:** Sandflies and some mosquito species. The virus is not contagious. • **Symptoms:** Flu-like symptoms which can progress to seizures and encephalitis. • May then lead to mortality. Susceptibility largely limited to children below 15 years.
Zika Virus	• Single-stranded RNA virus. Primarily transmitted by infected Aedes aegypti mosquitoes (also vectors for dengue and chikungunya). • Can be transmitted from mother to foetus during pregnancy; through sexual contact, blood transfusion; organ transplantation. • Zika infection during pregnancy causes microcephaly (underdeveloped brain) and other congenital malformations.
Duchenne's Muscular Dystrophy	• Rare genetic disorder characterised by progressive muscle degeneration and weakness due to the alterations of a protein called dystrophin (dystrophin helps in wear and tear and regeneration of muscles). • As a result, the body cannot produce dystrophin, which weakens the muscles, and patients become wheelchair-bound in early teens and die prematurely. • **Cause:** Most cases of DMD are inherited as an X-linked recessive trait (passed on through the mother, who is a carrier). The disease primarily affects boys, but in rare cases, it can affect girls.

Guillain-Barré syndrome	• Serious autoimmune disorder that affects the peripheral nervous system. Affects nerves responsible for muscle movement, pain, temperature, and touch sensations. • Initially presents weakness, tingling, and numbness in the limbs, which can progress to paralysis lasting 6-12 months or longer. • Caused by: Bacterial and viral infections, spread through contaminated food and water. Can be transmitted through airborne particles. • Self-limiting and most patients recover fully with treatment. Impacts people with weak immunity.
Kala Azar or Visceral leishmaniasis	• Slow progressing Neglected Tropical Disease and zoonotic infection caused by a protozoan parasite of genus Leishmania. • Transmitted by the sand fly found in moist mud and sand close to livestock. • The parasite primarily infects the reticuloendothelial system and may be found in abundance in bone marrow, spleen and liver. • It is the second-largest parasitic killer in the world after Malaria.
Osteoporosis	• Bones become porous (less dense), brittle and fracture can occur with a minor fall. • Prevalence is higher in elderly females than males as Estrogen hormone (protects bones) in women decreases sharply after menopause. • **Injectable hydrogels** can provide localised treatment of osteoporosis.
Idiopathic Pulmonary Fibrosis	• Chronic, progressive lung disease characterised by scarring (fibrosis) of the lung tissue. This leads to irreversible loss of lung function. • IPF specifically targets interstitium (tissue surrounding air sacs). As a result, the alveoli walls get thicker, which makes it difficult for oxygen to move into the bloodstream. • This leads to persistent breathlessness, fatigue, and reduced quality of life.
Kyasanur Forest Disease/ Monkey fever	• Tick-borne viral haemorrhagic fever caused by a virus. • Endemic to the South-western part of India (first noticed in Kyasanur Forest area in Karnataka). • **Transmission:** Primates (Monkeys) come in contact with infective ticks. Human beings who visit the endemic forest area can contract the disease.
Lumpy Skin Disease	• Highly contagious disease caused by lumpy skin disease virus that primarily affects cattle. • **Transmission:** Biting insects such as flies and mosquitoes and possibly ticks. Also spread by direct contact to skin lesions, saliva, nasal discharge, milk, or semen of infected animals. Not a zoonotic virus i.e., cannot spread to humans via milk consumption. • **Symptoms:** Formation of nodules/lumps on skin, head, neck, back, and genitalia, high fever, sharp drop in milk yield, loss of appetite, thinness or weakness in animals.
Foot and Mouth Disease	• Highly contagious viral vesicular disease of cloven-hoofed animals such as cattle, buffaloes, sheep, goats and pigs etc. • **Symptoms**: Abscesses and ulcers to the mouth and foot that prevent the animal from eating and walking, reduced milk yield, infertility, reduced working capacity. • Human infection is rare, and is not transmitted to humans via meat consumption.

	• The Department of Animal Husbandry and Dairying aims to eradicate FMD by 2030 through vaccination of livestock.
Sickle Cell Anaemia (SCD)	• Inherited/genetic blood disorder that affects haemoglobin, the protein in RBCs that carries oxygen to all parts of the body. • Healthy RBCs are soft and round. In SCD, the haemoglobin is abnormal, which causes the RBCs to become hard and sticky and look like a "sickle." • These rigid, sticky cells die early and often get stuck in blood vessels, clogging the flow of blood. As a result, different parts of the body do not get the oxygen they need. • This can cause pain and other serious health problems such as infection, acute chest syndrome and stroke. 
Haemophilia	• Rare inherited bleeding disorder that impairs the body's ability to make blood clots. • People with haemophilia have lower levels of clotting factors (Factor VIII), which can lead to excessive bleeding, even after a minor injury. • X-chromosome linked recessive disorder. More common in males than in females (Females are carriers of haemophilia genes). • Treatment: Replacing deficient clotting factors through clotting factor infusions. • The Department of Biotechnology has successfully executed a **first-in-human gene therapy** using **lentiviral vectors** (viral vectors used to transfer genetic material into cells for gene therapy) to treat Haemophilia-A. • Lentiviral vector is used to introduce a normal copy of Factor VIII gene into autologous haematopoietic stem cells (HSCs). These modified HSCs generate blood cells capable of producing functional Factor VIII, thus terminating the need for repeated infusions.
Thalassemia	• Inherited blood disorder characterised by abnormal haemoglobin production, which in turn, leads to Anaemia. • Results from a mutation in one or more of the genes that make haemoglobin. It is passed down from one or both parents through their genes. • **Treatment:** Bone marrow transplantation (very costly). Mainstay treatment is repeated blood transfusions every two to three weeks to survive.

	
Gaucher's disease	• Rare genetic disorder caused by deficiency of an enzyme (glucocerebrosidase) that breaks down fatty substances called lipids. • Due to enzyme deficiency, lipids accumulate in certain organs and tissues like liver, spleen, bone marrow, and sometimes the brain. • Caused by mutation in the GBA gene. Autosomal recessive disorder, i.e., defective gene copy must be inherited from both the parents. • Treatment include- enzyme replacement therapy.
Alzheimer	• Progressive neurological disorder that causes the brain to shrink (atrophy) and brain cells to die, which primarily affects memory (dementia) and cognitive function. • Caused by abnormal build-up of proteins in and around brain cells, which either form plaques (Amyloid protein) or tangles (Tau protein) around brain cells.
Down's syndrome	• Genetic condition caused by the presence of an extra copy of chromosome 21. • People with Down's syndrome have 47 chromosomes instead of the usual 46. • Lifelong condition that affects physical and intellectual development with no cure.
Human immuno-deficiency virus (HIV)	• RNA virus that damages the immune system. HIV primarily targets CD4 cells (type WBCs), essential for the proper functioning of the immune system. • If left untreated, HIV can lead to AIDS (acquired immunodeficiency syndrome). • **Transmission:** Contact with certain bodily fluids (infected blood, semen, or vaginal fluids) of a person with HIV. Can transmit from an HIV-positive mother to child during childbirth, breastfeeding, or pregnancy. • Not transmitted by insect vectors (HIV can only survive in human blood).
Rabies	• Vaccine-preventable zoonotic disease caused by a RNA virus which infects the central nervous system of mammals, including humans. • **Transmission:** Bite of an infected animal (dogs, cats, bats, monkeys, foxes), contact with saliva or other bodily fluids of infected animal. • **Symptoms**: Fever and headache, hallucinations, paralysis and hydrophobia. • **Treatment**: Once symptoms appear, no cure for rabies (100% fatal).
Polio	• Highly infectious disease caused by a single-stranded RNA virus. • **Transmission:** Faecal-oral route (contaminated water sources or poor sanitation. It can also spread through direct contact with infected individuals.

	• The virus multiplies in the intestine and invades the nervous system and can cause total paralysis. Mainly affects children under 5 years of age. • **Prevention:** Polio vaccines - Oral polio vaccine (OPV) and inactivated polio vaccine. • Vaccine-derived polio (very rare) occurs when the weakened (attenuated) strain of poliovirus used in the OPV mutates and regains the ability to cause paralysis.
Influenza	• Influenza (flu) is a highly-contagious respiratory illness caused by influenza viruses. • Influenza viruses are of four different types: A, B, C and D. • **H3N2** (HongKong Flu) and **H1N1** (Swine Flu) viruses are subtypes of Influenza A virus. Influenza A is associated with severe respiratory illness and deaths in humans. • **H5N1** (Avian influenza/ Bird Flu) is also a subtype of the Influenza A virus that primarily infects birds but can also infect humans and other mammals.
Tuberculosis (TB)	• Airborne communicable disease caused by bacteria Mycobacterium tuberculosis. • Can cause- Pulmonary TB (infection in lungs, 80% cases) and extra-pulmonary TB which can infect brain, uterus, stomach, mouth, kidneys and bones. • **High-risk groups:** People with weak immunity like those infected with HIV, Under-nutrition, Diabetes, Smoking and Alcohol consumption. • India has more than 25% of the total TB patients in the world. • **Treatment:** DOTS strategy (Directly Observed Treatment Short Course) is a WHO-recommended cost-effective strategy.
Malaria	• Potentially life-threatening disease caused by plasmodium parasites transmitted through the bite of infected female Anopheles mosquitoes. • Mosquirix is the world's first vaccine against the parasitic disease. • Currently, Malaria is a notifiable disease in 33 (States and UTs) in India. (Notifiable disease is required by law to be reported to government authorities, any failure to notify would be a criminal offence). • India has the vision to be malaria-free by 2027 and to eliminate it by 2030.
Leprosy/ Hansen's disease	• Chronic infectious disease caused by a bacteria Mycobacterium leprae. • Primarily affects skin (causes severe, disfiguring skin sores) and peripheral nerves (damage in arms, legs). • Loss of sensation in the affected area. Left untreated, it may cause progressive and permanent disabilities. • Contagious. Curable with multi-drug therapy (MDT), recommended by WHO. MDT is a combination of three drugs: dapsone, rifampicin, and clofazimine. • National Strategic Plan for Leprosy (2023-27) to achieve zero transmission of leprosy by 2027.
Disease X	• Not a specific disease but name given to a potential novel infectious agent currently unknown but could pose a serious microbial threat to humans in the future.

➔ MISCELLANEOUS

35. NITROGEN HYPOXIA

- Hypoxia is a form of asphyxiation, i.e., a state in which oxygen is not available in sufficient amounts.
- **Nitrogen Hypoxia** is a form of capital punishment where the culprit is forced to inhale pure nitrogen via a respirator mask. This will increase the concentration of nitrogen in his body to lethal levels.
- The nitrogen will essentially squeeze any oxygen out of his respiratory system, depriving tissues and organs of oxygen, causing low blood flow and, ultimately, death.

NITROGEN IMBALANCE

- The air around us (the atmosphere) is made up of about 78% nitrogen and only 21% oxygen. The rest consists of water vapour, argon, neon, helium, hydrogen and xenon.
- At a concentration of 78% in the atmosphere, nitrogen is safe to breathe. But it grows dangerous and potentially fatal once levels of nitrogen reach 80% or more.

36. CLOSTEBOL

- An **androgenic steroid** which is a synthetic/modified form of testosterone.
- Has anabolic effects, i.e., promotes muscle growth, protein synthesis, and red blood cell production. Frequently used in sports to increase physical performance of athletes.
- Because of medical and ethical reasons, the use of clostebol is banned by the International Olympic Committee.

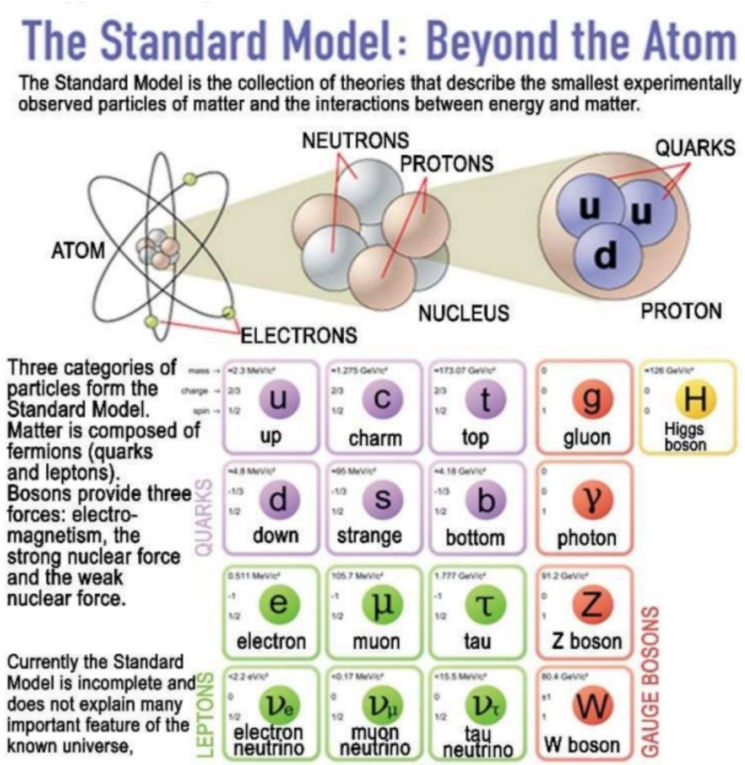
Click here for INDEX

2 CHAPTER SPACE TECHNOLOGY

➔ BASIC PHYSICS AND ASTRONOMY

1. STANDARD MODEL OF PHYSICS

- The fundamental questions that have intrigued humans for long are "What is the world made of?" and "What holds it together?".
- The Standard Model is an attempt to explain these questions. The Model describes the **behaviour of six types of quarks, six types of leptons, three fundamental forces** (Strong force, Electromagnetic force & Weak force) and their **four associated particles** (Bosons), plus the Higgs boson.
- As per the Model:
  - All the known matter is made up of **fundamental particles called quarks and leptons.**
  - These particles interact with each other in accordance with rules known as the **'fundamental forces'**.
- **Limitations:** Currently, the Model is incomplete and does not explain: a) Gravitational force. (Graviton, a force-carrying particle for gravity, has not been discovered yet). b) existence of dark matter and dark energy.



2. FUNDAMENTAL PARTICLES AND FORCE CARRIERS

1. Quarks:

- Quarks are elementary particles that are considered fundamental constituents of matter.
- They combine to form composite particles known as **hadrons** (such as protons and neutrons), which are the building blocks of atomic nuclei.
- There are 6 principal quarks, and they interact through the exchange of other elementary particles called gluons.
- A quark exhibits confinement i.e., quarks are not observed independently but always in combination with other quarks. This makes determining their properties (mass, spin, and parity) impossible to measure directly.

2. Leptons:

- There are 6 known leptons (fundamental particles) - Electron, Muon, Tau, three types of Neutrinos.
- **Note: Fermions** (Quarks + Leptons) are **fundamental particles** which have half-integer spin. Fermions are the fundamental building blocks of matter. E.g., Electron, Proton.

SEMI-DIRAC FERMION

- **Semi-Dirac Fermion** is a quasi-particle (a collection of particles) recently observed in a crystal of zirconium silicon sulphide (ZrSiS).
- Exhibit unique behaviour where they are massless when moving in one direction, and possess mass when moving in a perpendicular direction.
- **Potential applications:** quantum computing, high-end electronics, batteries and sensors.

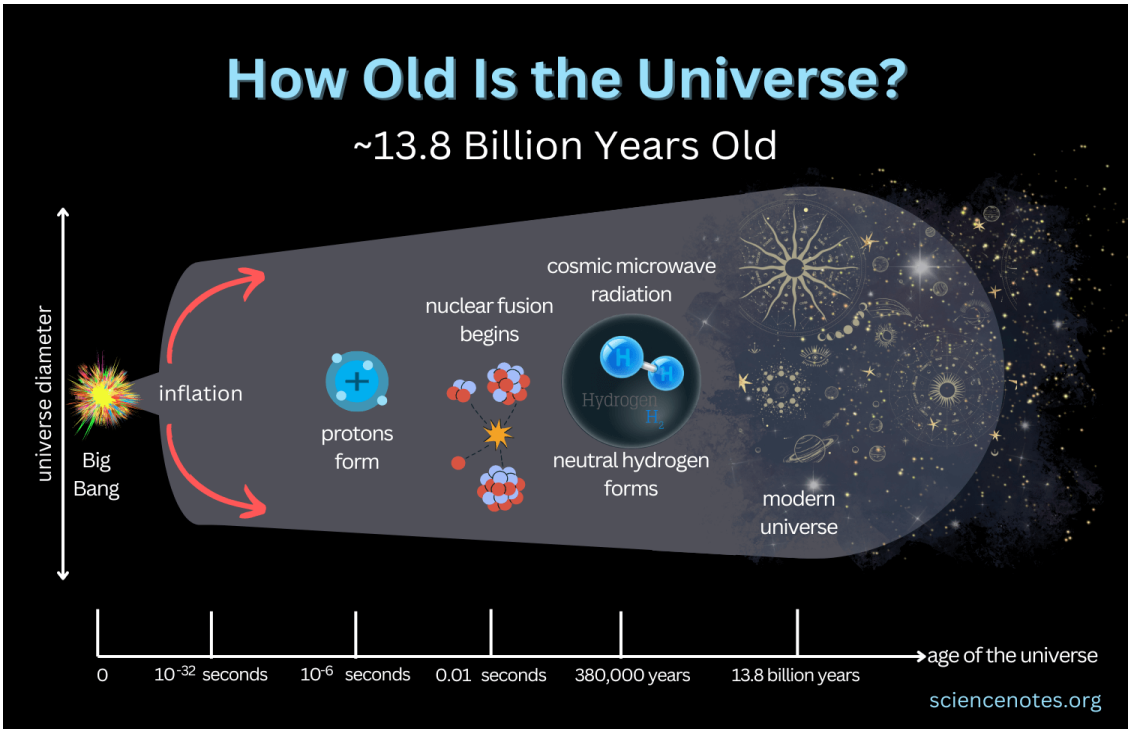
3. Bosons:

- Boson is a collective name given to particles that carry fundamental forces. Each fundamental force has its own corresponding **force carrier** (boson).
  - Strong force is carried by gluons.
  - Electromagnetic force is carried by photons.
  - W and Z bosons are responsible for the Weak force.
  - Graviton is a suggested force carrier for Gravitational force (Graviton is still not discovered).

➔ EVOLUTION OF UNIVERSE

3. BIG BANG THEORY

- Big Bang theory is the prevailing cosmological model for the universe's origin and evolution.
- **Origin from a Singularity**: Approximately 13.8 billion years ago, all the matter and energy in the universe were created from an incredibly hot, dense state called a Singularity in an enormous explosion- "Big Bang".



- **Cosmic Inflation (Rapid expansion):** In a fraction of a second after the Big Bang, the universe underwent a period of rapid and exponential expansion from a single point.
- **Cooling and Particle formation:** As the universe expanded and cooled, fundamental forces and particles formed. The first fraction of a second witnessed the creation of elementary particles like quarks and gluons.
- **Big Bang nucleosynthesis (light element formation):** After a few minutes, protons and neutrons formed, allowing for the creation of the light elements - Hydrogen, Helium - through a process called Big Bang nucleosynthesis.
- **Formation of stars and galaxies**: Over billions of years, gravity caused the light elements to clump together, forming stars and galaxies. Over time, heavier elements were forged within stars and dispersed through stellar explosions, enriching the universe for future solar system formation.

Hubble's Law and Expanding Universe:

- Discovered by Edwin Hubble in 1929, the law states that galaxies farther away from us are receding at a faster velocity compared to closer galaxies. This recession velocity is proportional to the distance of the galaxy.
- The farther a galaxy is, the faster it is moving away from us. Thus, the universe is constantly expanding.

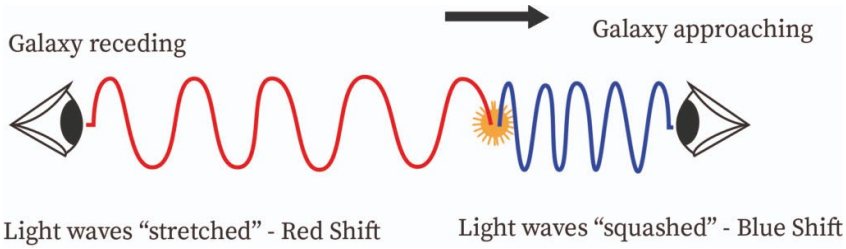
SUPPORTING EVIDENCE FOR BIG BANG THEORY

1. Cosmic Microwave Background Radiation:

- CMB is a form of electromagnetic radiation that permeates the entire universe and is almost uniform in all directions.
- It is considered to be the oldest light in the universe, and is often referred to as the faint afterglow heat radiation from a very hot, early universe.
- The expansion of the universe stretched CMB's wavelength, shifting it from high-energy photons to the microwave range (long wavelength) as we observe today. This expansion is indeed evidence for the Big Bang.

2. Redshift of Galaxies:

- Redshift is a phenomenon observed in the light emitted by distant objects, such as galaxies or quasars.
- When an object moves away from an observer, the light it emits gets stretched or "shifted" toward longer wavelengths or towards the red end of the electromagnetic spectrum. This is interpreted as a consequence of the universe's ongoing expansion.



3. Abundance of Light Elements:

- The observed abundance of light elements like hydrogen and helium closely matches the predictions of Big Bang nucleosynthesis.

4. GENERAL THEORY OF RELATIVITY (GTR)

- GTR is a fundamental theory of gravitation published by Albert Einstein in 1915.
- **According to the theory:**
  - Massive objects cause a curvature in space-time structure, which causes other objects to move along a curved path.

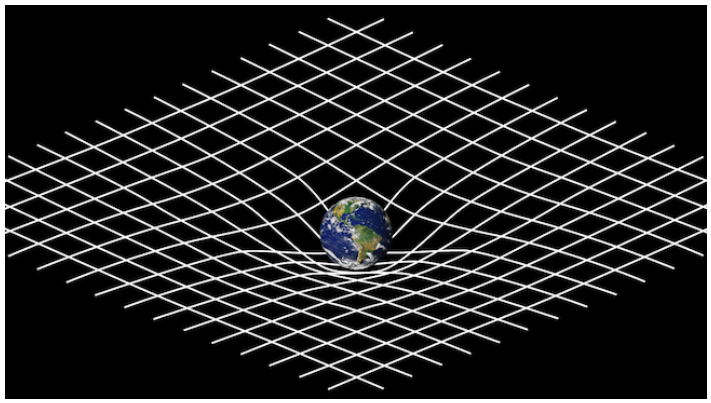
- Speed of light is constant in all inertial reference frames. (Speed of light remains the same for all observers, regardless of their position or motion within a gravitational field)

- GTR predicted the existence of gravitational waves, black holes, time dilation, gravitational lensing (light is deflected by objects with very strong gravity), and expansion of the Universe.

KEY PREDICTIONS OF GTR

1. Gravitational waves:

- Ripples in space-time caused by accelerating massive objects (similar to ripples in a water pond). The curvature of spacetime is directly proportional to the mass of the object causing the curvature, i.e., the greater the mass, the greater the curvature of spacetime it causes.
- The waves travel at the speed of light and squeeze and stretch anything in their path.
- Gravitational radiation is exceedingly difficult to detect because gravity is the weakest of the four fundamental forces. The waves were detected in 2015 by **LIGO (**Laser Interferometer Gravitational Wave Observatory).



2. Black holes:

- Regions of spacetime where gravity is so strong that not even light can escape. Black holes have been observationally confirmed. (Details later in the segment)

3. Time dilation:

- Time dilation refers to the idea that time is relative and runs/passes at different rates for different observers, depending on their relative motion or their positions in a gravitational field.
- Closer an object's velocity is to the speed of light, the more pronounced the time dilation effect becomes.
- GTR predicts that time runs slower in stronger gravitational fields. This has been experimentally verified using high-precision atomic clocks, observations of Quasars etc.

5. IMPORTANT PARTICLES YOU OFTEN HEAR

1. Higgs Boson:

- In 1964, Peter Higgs gave a theory related to how subatomic particles (building blocks of matter) get their mass. He suggested that just after the Big Bang (in the early stages) the fundamental particles did not have any mass. As the universe cooled, **an invisible force field** got formed, which was termed as the Higgs Field.
- The associated particles with the Higgs field are the **Higgs Bosons**. Any particle that interacted with the Higgs field got mass, and those particles that were left out of the Higgs field remained massless.
- As these Higgs bosons grant mass to particles like electrons, quarks, etc. they were termed as the **'God particle'.** Photon (the particle of light) does not interact with the Higgs field, and therefore has no mass.
- Higgs boson was finally discovered by the Large Hadron Collider in 2012. (Nobel in Physics 2013- Peter Higgs)

2. Neutrinos:

- Neutrinos belong to a group of fundamental particles called leptons in the Standard Model.
- Have no electric charge and very little mass (nearly massless).

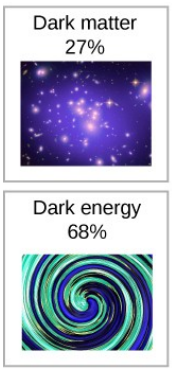
- Very rarely interact with matter and that is why they are called "ghost particles". This means they can travel through vast distances, including entire planets, almost undetected.
- **Source of Neutrinos:** Stars, Supernovae, Galaxies, Nuclear reactions.

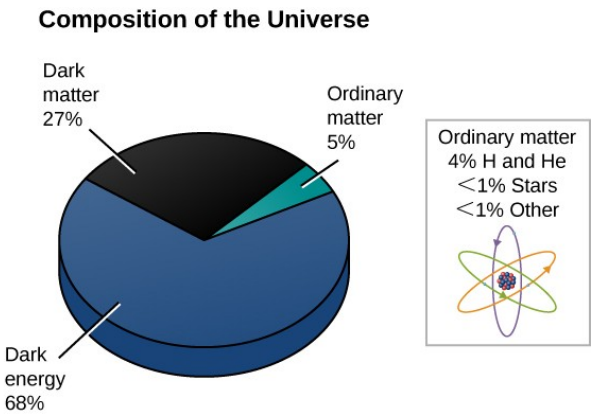
6. DARK MATTER AND DARK ENERGY

- Dark matter and dark energy together make up **95% of the Universe.**
- Only the remainder (5%) is composed of **fermionic matter,** i.e., things on the Earth, planets, stars, etc.

DARK MATTER

- Dark matter is completely invisible. It does not interact with matter, emits no light or energy and thus cannot be detected directly by conventional sensors and detectors.
- Scientists are confident it exists, because of the gravitational effects it has on galaxies and galaxy clusters.





DARK ENERGY

- The existence of dark energy was theorised 25 years ago, when a team of researchers found that the expansion of the Universe was speeding up or accelerating, instead of slowing down due to gravity (inwards pulling force). This is hypothesised to be happening due to a mysterious form of energy called dark energy.

CHARACTERISTICS OF DARK ENERGY

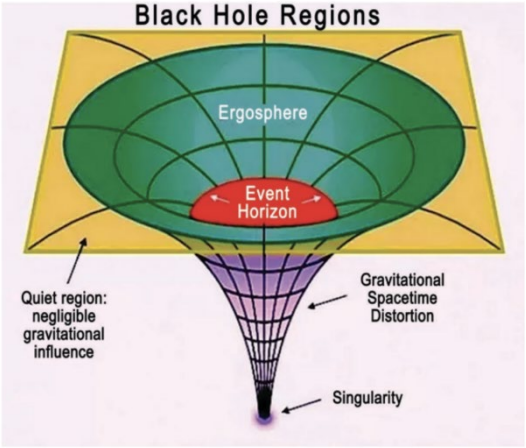
- Dark energy has been hypothesised as a repulsive force or anti-gravity, i.e., while gravity tends to make objects attract, dark energy would pull them apart by increasing the space between them. Thus, dark energy has an expansionary effect.
- As our universe is expanding, it indicates that dark energy has a greater abundance than dark matter.
- Dark energy is a property of space, so it does not get diluted as space expands.
  - Normally, as the universe expands, the density of mass and radiation in it decreases.
  - However, the density of dark energy remains constant throughout. This means the dark energy in the universe is ever increasing, in order to keep the energy-density constant. Thus, dark energy should be energy inherent in the fabric of space itself.

7. BLACK HOLES

- Regions of spacetime where gravity is so strong that nothing, including light and other electromagnetic waves, has enough energy to escape.
- **Formation:** A black hole forms when a massive star (at least three times the mass of our Sun), exhausts its fuel, explodes in a supernova, and collapses under gravity into an incredibly dense core called a singularity.
- **Types:**
  - **Stellar Black Hole:** Formed by the collapse of a single massive star at the end of its life cycle. Mass typically ranges between 3 to 100 times the mass of our Sun.
  - **Intermediate Black Hole**: Formed through merger of stellar black holes or collapse of massive star clusters. Mass between 100 and 1,00,000 times that of the Sun.
  - **Supermassive Black Hole:** Found at the centres of most galaxies. Mass ranges from millions to billions of times the sun's mass.

ARE BLACK HOLES OBSERVABLE?

- Black holes are not directly observable with telescopes. However, their presence can be inferred through their effects on surrounding matter and the gravitational waves they produce.
  - **E.g.,** If a black hole passes through a cloud of interstellar matter or if a star passes close to a black hole, it will draw matter inward in a process known as **accretion**. As the attracted matter accelerates and heats up, it emits electromagnetic radiation into space. This reflects the presence of black holes.
  - Merger of two blackholes produces powerful gravitational waves. The detection of these waves (through LIGO) can confirm the existence/ location of the black holes.



TERMS RELATED TO BLACK HOLES

Singularity:

- Point of infinite density and infinitesimal size at the centre of a black hole. It is a region of extreme spacetime curvature, where the fabric of space and time becomes highly distorted.

Event Horizon (a point of no return):

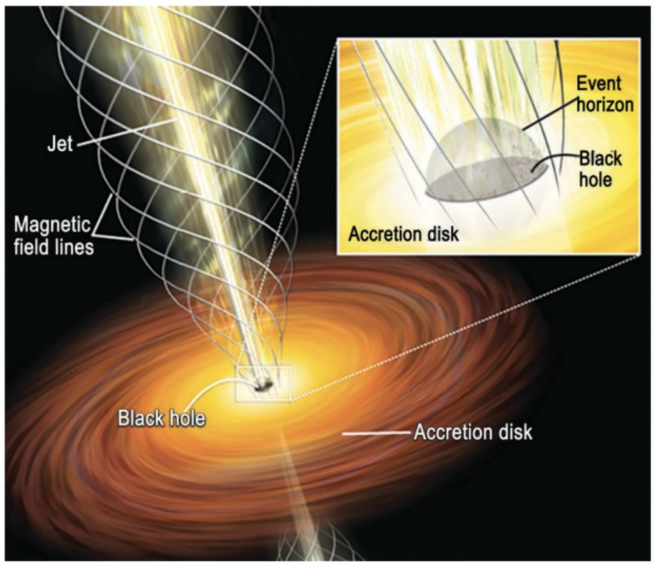
- Boundary around a black hole (around the singularity). Once anything (matter, energy, light) crosses this boundary, it can not escape unless it travels faster than the speed of light (which is impossible).

Ergosphere:

- The Ergosphere is a bigger sphere, outside the event horizon of a black hole, where matter can enter and then return (escape the black hole's gravitational pull), if the matter is moving with speeds very close to the speed of light. Rotating (Kerr) black holes have an ergosphere.

Accretion disc:

- A flat, rotating structure of matter (such as gas, dust, or other material) that forms around a black hole.
- The material in the accretion disc spirals inwards due to the gravitational attraction of the black hole.
- As the matter spirals inward, it gets heated up due to friction, and emits various forms of electromagnetic radiation (including visible light, X-rays, gamma rays and radio waves).





SPAGHETTIFICATION

- Spaghettification refers to the effect of extreme gravitational pressure on any particle or body of matter, when exposed to the extreme forces of the black hole.
- When matter/body comes too close to the event horizon, it is stretched into long thin shapes (stretching it like spaghetti/pasta).

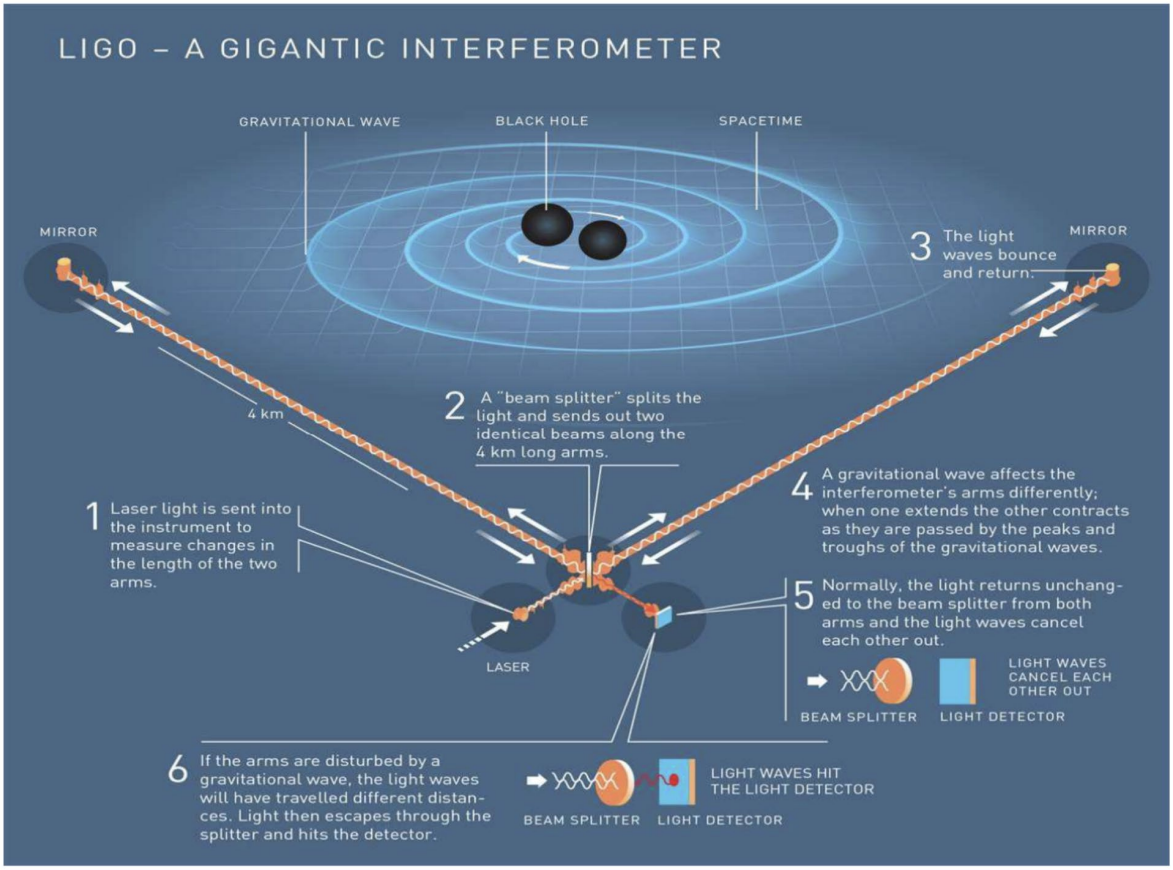
BLACK HOLE MASS GAP

- Gap between the mass of the heaviest Neutron star (maximum mass around 2.2 solar masses) and the lightest Black hole (around 5 solar masses) is called Black Hole Mass Gap.
- At the boundary between neutron stars and black holes there is always the possibility that some new, as yet unknown, astrophysical object might exist.
- Recently, an object has been detected in the star cluster NGC 1851, whose mass (2.09-2.7 times solar Sun) falls within the black hole mass gap.

➔ UNVEILING THE UNIVERSE'S SECRETS

8. LASER INTERFEROMETER GRAVITATIONAL-WAVE OBSERVATORY (LIGO)

- LIGO is an observatory designed to detect gravitational waves using laser interferometry.
- LIGO consists of two widely-separated interferometers within the United States—one in Hanford, Washington and another in Livingston, Louisiana—operated in unison to measure minute ripples in space-time caused by gravitational waves.



Working:

- LIGO comprises two 4-km-long vacuum chambers, built perpendicular to each other. Highly reflective mirrors are placed at the end of the vacuum chambers.
- Light rays are fired simultaneously in both vacuum chambers. They hit the mirrors, get reflected, and are captured back.
  - In normal circumstances, the light rays in both chambers would return simultaneously.
  - But when a gravitational wave arrives, one of the chambers gets a little elongated, while the other one gets squeezed a bit. In this case, light rays do not return simultaneously, and there is a phase difference. The presence of a phase difference marks the detection of a gravitational wave.

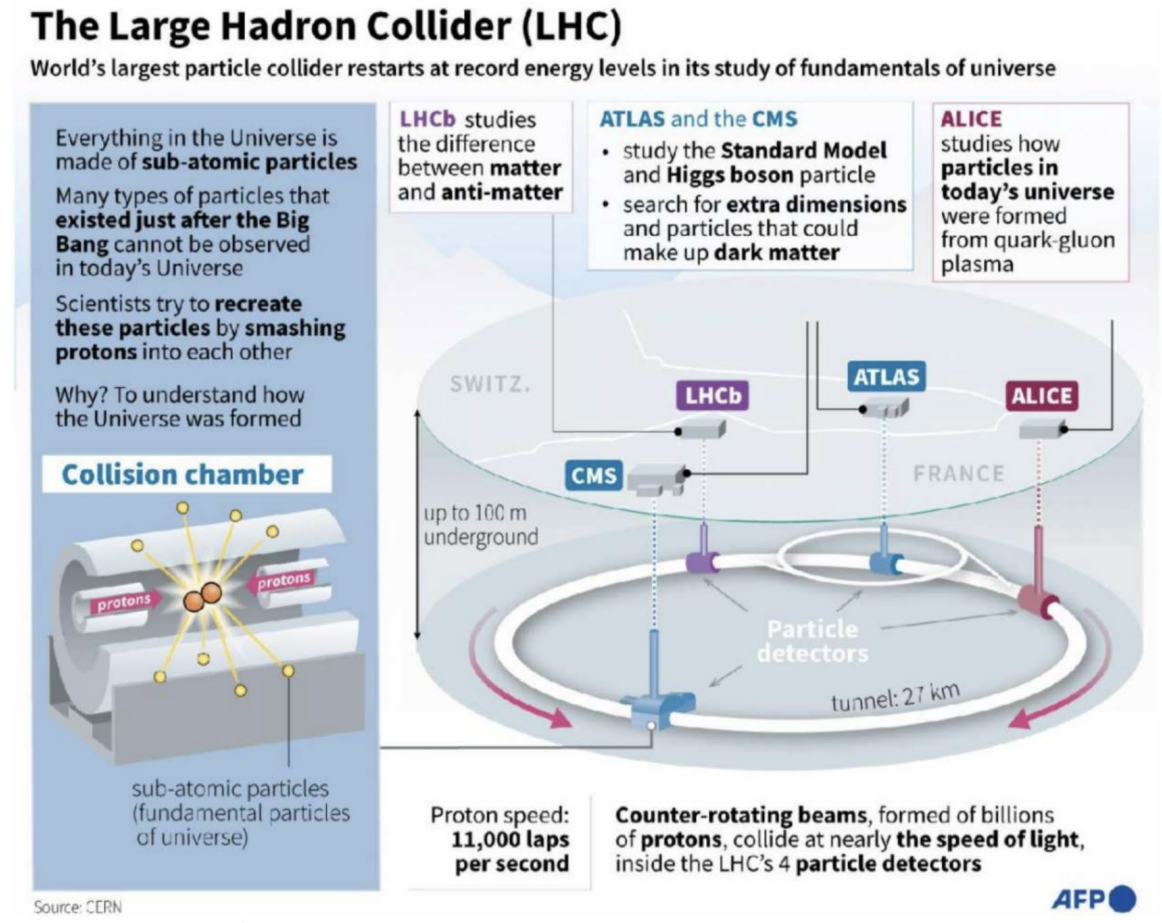
Significance:

- Direct measurement of gravitational waves to study their properties; allows to study mergers of black holes and neutron stars; valuable insights into the early universe; advances understanding of fundamental physics.

LIGO INDIA PROJECT

- In 2023, the government approved construction of the gravitational-wave detector project in India.
- Estimated budget Rs 2,600 crores. Expected to be built by 2030.
- **Location:** Hingoli district, Maharashtra.
- While two LIGOs can detect gravitational waves, a third observatory is required for better triangulation of the source's location (source of gravitational waves).

9. LARGE HADRON COLLIDER (LHC)



- LHC is the world's largest and most powerful particle accelerator.
- **Built by:** European Organisation for Nuclear Research (CERN).
- It lies in a tunnel 27 kilometres in circumference and as deep as 175 metres beneath the France-Switzerland border near Geneva.
- Inside the LHC, two high-energy particle beams of protons are directed at each other at nearly the speed of light and made to collide in the 27-kilometre accelerator ring. The proton beam is guided around the accelerator ring by a strong magnetic field maintained by superconducting electromagnets.
- These high-energy collisions recreate conditions similar to the Big Bang, and generate new particles. Using detectors scientists study their properties and interactions to understand fundamental particles, dark matter and possibly new physics beyond the Standard Model.
- The ATLAS and CMS detectors of LHC discovered the Higgs boson in 2012.

Hadrons: Subatomic particles composed of two or three fundamental particles known as quarks, which are held together by strong Nuclear force. **E.g.,** Protons, Neutrons.

FUTURE CIRCULAR COLLIDER

- CERN plans to build a largest particle collider in a tunnel **91 kilometres** in circumference.
- **Location:** Built underground beneath the French and Switzerland borders.
- The proposed supercollider will be three times larger than LHC, and reach energies 8 times higher than LHC.

10. ICECUBE NEUTRINO OBSERVATORY

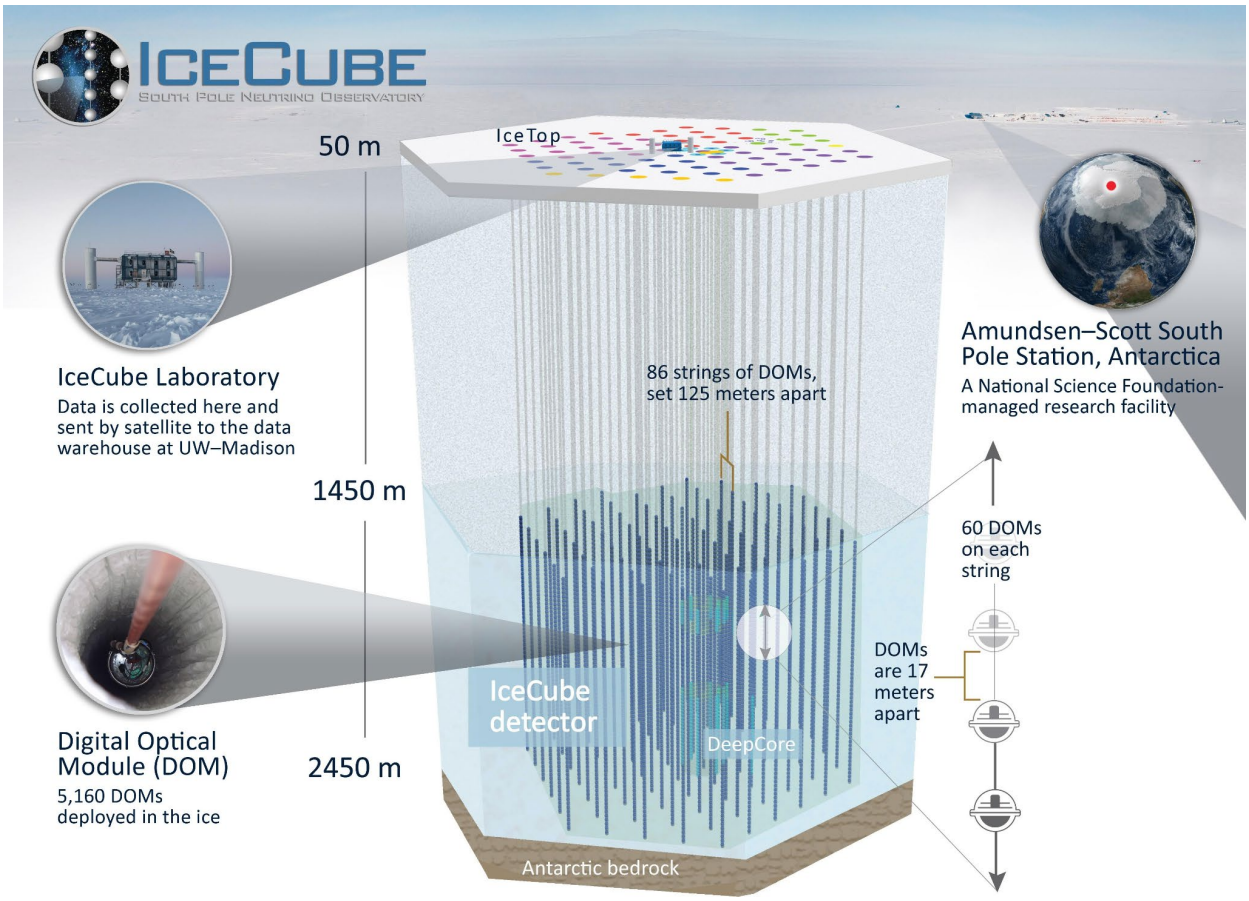
- The world's biggest neutrino telescope. Cubic kilometres in size.
- **Location**: Amundsen–Scott South Pole Station, Antarctica.
- Contains over 5000 Digital Optical Modules (DOMs) deployed at depths between 1.45 km and 2.45 km beneath the Antarctic ice. DOMs detect the Cherenkov radiation, when neutrinos interact with ice.

WORKING

- Neutrinos rarely interact with matter. Most of the neutrinos would pass through the detector unnoticed.
- In very rare cases, a neutrino will interact with an atom in the ice. This collision will produce secondary particles (like muons etc.).
- These secondary particles travel faster in the water than the speed of the light in the water, thus producing a faint bluish glow known as Cherenkov radiation.
- The light detectors (DOMs) will detect the Cherenkov radiation and send a signal to the surface.
- By studying the pattern of Cherenkov radiation, scientists can reconstruct/ study the original energy of the neutrino and its direction.

WHY STUDY NEUTRINO?

- Insights into the universe's evolution: Neutrinos travel vast distances with minimal interaction, and thus carry information about the early universe, moments after the Big Bang.
- Studying their properties might shed light on the Higgs mechanism and mass generation in general.
- There are discrepancies between Standard Model's predictions (Neutrino is massless) and the observed behaviour of neutrinos (non-zero mass). Studying these anomalies could lead to the discovery of new physics beyond the Standard Model.

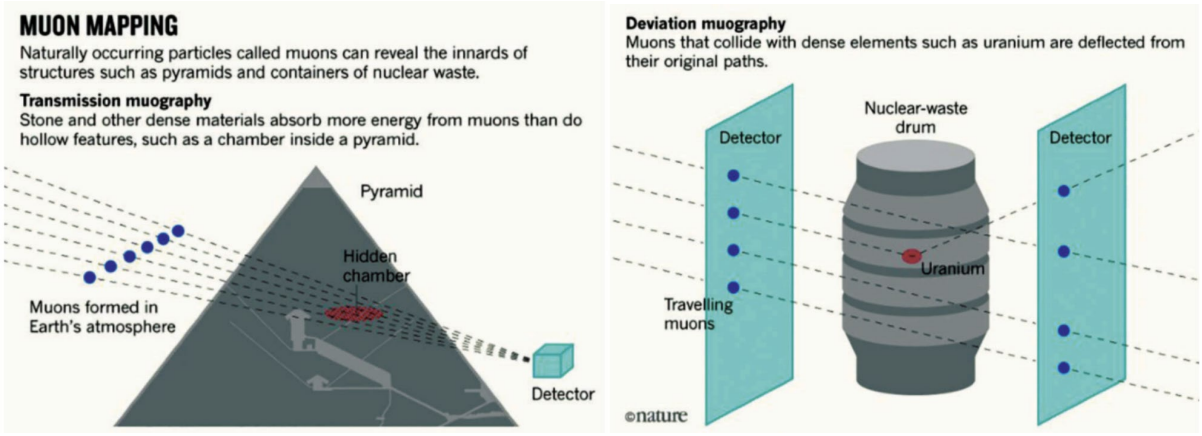


IMPORTANT OBSERVATORIES	DESCRIPTION
Indian Neutrino Observatory	• Proposed underground laboratory at Bodi West Hills, Theni district, Tamil Nadu. • TN has opposed the project citing environmental concerns- as the site is near the Periyar Tiger Reserve and close to Mathikettan Shola National Park, Kerala. • It will be at a depth of 1.3 km under a mountain. Concerns are raised regarding rock bust and roof collapse due to vertical stress.
Cubic Kilometre Neutrino Telescope (KM3NeT)	• European neutrino detector under-construction at the bottom of the Mediterranean Sea (off the coast of France, Italy, and Greece). • KM3NeT will be made up of more than 6,000 light detectors (DOMs) inside the ocean. It will cover about a cubic kilometre of sea. • Recently, KM3NeT detected an ultra-high energy neutrino (220 petaelectronvolts), having 30 times more energy than any previously detected neutrino.
Tropical Deep-sea Neutrino Telescope (TRIDENT)	• China is planning to build the world's largest neutrino observatory in the South China Sea, near the equator. • It would span a staggering 7.5 cubic kilometers.

11. MUON TOMOGRAPHY OR MUOGRAPHY

- Muons are elementary particles (a type of leptons) constantly produced in the Earth's upper atmosphere by cosmic ray interactions. They are about 200 times more massive than electrons.

- Muons do not lose much energy as they travel, which allows them to penetrate more deeply into materials than X-rays or other forms of radiation.



MUON TOMOGRAPHY

- A technique used to image the interior of dense objects or structures using cosmic ray muons.
- When muons pass through matter, their trajectory is affected by the density and composition of the material.
- By measuring the paths of muons as they pass through an object from different angles, it is possible to create a 3D image of the object's internal structure.

APPLICATIONS

- Non-invasive imaging technique (does not require any drilling or excavation). Image interior of archaeological sites (pyramids in Egypt); volcanoes to monitor for potential eruptions.
- Inspect integrity of spent fuel rods and other components of nuclear reactors.
- Used in customs security to detect contraband and other illicit materials in shipping.

➔ STARS AND STELLAR PHENOMENA

12. SUPERNOVA

- Incredibly powerful stellar explosions that occur when a massive supergiant star reaches the end of its life.
- These explosions release an astonishing amount of energy, up to 10^44 joules.

TYPES

TYPE I:

- Occur in binary star systems where one of the stars is a white dwarf (a dense, Earth-sized remnant of a star that has exhausted its nuclear fuel and collapsed under its own gravity).
- The white dwarf accretes matter from its companion star until it reaches a critical mass, known as the **Chandrasekhar limit** (1.4 times the mass of the Sun).
- The pressure and temperature at the core of the white dwarf become so high that carbon and oxygen nuclei in the core begin to fuse rapidly into heavier elements, primarily iron.
- The fusion process proceeds at an ever-increasing rate, leading to a thermal runaway/uncontrolled nuclear reaction that releases an immense amount of energy, causing the white dwarf to explode.

TYPE II:

- A star is a delicate balance between two forces: the outward energy and pressure created by nuclear fusion and the inward gravitational force, as a result of the star's large mass.

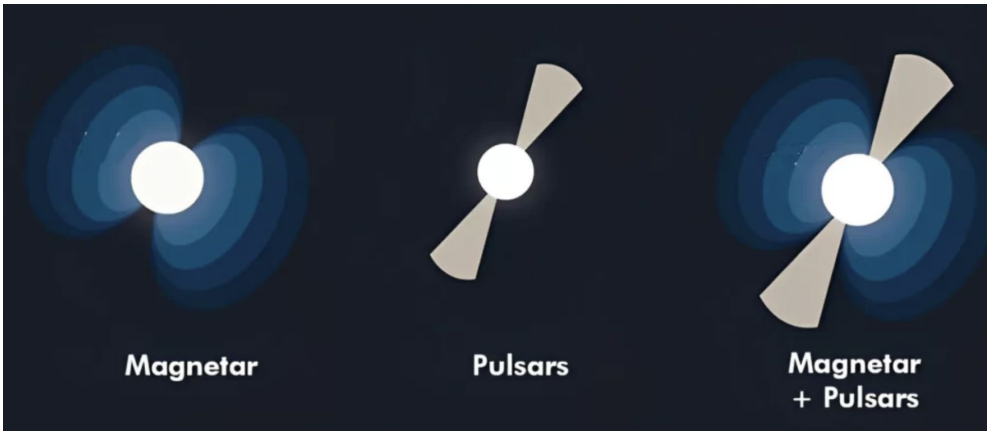
- Core-collapse supernovae occur when massive stars (at least 8 times the mass of our Sun) reach the end of their lives. As the core runs out of nuclear fuel, gravity starts to gain the upper hand, and the star collapses under the force of gravity.
- The core collapse triggers a massive shockwave that eventually blows off the outer layers of the star. It leads to the formation of a **neutron star** (or a black hole, if the star is extremely massive).

SIGNIFICANCE

- Supernovae are responsible for creating and dispersing heavy elements like iron, gold, and uranium into space, essential for the formation of stars, planets and life.
- Supernovae have a consistent peak brightness and are used as "standard candles" in cosmology to measure cosmic distances.

13. NEUTRON STARS

- Incredibly dense remnants (leftover core) of supermassive stars that have exploded as supernovae.
- Their immense gravity causes protons and electrons together in the core to merge to form neutrons.
- **Size:** Typically only 15-30 kilometres in diameter.
- **Mass:** 1.4 to 2.2 times our Sun (densest known stellar objects; second only to black holes)
- **Fast Spinners:** Rotate incredibly fast, with periods ranging from milliseconds to a few seconds. This rapid rotation is a result of the conservation of angular momentum.
- They possess intense magnetic fields, which can accelerate particles and can produce beams of electromagnetic radiation (radio waves, X-rays, and gamma rays).



TYPES OF NEUTRON STARS

(a) Pulsars: Emits an electromagnetic beam out of their poles. The difference between pulsars and neutron stars is that other neutron stars do not shoot a pulse (beam) as strong as pulsars.

(b) Magnetars: Have powerful magnetic fields (in the range of 10^{15} gauss) as compared to other neutron stars. They are the most powerful known magnetic objects in the Universe.

14. FAST RADIO BURSTS

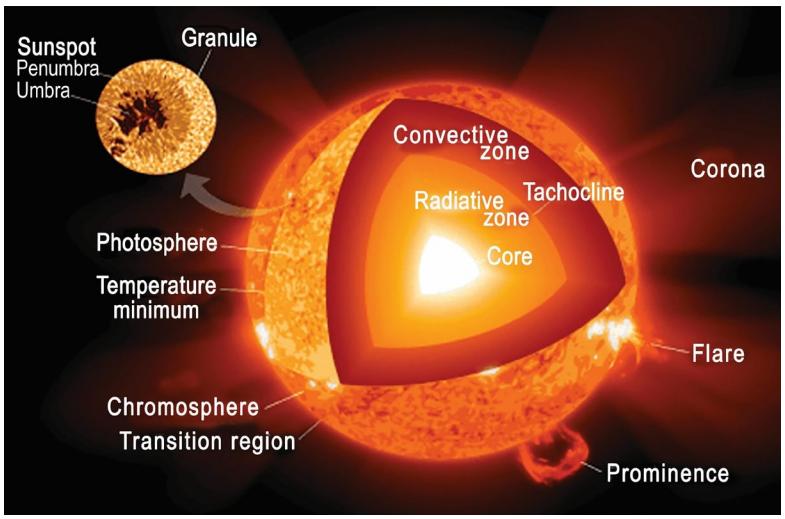
- Bright and brief bursts of electromagnetic radiation seen in radio-wave frequencies (usually last a thousandths of a second).
- Exact cause of FRB is unknown but the sources include magnetars, colliding neutron star binaries, and merging white dwarfs.

15. QUASARS

- Quasars are not actually stars but active galactic nuclei (AGN) located at extreme distances from Earth.
- They are the bright cores of distant galaxies powered by supermassive black holes. The intense gravitational forces exerted by black holes cause the surrounding material to heat up and emit enormous amounts of energy, including visible light and radio waves.
- They are the most luminous objects (equivalent to the combined light of hundreds of billions of stars) in the known universe, outshining entire galaxies. As a general rule, the most luminous quasars indicate the fastest-growing supermassive black holes.
- The nearest quasar is Markarian 231, which lies about 600 million light-years from Earth.

16. SUN

- **Composition:** Hydrogen (about 74%), Helium (24%) and other elements in trace amounts.
- **Energy Source**: Nuclear Fusion (Hydrogen atoms combine to form Helium in its core).
- **Size and Mass:** Approximately 109 times Earth's diameter, and about 333,000 times Earth's mass.
- **Distance:** Average distance from Earth to the Sun is known as one astronomical unit (AU). 1 AU is about 150 million kilometres.



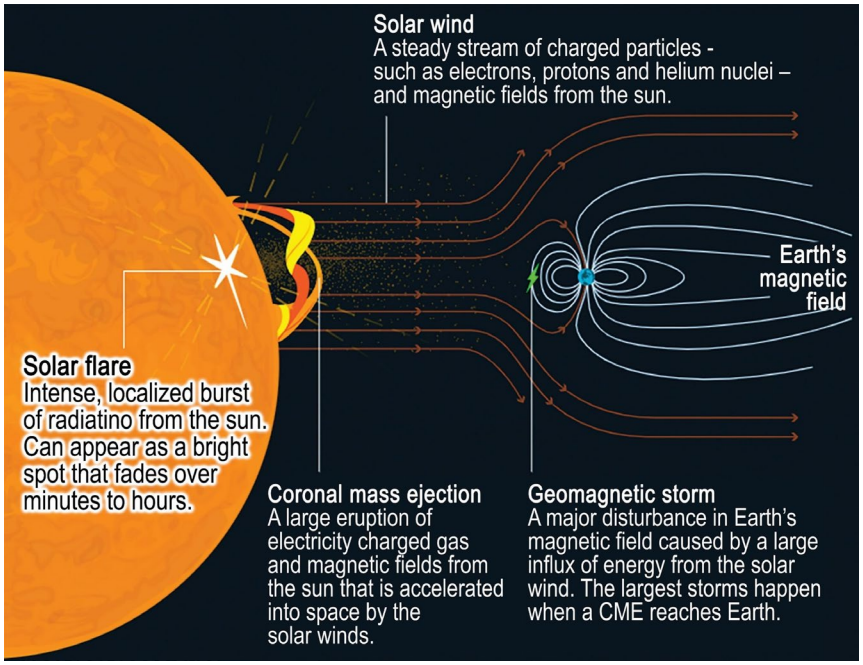
LAYERS OF THE SUN

- **Core:** Innermost region where nuclear fusion occurs.
- **Radiative Zone**: Above the core, energy is transported outward by photons through this layer.
- **Convective Zone**: This layer is closer to the surface and transports energy through the movement of hot gas (plasma) cells.
- **Photosphere**: Visible surface of the Sun, where most of its energy is radiated into space as visible light.
- **Chromosphere**: Thin layer between photosphere and corona. (named for its reddish colour, most easily observed during a total solar eclipse)
- **Corona**: Outermost and extremely hot layer of Sun's atmosphere, visible as a halo during solar eclipses.

KEY TERMS

- **Heliosphere:** Vast, bubble-like region of space dominated by the Sun's solar wind and magnetic field emanating from the Sun. It extends far beyond the orbit of Pluto and protects the solar system from harmful cosmic rays.
- **Solar Magnetic Field:** Sun has a complex magnetic field generated by the motion of charged particles in its interior. This field plays a crucial role in the behaviour of the Sun, including the formation of sunspots, solar flares, and CMEs.
- **Sunspots:** Temporary spots on the Sun's photosphere that appear darker than the surrounding areas. They are cooler parts of the Sun's surface caused by massive changes in its magnetic field.
- **Solar Wind:** Created by the outward expansion of plasma (a collection of charged particles) from the Sun's corona. This plasma is continually heated to the point that the Sun's gravity can not hold it down. It then travels along the Sun's magnetic field lines that extend radially outward.

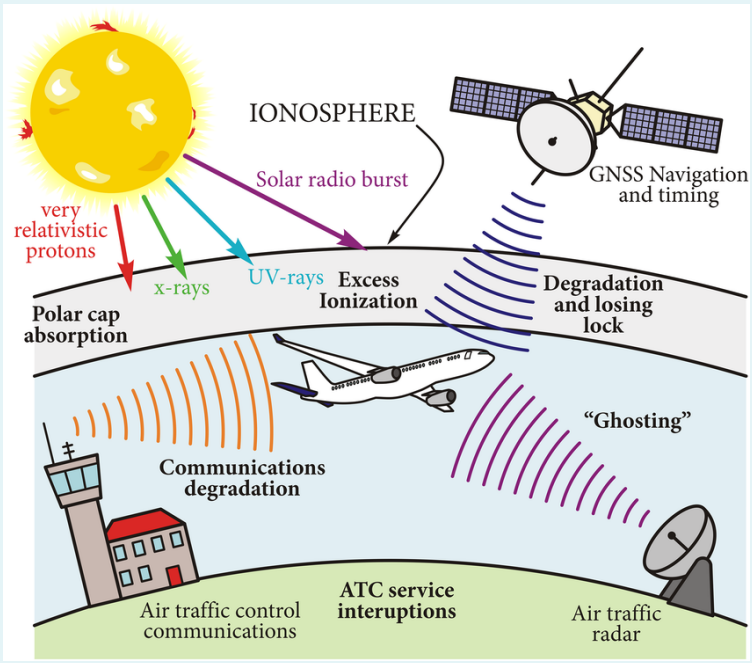
- **Solar Flares:** Massive explosion on the Sun's surface that releases intense bursts of radiation across almost the entire electromagnetic spectrum (from radio waves to X-rays). Solar flares can disrupt radio communications, satellite operations, and power grids on Earth.



- **Coronal Mass Ejections:** Large expulsions of plasma and magnetic fields from the sun's corona into space. When they interact with Earth's magnetic field, they can cause geomagnetic storms, which can impact power grids, GPS systems, communication networks and orbiting satellites and trigger Aurora displays.
- **Picoflare jets:** Relatively small jets of charged particles expelled intermittently from the sun's corona at supersonic speeds for 20 to 100 seconds. These charged jets are a source of the solar wind. These jets carry approximately one-trillionth the energy of the largest flares produced by the sun. ("Pico" denotes 10^{-12} or one trillionth of a unit). It has significant effects on the large-scale solar system, as well as on Earth's magnetic field and poses risks to satellites.
- **Geomagnetic storms:** A geomagnetic storm is a disturbance in the Earth's magnetic field caused when a solar wind shock wave or cloud of the magnetic field interacts with the Earth's magnetic field.

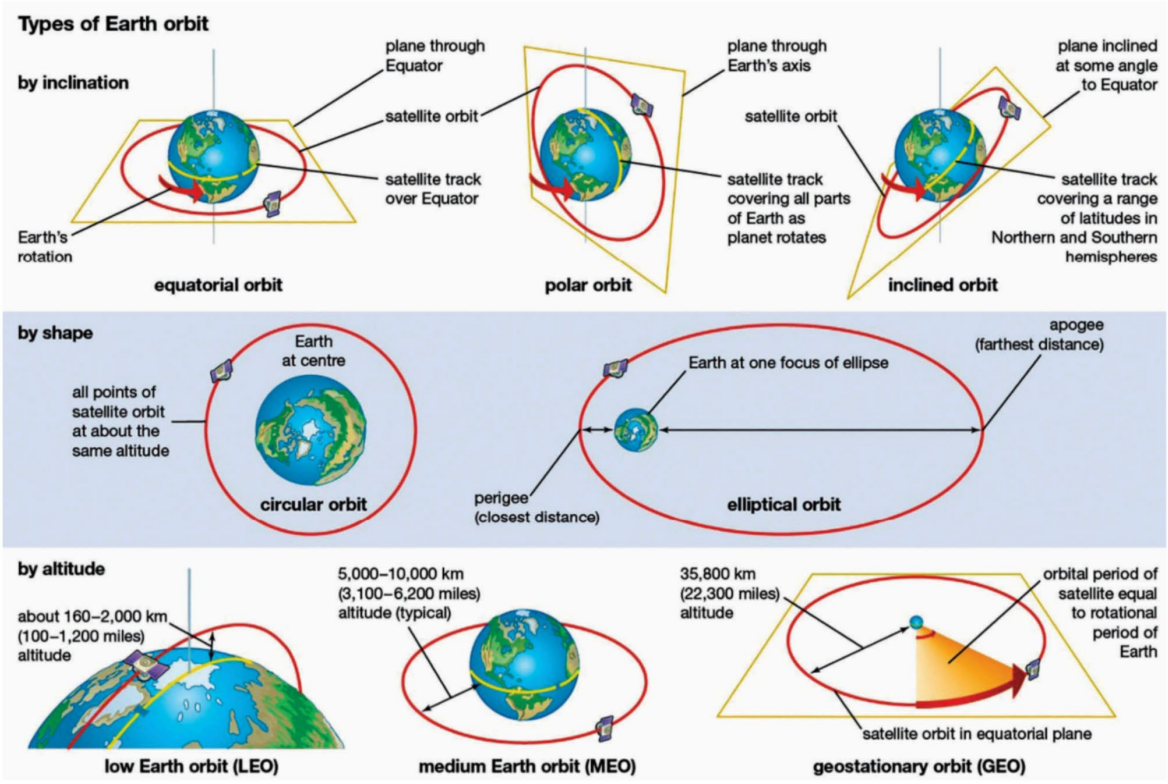
IMPACTS OF SOLAR FLARES ON EARTH

- Strong flares can **trigger coronal mass ejections**. This can **disrupt Earth's magnetosphere** and **result in geomagnetic storms.**
- Such geomagnetic storms can **lead to auroras** closer to the equator than is possible during calm conditions.
- Emission of fast-moving charged particles can **endanger astronauts** and **Earth-orbiting spacecraft.**
- **Impacts satellites** and thus disrupts GPS navigation, Satellite Television, Air Traffic Communication systems.
- **Minor to extensive radio blackouts** (mainly impacts high frequency radio communications) on the side of Earth facing the sun.
- Damage Power grids and cause **electrical blackout**.



➔ SPACE TECHNOLOGY AND MISSIONS

17. TYPES OF ORBITS



1. LOW EARTH ORBIT (LEO):

- **Altitude:** Between 200 km and 2000 km above Earth.
- Satellites in this orbit take approximately 90 minutes to circle Earth.

- **Utility:** Satellite imaging, location of International Space Station.

2. MEDIUM EARTH ORBIT (MEO):

- **Altitude:** 2,000 to 36,000 km above the Earth's surface.
- **Utility:** Satellite navigation, Communication, and Earth observation. **E.g.,** Global Positioning System.

3. GEOSTATIONARY EARTH ORBIT (GEO):

- GEO is located at an altitude of 35,786 km above Earth.
- Satellites in GEO circle Earth above the equator from West to East following Earth's rotation – taking 23 hours 56 minutes and 4 seconds – by travelling at the same rate as the Earth. This makes satellites in GEO appear to be 'stationary' over a fixed position from the Earth.
- **Utility:** Telecommunication satellites, weather monitoring satellites to continually observe specific areas.

4. POLAR ORBIT:

- An orbit that passes over or near the Earth's poles and covers the entire surface of the Earth over a period of time. A type of LEO.

5. SUN-SYNCHRONOUS ORBIT (SSO):

- A particular kind of polar orbit. Satellites in SSO are synchronous with the Sun i.e., they are synchronised to always be in the same 'fixed' position relative to the Sun. Hence, the satellite will always observe a point on the Earth as if constantly at the same time of the day.
- Satellites in SSO can be used to investigate how weather patterns emerge, monitor emergencies (like forest fires, floods), accumulate data on deforestation & rising sea levels.

6. TRANSFER ORBITS:

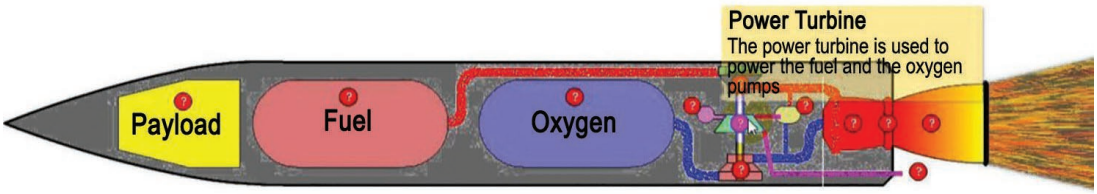
- A special kind of orbit used to transfer a satellite from one orbit to another, typically used for interplanetary missions.
- When satellites are launched from Earth and carried to space with launch vehicles, they are not always placed directly in their final orbit.
- Satellites are instead placed on a transfer orbit from where, by using relatively little energy, the satellites can move to the destination orbit.

18. ROCKET PROPULSION

- A rocket has to clear earth's atmosphere to travel in space or go into an orbit.
- The minimum height for a satellite to go into the Earth's orbit is approximately 200 km.
- At a certain initial speed, as the satellite tries to go off at a tangent to the earth, the Earth's gravity pulls it back.

PRINCIPLE BEHIND ROCKET PROPULSION: NEWTON'S 3RD LAW OF MOTION

- The only means available for any object to be put in orbit is rocket propulsion.
- Newton's third law of motion governs the working of a rocket engine, viz., For every action, there is an equal and opposite reaction.
- The mass of gas escaping through a rocket's nozzle gives a push or commonly called thrust to the rocket to fly in the opposite direction.



PERFORMANCE METRIC OF A ROCKET: THRUST AND SPECIFIC IMPULSE

- Power generated by the rocket engine is balanced by the thrust in the opposite direction on the rocket itself, resulting in pushing the rocket at a certain initial velocity. More the power of the engine, more the thrust, more is the initial velocity.
- Efficiency of a rocket is expressed in terms of specific impulse (the amount of thrust derived from one kilogram of propellant (rocket fuel) in one second of engine operation). Specific impulse depends on two things: quality of fuel used and performance of the engine. Higher the specific impulse, the higher is the push to the rocket.

19. LAUNCH VEHICLES OF ISRO

1. POLAR SATELLITE LAUNCH VEHICLE

- Workhorse of ISRO known for its reliability, versatility, and cost-effectiveness since 1994.
- **Stages**: Four-stages launch vehicle
  - The first stage is powered by a solid rocket motor (strap-on motors to provide additional thrust to the rocket to overcome air resistance).
  - Second stage uses a liquid propulsion system (Vikas Engine)
  - Third stage is a solid rocket motor
  - Fourth stage is a liquid-fueled engine
- PSLV can deliver payloads of up to:
  - 3250 kg to Low Earth Orbit
  - 1600 kg to Sun Synchronous Orbit
  - 1400 kg to Geosynchronous Transfer Orbit
- **Successful launches:** Chandrayaan-1 Mission (2008), Mars Orbiter Mission/Mangalyaan (2013), 104 satellites at one go (2017).

2. GEOSYNCHRONOUS SATELLITE LAUNCH VEHICLE

- **Stages:** Three-stage launch vehicle
  - **First Stage:** Uses four strap-on solid boost motors (HS200).
  - **Second Stage**: Liquid core stage (L110) powered by two Vikas liquid engines.
  - **Third Stage:** Cryogenic Upper Stage with cryogenic engine using liquid Hydrogen and liquid Oxygen for high specific impulse and efficiency.
- GSLV can carry more than 2,200 kg to geostationary orbits, and over 6,000 kg to LEO.
- **Issues:** GSLV has a patchy track record, has flown 16 times of which 4 are unsuccessful (high failure rate for any rocket). The problems have mainly been with the cryogenic engine used by GSLV that is reverse-engineered on a Russian design.

3. LAUNCH VEHICLE MARK-3

- LVM3 (previously referred to as Geosynchronous Satellite Launch Vehicle Mark III, GSLV Mk III) is a three-stage medium-lift launch vehicle developed by ISRO.
- **Stages:** Three-stage launch vehicle
  - **First stage**: Solid fuel S200 stage.
  - **Second stage:** Liquid fuel L110 stage. (Vikas Engine)
  - **Third stage:** Cryogenic fuel C25 stage (uses 25 tonnes of a mixture of liquid Hydrogen and Liquid oxygen). This upper stage (CE-20 cryogenic engine) is developed entirely in India.
- **Payload capacity** (one of the most powerful rockets in ISRO's fleet):
  - 4,000 kilograms to geosynchronous transfer orbit (GTO).

- 10,000 kilograms to low Earth orbit (LEO).
- Primarily designed to launch communication satellites into geostationary orbit.
- **Successful launches:** Chandrayaan-2, Chandrayaan-3. Human-rated LVM-3 is due to launch crewed missions under Gaganyaan Mission.

4. SMALL SATELLITE LAUNCH VEHICLE

- New small satellite launch vehicle developed by ISRO to launch small satellites to LEO on 'launch-on-demand' basis.
- Three-stage Launch Vehicle configured with three Solid Propulsion Stages and liquid propulsion based Velocity Trimming Module (VTM) as a terminal stage.
- Can launch Mini, Micro, or Nanosatellites (10 to 500 kg mass) up to 500 km in LEO.
- Capable of multiple orbital drop-offs i.e., launch multiple microsatellites in one launch.
- SSLVs will cost 1/10th of a PSLV and will need only 72 hours for launch in comparison to 45 days for PSLV.

5. NEXT GENERATION LAUNCH VEHICLE

- Three-stage partially reusable heavy-lift launch vehicle, under development by ISRO (expected by 2032).
- **Reusability**: Will have a reusable first stage, which can be utilised 15 to 20 times.
- Height: 91 meters, significantly larger than the 43-meter LVM3.
- **Fuel**: Semi-cryogenic propulsion (refined kerosene as fuel with liquid oxygen (LOX) as oxidiser) for booster stages.
- Maximum payload capability: 30 tonnes (30,000 kg) to Low Earth Orbit.
- **Utility:** Deployment of communication and earth observation satellite constellations to LEO. Launch Bharatiya Antariksh Station, Indian crewed lunar missions by 2040 and other interplanetary exploration.
- Higher payload capacity; modular green propulsion system; multiple reuses will reduce operational costs.

➔ ROCKET FUEL TECHNOLOGIES

20. VIKAS ENGINE

- Family of liquid-fueled rocket engines.
- Hypergolic propellant - ignites spontaneously on contact.
- **Fuel:** Dimethyl Hydrazine
- **Oxidiser:** Nitrogen Tetroxide (N_2O_4)
- **Applications:** Powers various stages of India's launch vehicles:
  - Second stage of the PSLV
  - Second stage and strap-on boosters of GSLV Mark II
  - Core stage (L110) of LVM3

21. LOx METHANE ENGINE

- The Liquid Propulsion Systems Centre of ISRO is developing Lox methane-powered rocket engines.
- The 'LOx methane' engine uses methane as fuel and liquid oxygen as oxidiser.

Advantages:

- Can be synthesised in space (Methane can be synthesised using water and carbon dioxide in space).
- It is non-toxic. (Di-Methyl Hydrazine and Nitrogen tetroxide is said to be highly toxic)

- Higher specific impulse
- Less bulky and easy to store
- Does not leave a residue upon combustion.

22. ION ROCKETS

- Ion rockets are the rockets of the future for deep space exploration.
- They are much more efficient than conventional rockets that use chemical fuels.
- While chemical fuels generate velocities of up to 2 to 3 km/s, ion rockets can achieve velocities around 20-80 km/s.
- Ion rockets use electric propulsion systems, typically ions, to generate thrust and propel rockets.
- The ions are accelerated using an electric field and expelled at high velocities, resulting in a continuous thrust.
- Small scale ion propulsion is used in a number of missions including NASA's Dawn mission and Deep Space mission, ESA's LISA Pathfinder and Japan's Hayabusa Mission.

23. ISRO's CRYOGENIC ENGINE (CE-20)

- CE-20 is an indigenous cryogenic engine developed by ISRO.
- **Fuel:** Liquid hydrogen and liquid oxygen as propellants stored at extremely low temperatures (around -253°C for hydrogen and -183°C for oxygen).

Advantages:

- Cryogenic propellants have a high energy density, high fuel efficiency and high specific impulse (more thrust per kilogram of propellant). This allows rockets to carry less fuel, reduces their overall weight and can carry heavier payloads or travel further.
- Cryogenic engines are throttleable (ability to vary/adjust their thrust levels during flight). This capability is essential for precise control during ascent, orbit insertion, manoeuvring, and controlled reentry of a rocket.

Challenges:

- They require complex and expensive infrastructure to store and handle extremely cold propellants.
- The initial development process of cryogenic engines and maintenance is expensive.

➔ ROCKET LAUNCH STATIONS IN INDIA

24. CONSIDERATIONS FOR SELECTING ROCKET LAUNCH SITE IN INDIA

- **Proximity to Equator:** Satellites launched from near the equator and towards the eastward direction receive an initial velocity boost due to the Earth's rotation. At the equator, Earth's surface rotates at approximately 1,670 km/h. The initial boost helps in cutting down the fuel requirement to reach the orbit.
  - However, this initial boost can only benefit satellites in geo-stationary orbit or which circle the Earth parallel to the equator (usually communication satellites).
  - Polar satellites are launched in southward or northward direction, and therefore cannot take advantage of the Earth's rotation.
- **Location on Eastern coast:** In case of any launching failure, the debris falls harmlessly in Bay of Bengal or Indian Ocean and not on the main hinterland.

25. LAUNCH STATIONS OF ISRO

1. SATISH DHAWAN SPACE CENTRE (SHAR):

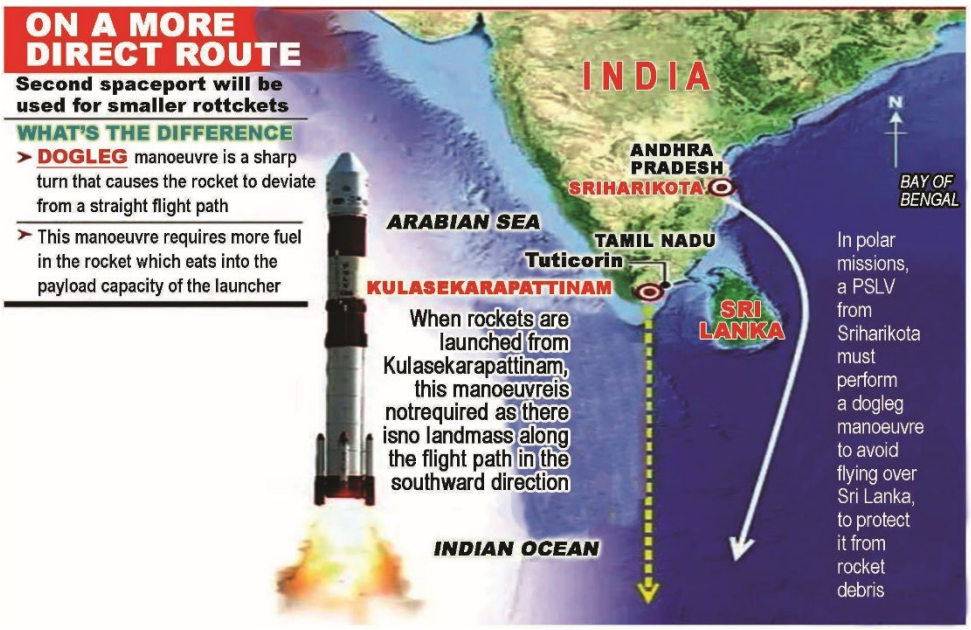
- India has SHAR in Sriharikota, Andhra Pradesh, spread over 145 sq. km.
- SHAR has **two launch pads** for PSLV and GSLV flights (heavier satellites).

ISRO'S THIRD LAUNCH PAD (TLP)

- TLP will be built at Satish Dhawan Space Centre, Sriharikota. Estimated duration: 4 years (till 2029).
- TLP is designed to have a universal configuration, such that it can support:
  - NGLV (Next Generation Launch Vehicle) and scaled-up configurations of NGLV.
  - LVM3 vehicles with semicryogenic stages.
  - Act as a standby launch pad for Second Launch Pad at Sriharikota.
- TLP will be co-located with the second launch pad, maximising use of existing infrastructure.

2. KULASEKARAPATTINAM SPACEPORT:

- India's new spaceport for SSLV is under development at Kulasekarapattinam.
- **Location:** Thoothukudi district, Tamil Nadu.
- **Will launch:** Smaller payloads like Nano and Microsatellites.
- **Project cost:** ₹950-crores, expected to take nearly two years to complete.
- Kulasekarapattinam gives the benefit of launching straight in southward direction, i.e., dogleg manoeuvre is not required, thereby saving the rocket's fuel and improving payload capability.

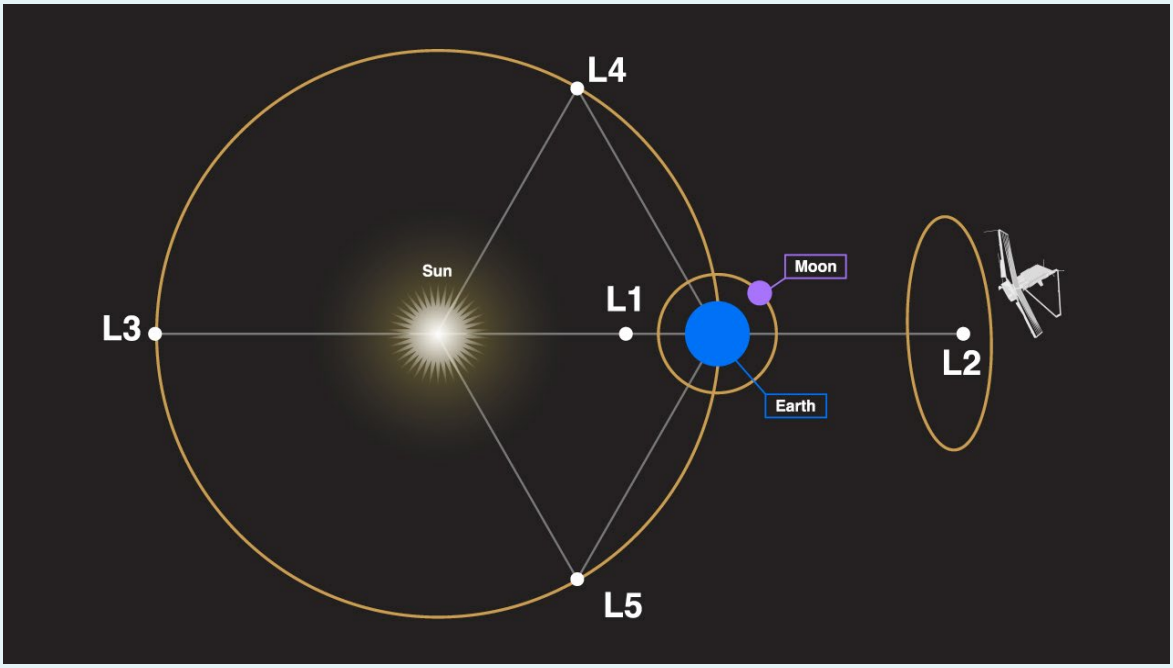


➔ IMPORTANT SPACE MISSIONS OF ISRO

26. ADITYA L1 MISSION

- India's first solar mission dedicated to study the Sun launched by ISRO in 2023.
- **Location:** Lagrange point 1 (located at a distance of about 1.5 million km from the Earth).

- **Objectives:** To study Solar upper atmospheric dynamics (photosphere, chromosphere, and Corona), coronal heating mechanism, origin of coronal mass ejections, and flares etc.
- **Payloads:** Weighing 1475 kg, it will carry seven payloads, including:
  - **Solar Ultraviolet Imaging Telescope** (SUIT): To image the Sun in 200-400 nanometre (nm) of Ultraviolet band. SUIT's imager will continuously record the entire disk of the Sun. Images of various layers could improve understanding of the Sun's immediate atmosphere.
  - **Visible Line Emission Coronagraph** (VELC) is the Corona/Imaging and Spectroscopy payload to focus on Sun's Corona.
  - **Solar Low Energy X-ray Spectrometer** (SoLEXS) and **High Energy L1 Orbiting X-ray Spectrometer** (HEL1OS) will study X-ray flares emitted by the Sun over a wide range of X-ray energy range.
  - **Aditya Solar wind Particle Experiment** (ASPEX), during the cruise phase, will turn on and start performing in-situ measurements of solar particles and ions.
  - **Plasma Analyser Package for Aditya** (PAPA) will study solar wind and energetic ions.
  - **Magnetometer** (MAG) to measure low-intensity interplanetary magnetic field in space.



  - **Lagrange points** are positions in a moving two-body system where the combined gravitational forces of two large bodies (such as the Sun and Earth or Earth and the Moon) acting on a third body of smaller mass cancel each other out.
  - While absolute neutralisation is not achievable due to the influence of other celestial bodies (such as Moon, Mars, and Venus etc.), they provide a stable position where a spacecraft may be "parked" to make observations.

There are **five Lagrange points** located along the line connecting the two larger celestial bodies.

  - **L1 point:** Located roughly 1.5 million kilometres from Earth. It provides an uninterrupted view of the Sun (the satellite can view the Sun without eclipses).
  - **L2 point:** Provides an unobstructed view of the Universe and is stable enough for long-duration observations. Location for space-based observatories (James Webb Space Telescope).
  - **L3 point:** Lies behind the Sun, opposite to Earth. Offers the potential to observe the far side of the Sun. Possible location for a future space-based observatory.
  - **L4 and L5 point:** Stable locations but are relatively farther from Earth than L1.

27. GAGANYAAN MISSION

- ISRO's ambitious **human spaceflight mission** which will carry 3 astronauts (Vyomanauts) to Low Earth Orbit (400 kilometres) for a 3-days mission, and bring them back safely to Earth by landing in Indian seawaters.
- Crewed flight is planned for 2026 on the Human Rated LVM3 as the launch vehicle.
- **Human Space Flight Centre** will coordinate the programme, and will be responsible for the implementation of the project.
- Once completed on schedule, India will become the world's fourth nation to conduct independent human spaceflight after Russia, United States and China.

COMPONENTS OF GAGANYAAN

Crew Module

- A crew module and service module. The crew members have been shortlisted by the IAF and ISR.
- The crew will perform micro-gravity and other scientific experiments.

Crew Module Atmospheric Re-entry Experiment (CARE)

- Gaganyaan would return back to Earth. While re-entering Earth's atmosphere, the spacecraft needs to withstand very high temperatures created due to atmospheric friction.
- A prior critical experiment was carried out in 2014 along with LVM3, when the CARE capsule successfully demonstrated that it could survive atmospheric re-entry.

Crew Escape System- PAT

- The Crew Escape System is an emergency accident avoidance measure.
- In July 2018, ISRO completed the first successful flight 'pad abort test' or Crew Escape System.

Environmental Control & Life Support System ECLSS: ECLSS will:

  - Maintain steady cabin pressure and air composition
  - Remove carbon dioxide and other harmful gases
  - Control temperature and humidity
  - Manage parameters like fire detection and suppression

Vyom Mitra: ISRO's female humanoid robot that will test-flight Gangayaan.

Space-borne Assistant and Knowledge Hub for Crew Interaction (SAKHI) App: A digital platform integrated to the space suits of astronauts to assist astronauts regarding technical documents and training manuals. Monitor vitals (blood pressure, heart rate and oxygen saturation), hydration level, sleep patterns. Maintain mission log and keep crew connected with the onboard computer and ground-based stations on Earth.

28. CHANDRAYAAN SERIES

CHANDRAYAAN-1

- India's first moon mission launched in 2008. **Launch vehicle**: PSLV
- The spacecraft orbited around the Moon at a height of 100 km. It mapped the Moon in infrared, visible, and X-ray light from lunar orbit, and used reflected radiation to prospect for various elements, minerals, and ice.
- It released an impactor (**Moon Impact Probe- MIP**) that studied the thin lunar atmosphere before crashing on the Moon's surface.
- It gave definitive proof of the presence of **water ice** in the Moon's atmosphere and surface.

CHANDRAYAAN-2

- Second moon mission launched in 2019. **Launch vehicle**: LVM-3
- The spacecraft consisted of an orbiter, a lander, and a rover.

- The orbiter circled the Moon in a polar orbit at a height of 100 km, and has a planned mission lifetime of seven and a half years.
- Vikram lander planned to demonstrate soft landing on the moon, however, it crashed due to a software error. Despite it, 90-95% of the mission objectives have been accomplished.
- It has enriched understanding of the Moon's evolution and mapped minerals and water molecules in the Polar Regions.

CHANDRAYAAN-3

- Third moon mission launched in 2023. **Launch vehicle**: LVM3
- The spacecraft consisted of Vikram lander and Pragyan rover. The Vikram lander touched down on the Moon on August 23, 2023.
- It made one of the closest approaches to the moon's South Pole. The landing site is located about 600 kilometers from the South Pole of the Moon, on the near side of the Moon.
- The rover operated for **one lunar day** (approximately 14 Earth days). It aimed to collect lunar samples, do in-situ experiments and send data to Vikram lander, to transmit it back to Earth.
- India became the **fourth country** to land spacecraft on the Moon after the US, Russia, and China.
- **The Virtual Launch Control Centre** at the Vikram Sarabhai Space Centre played a vital role in continuous real-time monitoring of the launch activities from SHAR.
- The International Astronomical Union (IAU) has approved the name **"Statio Shiv Shakti"** for the landing site of Chandrayaan-3's Vikram lander.

PAYLOADS IN CHANDRAYAAN-3

Propulsion module: Spectro-polarimetry of Habitable Planet Earth (SHAPE) to gather data on the polarisation of light reflected by Earth to assist with exoplanet searches.

Lander payloads:

- Chandra's Surface Thermophysical Experiment (ChaSTE) to measure thermal conductivity and temperature on the surface.
- Instrument for Lunar Seismic Activity (ILSA) to detect Moonquakes.
- Radio Anatomy of Moon Bound Hypersensitive ionosphere and Atmosphere (RAMBHA) to measure the density of near-surface plasma, encompassing ions, and electrons, and monitor its temporal variations.
- Langmuir Probe to estimate the density and variation of plasma, or superheated gas, in the Moon's environment.
- Laser Retroreflector Array (from NASA) to measure distances using laser ranging to understand the dynamics of the Moon system.

Rover payloads:

- Alpha Particle X-ray Spectrometer (APXS) to look for elements in the lunar soil and rocks
- Laser Induced Breakdown Spectroscopy (LIBS) to examine the chemical and elemental composition of the lunar surface.

CHANDRAYAAN-4

- Planned lunar sample return mission expected to launch around 2027.
- **Aim**: To collect samples from the moon's surface and bring them back to the Earth.
- **Landing site:** Planned near Statio Shiv Shakti (landing site of Chandrayaan-3), near south pole region.
- **Design:** The spacecraft would comprise five modules onboard two separate LVM-3, and will be docked in the Earth orbit. In the final lunar orbit, the modules will undock. The samples will be collected and brought back to Earth.

FACTS ABOUT MOON

- **Lunar soil** contains several elements such as iron, silicon, potassium, manganese, and magnesium. However, it does not contain organic matter such as microbes and insects. Lunar soil is hydrophobic and is exposed to extremely unhealthy amounts of solar radiation.
- **The Moon is tidally locked** to Earth, i.e., it rotates on its axis in the same time it takes to orbit Earth (27.3 days). That is why we always see the same side of the Moon from Earth, while the other side (far side) faces away from Earth.
- **The far side of the Moon** has a dramatically different landscape as compared to the near side (face of the Moon visible from Earth). It has a thicker crust by almost 20 km. The far side has almost no characteristic dark spots. (Dark spots are large basaltic plains formed from ancient volcanic eruptions on the Moon)
- **Chang'e 4 mission** (China, 2019) is the first mission to have landed on the far side of the Moon.
- **Chang'e 6 mission** (2024) landed on the Moon's far side, and returned samples to Earth, marking the first-ever sample return from the region.

Reason for increased interest in the South Pole of the Moon:

- Water ice has been detected at both poles of the Moon, but the South Pole has more area in permanent shadow and colder temperatures, having higher possibility of presence of water.
- Due to extremely cold temperatures, anything trapped in the South Pole would remain frozen without undergoing much change. The rocks and soil, here, could provide clues to the early solar system.
- The South Pole is located in the South Pole-Aitken basin, which is a huge crater. There may be material from the deep crust and upper mantle of the Moon on or near the surface.

29. SHUKRAYAAN MISSION

- ISRO's planned orbiter to Venus, expected by 2028. India's second interplanetary mission after the Mars Orbiter Mission (Mangalyaan) launched in 2013.
- **Objective**: To study the structure, composition, and dynamics of Venus's atmosphere; Investigate surface processes and subsurface stratigraphy; Explore solar wind interactions with the Venusian ionosphere.
- The mission will place a spacecraft in orbit around Venus. The orbiter is expected to carry instruments like synthetic aperture radar, infrared and ultraviolet cameras, and sensors that will study Venus's ionosphere.
- The mission will also see India perform aero-braking for the first time. Aero-braking is a technique used to reduce a satellite's orbit by using atmospheric drag instead of relying solely on fuel-powered engines.

Venus:

- Venus has a solid surface by virtue of being one of the four inner planets besides Mercury, Earth and Mars. It is nearly the same size as the Earth.
- Its atmosphere is composed of 95% carbon dioxide (high greenhouse effect), making it the hottest planet in our Solar system.
- Volcanic eruptions on Venus release sulphur dioxide (SO2) which interacts with the atmosphere to produce hot sulfuric acid clouds that envelop the planet.
- About 80% of the surface of Venus is composed of flat plains of volcanic origin.
- Its rotation period is longer than its orbital period. (Rotation on its own axis - 243 days, Orbital period around the sun - 224.7 days)
- It has retrograde rotation, i.e., Venus spins on its axis from east to west.

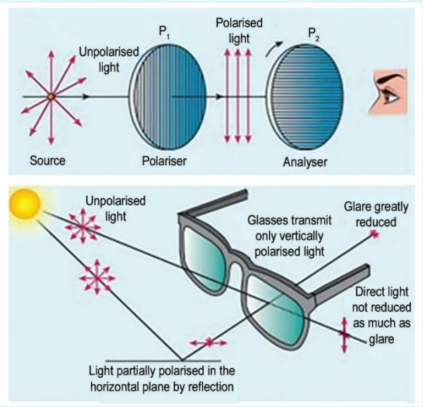
- Due to the slow rotation of Venus, it has no global magnetic field. (Earth's magnetic field is due to rotation of iron core)

30. X-RAY POLARIMETER SATELLITE (XPOSAT) MISSION

- India's first dedicated satellite to study the polarisation of cosmic X-rays and dynamics of bright astronomical X-Ray sources like pulsars, black holes and neutron stars.
- World's second polarimetry mission using X-Ray. The first is NASA's imaging X-ray Polarimetry Explorer launched in 2021.
- **Launched in:** 2024 (lifespan: 5 years)
- Designated for observation from LEO (~650 km altitude)
- **Carries two scientific payloads:**
  - **POLIX** (Polarimeter Instrument in X-Rays) will observe about 40 bright astronomical sources, and measure the degree and angle of polarisation, in the medium X-ray energy band (8-30keV).
  - **XSPECT** (X-Ray Spectroscopy and Timing) will observe a variety of sources like X-ray pulsars, black hole binaries, low-magnetic field neutron stars, active galactic nuclei, and conduct fast timing and high spectroscopic resolution in the soft X-ray energy band (0.8-15 keV).

WHAT IS POLARISATION?

- Light travels in the form of transverse electromagnetic waves. The light emitted by sources like the sun, bulb, candle etc. has vibrations in several planes (oscillates in all directions), and it is called unpolarised light.
- **Polarisation** is the phenomenon of restricting the vibration of light waves to one specific direction/plane. **E.g.,** Polarised sunglasses have a special filter that allows only light with a specific polarisation direction to pass through, which helps to reduce glare and improve visibility by blocking unwanted polarisations.



SIGNIFICANCE OF X-RAY POLARISATION MEASUREMENTS

- In space, X-rays get polarised due to multiple reasons. **E.g.,** X-rays can be polarised when subjected to strong magnetic fields or due to interactions with material present around black holes etc.
- Polarisation measurements of X-rays (angle and degree of polarisation) emanating from sources like magnetars, neutron stars, black holes etc. can provide information about the bright X-ray emitting sources (their composition, temperature, and density) and complex processes they undergo.
- Can help in deducing the orientation and strength of magnetic fields in celestial objects.
- Can help to understand the behaviour of high-energy particles in extreme environments.

31. NASA-ISRO JOINT SATELLITE – NISAR

- Joint active remote sensing satellite project between NASA and ISRO. Expected to launch in 2025.
- Low Earth Orbit earth observation satellite to be placed in sun-synchronous orbit (747 km).
- Can scan nearly all the planet's land and ice surfaces twice every 12 days to collect data.
- First radar imaging satellite to use **dual frequencies.** Consists of both L-band and S-band synthetic aperture radar (SAR) instruments.

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- **L-band SAR** operates at a frequency of around 1-2 GHz. Lower frequency (higher wavelength) of L-band SAR allows it to penetrate through vegetation and soil, making it useful for monitoring changes in forest cover, soil moisture etc.
- **S-band SAR** operates at a frequency of around 2-4 GHz. S-band SAR has a higher resolution than L-band SAR. Typically used for applications where higher resolution is required, such as monitoring changes in urban areas or coastal zones.

APPLICATIONS:

- Earthquakes and volcanology
- Distribution of carbon stocks in terrestrial biomass.
- Vulnerability of wetlands by studying the extent of inundation
- Geomorphology, movement of sea ice and hydrology
- Movement of glaciers and ice sheets thereby monitoring climate change.

32. SPADEX MISSION

- ISRO's SPADEX (Space Docking Experiment) was launched in 2024 onboard PSLV. The PSLV carried two satellites named Chaser (SDX01) and Target (SDX02), each weighing 220 kg.
- **Aim**: To demonstrate the satellite docking in Space (in a low-Earth circular orbit).
- The mission also leveraged PSLV's fourth stage (POEM-4) to host 24 scientific experiments.

SUCCESSFUL SPACE DOCKING BY ISRO:

- The two satellites were brought within a distance of 3 metres from each other in orbit, their extended ring was joined with each other, retracted, and locked in space (at an altitude of 470 kms). ISRO gave commands to the two satellites as one composite object. The satellites were successfully undocked (separated), later.
- India became the fourth country after the United States, Russia, and China to demonstrate such capability.
- **Significance:** Paves the way for smooth conduct of ambitious future missions including the Bharatiya Antriksha Station, Chandrayaan 4 and Gaganyaan.

33. PSLV ORBITAL EXPERIMENTAL MODULE (POEM)

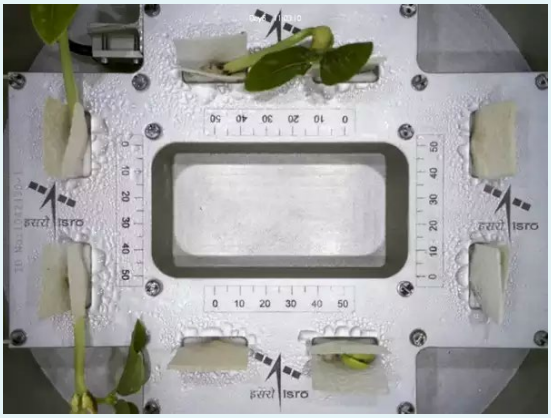
- POEM is an experimental mission or platform to perform in-orbit experiments using the final, and otherwise discarded, stage of ISRO's PSLV.
  - PSLV is a four-stage rocket where the first three spent stages fall back into the ocean, and the final stage (PS4) — after launching the satellite into orbit — ends up as space junk.
  - In POEM, the **spent final stage is utilised as a stabilised platform** to perform experiments.
- POEM has a dedicated Navigation Guidance and Control (NGC) system which will act as the platform's brain for attitude stabilisation with specified accuracy.
- POEM derives its power from solar panels mounted around the PS4 tank, and a Li-Ion battery. It will navigate using four sun sensors, a magnetometer, gyros & NavIC.

- **POEM 4.0** leveraged the fourth stage of PSLV-C60 mission (SpaDeX) to host 24 scientific experiments.
- **Compact Research module for Orbital Plant Studies** (CROPS) was one of the payloads.

CROPS EXPERIMENT

- **Aim:** To study the germination and growth of cowpea seeds in a controlled environment.
- CROPS is an automated platform designed to cultivate and research seed germination and plant survival in the microgravity environment of space.

- Eight cowpea seeds are placed in a regulated closed-box setting with precise temperature control and advanced monitoring equipment to observe plant development continuously.
- **Success:** The batch of cowpea seeds have sprouted and unveiled their first leaves.
- **Significance:** Advances development of space agriculture, essential for self-sufficient space habitats.



34. ANALOG SPACE MISSION

- ISRO launched its first analog or simulated space mission at Leh in Ladakh.
- The mission includes a compact, inflatable habitat named **Hab-1** designed to mimic environments on Mars and the Moon. The mission is spearheaded by ISRO's Human Spaceflight Centre.
- **Analog Space Missions** are field tests in locations that have physical similarities to the extreme space environments.
- Tests include: Testing new technologies, robotic equipment, vehicles, habitats, communications, power generation, mobility, infrastructure, and storage.
- Behavioural effects: such as isolation and confinement, team dynamics, menu fatigue etc. on the astronauts.
- **Significance:** Provide data about strengths, limitations, and the validity of planned human-robotic exploration operations.

35. RE-ENTRY OF SATELLITE AND SPACE DEBRIS

CONTROLLED RE-ENTRY ATTEMPT BY ISRO

- ISRO attempted the controlled re-entry of a decommissioned weather satellite (Megha Tropiques-1) to clear out space debris.
- After a series of 20 manoeuvres over eight months, the orbit of the satellite was lowered such that it re-entered the dense atmosphere of Earth and burnt up over the Pacific Ocean. (Upon re-entry, the atmospheric friction causes satellites to heat up to extremely high temperatures).
- Controlled re-entry is possible only for satellites in LEO. These manoeuvres, however, are not usually attempted because fuel reserves have to be maintained in the satellite after mission life is over.
- For attempting to bring down a satellite from geo-stationary or geosynchronous orbit, a huge fuel reserve would be needed, thus making the satellite heavier and costlier at launch.

SPACE DEBRIS: Human-made objects in space that are no longer in use, such as discarded rocket stages, defunct satellites etc. The debris can remain in orbit for years or decades, increasing the risk of collisions with active spacecraft.

KESSLER'S SYNDROME: Phenomenon in which the density of objects in the Low Earth Orbit grows so high that collisions between two objects could cause a massive cascade of space junk and trigger further collisions.

PROJECT NETRA:

- Initiative by ISRO, an early warning system in space to detect debris and hazards to Indian satellites and thus gaining capability in space situational awareness (SSA).

- NETRA will use telescopes, radars, data processing units, and other tools to spot and track objects as small as 10 cm, up to a range of 3,400 km.

36. IMPORTANT GLOBAL SPACE MISSIONS

S. No.	Space Missions	Description
1.	**Artemis Mission**	• Joint mission of NASA, ESA, JAXA and Canadian Space Agency. • **Aim:** To land the first woman and person of colour on the Moon, create a lunar base, and lay the groundwork for a future trip to Mars. • **Artemis 1:** An uncrewed test flight of the Orion spacecraft around the Moon (launched in 2022). • **Artemis 2:** Fly astronauts around the Moon and back (lunar flyby) without landing, planned in 2026. • **Artemis 3**: Land the first humans near the lunar south pole, planned in 2027. • **Artemis 4:** Aims to deliver a crew to the Lunar Gateway, a space station in lunar orbit in 2028.
2.	**Jupiter Icy Moons Explorer (JUICE) Mission**	• **Launched by:** European Space Agency in 2023. • Mission spacecraft to study Jupiter and its three largest icy moons — Europa, Ganymede, and Callisto. • Particular emphasis on Ganymede as a planetary body and potential habitat. • The spacecraft will arrive at Jupiter in 2031. • After a series of visits to Europa and Callisto, Juice will enter the orbit of Ganymede in 2034.
3.	**OSIRIS REx (Origins, Spectral Interpretation, Resources Identification and Security-Regolith Explorer)**	• Asteroid study and sample return mission by NASA. • **Launched in** 2016 to study asteroid Bennu, and returned to Earth with a sample for analysis in 2023. • Sample analysis may reveal clues about the birth of our solar system (4.5 billion years ago) and genesis of life on Earth. **About Bennu:** • Small Carbon-rich asteroid. More than 4.5 billion years old. • Little less than 500 metres in depth. Classified as a "near-Earth object" because it passes relatively close to Earth every six years. **#Breakthrough**: Samples collected from Bennu have amino acids and nucleobases (fundamental components of proteins and genetic material, respectively). It supports the hypothesis that asteroids can deliver life's building blocks to planets.
4.	**Psyche Mission**	• **Launched by:** NASA in 2023

		• To explore a 225-kilometre metallic asteroid called Psyche located between Mars and Jupiter. • Psyche is believed to be primarily composed of iron-nickel. • Studying it could provide invaluable insights into the composition of Earth's core. • The Psyche spacecraft also hosts NASA's Deep Space Optical Communications (DSOC) experiment.
5.	**Double Asteroid Redirection Test (DART)**	• Collaboration between NASA and the Johns Hopkins University Applied Physics Laboratory. • Successfully crashed a spacecraft into asteroid Dimorphos in 2022. • First ever test of a planetary defence technique involving a kinetic impactor to deflect an asteroid. • Demonstrated the capability of redirecting the trajectory of an asteroid by impacting it with a spacecraft.
6.	**PROBA** **(World's first precision formation flying mission)**	• **Initiative of:** European Space Agency; **Launched by**: ISRO. • **Aim:** To demonstrate precision formation flying between two satellites and study the Sun's corona (the outer layer of the Sun's atmosphere). • Placed in: Highly elliptical orbit that extends up to 60,000 km from Earth. • Proba-3 has two satellites- Coronagraph spacecraft and the Occulter spacecraft flying in a parallel formation at distances of 144 metres from each other for six hours a day. • The two satellites will work in tandem to study the Sun's corona and measure its energy output with unprecedented precision.
7.	**PUNCH (Polarimeter to Unify the Corona and Heliosphere)**	• NASA's solar mission to study the solar corona (the outermost layer of the Sun's atmosphere). • **Aim:** To better understand the solar atmosphere and solar wind. • PUNCH consists of four identical satellites in LEO that will continuously image the Sun's corona and provide origins of solar flares, etc. • The mission uses polarisation of light to study the Sun's corona in 3D. When particles (such as electrons) scatter sunlight, the waves of light become aligned in a particular direction (this is polarised light). • PUNCH will measure the light using polarising filters similar to polarising sunglasses, allowing scientists to make a 3D map of the corona and inner solar system.

37. PRIVATE SPACE EXPLORATION MISSIONS

AXIOM MISSION SERIES

- Privately-funded and operated crewed mission to the International Space Station **(ISS).**
- **Launched by:** Axiom Space (US) onboard SpaceX Crew Dragon spacecraft; Falcon 9 rocket.
- **Axiom Mission 1 (2022)**: World's first commercially crewed private spaceflight to ISS.
- Axiom Mission 2 (2023); Axiom Mission 3 (2024); Axiom Mission 4 (2025)

- **Shubhanshu Shukla (**IAF officer and ISRO astronaut) has been named the pilot for Axiom-4.

POLARIS DAWN MISSION

- 2024: Private human spaceflight mission onboard SpaceX Crew Dragon spacecraft; Falcon 9.
- The crew travelled further into space than any humans since the **Apollo 17 mission** (1972).
- Passed through parts of the Van Allen radiation belt to study the health effects of space radiation and spaceflight on the human body.

ODYSSEUS

- 2024: First commercial spacecraft to reach the Moon. Initiative of US-based **Intuitive Machines.**
- Short-lived as the spacecraft hard-landed on the slope of a crater, and damaged onboard instruments.

BLUE GHOST MISSION-1

- 2025: Uncrewed Blue Ghost lunar lander successfully **soft-landed on the Moon**'s Mare Crisium region (a prominent lunar basin visible from Earth).
- Part of NASA's Commercial Lunar Payload Services (CLPS) initiative.
- Initiative of US-based Firefly Aerospace. Launched using SpaceX Falcon 9 rocket.
- Has 10 scientific and technological payloads, and a lander to operate 1 lunar day.

Note:

- **Crew Dragon**: spacecraft manufactured by SpaceX as part of NASA's Commercial Crew Program. It can carry up to seven astronauts.
- **SpaceX Falcon 9 rocket**: partially reusable two-stage rocket that can propel both satellites and astronauts into a variety of orbits.

38. INTERNATIONAL SPACE STATION

- Modular space station (habitable artificial satellite) and the single largest man-made structure in LEO.
- **Launched** in 1998. Multinational collaboration involving five space agencies: NASA (United States), Roscosmos (Russia), JAXA (Japan), ESA (Europe), CSA (Canada) .
- It circles the Earth in roughly 92 minutes and completes 15.5 orbits per day, hosting a maximum of seven astronauts.
- Serves as a microgravity and space environment research laboratory in which scientific research is conducted in astrobiology, astronomy, meteorology, physics, and other fields.
- **Note:** ISRO is working to set up Bhartiya Antariksh Station by 2035.

39. IMPACTS OF SPACE TRAVEL ON ASTRONAUT'S HEALTH

- American astronauts Sunita Williams and Butch Wilmore have returned to Earth onboard the SpaceX Dragon spacecraft, after approximately nine months (286 days) from the ISS.
- The astronauts planned an eight-day mission aboard Boeing's Starliner spacecraft to validate the capability of Boeing's CST-100 Starliner in taking crewed missions to the ISS. Starliner spacecraft experienced several technical problems, including thruster malfunctions and Helium leaks in the thrusters.

FACTORS IN SPACE IMPACTING ASTRONAUTS

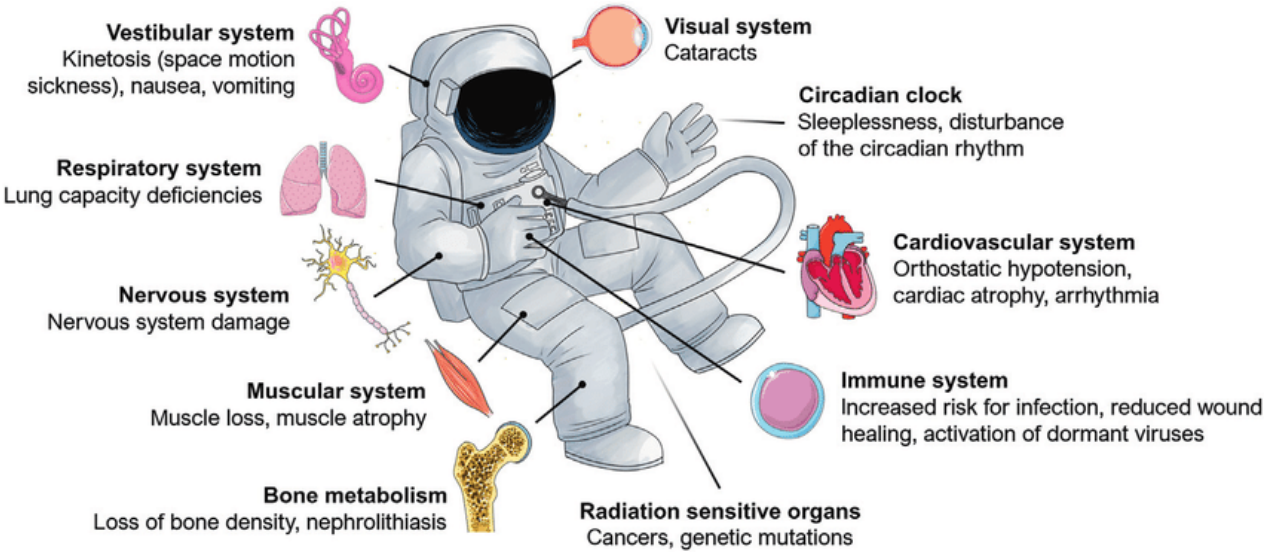
Space travel, particularly prolonged stay in space, poses significant challenges to human health.

- **Radiation exposure** (unlike earth's magnetic field that shields from space radiation, astronauts are exposed to high-energy radiation permeating the cosmos).
- **Microgravity** (without gravity bodily fluids shift upward and thus increase intracranial pressure).

- **Confined environments** (Isolation and limited social interaction and exposure to natural stimuli).
- **Time** (symptoms become more pronounced with increased stay duration in space).

IMPACTS INCLUDE

- Increase in DNA methylation and alteration in gene expression (the process of turning genes on and off).
- Risk of DNA damage, increased cancer risk, neurodegenerative effects and immune system dysregulation.
- Spaceflight-Associated Neuro-Ocular Syndrome (SANS): accumulation of fluid at the back of the eye resulting in facial swelling and vision loss.
- Bone and muscle degeneration; psychological stress, sleep disturbances, and mood disorders.



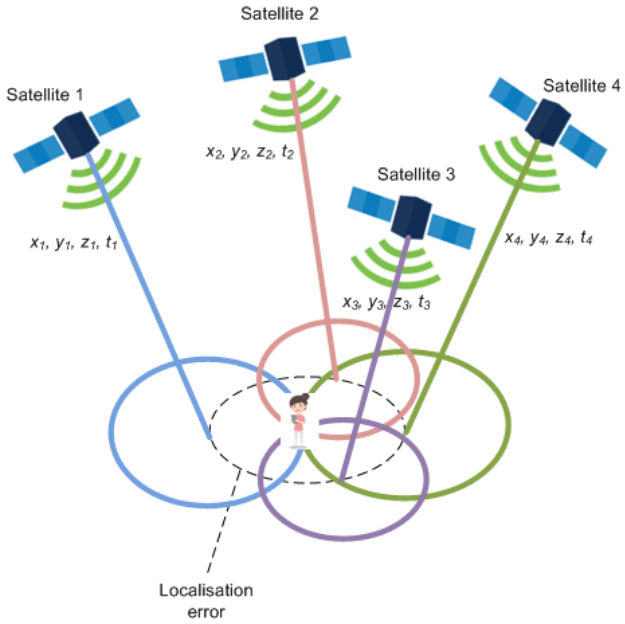
40. SATELLITE-BASED NAVIGATION SYSTEM

- A satellite-based navigation system consists of a network of satellites and receiving devices that allows users to determine their precise location (in latitude, longitude, and altitude) and obtain accurate time information anywhere on Earth.
- Presently, there are **four global satellite-based navigation systems.**
  - **United States:** Global Position System (GPS)
  - **Russia:** Global Navigation Satellite System (GLONASS)
  - **China:** BeiDou Navigation Satellite System (BDS)
  - **European Union**: Galileo
- **Japan** has a four-satellite regional navigation system (Quasi-Zenith Satellite System).
- **India** has its own Indian Regional Navigation Satellite System- NavIC (Navigation with Indian Constellation).

HOW DOES GPS WORK?

- GPS receiver picks up signals from multiple GPS satellites that are in view of the receiver's location.
- Each satellite sends out radio signals at the speed of light that includes information about the satellite's location and the precise time the signal was transmitted.
- GPS receiver measures the time it takes for the signals to reach it from each satellite.
- Using the known locations of the satellites and the time it took for the signals to travel, the GPS receiver can calculate the distance between itself and each satellite.

- With the distance measurements from at least four satellites (trilateration), the GPS receiver can determine an object's precise 3-D position (latitude, longitude, and altitude).
- A GPS receiver can also provide additional information such as speed, direction, and time by continuously updating its position based on the signals received from the satellites.

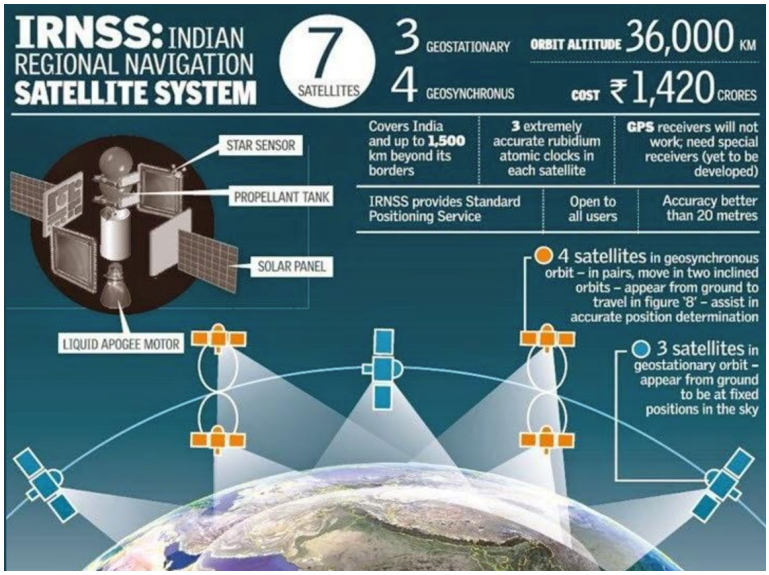


APPLICATIONS OF GPS (NAVIGATION SYSTEM) INCLUDE

- Terrestrial, aerial, and marine navigation **E.g.,** Location-based services in mobile devices, visual and voice navigation for drivers.
- Vehicle tracking and supply-chain management.
- Mapping, survey and geodetic measurements **E.g.,** Mapping of terrains, creating digital maps, and conducting land surveys, Terrestrial navigation aid for hikers and travellers.
- Precision Agriculture **E.g.,** Yield monitoring, variable rate application of fertilisers and pesticides.
- Meteorology **E.g.,** GPS receivers on weather balloons and satellites provide data on atmospheric conditions to monitor and predict weather patterns more accurately.
- Seismology: Monitor ground movements and tectonic plate activity; EWS; Disaster Management.
- Military Operations- GPS for navigation, target tracking, and coordination of operation.

41. NAVIGATION WITH INDIAN CONSTELLATION (NAVIC)

- NavIC (earlier known as Indian Regional Navigation Satellite System) is a satellite navigation system developed by ISRO.
- Consists of a constellation of 7 satellites (First-generation Satellites). Three satellites are located in the geostationary orbit and the remaining four are located in geosynchronous orbits.
- Can provide positioning, navigation, and timing services to users across India and the region extending up to 1,500 km around the country.
- Designed to provide position accuracy of better than 20 metres and timing accuracy of better than 50 nanoseconds.
- **NAVIC OFFERS TWO TYPES OF SERVICES:**
  - **Standard Positioning Service**: Available to all users and provides positioning accuracy of around 20 metres throughout the Indian region.
  - **Restricted Service**: Encrypted service primarily intended for authorised users such as the military, government agencies, and other security-sensitive applications.

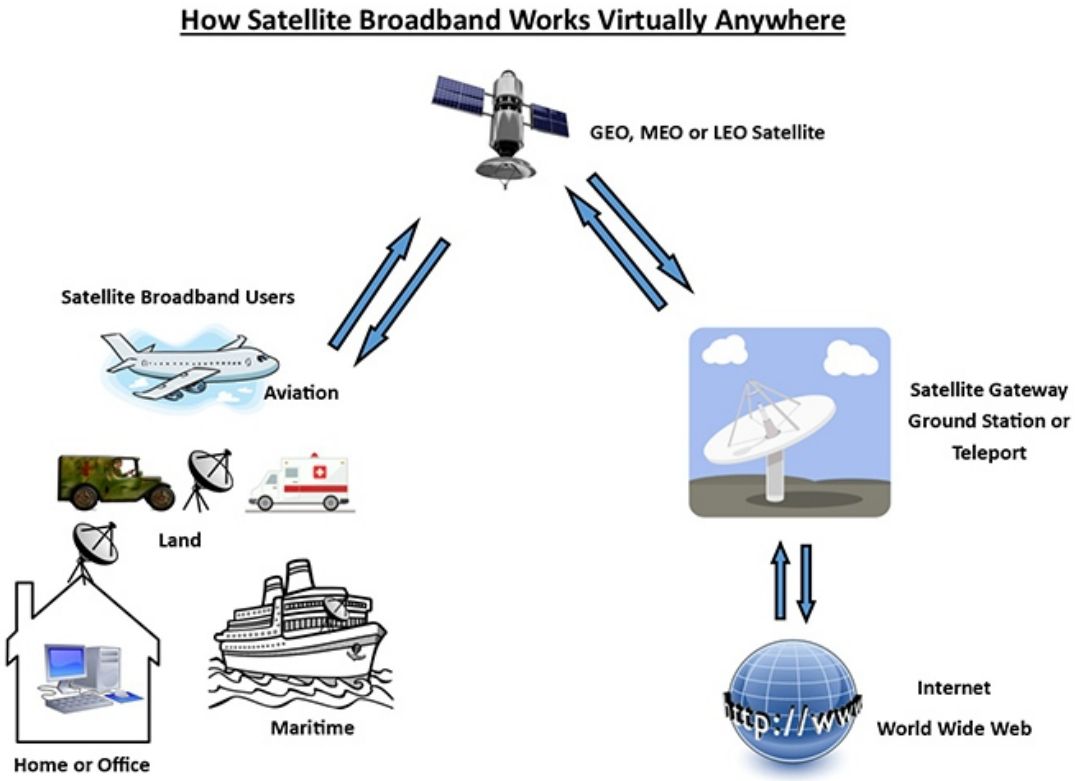


SECOND-GENERATION NAVIC SATELLITES

- ISRO launched the NVS-02 satellite (ISRO's 100th mission) aboard GSLV in January 2025. NVS-02 is the second satellite in the series of 2nd-generation navigation satellites, and the 9th satellite in NavIC.
- NVS-01 is the first 2nd generation satellite, launched in 2023.
- Existing satellites in NavIC use L5 (protected frequency specifically assigned to India) and S band frequency signals. 2nd-gen will send signals in a third frequency, L1, besides the L5 and S. This will enhance NAVIC's compatibility with other global navigation systems like GPS that also use L1.
- Rubidium atomic clock onboard with errors of <10 nanoseconds (accurate ranging for position determination).

42. SATELLITE INTERNET

- **Wireless internet** beamed down from satellites orbiting the Earth instead of optical fiber or cellular networks. **E.g.**, Starlink (SpaceX); Eutelsat (OneWeb).



S. NO.	ASPECTS	TERRESTRIAL BROADBAND INTERNET	SATELLITE INTERNET
1.	**Data Aggregation**	Happens on the ground (in base stations)	Happens in the space (via satellites)
2.	**Line of Sight Requirement**	No direct line of sight needed	Requires a clear line of sight to the satellite (via an antenna)
3.	**Latency**	Minimal latency	Higher latency due to time taken by signals to travel from space to Earth

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4.	**Coverage**	Most suitable for urban areas. High cost of deployment in remote/inaccessible areas.	Can cover remote and underserved areas.

43. STARLINK PROJECT

- Satellite network developed by the private spaceflight company SpaceX to provide **high-speed internet to remote locations.**
- **Location**: Starlink satellites are placed in altitude range between 350 km-1200 km in Low Earth Orbit.
- **Lifespan**: One satellite has a lifespan of approximately five years.
- **Constellation** consists of over 7000 satellites. SpaceX plans to have ~42,000 satellites.

FEATURES

- Starlink offers internet connectivity via a network of LEO satellites from space directly to the devices.
- Starlink satellites utilize an automated collision avoidance system to avoid crashing into other satellites in space.
- Satellites have an advanced end-of-life mechanism and ION propulsion system with argon thrusters. The satellites upon re-entering the Earth's atmosphere, disintegrate completely and burn.

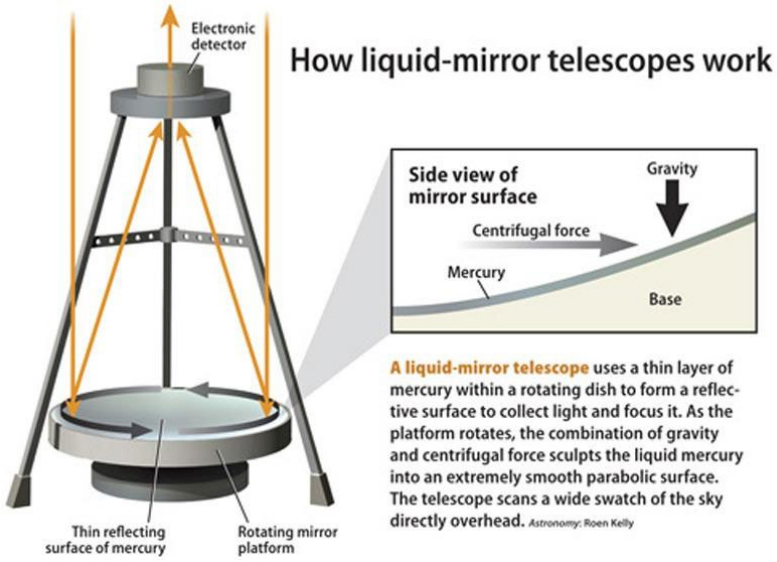
BENEFITS OF STARLINK

- Internet service to the remotest places; Lower latency (25-50 ms) compared to traditional geostationary satellite internet (600+ ms).
- Can be utilised by commercial fishing shippers, merchant vessels, off-shore oil drilling stations, deep sea research workers.
- Useful during natural disasters such as tsunamis, cyclones or earthquakes that can destroy physical infrastructure such as telecom towers.

Challenges: High cost; weather interference (cloud cover); privacy concerns; LEO satellites require larger constellations to provide global coverage.

44. IMPORTANT TELESCOPES

S. No.	Telescopes	Description
1.	**International Liquid Mirror Telescope**	• Asia's largest Liquid Mirror Telescope was inaugurated at Devasthal observatory in Uttarakhand. • Employs a 4-metre-diameter rotating mirror made up of a thin layer of liquid mercury, to collect and focus light. • Enhances India's capabilities to explore the mysteries of the deep celestial sky and astronomy.

		
2.	**Giant Metrewave Radio Telescope (GMRT)**	• **Location**: Near Pune, Maharashtra. • Array of 30 fully steerable parabolic radio telescopes, each 45 metres in diameter, spread over a 25-kilometre area. • One of the world's largest and most sensitive radio telescopes operating at metre wavelengths (low frequencies).
3.	**Square Kilometre Array (SKA) project**	• Group of radio telescopes being built in South Africa and Australia in two phases. The first phase is expected to begin operations by 2029. • Capable of detecting faint radio signals from extreme distances, with eventually over a square kilometre (one million square metres) of collecting area. • Will operate in two different ranges of radio frequency. ○ South-African array will scan for mid-frequency signals, between 350 MHz and 15.4 GHz. ○ Australian telescopes will work in the low-frequency range of 50-350 MHz. • India officially gained full membership in the SKA Project in January, 2024.
4.	**Thirty Meter Telescope (TMT)**	• International project which aims at building a 30-metre diameter telescope at Mauna Kea, Hawaii, USA. • TMT has a segmented mirror (made up of 492 individual segments) which makes it three times as wide as the world's largest existing visible-light telescope. • Precisely aligned, these segments will work as a single reflective surface of 30 m diameter. • Upon construction, TMT would be one of the world's most advanced and capable ground-based optical and infrared observatory. • Joint effort of institutions in Canada, China, India, Japan, USA.

5.	**Euclid Space Telescope (EST)**	• Launched in 2023 by the European Space Agency. • **Location:** Lagrangian Point 2 (1.5 million km away from Earth) • To create the largest cosmic 3D map of the universe to better understand the distribution of dark matter and reveal the influence of dark energy in the early universe. • The European Space Agency has unveiled the first images captured by EST. (picture of a group of thousands of galaxies)
6.	**Event Horizon Telescope (EHT)**	• Network of 8 radio telescopes located in Hawaii, Arizona, Chile, Mexico and Spain, and at the South Pole. • It is synchronised in such a way that, in effect, they form a radio telescope of the size of the earth itself. • In 2019, the 1st ever picture of a black hole was captured by EHT. • In May 2022, the supermassive blackhole at the centre of the Milky way galaxy was captured.
7.	**Hubble Space Telescope**	• **Location:** Low Earth orbit (540 km) • Large optical/ultraviolet telescope for deep sky observations • **Launched:** 1990 by NASA and ESA.
8.	**James Webb Space Telescope (JWST)**	• **Launched:** 2021 by NASA. • **Location:** Orbiting the Sun at the L2 Lagrange point (1.5 million km from Earth) • Largest infrared telescope ever built (6.5 metre primary mirror). • Detects near-infrared and mid-infrared wavelengths to observe faint and distant objects.
9.	**SPHEREX (Spectro-Photometer for the History of the Universe, Epoch of Reionisation, and Ices Explorer)**	• Telescope equipped with infrared detectors capable of distinguishing 102 colors invisible to the human eye. • **Location**: Sun-synchronous Low Earth Orbit • **Mission:** To create a 3D map the entire sky in optical as well as near-infrared light. Provide data on over 450 million galaxies and more than 100 million stars in the Milky Way. • Understand the formation and evolution of galaxies, investigate rapid expansion of the universe (cosmic inflation), and search for water and organic molecules in interstellar clouds across the Milky Way.

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3 CHAPTER NUCLEAR TECHNOLOGY

1. BASICS OF NUCLEAR ENERGY

- An atom is made up of a Nucleus (with Protons and Neutrons) and electrons revolving around the nucleus.
- Mass of the atom is concentrated in the nucleus.
  - The number of protons (Z) in an atom determines the atomic number of an element.
  - The total number of Protons and Neutrons, called Nucleons, is called the Atomic Mass Number (A).
- Atoms can be stable or unstable, and the instability is caused by the neutrons in their nucleus.
- All elements with atomic numbers greater than 83 are considered radioactive, including Uranium (92), Plutonium (94), and Thorium (90).
- These elements have unstable atomic nuclei, and over time, they decay by releasing energy and radiation to reach a more stable configuration.
- They release energy in the form of three types of radiations: **alpha, beta, and gamma.**

THREE TYPES OF RADIATIONS

Feature	Alpha (α) Radiation	Beta (ß)Radiation	Gamma (γ) Radiation
Composition	Helium nucleus (2 protons, 2 neutrons)	Beta-minus (electron). Beta-plus (positron)	High-energy electromagnetic wave (photon)
Charge	Positive (2+)	Beta-minus (negative –), Beta-plus (positive +)	Neutral (no charge)
Mass	Largest and heaviest	Lighter than alpha, heavier than gamma	No mass (pure energy)
Penetrating Power	Lowest	Medium	Highest
Stopped by	Paper, skin	Aluminum sheet	Concrete, human tissue
Ionising Power	High	Medium	Low

RADIOACTIVE HALF-LIFE

- For a given radioisotope, the radioactive half-life is the time for half the radioactive nuclei in any sample to undergo radioactive decay.
- After two half-lives, there will be one fourth the original sample, after three half-lives one eighth the original sample, and so forth.

2. APPLICATIONS OF RADIOACTIVITY

1. **Radioisotope Thermo-electric Generator:** A radioactive material is used which when decays produces heat. This heat is in turn used by a generator to produce electricity. **E.g.,** the New Horizon spacecraft which went to Pluto uses this kind of device.
2. **Medical Imaging:** Radioactive isotopes are used in medical imaging techniques such as X-rays, CT scans, PET scans, and MRI scans.
3. **Radiation Therapy:** Radioactive isotopes are used to treat various types of cancer through radiation therapy.

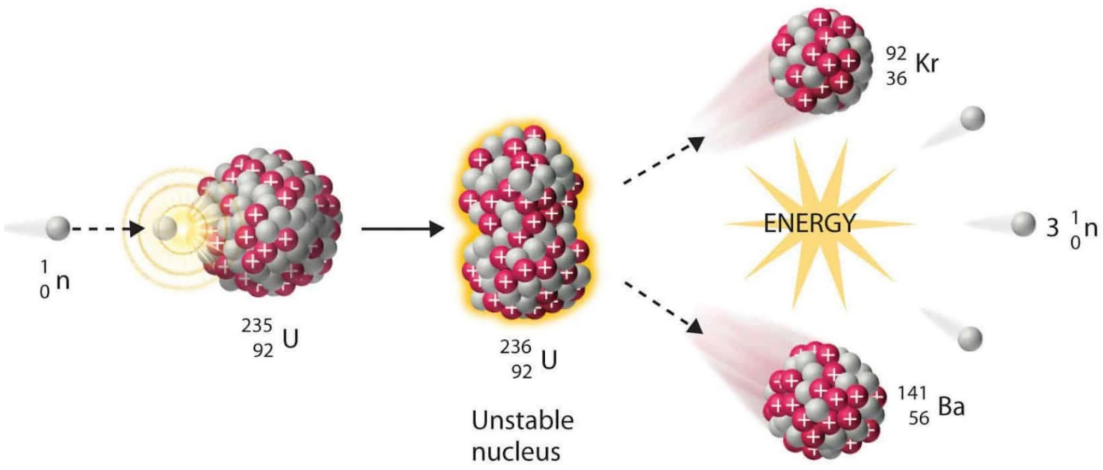
4. **Smoke Detectors:** Smoke detectors use a small amount of radioactive material to detect smoke and trigger an alarm.
5. **Industrial Radiography:** Radioactive isotopes are used in industrial radiography to test the integrity of metal structures such as pipelines and oil rigs.
6. **Carbon Dating:** Radioactive isotopes are used in carbon dating to determine the age of ancient fossils and artefacts.
7. **Nuclear Power:** Radioactive isotopes are used to generate electricity in nuclear power plants.
8. **Food Irradiation:** Radioactive isotopes are used to sterilise and preserve food products, preventing spoilage and disease.
9. **Geological Dating:** Radioactive isotopes are used to determine the age of rocks and minerals in geology.
10. **Sterilisation:** Radioactive isotopes are used to sterilise medical equipment, surgical instruments, and other devices to prevent the spread of infection.

RADIOACTIVE DATING

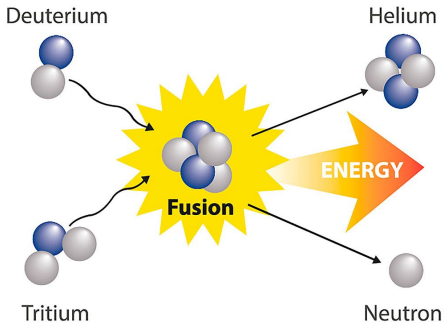
- Carbon-14 is produced in the atmosphere due to the interaction of cosmic rays with Nitrogen-14.
- We all have a certain amount of C-14 in our bodies (23% of the human body) as do plants and all living beings.
- Each gram of radioactive C-14 in our bodies decay at the rate of 12 per minute. (Means it becomes 1/2, 1/4, 1/8, 1/16 etc. every 12th minute.) The decayed C-14 is further replaced with atmospheric C-14 as we eat.
- When an organism dies, the remaining C-14 gradually turns into Carbon-12.
- **C-14 dating** measures the ratio of C-14 to C-12 to determine how many years have passed since an organism died.
- Carbon-14 dating can be used to determine/estimate the age of organic remains between 500 years and approximately 60,000 years.

3. NUCLEAR PROCESSES

- **Nuclear Fission:** It involves disintegrating a heavy atom's nucleus, such as uranium or plutonium, into two or more smaller nuclei. **E.g.,** A high-energy neutron collides with the nucleus of a U-235 atom, causing it to split into two smaller nuclei, additional neutrons (usually 3) and enormous energy.



- **Nuclear Fusion:** Process of combining two or more nuclei to form a heavier nucleus and release high-energy radiation. This process occurs naturally in stars, where the high temperature at the core allows for the nuclei to overcome the repulsive electromagnetic force and fuse together. **E.g.,** Combination of hydrogen nuclei to form helium, releasing energy in the form of neutrinos, gamma rays, and positrons.



Parameter	Nuclear Fission	Nuclear Fusion
Definition	Splitting of a large atom into smaller ones	Fusing of two or more tighter atoms into a larger one
Natural occurrence	Spontaneous fission of some elements occurs at a slow rate	Occurs in stars
By-products	Highly radioactive particles	Neutrons (can induce radioactivity)
Conditions	High-speed neutrons required	High density & high temperature
Energy Requirement	Relatively low	Extremely high
Energy Released	Millions of times more than chemical reactions	Three to four times more than fission (can vary depending on isotopes)
Nuclear weapon	Fission bomb {atomic bomb)	Fusion bomb (hydrogen bomb) uses fission to trigger fusion
Energy production	Used in nuclear power plants (current technology)	Experimental technology for power production
Fuel	Uranium (primary fuel)	Hydrogen isotopes (Deuterium and Tritium)

4. NUCLEAR FUEL: FISSILE AND FERTILE

FISSILE MATERIAL: READY FUEL FOR FISSION

- Fissile isotopes readily undergo fission when hit by neutrons, releasing a tremendous amount of energy. **E.g.,** Uranium-235 (U-235), Uranium- 233 (U-233) and Plutonium-239 (Pu-239).
- However, fissile materials like U-235 make up a small portion of Natural Uranium (less than 1%). Naturally occurring Uranium is composed of three major isotopes:
  - **Uranium-238** (99.284% natural abundance)
  - **Uranium-235** (0.711%)
  - **Uranium-234** (0.005%)
- U-235 atoms is the only nuclide existing in nature (in appreciable amount) that is fissile with thermal neutrons.

URANIUM ENRICHMENT

- Since naturally occurring Uranium does not have a high enough concentration of U-235, Uranium enrichment is necessary to create an effective nuclear fuel out of mined Uranium.

- In enriched Uranium, the percent composition of U-235 has been increased through the process of isotope separation.
- Enriched Uranium is a critical component for both civil nuclear power generation and military nuclear weapons.
  - Typically, Uranium used for power generation is enriched to levels between 3% and 5% U-235.
  - Weapon-grade uranium is considered to have been enriched above 90% U-235.

FERTILE MATERIAL: BREEDING POTENTIAL

- Though not directly fissile by slow neutrons, fertile isotopes can absorb neutrons and transmute into fissile materials through radioactive decay.
  - **Nuclear transmutation** often refers to the conversion of one nuclide into another within the fuel or target material. This process can involve the absorption of a neutron, the emission of a neutron, or the capture and subsequent decay of a particle.
- **Common fertile materials** include Uranium-238 (U-238) and Thorium-232 (Th-232).
  - Uranium U-238 cannot by itself make the reactor achieve criticality, so it has to be converted to fissile plutonium (Pu-239) in a nuclear reactor.
  - Thorium Th-232 is a fertile material that has to be converted to fissile U-233.

CRITICALITY OF NUCLEAR REACTOR

- **Nuclear reactor is said to be critical** when the number of neutrons produced in Nuclear fission reactions equals the number of neutrons lost through absorption, leakage, and other processes. This equilibrium state allows for a sustained and controlled chain reaction.
- **Supercriticality** refers to a state where the rate of neutron production exceeds the rate of neutron losses, causing the overall neutron population to increase rapidly. This can lead to an uncontrolled chain reaction and a dangerous power surge.

5. KEY COMPONENTS OF A NUCLEAR REACTOR

NUCLEAR FUEL

The core of a nuclear reactor houses the fuel (typically enriched U-235) pellets contained in long, thin metal tubes called fuel rods.

MODERATOR

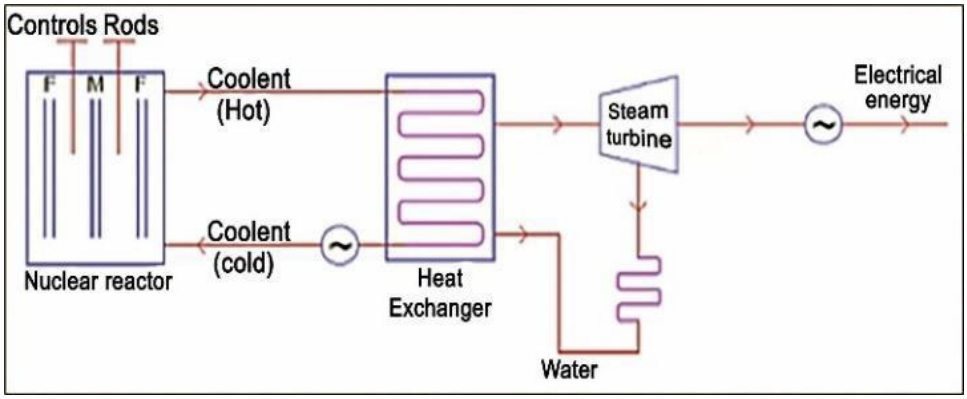
- Located within the reactor core, the moderator slows down neutrons released from fission.
- These neutrons are initially high-energy (fast neutrons). For a sustained chain reaction, neutrons need to be slowed down to thermal neutrons, which are more likely to cause fission in other U-235 atoms.
- **E.g.,** Ordinary water (light water reactors), heavy water (deuterium oxide), and graphite.

CONTROL RODS

- Control rods are made of materials that readily absorb neutrons.
- They are inserted or withdrawn from the core to control the rate of fission (regulate the number of neutrons available to cause fissions) and the power output of the reactor.

COOLANT

- Coolant circulates throughout the reactor core, absorbing immense heat produced by the ongoing nuclear chain reaction to prevent the reactor from overheating, which could lead to a meltdown. **E.g.,** Water, Liquid Sodium.



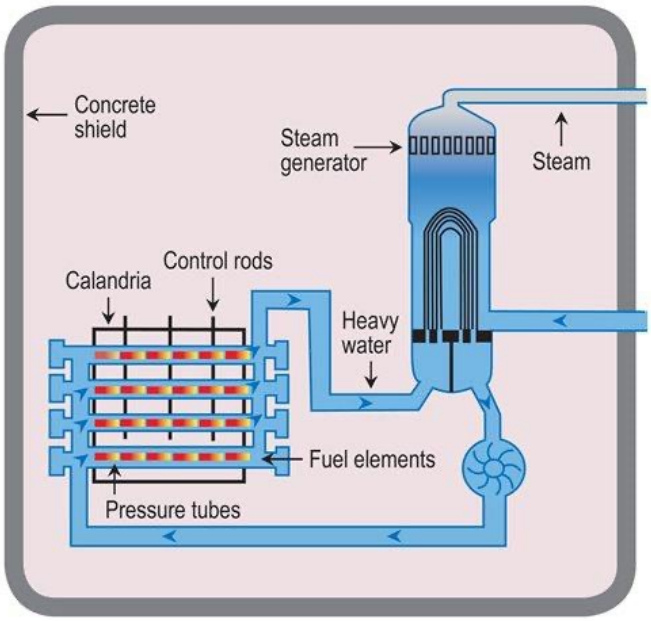
The heat generated by the chain reaction of nuclear fission produces steam. The steam is used to move a turbine which generates electricity.

Note: Presently, India operates 24 nuclear reactors with an installed capacity of 8180 MW. These include- 20 Pressurised Heavy Water Reactors (PHWRs) and 4 Light Water Reactors. NPCIL has 8 more reactors under construction with a total capacity of 6800 MW.

➔ IMPORTANT TYPES OF NUCLEAR REACTORS IN INDIA

6. PRESSURISED HEAVY WATER REACTOR (PHWR)

- **Fuel:** Natural uranium (unenriched)
- **Moderator and Coolant:** Heavy water as both moderator and coolant.
- **Cooling System:** Uses a combination of heavy water and light water to cool the reactor. Heat is transferred to a secondary loop, which then generates steam to drive turbines.
- **Control Rods:** Boron or cadmium control rods.
- **Fuel requirement:** Annual requirement of fuel (UO2) of a 700 MW PHWR (at 85% Capacity Factor) is about 125 tons.
- **Advantages:** Use natural uranium fuel, produce less high-level radioactive waste, and operate at lower pressures compared to some other reactor types.



7. LIGHT WATER REACTOR (LWR)

- **Fuel:** Low-enriched Uranium

- **Moderator and Coolant:** Light water (ordinary H2O)
- **Cooling System:** Employs a pressurised water system to transfer heat from the reactor core to a secondary loop, which generates steam for turbines.
- **Control Rods:** Boron or other neutron-absorbing materials.
- **Fuel requirement:** Annual requirement of fuel (UO2) of a 1000 MW LWR (at a capacity factor of 90%) is about 25 tons.
- **Advantages:** Have a proven track record for safety and reliability. However, they require enriched uranium and produce more high-level radioactive waste compared to PHWRs.
- LWRs include both Boiling Water Reactors (BWRs) and Pressurised Water Reactors (PWRs/ VVERs).

8. FAST BREEDER REACTOR (FBR)

- Prototype Fast Breeder Reactor (PFBR) is being implemented by Bharatiya Nabhikiya Vidyut Nigam Limited (BHAVINI).
- **Fuelled by**: Mixed Oxide (MoX) Fuel (It generally contains between 3% and 5% Plutonium Oxide blended with 95–97% natural or depleted Uranium Oxide).

9. SMALL MODULAR REACTORS (SMR)

- Advanced nuclear reactors with a power generation capacity ranging from less than 30 MW to 300 MW.
- They are characterised by smaller size, modular design, and enhanced safety features.
  - **Small** – a fraction of the size of a conventional nuclear power reactor.
  - **Modular** – possible for systems and components to be factory-assembled and transported as a unit to a location for installation.
  - **Reactors** – harnessing nuclear fission to generate heat to produce energy.
- **Applications:** Electricity generation in remote regions, energy requirements for industrial processes, water desalination, nuclear submarines etc.

Advantages:

- Adaptable: can be scaled up or down to supply more or less power.
- Only need to refuel every 3-7 years, as opposed to every 1-2 years for conventional nuclear plants.
- Extensive use of passive safety features to shut down and cool reactors under abnormal circumstances, reducing the risk of catastrophic failures.
- Lower-capital requirements, can complement renewable energy sources and stabilise the grid.

Challenges:

- Higher cost per unit of electricity production in SMRs due to supply-chain issues and the absence of economies of scale.
- SMRs are inferior to conventional reactors with respect to radioactive waste generation and disposal options and require spent fuel storage & disposal facilities.

10. THREE-STAGE NUCLEAR PROGRAMME OF INDIA

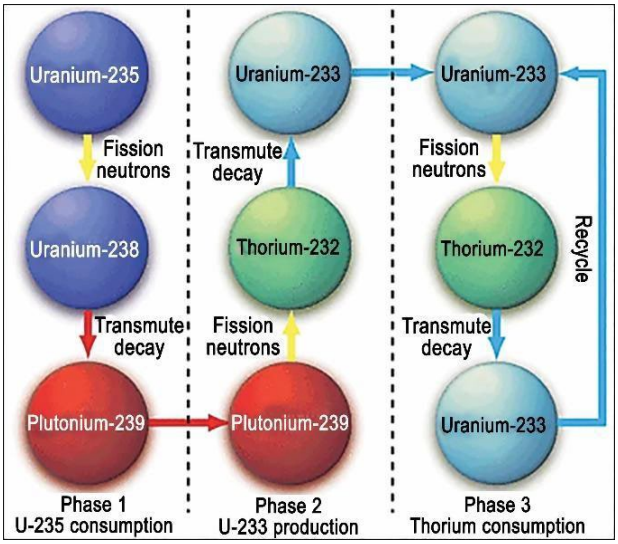
- The Department of Atomic Energy's 3-stage nuclear programme is a strategy for a sustainable, continuous, reliable, and self-dependent supply of nuclear energy.
- The programme is developed with the aim of utilising the vast Thorium reserves of India. India has a limited availability of Uranium reserves (about 2% of global Uranium reserves).
  - India has about 25% of global thorium in the form of monazite sands along its coastline.

- They are found in coastal and inland placer sands on the beaches of Kerala, Tamil Nadu, Odisha, Andhra Pradesh, Maharashtra, and Gujarat, and in the inland riverine sands of Jharkhand and West Bengal.

1ST STAGE: PRESSURISED HEAVY WATER REACTORS (PHWRS)

- The first stage includes the setting up of PHWRs and associated fuel cycle.
- Based on natural Uranium that contains 99.3% U-238 and 0.7% U-235.
- U-235 being fissile sets up the chain reaction and U-238 being non-fissile gets converted to Pu-239 as a by-product (spent fuel) which is in turn used in the Fast Breeder Reactors in the 2nd stage.

2ND STAGE: FAST BREEDER REACTORS (FBR)



- FBR uses the Uranium-Plutonium Mixed Oxide (MOX) fuel.
- The reactors use a mix of Plutonium-239 (byproduct/radioactive waste produced in the 1st stage) and Uranium-238, as part of their MOX fuel.
  - The U-238 surrounding the fuel core undergoes nuclear transmutation to produce more fuel (Pu-239).
  - Since the amount of Plutonium produced inside the reactor is more than that initially used, it is called the Breeder reactor.
- The reactor uses fast neutrons without the need for a moderator to slow them down. Fast neutrons are more efficient at inducing fission in certain isotopes, such as plutonium-239.
- Further, FBR can aid in conversion of fertile Thorium 232 to fissile Uranium 233 that will be used in the 3rd stage.

Significance of FBR:

- Significant reduction in nuclear waste generated as FBR utilises/reprocesses the radioactive waste (Plutonium-239) as fuel from the first stage.
- Advanced reactor with inherent passive safety features ensuring a prompt and safe shut down of the plant in the event of an emergency.
- Both the capital cost and the per unit electricity cost is comparable to other nuclear and conventional power plants.
- Stepping stone for the third stage of India's Nuclear Programme.

3RD STAGE: THORIUM-BASED REACTORS

- The third stage is focussed on building Advanced Heavy Water Reactor to utilise India's vast Thorium reserves.
- In the thermal breeder reactors, Thorium-232 will be converted into Uranium-233, by transmutation.

To showcase the practicality of this phase, a 40 KW Thorium based KAMINI reactor is already operational in Kalpakkam, Tamil Nadu.

CIVIL LIABILITY FOR NUCLEAR DAMAGE ACT (CLNDA), 2010

- Aim: To provide a quick compensation mechanism for victims of a nuclear accident.
- The Act establishes a **strict and no-fault liability** for nuclear plant operators, meaning they are liable for damage regardless of fault.
- However, the operator of the nuclear installation, after paying the compensation for nuclear damage shall have the **right to recourse** where-

- The nuclear incident has resulted as a consequence of an act of supplier or his employee, which includes supply of equipment or material with patent or latent defects or sub-standard services.
- The nuclear incident has resulted from the act of commission or omission of an individual done with the intent to cause nuclear damage.

- The operator will have to maintain a financial security to cover its **maximum liability of ₹1,500 crore** for civil nuclear damage and requires the operator to cover liability through insurance or other financial security.
- In case the **damage claims exceed ₹1,500 crore,** the gap will be bridged by the Central Government. The government liability amounts to the rupee equivalent of 300 million Special Drawing Rights (SDRs) or about ₹2,100 to ₹2,300 crore.

ATOMIC ENERGY ACT, 1962

- The Act provides for the development, control and use of atomic energy for the welfare of the people of India and for other peaceful purposes.
- Nuclear Power Corporation of India (NPCIL) holds the authority for activities related to nuclear energy, including its production, development, use, and disposal. The Act restricts private companies from owning and operating nuclear power plants in India.
- The 2015 amendment to the Act, allows NPCIL to form joint ventures with other public sector units (PSUs) to secure funding for new projects. However, this does not extend to private or foreign companies.
- Currently, private companies can participate in specific areas like supplying components and reactors, but not owning or operating plants. Discussions are ongoing about allowing Public-Private Partnerships (PPPs). This would require amendment to the Act.

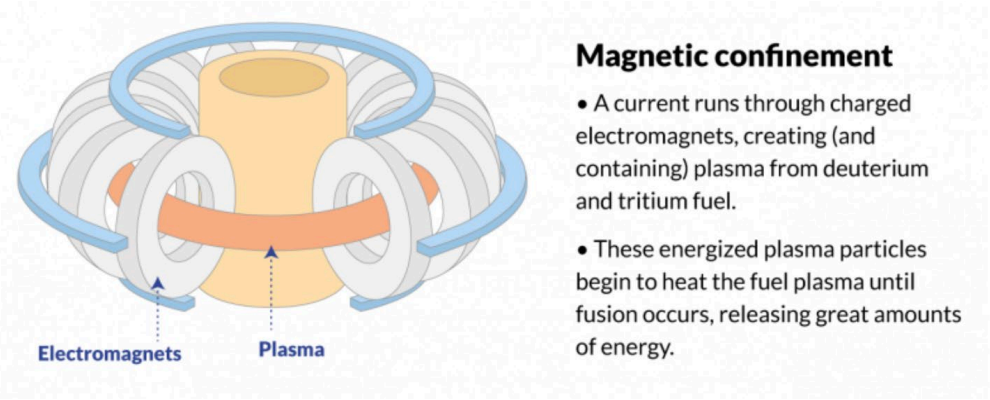
11. FUSION REACTORS

- Fusion reactors is increasingly seen as the future of energy security due to following factors:
  - Abundance of fuel (Hydrogen in the form of water in oceans).
  - Clean source of energy as it involves no release of carbon dioxide.
  - Elimination of risk from nuclear waste.
- Two main factors to achieve fusion reaction are fuel and conditions for fusion.
  - A typical fusion reactor uses hydrogen as a fuel that is abundant in the water of the oceans.
  - However, the main problem in fusion is that the hydrogen nuclei repel each other.
- The electric repulsion of two hydrogen nuclei can be overcome by heating the hydrogen to temperatures of millions of degrees C. This is what happens in a typical hydrogen bomb.
- However, the challenge for building a fusion reactor is that such high temperatures leads to high pressure posing the problem of explosion. This problem of explosion is currently being addressed in three ways:
  1. The first is to make the hydrogen work at a very low density, so the pressure will not get high. This is the approach used in the **Tokamak approach**.
  2. The second method is to let the hydrogen explode, but to keep the explosions small. This is done in the **laser method**.
  3. The third way to achieve fusion is by keeping the hydrogen cold. This is called **cold fusion**.

12. TOKAMAK APPROACH FOR MAGNETIC CONFINEMENT

- Tokamak is a specific type of **magnetic confinement device**. It utilises powerful magnetic fields to confine extremely hot plasma, keeping it stable and away from the physical contact with reactor walls.

- Inside the Tokamak, light isotopes of hydrogen (Deuterium and Tritium) are heated and compressed to undergo nuclear fusion to release enormous amounts of energy.



13. INTERNATIONAL THERMONUCLEAR EXPERIMENTAL REACTOR (ITER)

- ITER is the largest fusion reactor that works on the basis of the Tokamak approach, launched in 1985.
- **Location:** Saint-Paul-les-Durance, southern France.
- **Objective:** To demonstrate the scientific and technological feasibility of controlled fusion for future commercial power generation.
- ITER is a joint collaboration of 35 countries, members include- India, China, Japan, South Korea, Russia, United States and European Union.
- ITER is designed to produce 500 MW of fusion power from 50 MW of input heating power.
- According to its current timeline, it would begin deuterium-tritium fusion reactions by 2039, producing 500 MW of fusion power.

14. EXPERIMENTAL ADVANCED SUPERCONDUCTING TOKAMAK (EAST)

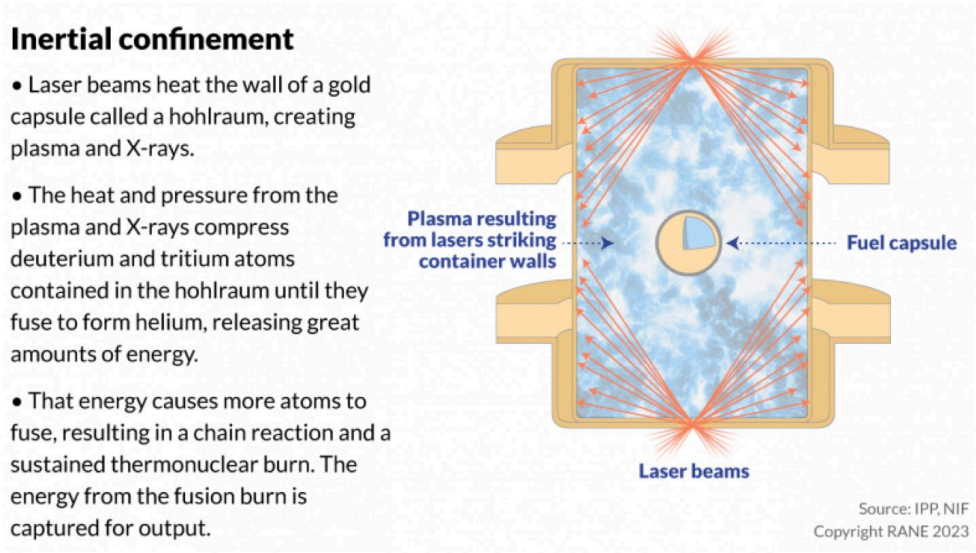
- EAST is an experimental nuclear fusion reactor located in China that works on the Tokamak approach.
- The reactor was able to sustain/maintain its operational state for more than 1,066 seconds, or over 17 minutes, which is a new record. Real-life electricity-generating fusion reactors would require this state to be maintained for days at a stretch.
- **Significance:** Maintaining plasma in a steady state of confinement is a major step towards realising a fusion-based nuclear reactor.

15. LASER-IGNITED FUSION

- Laser-ignited fusion, or inertial confinement fusion, uses powerful lasers to heat and compress fuel pellets, creating conditions for nuclear fusion reactions.

WORKING

- A tiny capsule containing isotopes of hydrogen, deuterium (D) and tritium (T), is suspended inside a cylindrical x-ray "oven" called a hohlraum.



- The hohlraum is heated by powerful lasers to temperatures of >3 million degrees Celsius. This generates X-rays that ablate (heat and blow off) the surface of the target capsule.
- This causes a rocket-like implosion that compresses and heats the fuel to extreme temperatures and densities until the hydrogen atoms fuse, creating helium nuclei (alpha particles) and releasing high-energy neutrons and other forms of energy.
- **National Ignition Facility** (NIF), the US, is the world's largest and most energetic laser system.

16. COLD FUSION

- Traditional hot fusion requires extremely high temperatures (millions of degrees Celsius) to overcome the electrostatic repulsion between positively charged atomic nuclei (protons).
- Cold fusion proposes achieving fusion at much lower temperatures (possibly room temperature and pressure) by creating an environment where this repulsion is somehow negated.

PROPOSED MECHANISM

- It suggests that introducing a negatively charged particle to a hydrogen nucleus (proton) could neutralise its positive charge.
- This would allow two hydrogen nuclei to get close enough for the attractive nuclear force to overcome the remaining repulsion and initiate nuclear fusion.
- Metals like palladium, zirconium, and nickel, known as "hydrogen absorbers" or "soakers," can potentially facilitate this process.

Click here for INDEX

4 CHAPTER INTERNET AND COMMUNICATION TECHNOLOGY

1. SEMICONDUCTORS

A semiconductor material has an electrical conductivity value falling between a conductor (such as metallic copper) and an insulator (such as glass). Lattice structure and atomic structure of constituent elements decide whether a particular material will be insulator, metal or semiconductor.

CLASSIFICATION OF SEMICONDUCTORS

I. Based on material:

(i) Elemental semiconductors: Silicon (Si) and (Ge)

(ii) Compound semiconductors:

a. Inorganic: CdS, GaAs, CdSe, InP etc.

b. Organic: Anthracene, Doped pthalocyanines etc.

c. Organic polymers: Polypyrrole, Polyaniline, Polythiophene etc.

Most of the currently available semiconductor devices are based on elemental semiconductors Silicon or Germanium (Ge) and compound inorganic semiconductors.

However, after the 1990s, a few semiconductor devices using organic semiconductors and semi-conducting polymers have been developed.

II. Based on purity:

(i) **Intrinsic semiconductor**: They are pure semiconductors with no impurities. They have no or zero conductivity at very low temperatures. However, as temperature rises, the conductivity of these materials increases.

(ii) **Extrinsic semiconductor:** When a small quantity of impurity is added to pure semiconductor, the conductivity of the semiconductor is increased manifold. The deliberate addition of a desirable impurity is called doping and the impurity atoms are called dopants.

SHAKTI-BASED SEMICONDUCTOR CHIP

- IIT-Madras and ISRO have developed an indigenous aerospace quality Shakti-based semiconductor chip.
- The chip is named **IRIS** (Indigenous RISC-V Controller for Space Applications).
- It is based on the SHAKTI microprocessor. SHAKTI systems use RISC-V (an open-source processor technology) and are supported by the Ministry of Electronics and Information Technology under the 'Digital India RISC-V' (DIRV) initiative.
- IRIS is the third SHAKTI chip successfully fabricated in India, after RIMO in 2018 and MOUSHIK in 2020.
- Utility: IRIS can be used for applications on launch vehicles, ground stations and Internet of Things (IoT) and industrial IoT applications.

DIRV INITIATIVE

Aim: To promote indigenous development of microprocessor-based products that offer best-in-class security and visibility for users adopting RISC-V technology.

RISC-V

- RISC-V is an open standard Instruction Set Architecture (ISA) based on established RISC principles. Each computer hardware will support a particular ISA.
- Unlike most other ISA designs, RISC-V is provided under open-source licenses that do not require fees to use. RISC-V can be extended or customized for a variety of hardware or application requirements.
- India has developed two series of microprocessors: SHAKTI series of microprocessors by IIT Madras & VEGA microprocessors by C-DAC.

2. LIGHT-EMITTING DIODE

- LED is an electric component or a semiconductor device that emits light when an electric current flows through it.
- LEDs can produce all three primary colours (red, green, and blue) and different LEDs can be combined on a display board to produce a large variety of colours.

ADVANTAGES

- More efficient than incandescent bulbs and fluorescent lamps, as they convert a higher percentage of electrical energy into light.
- Compact and can be made in very small sizes.
- Much longer lifespan and durability compared to conventional bulbs.
- Light up instantly to their full brightness without any warm-up time, and are particularly useful in applications where quick and frequent switching is required.

3. MICROLED DISPLAY TECHNOLOGY

- MicroLEDs are self-illuminating diodes that have brighter and better colour reproduction than Organic Light Emitting Diode (OLED) display technology.
- MicroLEDs are made up of numerous microscopic LEDs which self-illuminate per display pixel i.e., they produce colours without backlighting or colour filters. It is called a MicroLED panel because its pixels are very small and measured in micrometres.
- While OLEDs use tiny sub-pixels made from organic emissive materials, Micro-LEDs use an inorganic LED structure.

ADVANTAGES OVER OLED

- Increased brightness and better colour reproduction
- Longer lifespan
- Lower power consumption
- Low response time
- Limitless scalability as they can be freely resized in any form.

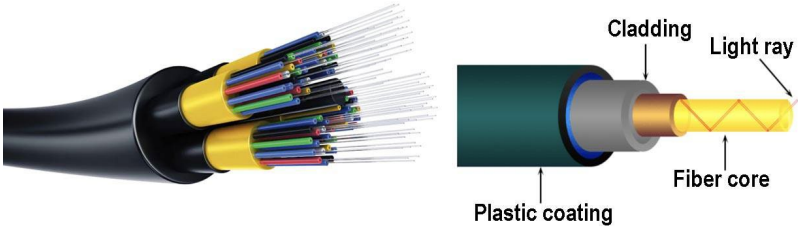
DISADVANTAGE

High costs and technical difficulties in its development.

4. OPTICAL FIBRES

- Thin, flexible, and transparent fibre made of glass (high quality silica) or plastic.
- It is used to transmit information, such as text, images, videos, etc. encoded as digital information or electromagnetic waves/light pulses almost at the speed of light.

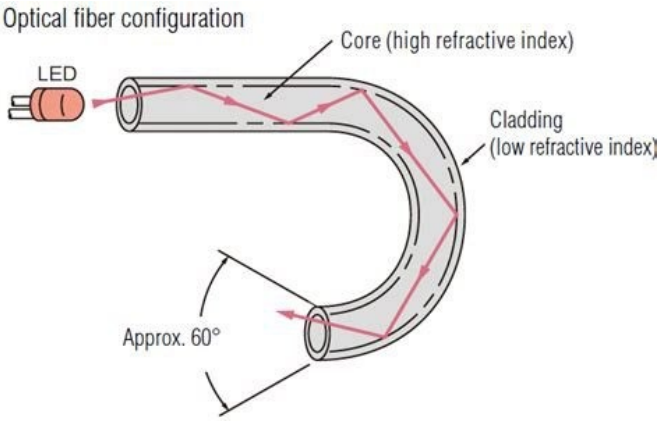
- Optical fibres utilise the phenomenon of total internal reflection for transmission.



Structure:

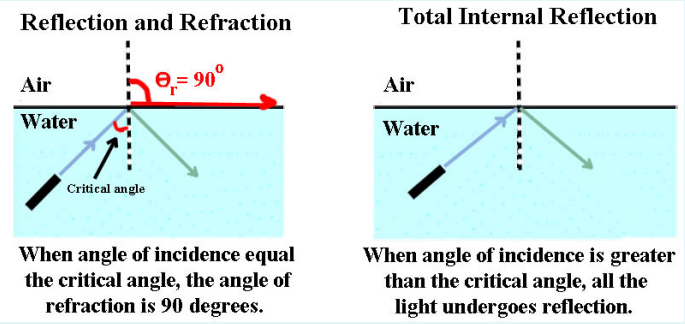
An optical fibre consists of:

- **Core**, which is the innermost part where light travels.
- **Cladding**, which surrounds the core and helps contain the light within the core by reflecting it back into the core.
  - Core and cladding are usually made of materials with different refractive indices, allowing for the total internal reflection of light within the core.
  - The refractive index of the core is higher (denser medium) than the refractive index of the cladding (rarer medium).



TOTAL INTERNAL REFLECTION (TIR)

- TIR occurs when light travelling from a rarer medium to a denser medium, strikes the boundary of the medium at an angle greater than the critical angle.
- If the angle of incidence is greater than the critical angle, all the light is reflected back into the original medium; none of it is refracted into the second medium.
- This phenomenon is crucial in optical fibres, where it allows light signals to be transmitted over long distances by repeatedly undergoing TIR within the core of the fibre.



Advantages:

- Long distances with minimal loss of signal quality (low signal attenuation).
- Optical waves have high bandwidth, thus allowing a high data-transmission rate, up to several terabits per second in a single fibre.
- Fibre cables are immune to electromagnetic interference and less sensitive to external perturbations such as lightning and bad weather, unlike copper cables.

Applications:

- **Telecommunications**: Used to connect telephone lines, cable television systems, and internet service providers.
- **Medical imaging**: Used in a variety of medical imaging applications, such as endoscopy, ultrasound, and laser surgery.
- **Sensors:** Used in a variety of sensors, such as temperature sensors, pressure sensors, and chemical sensors.
- **Lighting:** Used in a variety of lighting applications, such as decorative lighting, stage lighting, and outdoor lighting.
- **Military:** Used in a variety of military applications, such as communication, surveillance, and targeting.

5. INTERNET

- **Internet:** Global network of interconnected computers and devices that use standardised communication protocols to exchange information.
- **Internet Service Providers** (ISPs): Companies or organisations that provide access to the Internet. They offer various types of connections such as dial-up, broadband, and wireless.
- **World Wide Web (WWW):** An information system on the Internet that allows users to access and navigate through web pages using hyperlinks.

EVOLUTION OF WORLD WIDE WEB

THREE STAGES OF INTERNET CONSUMPTION			
	Web1	**Web2**	**Web3**
Time period	**1990-2005**	**2005-till date**	**2021 - under development**
Where data is stored	Server's file system	On-premises/Cloud	Blockchain, distributed across multiple networks
Examples	Static web pages	User generated content like Social media, and web applications like e-commerce etc.	NFTs, cryptocurrency transaction
Who owns data	Companies running the web pages	Companies that host application, cloud service providers	No one owns the data
Transacting	No transaction possible	Payment gateways for currency transactions	Transaction happens using crypto tokens

- **Web 1.0 [1990 – 2000]:** It is regarded as the first generation of the World Wide Web. Also known as Syntactic web or read only web. Mostly, Web 1.0 was limited to searching info and reading what's already there. There was little user interaction or content contribution. It was disorganised and overwhelming, and soon it came to be dominated by AOL, CompuServe, early Yahoo and other portals. These online service providers were gateway to Web 1.0. (HTML 1.0)
- **Web 2.0 [from mid-2000s]:** This phase was characterised by enhanced user experience and made the internet interactive. Also known as Social Web or read-write web. It enabled users to participate in content creation on social networks, blogs, sharing sites and more. Search engines (Google) and social media platforms (Facebook, Twitter) driven by user-generated content disrupted media, advertising and retail industries. Web 2.0's business model relies on user participation to create fresh content and resultant data being sold to third parties for marketing. Facilitated by HTML 2.0.
- **Web 3.0 [yet to arrive]:** Next stage of web evolution. It would make the internet more intelligent, or process information with near-human-like intelligence through the power of AI systems.

6. WEB 3.0

- Web 3.0 is the next version of the internet where services will run on blockchain.
- It is a decentralised internet that runs on a public blockchain, which is also used for cryptocurrency transactions.
- It will be permissionless and democratic. **For instance**: Twitter will not be able to censor posts and Facebook will not be able to maintain a database of billions of users.
- In a Web 3.0 universe, people will control their own data and will be able to move around from social media to email to shopping using a single personalised account, creating a public record on blockchain of all that activity.
- All data will be interconnected in a decentralised way, unlike current generation of internet (Web 2.0), where data is mostly stored in centralised repositories.
    - Key features of Web 3.0 are: Ubiquity, Semantic Web, Artificial Intelligence and 3D Graphics.
    - Examples of Web 3.0: Most recent examples of Web 3.0 are the NFTs or non-fungible tokens.

NEED FOR WEB 3.0

- **Loss of privacy:** Data is stored in servers of companies that people interact with. Intermediaries become custodians of user data and profit from. For such companies, more time consumers spend creating content, more data the company can collect, helping it to improve its AI algorithm and its advertising engine, a key revenue model for the company. This gives rise to issues of privacy, wherein user data is shared for profit without their consent.
- **Data ownership:** Only centralised repositories are the ones that own user data and profit from it. In Web 3.0, users can own and be properly compensated for their time and data.
- **Plagiarism:** It is very easy to copy original content and build a following around it on social media. Plagiarism makes it harder for creators to get compensated. Web3 might help address that issue as the transparent nature of blockchain makes it easy for anyone to track the originator of content.

KEY DIFFERENCES BETWEEN WEB 2.0 AND WEB 3.0

- Any information that users share on Web 2.0 is stored with a **cloud service provider** whereas in Web3, all services are built on top of a **blockchain**.
- The cloud is controlled by internet giants and is centralised. In blockchain, data is distributed across networks and no single entity owns the information.

METAVERSE AND WEB 3.0

Metaverse is about creating digital avatars and interacting with others in virtual spaces. It does not have to be on a blockchain. The whole point of Web 3.0 is decentralisation.

7. DARK WEB

The total web content on the internet is broadly classified into three broad categories: Surface web, Deep web and Dark web.

SURFACE WEB

- Usual search engines such as Google, Yahoo & Bing etc. can 'look for' and extract content and present it in the form of a website/webpage. For this, web pages are 'indexed' by search engines.
- Only about 10-15% of web content is present on surface web accessible by common searches.

DEEP WEB

- Term used for all those content or web pages that are there on the internet but are not indexed by search engines and therefore not discernible by conventional search engines.

- Thus, webpages on the deep web do not show up in conventional search engines like Google, Yahoo, and Bing etc. About 75-80% of the web content/web pages are on the deep web.
- Common examples of web content on the deep web include financial data, bank account details, emails, personal data etc. that are password protected.

DARK NET/DARK WEB

- Part of the deep web that is intentionally hidden to provide anonymity.
- **Key features:**
  - No webpage indexing by surface web search engines.
  - Virtual traffic tunnels via randomised network infrastructure.
  - Inaccessible by traditional browsers due to its unique registry operator.
  - Further hidden by various network security measures like firewalls and encryption.
- To do this, Dark Net uses a specialised network of computers called relays through which the information passes. Commonly, information on DarkNet passes through at least 3 relay computers between the source and destination.
- In addition, dark net uses network technology that hides the locations of these relay computers (IP address) to ensure anonymity of the users.
- Dark net can be accessed through TOR Browser (Anonymity Network).

THREATS OF DARK WEB

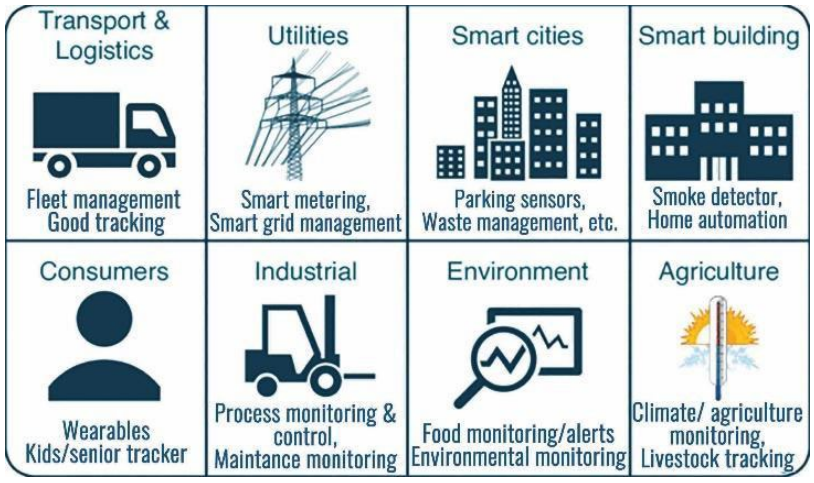
- **Is using the dark web legal?** Using DarkNet is per say is not illegal. By virtue of its ability to provide anonymity, dark net is used by human right activists, free internet activists, media personnel etc. in countries where there are severe restrictions, censorship on internet usage like that in China, Iran, Saudi Arabia etc. Also used by whistleblowers to maintain anonymity.
- **Threats of Dark Web:** Malicious software, Scam, Identity theft monitoring, Illegal sales of COVID-19 vaccines, Drug trafficking, Bitcoin laundering, Drug trafficking

8. INTERNET OF THINGS (IOT)

- IoT refers to a network of physical devices embedded with sensors, software, and other technologies that connect and exchange data with other devices and systems over the internet or other communication networks.
- These devices can communicate and interact with other devices over the Internet, and they can be remotely monitored and controlled by computers and smartphones.

APPLICATIONS OF INTERNET OF THINGS

- **Agriculture:** Sensing for soil moisture & nutrients, controlling water usage for plant growth & determining custom fertiliser are uses of IoT.
- **Energy utilisation**: Smart Grids will be able to detect sources of power outages, can automatically take inputs of solar panel, making possible distributed energy system.
- **Smart cities:** Cellular communication enabled Smart municipal bins, Parking sensors etc.



- **Healthcare:** Personalised analysis of an individual's health and tailor-made strategies to combat illness will be possible. Enhanced patient monitoring and better health outcomes.
- **Manufacturing:** IoT intelligent systems enable rapid manufacturing of new products, dynamic response to product demands, and real-time optimisation of manufacturing production and supply chain networks, by networking machinery, sensors, and control systems together.
- **Environmental monitoring:** Assist in environmental protection by monitoring air or water quality, atmospheric or soil conditions. It can even include areas like monitoring the movements of wildlife and their habitats.
- **Supply chain:** By placing RFID tags on individual products, the exact location of single items in a large warehouse can be shared, thus saving search time, streamlining infrastructure, and lowering labour costs.

9. EVOLUTION OF MOBILE NETWORKS

- Mobile phones communicate through ground-based cellular networks. Cellular networks are divided into 'communication cells' with which our mobile phones and mobile devices communicate.
- Mobile communication involves transmitting voice or data using wireless radio transmission.
- The first mobile systems were based on analogue transmission called 1G.
- The second-generation mobile systems were based on digital transmission.
- Initially, only voice was carried over the network.The commonly used standards for voice communication were GSM and CDMA.
- These days, most mobile communications use Long Term Evolution communication– or LTE which allows us to communicate with voice and data simultaneously over the same network.
- The rules for carrying voice or data in a network are defined under the standards for mobile network communications often seen as 2G, 3G, 4G and 5G.

1G

- Mobile phones began with 1G technology in the 1980s.
- 1G is analog technology that supports only voice communications.
- The maximum speed is 2.4 Kbps.

2G

- First launched in 1991.
- For the 1st time radio signals became digital rather than analog.
- 2G phones are used for data also along with voice.
- Thus, 2G telephone introduced call and text encryption, SMS, picture messages, and MMS.
- Maximum speed under 2G networks with General Packet Radio Service (GPRS) was 50 kilobits per second.
- With Enhanced Data Rates for GSM Evolution (EDGE) the speed went up to 1mbps

GPRS

- Launched in 2000
- Bridge between 2G and 3G.
- It marks the coming of data transmission besides voice communication.
- General Packet Radio Service enabled mobile devices to send and receive emails and pictures.
- GPRS used EDGE and GSM standards for both voice and data transmission.
- GPRS had operating speeds of up to 115 kbit/s.
- It increased to a maximum of 384 kbit/s by using EDGE.

3G

- The introduction of 3G networks in 1998, ushered in faster data-transmission speeds.
- Maximum speed of 3G is estimated to be around 2 Mbps for non-moving devices and 384 Kbps in moving vehicles.
- Further a 3G phone cannot communicate through a 4G network, but a 4G phone can communicate through a 3G or even 2G networks.
- Under both 2G to 3G technologies, data and voice transmission over the different networks using GSM or CDMA technology.

4G

- 4G uses LTE which allows us to communicate with voice and data simultaneously over the same network.
- Applications include amended mobile web access, IP telephony, gaming services, high-definition mobile TV, video conferencing, 3D television, and cloud computing.
- Max speed of a 4G network when the device is moving is 100 Mbps or 1 Gbps for low mobility communication like when stationary or walking.

	1G	2G	3G	4G
Period	**1980-1990**	**1990-2000**	**2000-2010**	**2010-(2020)**
Bandwidth	150/900MH2	900MHz	100MHz	100MHz
Frequency	Analog signal (30 Khz)	1.8GHz (digital)	1.6-2.0 GHz	2-8 GHz
Data rate	2kbps	64kbps	144kbps-2Mbps	100Mbps-lGbps
Characteristic	First wireless communication	Digital	Digital broadband, increased speed	High speed, all IP
Technology	Analog cellular	Digital cellular (GSM)	CDMA, UMTS, EDGE	LTE, WiFi

10. 5G TECHNOLOGY

- Fifth generation (5G) of long-term evolution (LTE) mobile broadband networks is the most recent update.
- The next generation cellular technology will provide faster and more reliable communication with ultra-low latency (Latency is gap time or transmission time for a packet of data).
- The Steering Committee constituted for identifying the 5G deployment roadmap for India recently submitted a report titled 'Making India 5G Ready'. As per a government panel report with 5G data speed would be 2-20 Gbps.

FEATURES OF 5G

- **High data rates** (1Gbps for hotspots, 100Mbps download and 50Mbps upload for wide-area coverage)
- **Massive connectivity** (1 million connections per square kilometre)
- **Ultra-low latency** (1 millisecond)
- **High reliability** (99.999% for mission critical 'ultra-reliable' communications).
- **Mobility at high speeds** (up to 500 km/hr i.e., high-speed trains).

APPLICATIONS IN INDIA

- Will enhance infrastructure efficiencies like **'vehicle platooning'.** Platooning can double vehicle density in roads promoting efficient and safer use of the limited road infrastructure.

- In manufacturing, 5G will enable use of robotics for precision manufacturing.
- 5G can also enable better logistics to track goods from raw materials to product delivery.
- In agriculture, 5G can enable improvement in the entire value-chain, from precision farming, smart irrigation, improved soil and crop monitoring to livestock management.
- In the energy sector, 'smart grids' and 'smart metering' can be efficiently supported enabling growth of alternate energy technologies.
- In healthcare, 5G can enable more effective tele-medicine delivery, tele-control of surgical robotics and wireless monitoring of vital statistics.
- 5G will be used in key government projects such as smart cities and Digital India.

Feature	4G (LTE)	5G (Current Status)	6G (Vision)
Deployment Status	Widely deployed	Limited rollout (October 2022 launch)	Under development (Bharat 6G project initiated)
Speed	Up to 100 Mbps (typical)	Up to 1 Gbps (limited areas)	Up to 1Tbps (theoretical)
Latency	50-100 milliseconds	10-20 milliseconds (limited areas)	Under 1 millisecond (theoretical)
Spectrum Availability	Below 6 GHz bands	Below 6 GHz bands (limited), mmWave auctions awaited	Below 6 GHz bands, mmWave, Terahertz (future)
Applications	Basic mobile broadband, social media, video streaming	Early adoption for enhanced mobile broadband, potential for future VR/AR	Focus on R&D for ultra-high-speed data, mission-critical applications (long term)

5G SPECTRUM

5G primarily operates in three bands, namely the low, mid, and high-frequency spectrums, each of which has its own set of benefits and drawbacks.

LOW BAND SPECTRUM

- It has a lot of potential in terms of coverage and internet and data transfer speed; however, the maximum speed is just 100 Mbps (Megabits per second).
- The low band spectrum may not be ideal for specialised requirements of the business; thus, Telcos may utilise and deploy it for commercial cell phone customers who may not have need for extremely high-speed internet.

MID-BAND SPECTRUM

- It has faster speeds than the low band, however it has restrictions in terms of coverage area and signal penetration.
- This band might be utilised by companies and specialised production units to create captive networks that can be tailored to their specific demands.

HIGH-BAND SPECTRUM

- It has the fastest speed of the three bands, but its coverage and signal penetration intensity are severely restricted.
- Internet speeds in the 5G high-band spectrum have been tested to reach 20 Gbps (gigabits per second), although the greatest internet data speed in 4G has been reported at 1 Gbps in most circumstances.

MILLIMETRE (MM) WAVE BAND

- The millimetre wave band, or mmWave, is a portion of the radio frequency spectrum that spans 24 to 100 GHz.

- As the name implies, this spectrum has a short wavelength and is more likely to give higher speeds and shorter latencies.
- Data transport becomes more efficient and smoother, as existing networks are optimised for lower frequency bandwidths.

SIGNIFICANCE OF THIS MM BAND

- Lower frequency bands may be used to deliver 5G services.
- They can reach longer distances and have been demonstrated to perform well even in congested metropolitan areas.
- However, when it comes to data rates, these bands fall short of the maximum capacity required for a real 5G experience.

So, mmWave is that key component in the 5G jigsaw puzzle for mobile service providers.

11. WI-FI AND LI-FI

Wi-Fi (Wireless Fidelity) and Li-Fi (Light-Fidelity) are two different wireless communication technologies that enable data transmission, i.e., offer a way to connect to the internet without physical cables.

S. No.	Parameters	Wi-Fi	Li-Fi
1.	**Medium of transmission**	Relies on radio waves to transmit data wirelessly between devices	Transmits data using visible light or ultraviolet light. LEDs act as transmitters, and a photodetector receives the data.
2.	**Range**	Typically up to 32 metres	Typically around 10 metres
3.	**Frequency**	Radio frequency spectrum, between 2.4 GHz and 5 GHz	Visible light spectrum or UV light spectrum
4.	**Advantages**	Wider range	Faster speed (upto 100 Gbps theoretically), high security
5.	**Disadvantages**	Susceptible to radio-frequency interference causing latency	Shorter range, Requires a direct line of sight communication between the transmitter and receiver.

12. WIFI 6.0

- Next generation standard in WiFi technology. Also known as “AX WiFi."
- It was built in response to the growing number of devices in the world of IoT (Internet of Things) due to faster data transmission rates.

WI-FI CERTIFIED 6 NETWORKS ENSURE

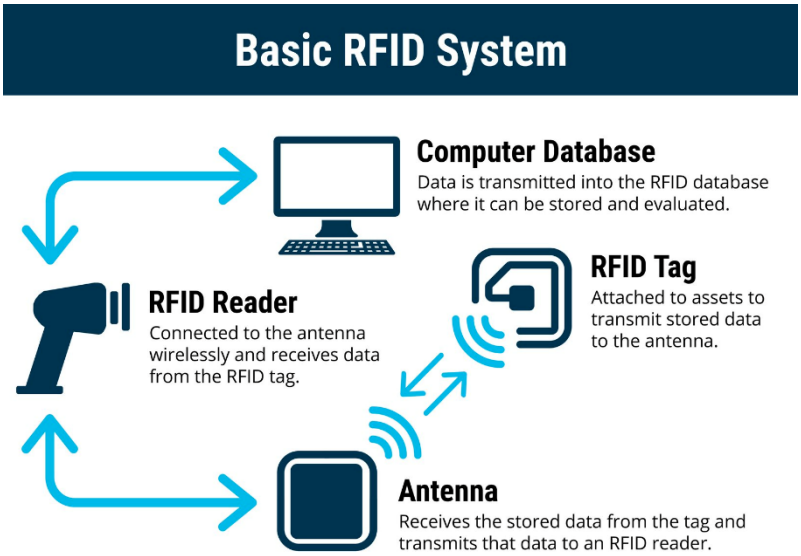
- Each connected device performs at an optimum level, in locations with hundreds or thousands of connected devices.
- Highest standards for security and interoperability.
- Lower battery consumption.
- Increased bandwidth to deliver greater performance with lower latency.

13. NEAR FIELD COMMUNICATION TECHNOLOGY

- A short-range wireless connectivity technology that allows NFC-enabled devices to communicate with each other and transfer information quickly with a single touch.
- NFC transmits data through electromagnetic radio fields between two devices typically requiring a distance of **4 cm or less.**
- Both devices must be equipped with NFC technology and be physically contacting or within a few centimetres of one another for data transmission.
- **Applications:** Contactless banking cards, contactless public transportation tickets, wireless charging, Vending machines, Parking metres, NFC-enabled wristbands (E.g., track patient data in the hospital setting), Inventory and sales monitoring etc.
- **Limitations of NFC:** Can work only on limited distances. Offers exceptionally low information or data transfer rate of a few Kbps.

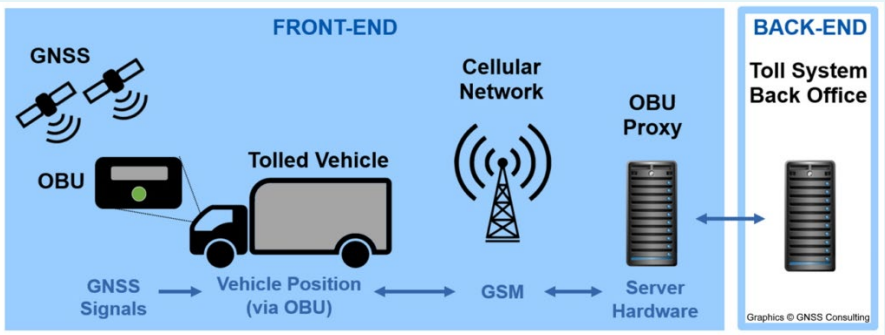
14. RADIO FREQUENCY IDENTIFICATION (RFID) TECHNOLOGY

- RFID technology refers to a wireless system comprising two components: **tags and readers.**
- **A reader** is a device that has one or more antennas that emit radio waves and receive signals back from the RFID tag. Readers can be mobile so that they can be carried by hand, or they can be mounted on a post or overhead.
- **RFID Tag** is embedded on the windscreens of the vehicles for transporters. It can store a range of information from one serial number to several pages of data. The tags use radio waves to communicate their identity and other information to nearby readers.



SATELLITE-BASED TOLL COLLECTION

- The central government has plans to implement satellite-based toll collection on highways in India in 2024.
- This will replace the **radio frequency identification-based FASTag system** that was rolled out in 2016 and made mandatory from January 2021.



- **Significance**: Efficiency in toll collection and barrier-free movement of vehicles.

Working Mechanism:

- **On-board unit:** Every vehicle registering for the Global Navigation Satellite System (GNSS) of toll collection will be equipped with an on-board unit (OBU) that will be linked to a constellation of satellites via a GPS receiver. The OBU transmits the vehicle's location and identification information to the toll collection system. The OBU will also be linked with a wallet from which the toll amount will be automatically deducted.
- **Satellite tracking:** Satellites track the movement of the vehicle and determine the distance travelled on tolled roads.
- **Toll calculation:** The toll collection system calculates the toll based on the actual length of the highway traversed by the vehicle and the vehicle's class (**E.g.,** car, truck).
- **Automatic payment:** Toll amount is automatically deducted from the driver's prepaid account or linked credit card.

15. FREE SPACE OPTICAL COMMUNICATION

- Free Space Optical (FSO) communication refers to the transmission of optical signals through free space, such as the atmosphere or vacuum, using line-of-sight (LOS) communication between transceivers.
- It is typically used for short-range communication links, such as within buildings or across relatively short distances.

ADVANTAGES

- FSO communication offers a high data rate to meet the tremendous increasing demand of broadband traffic mostly driven by Internet access and HDTV broadcasting services.
- Compared to fibre optics technology, FSO offers much more flexibility in designing optical network architectures at very high speeds, at tens and hundreds of Gbit/s rates.
- Both point-to-point, point-to-multipoint, multipoint-to-point, and multipoint-to-multipoint FSO communications are possible, depending on the different scenarios of establishing optical links.

LIMITATIONS

- However, FSO communication is affected by atmospheric effects (fog, rain, turbulence), which limits sensitivity and achievable data rates with acceptable BER (Bit Error Rate).

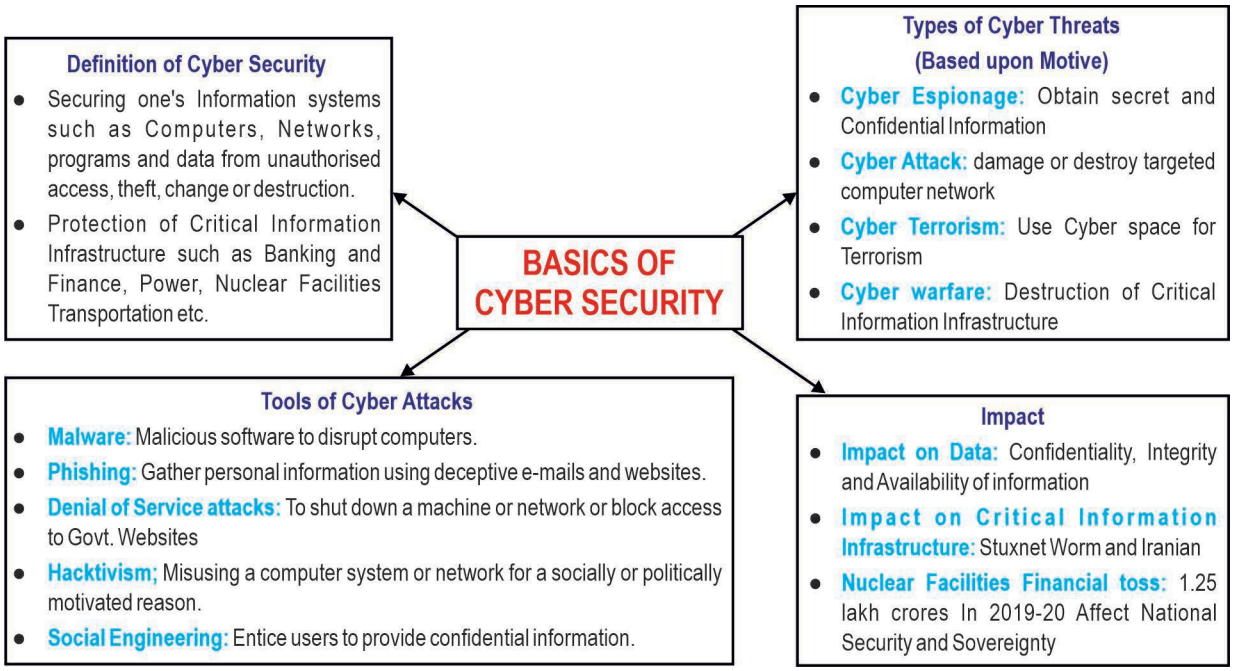
16. DEEP SPACE OPTICAL COMMUNICATION

- DSOC is NASA's first demonstration of optical communication beyond the Earth-Moon system. DSOC uses near-infrared signals to transmit data, instead of radio waves.
- It promises data rates at least 10 times faster than the current radio frequency communication.
- **Primary objective** is to enable high-data-rate communications offering the potential to transmit high-definition images, video streaming and increased scientific data volume during space missions.
- A recent demonstration of DSOC technology was done from NASA's Psyche spacecraft. The DSOC system on board Psyche successfully transmitted a 15-second ultra-high definition video from a distance of 31 million kilometres at a rate of 267 Mbps.
- This paves the way for future missions that will require high-bandwidth communications from deep space.

17. CYBER SECURITY THREATS

India has been victim to the Cyber-attacks number of times in the recent past:

- **2017:** WannaCry and Petya Ransomware
- **2018:** Aadhaar Software hacked and Aadhaar details of the people leaked online.



TYPES OF CYBER ATTACKS

1. Malware: Malicious software to disrupt computers. It can include Virus, Spyware, Trojans etc.

TYPES OF MALWARES:

- **Virus**: Most common type of malware. It can replicate itself and spread to other computers.
- **Worm:** Malware that can self- replicate without a host program. Worms typically spread without any human interaction or directives from the malware authors.
- **Trojan**: Malicious program that is designed to appear as a legitimate program. Once activated following installation, trojans can execute their malicious functions.
- **Spyware:** Malware designed to collect information and data on users and observe their activity without users' knowledge. **E.g.,** Pegasus.
- **Ransomware**: Malicious software injected into the user's computer to limit the access of the system and encrypt the data. Cyber criminals demand money/cryptocurrency in lieu of an encryption key. **E.g.,** *WannaCry*.

2. **Denial of Service attacks:** An attack meant to shut down a machine or network, making it inaccessible to its intended users.
3. **Crypto jacking**: Unauthorised use of a computer or device's processing power to mine cryptocurrencies, often without the owner's consent or knowledge.
4. **Hacktivism:** Misusing a computer system or network for a socially or politically motivated reason. E.g., block access to the government's website, unblock the sites which have been blocked by the Government.
5. **Social Engineering:** Manipulation and exploitation of human psychology to deceive individuals or gain unauthorised access to confidential information (passwords, bank information etc.). These attacks often exploit traits such as trust, helpfulness, curiosity, or fear. Common techniques used:
   - **Impersonation**: Impersonating friends, relatives in distress to seek money.
   - **Phishing:** Pretending to be legitimate entities, such as banks or organisations, to deceive individuals into revealing sensitive information.
   - **Baiting:** Installing malware on pretext of enticing offers like free software upgrade.

6. **Advanced Persistent Threat:** Unauthorised access to a system or network for an extended period without being detected, focussed on stealing data.
7. **Distributed Denial of Service (DDoS) attack:** Malicious attempt to disrupt the normal functioning of a targeted server, service, or network by overwhelming it with a flood of internet traffic. Unlike a Denial of Service (DoS) attack, which typically involves a single source, a DDoS attack leverages multiple compromised systems, often infected with malware, to generate the traffic.

MALWARE-AS-A-SERVICE (MAAS)

- Type of cybercrime model where malware is offered for sale or rent by cyber criminals as a service to other hackers or malicious actors on the internet.
- MaaS operates in a similar way to legitimate Software as a Service (SaaS) models, where software (malware) is provided on a subscription or pay-per-use basis.

18. DARK PATTERNS

- Any practices or deceptive design patterns using user interfaces on any platform (website, mobile applications etc.) designed to mislead or trick users to do something they originally did not intend or want to do.
- The Central Consumer Protection Authority established under Section 10 of Consumer Protection Act, 2019, has classified them as 'unfair trade practice' and notified **'Guidelines for Prevention and Regulation of Dark Patterns, 2023'** to curb dark patterns across various platforms offering goods or services in India.

S. No.	Dark Patterns	Definition
1.	False Urgency	Urgency to pressure consumers into making a purchase or taking an action.
2.	Basket Sneaking	Add additional products or services to the shopping cart without user consent.
3.	Confirm Shaming	Involves guilt as a way to make consumers adhere to a particular choice.
4.	Forced Action	Forcing consumers into taking an action they may not want to take, such as signing up for a service in order to access content.
5.	Subscription Trap	Makes it easy for consumers to sign up for a service but difficult for them to cancel it, often by hiding cancellation options or requiring multiple steps.
6.	Drip Pricing	Hiding additional costs from consumers, until they are already committed to making a purchase.
7.	Rogue Malwares	Using ransomware or scareware to mislead or trick users into believing there is a virus on their computer, to convince them to pay for a fake malware removal tool that actually installs malware on their computer.

19. END-TO-END ENCRYPTION

- Encryption is a way to protect data from unauthorised third-party access. It works by transforming the data into a secret code that only the intended recipient can decipher.
- There are two main types of encryption**:**
  - **Symmetric Encryption** uses the same key to encrypt and decrypt the data.

- **Asymmetric Encryption** uses a pair of keys: one public and one private. The public key can be shared with anyone, but the private key must be kept secret.

END-TO-END ENCRYPTION (E2EE)

- Security measure that protects data as it is transferred between end locations. The sender and receiver can read any exchanged messages, while preventing unauthorised third parties (including service providers and government entities) from intercepting and reading the content.
- **Working:** E2EE generally utilises asymmetric encryption, which uses a pair of keys – public and private – to secure communications. The public key is used to encrypt data, while the private key decrypts it.

20. OPEN-SOURCE SOFTWARE

- Software that is distributed with its source code, making it available for use, modification, and distribution with its original rights. E.g., Linux, Mozilla Firefox, VLC media player, SugarCRM, etc.
- While the operating system of Apple's iPhones (iOS) is closed source, meaning it cannot be legally modified or reverse engineered, Google's Android operating system is open source.
- Many other solutions launched by the government including Digilocker, Diksha, Aarogya Setu, the Covid-19 vaccination platform CoWIN have also been built on top of open-source digital platforms.

ADVANTAGES OF OPEN-SOURCE SOFTWARE

- More affordable than proprietary counterparts, and give increased personal control to creators and users alike.
- By harnessing crowdsourcing, open-source software allows developers to benefit from accelerated innovation, quicker development processes and having more success troubleshooting when problems arise.
- Spur growth of new technologies like 5G/6G, microprocessor technology, Artificial Intelligence, Internet of Things, by building indigenous technology capabilities.

21. ARTIFICIAL INTELLIGENCE

- **Artificial Intelligence:** Capability of a machine to imitate intelligent human behaviour and perform complex tasks similar to how humans solve problems. **E.g.,** Perform cognitive tasks like thinking, perceiving, learning, problem-solving and decision-making.
- **Machine Learning (ML):** Machine learning techniques, including Artificial neural networks (ANNs), are used to achieve the goals of AI. ML is a subfield of AI, and ANNs are a specific type of machine learning algorithm that uses interconnected nodes to learn from data.
- **Artificial Neural Networks (ANNs):** Algorithms that are inspired by the structure and workings of human brains, and have the capability to identify and learn patterns in data. Breakthroughs in ANNs have enabled the development of LLMs and AI-tools like AlphaFold.
- **Natural Language Processing:** NLP involves the interaction between computers and humans using natural language. Through techniques like ML and deep learning, computers can understand, interpret, and generate human language in a meaningful way.
- **Artificial General Intelligence (AGI):** Machines capable of 'thinking' and 'acting' autonomously through a process of self-learning or artificial general intelligence (AGI).
- **Foundational AI:** Large-scale AI models that are trained on very large datasets and over which numerous specific applications can be built, including generative AI.
- **Generative AI:** AI models that use machine learning algorithms to create new and original content, such as images, text, code, audio, or even video with the help of natural-language prompts. **E.g.,** DALL-E for image generation, ChatGPT for text generation. Generative AI models can be LLMs or SLMs.

- **Large Language Models:** Foundational AI model trained with vast datasets with at least one billion or more parameters. LLMs have shown an exceptional proficiency to understand and interact in human languages in a meaningful way. **E.g.,** ChatGPT, Gemini, DeepSeek, Grok. India aims to build its own LLM by the end 2025.
- **Small Language Models**: SLMs are compact AI systems typically having fewer than 1 billion parameters (ranges from millions to a few billion parameters). Cheaper to run and maintain, and ideal for specific use cases.

GENERATIVE ADVERSIAL NETWORKS

- Generative AI uses generative adversarial networks (GANs). GAN is a class of AI algorithms that consist of two neural networks: generator and discriminator, that compete against each other.
  - The generator learns to create new samples (synthetic data) that resemble the training data.
  - The discriminator tries to distinguish between the generated samples and the real data.
  - This adversarial training process leads to the improvement of both networks over time, resulting in generation of increasingly realistic synthetic content. **E.g.,** Deepfakes.

UNIMODAL AI AND MULTIMODAL AI

- **Unimodal AI:** AI models that work with a single type of data input and output. In simpler terms, they specialise in one specific modality, either text, images, audio, or video.
- **Multimodal AI:** AI systems that can understand and process information from multiple sources, such as text, images, audio, video, etc. **E.g.,** Gemini.

NOBEL PRIZE IN PHYSICS 2024

- Awarded to U.S. scientist John Hopfield and British-Canadian scientist Geoffrey Hinton for **foundational discoveries and inventions that enable Machine Learning with Artificial Neural Networks.**
- Professor Hopfield of Princeton University in the U.S. is credited with developing the Hopfield network, a type of recurrent neural network.

APPLICATIONS OF AI

- **Education:** Personalised learning tools, intelligent tutoring systems. **E.g., AI Appu**
- **Healthcare:** AI-enabled medical imaging; personalised treatments; Computer-aided design of new drugs/proteins **(AlphaFold)**; A-enabled robots for surgical procedures. **E.g., AMRSense** (understand patterns of antibiotic resistance in real time.
- **Agriculture:** Crop yield/price prediction; AI-based sensors for spraying pesticides/herbicides, autonomous robots for harvesting crops, monitor crop health; diagnose pests/soil defects, nutrient deficiencies in soil etc.
- **Infrastructure:** AI-enabled smart sensors; intelligent infrastructure (self-driving cars); and optimising complex transportation hubs. **E.g., Adaptive Traffic Control System.**
- **Data Analysis**: Analyse vast amounts of data, identify patterns, and aid in data-driven decision-making.
- **Security:** Detecting cyber threats, identifying vulnerabilities, and conducting cyber attacks; monitor intrusion. **E.g., Trailguard AI**
- **Space exploration**: Development of autonomous spacecraft, smart habitats.
- **Service delivery**: AI-powered chatbots and virtual assistants; quick and accurate responses to inquiries, automating routine tasks.
- **Energy**: Smart grid management; resource management. **E.g.,** Google's DeepMind has leveraged Machine Learning (ML) to improve wind energy forecasting, enabling more accurate wind pattern predictions.

CAN A COPYRIGHT BE GIVEN TO A CONTENT GENERATED BY AI?

- The **Indian Copyright Act, 1957**, grants exclusive rights to creators (authors, artists, etc.) over their original works for a specific duration.
- India does not have any specific litigation in the context of text and data mining by AI. The Act currently recognises only human authors for copyright protection.

GLOBAL PARTNERSHIP ON AI (GPAI)

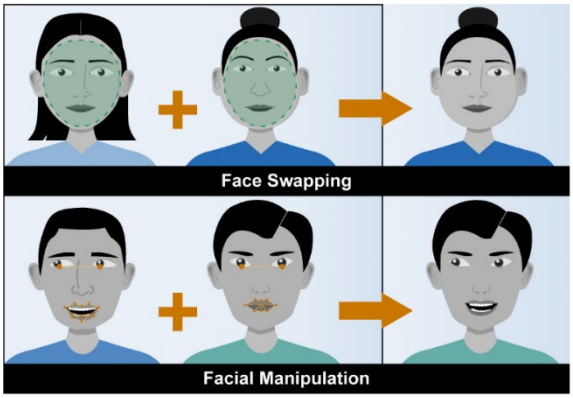
- Multi-stakeholder initiative which aims to bridge the gap between theory and practice on AI by supporting cutting-edge research and fostering international collaboration for responsible AI development.
- **Launched:** June 2020. India is a founding member of GPAI.
- **Members:** Presently, GPAI has 29 members, including Australia, Canada, France, Germany, India, United Kingdom, United States, European Union, among others.

NEW DELHI DECLARATION ON AI:

- In 2023, the New Delhi Declaration on AI was adopted to find a balance between innovation and the risks associated with AI systems.
- It aims to promote equitable access to critical resources for AI innovation including computing, high-quality diverse datasets, algorithms, software, testbeds, etc. and human-centred use of AI.

22. DEEP FAKES

- Synthetic media (image, video, audio) digitally created/altered to replace a person's face or body with others to deceive or manipulate.
- Deepfakes are generated by utlilising generative adversarial networks (GANs) and artificial intelligence.
- **Concerns:** Can be utilised for malicious activities like scams, identity theft, financial fraud, pornography, election meddling, social engineering, disinformation attacks.



Source: GAO. | GAO-20-379SP

- The law currently being invoked to tackle deep fakes is the **Information Technology Act, 2000.** Under section 66D of the IT Act, it is a punishable offence to use computer resources to cheat by personation.
- **Information Technology Rules, 2021** prohibit hosting any content that impersonates another person and requires social media firms to take down artificially morphed images when alerted. All content reported to be fake or produced using deep fake has to be taken down by social media intermediary platforms within 24 hours of the receipt of such a complaint.

23. NEUROMORPHIC COMPUTING

- Approach to computing that is inspired by the structure and function of the human brain.

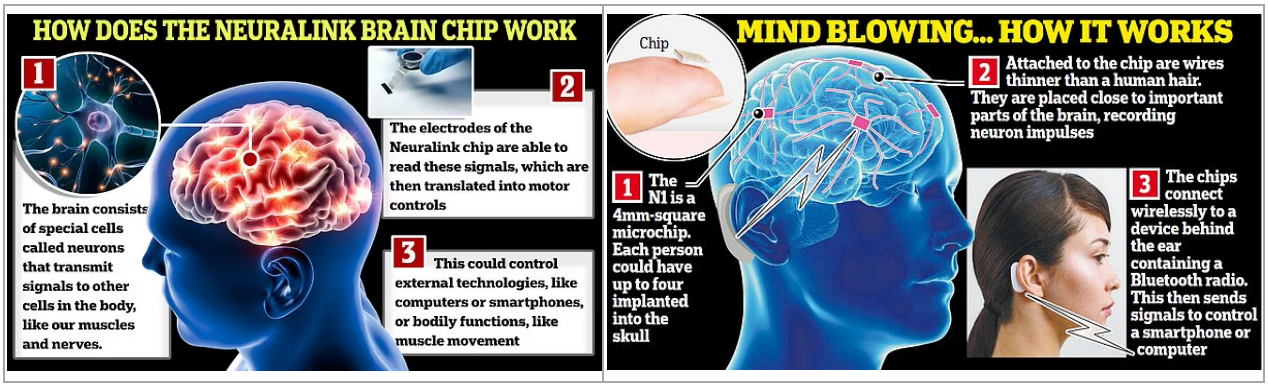
- The development of neuromorphic hardware aims at mimicking biological synapse that monitors and remembers the signal generated by the stimuli. (simply, aims to design computer systems that mimic how human brain processes information)

WORKING

- A neuromorphic computer/chip consists of Artificial Neural Networks composed of millions of artificial neurons (made from Silicon).
- These neurons pass signals to each other in layers, converting input into output through electric spikes or signals, based on the architecture of Spiking Neural Networks.
- This allows the machine (computer systems) to mimic the neuro-biological networks in the human brain and perform tasks efficiently such as visual recognition and data interpretation.

24. NEURALINK BRAIN CHIP IMPLANT

- As per the computer-brain interface company Neuralink, the first human has received a brain chip implant and is recovering well.
- Neuralink is working on linking the nervous system to computers, with efforts aimed at helping treat brain disorders, overcoming brain injuries, and other applications.
- **Aim:** To give people the ability to control a computer cursor or keyboard using their thoughts alone or "just by thinking".



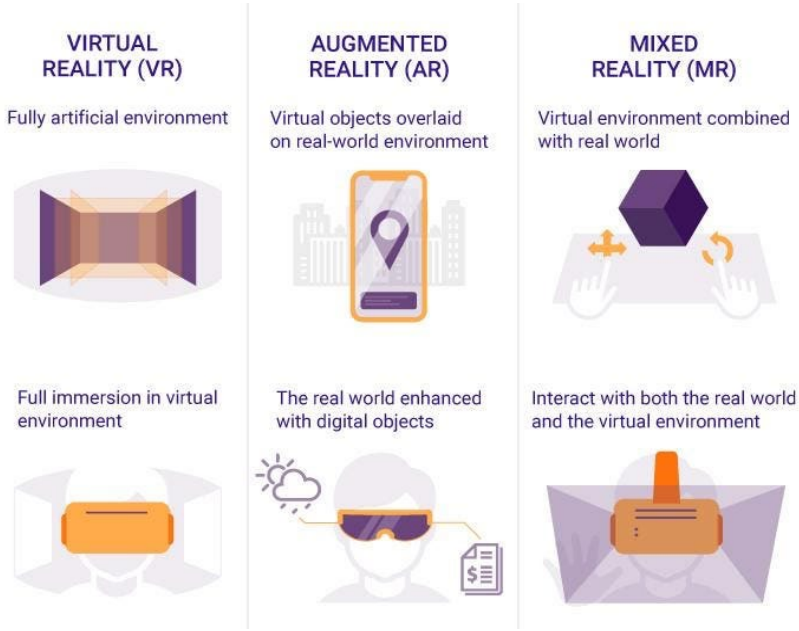
25. IMMERSIVE TECHNOLOGIES

Extended reality (XR): Umbrella term for immersive technologies that alter or add to our perception of the real world. Includes: Augmented Reality, Virtual Reality and Mixed Reality.

Augmented Reality (AR): Overlays digital elements onto the real world. AR does not replace the real world, but enhances it with virtual/digital information. **Devices:** Smartphones, tablets, or special AR glasses. **E.g.,** Pokemon Go (Game)

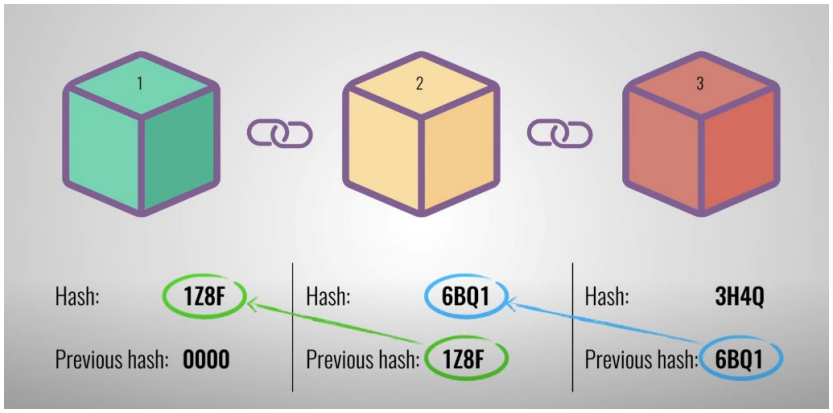
Virtual Reality (VR): Creates a completely immersive simulated environment and cuts off the user from the physical environment/reality. VR replaces the user's entire field of view with a computer-generated virtual world. **Devices:** Requires dedicated VR headsets and motion controllers.

Mixed Reality (MR): MR combines elements of both VR and AR. MR places virtual objects in the real world and enables users to interact with them as if they were physically present. This is typically achieved using specialised headsets that incorporate sensors and cameras that scan the environment and track the user's movements, making digital objects appear to exist in the real world.



26. BLOCKCHAIN TECHNOLOGY

- Decentralised, distributed ledger that records transactions across a network (multiple computers) in a way that makes them tamper-resistant and transparent.
- At its core, a blockchain is a chain of blocks, where each block stores data or digital information. (**E.g.,** In the case of Bitcoin, a block stores a record of cryptocurrency transfer).
    - Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data.
    - Once a block is added to the chain, it is extremely difficult to alter, hence, ensuring the integrity of the data.
    - The transactions are verified and added to the block through a process called consensus (proof of stake), which involves a network of participants known as nodes, who collectively validate the transactions.



KEY FEATURES

- **Decentralisation:** Blockchain operates on a peer-to-peer network, where each participant has a copy of the entire blockchain, ensuring transparency. All transactions recorded on a blockchain are visible to all participants in the network.
- **Security:** Utilises cryptography to secure transactions and control the transfer of assets. Each transaction is encrypted and linked to the previous transaction, forming a chain of blocks that cannot be easily altered.
- **Immutability:** Once a transaction is recorded on the blockchain, it is extremely difficult to change or delete. This immutability ensures the integrity of the data stored on the blockchain.
- **Smart Contracts:** Blockchain platforms often support the execution of smart contracts (self-executing contracts with predefined rules), eliminating the need for intermediaries.

POTENTIAL APPLICATIONS OF BLOCKCHAIN

- Transfer of land records (Property record management).
- Digital certificates management (Education, Death, Birth, Agreements, Sale Deeds)
- Pharmaceutical supply chain; Electronic Health Record Management
- e-Notary service (Blockchain enabled e-Sign solution)
- Farm insurance; Agriculture and other supply chains
- Identity management; Digital Evidence Management System
- Power distribution
- Duty payments; Microfinance for Self-Help Groups
- e-Voting
- Public Service Delivery
- IoT Device Management and Security

27. NON-FUNGIBLE TOKENS

- Digital assets (often in the form of digital art, music, videos, or other digital content) based on blockchain technology.
- Each NFT is assigned a unique and unchangeable code stored on the blockchain, providing a clear record of ownership and verifying the authenticity of the associated digital asset.
- Each NFT is distinct and scarce and cannot be duplicated or forged. This uniqueness adds value to the digital asset.
- Many NFTs utilize smart contracts, self-executing contracts with the terms of the agreement directly written into code. These contracts automate processes like royalties, ensuring creators receive a percentage of future sales.

28. DECENTRALISED FINANCE (DEFI)

- An emerging concept that leverages blockchain technology to offer financial products and services without relying on traditional intermediaries like banks or brokerages.
- DeFi applications are built on blockchains and rely on smart contracts that automate financial transactions according to predetermined rules. This eliminates the need for a trusted third party.
- **DeFi offers a variety of financial services, including:**
  - Users can borrow or lend cryptocurrencies directly to each other.
  - Interest-bearing accounts where users can deposit their crypto assets and earn interest.
  - Peer-to-peer insurance products where users can pool funds to cover losses.

Click here for INDEX

5 CHAPTER FRONTIER TECHNOLOGIES

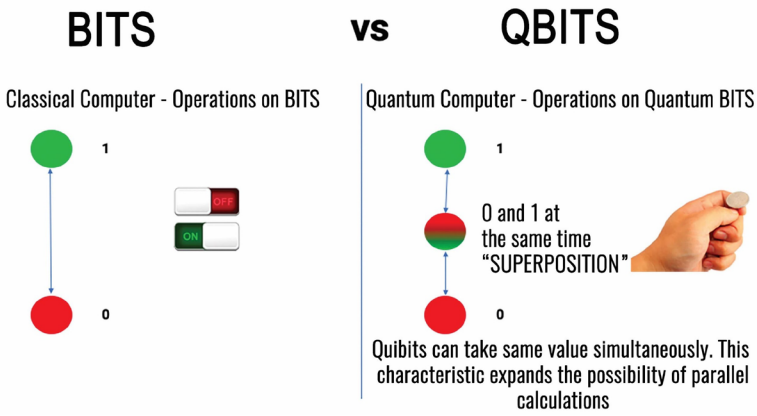
➔ QUANTUM TECHNOLOGIES

1. INTRODUCTION TO QUANTUM COMPUTERS

Quantum Mechanics is a field of physics that describes the behaviour of matter and energy at the smallest scales, typically at the level of atoms and subatomic particles.

QUANTUM COMPUTERS AND QUBITS

- Quantum computers use the properties of quantum physics to store data and perform computations.
- While classical computers use bits as the basic unit of information, that is a two-state system that can represent either 0 or 1 at a time (they just act like a light bulb to represent ON/OFF), quantum computers use qubits which represent 0, 1 and an intermediate state (superposition of 0,1) in between.
- Any particle that can be controlled and manipulated using quantum-mechanical phenomena can be used as a qubit. The information can be 'encoded' in some property of the particle (like an electron's spin) and then processed.



- **A qubit can be:**
  - a particle — like an electron, photon (E.g., Linear optical quantum computing uses photons as qubits).
  - a collection/group of particles
  - any quantum system engineered to behave like a particle.
- The power of a quantum computer scales exponentially (2^N) with the number of qubits, unlike classical computers, which scale linearly with the number of bits. This enables them to perform complicated large-scale calculations out of the reach of the classical and even the best supercomputers.

2. QUANTUM PHENOMENON

QUANTUM SUPERPOSITION

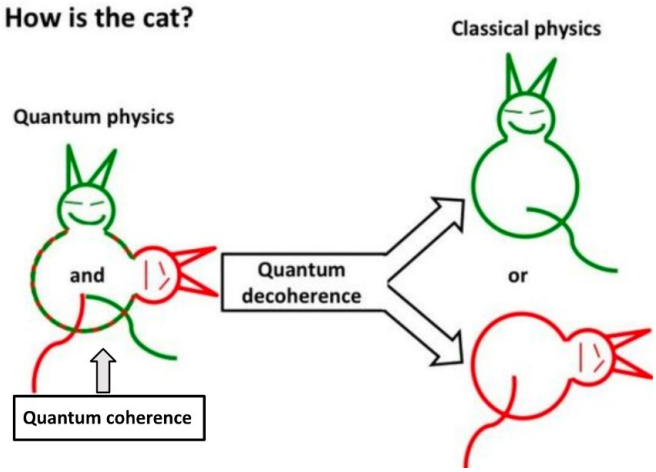
Qubits can simultaneously exist in more than one location or quantum state at one time while remaining as a single entity. Thus, superposition enables qubits to perform multiple operations simultaneously.

QUANTUM ENTANGLEMENT

- When two particles become entangled, the state of one particle becomes linked with the state of the other, regardless of the distance between them. This means that changes to the state of one particle instantaneously affects the state of the other.

- Hence, measuring the state of one entangled particle instantly determines the state of its entangled partner, even if they are vast distances apart.
- Entangled qubits can be manipulated collectively, allowing for the parallel processing of information.

QUANTUM COHERENCE



- Quantum mechanics allows qubits to exist in a superposition state, where they can be 0 and 1 simultaneously.
- Quantum coherence refers to the maintenance of the phase relationships between these superposed states, which is essential for performing calculations in quantum computers.

THE CHALLENGE

- Qubits are very fragile and susceptible to decoherence (slight disturbances in the surroundings may result in a change of the quantum state of the particle and can result in a change of the information). Thus, maintaining quantum coherence is difficult.
- Interaction between qubit and their environment can lead to decoherence. Decoherence can be caused by factors, such as noise, heat, and measurement. It causes qubits to collapse into one of the two states and lose quantum information.
- Maintaining quantum coherence requires extremely low temperatures, around -196°C (liquid nitrogen temperature). This makes building practical quantum computers challenging.

COHERENCE AT ROOM TEMPERATURE

- Recently, researchers were able to achieve quantum coherence at room temperature by embedding a light-absorbing molecule called a chromophore within a metal-organic framework (MOF), using Zirconium as the metal component.
- MOF's structure restricted the movement of the chromophore, minimising environmental disturbances. This allowed the chromophore to maintain quantum coherence for a brief period in the nanosecond range (billionths of a second) at room temperature.

CHROMOPHORE

A molecule in a given material that absorbs particular wavelengths of visible light, and emits colour as a result, and in doing so confers colour on the material.

METAL ORGANIC FRAMEWORK (MOF)

- A type of porous material that consists of a network of repeated molecular arrangements where the repeating structure has a metal atom or ion with organic molecules attached to its like tentacles. Each tentacle attaches to another metal atom and the structure repeats itself to make up the MOF.
- MOFs have various applications in fields such as gas storage, catalysis, and sensing, due to their high surface area, tunable structure, and functional properties. MOFs can also be used for quantum computing, as they can host qubits within their pores or on their surfaces.

QUANTUM SUPREMACY

- It is the point at which a quantum computer can complete a mathematical calculation that is beyond the reach of even the most powerful supercomputer.

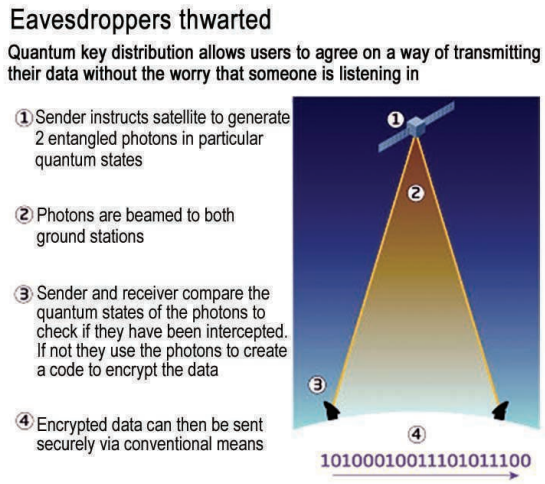
- In 2019, Sycamore (Google's quantum computer) claimed 'supremacy' because it reportedly took 200 seconds to perform a calculation that the world's fastest supercomputer, Summit, would have taken 10,000 years to accomplish.

3. APPLICATIONS OF QUANTUM TECHNOLOGY

- **Quantum Computers:** can solve complex mathematical problems exponentially faster than classical computers, particularly used for cryptography, optimisation, and quantum simulation.
- **Quantum Simulators**: can accurately model behaviour of complex systems like climate models, financial markets, molecular interactions at quantum level. This can be used to accelerate drug development, designing novel-materials with specific properties. Potential to revolutionise various sectors like health, meteorology (climate modelling), telecommunications, environment, logistics & finance etc.
- **Quantum Sensing:** Quantum sensors can measure physical quantities, like magnetic fields, gravity, time and biological processes with unprecedented precision. This presents advancements in navigation, geological exploration, medical diagnostics, brain-computer interfaces, and neuroimaging.
- **Quantum Cryptography and Communication:**
  - **Quantum key distribution** ensures theoretically unbreakable encryption, making communications secure against any computational attack.
  - **Post-Quantum Cryptography**: Breaking current encryption methods and developing alternative cryptographic algorithms secure against attacks from quantum computers.
  - **Quantum internet:** A network leveraging quantum entanglement could enable ultra-secure data transmission and quantum internet development.

QUANTUM KEY DISTRIBUTION

- QKD uses the laws of quantum physics to distribute secure keys between two parties which prevent the decryption of data, and thus, ensure secure communication.
- QKD involves sending encrypted data as classical bits over networks, while the keys to decrypt the information are encoded and transmitted in a quantum state using qubits.
- A key feature of the technology is its ability to detect and mitigate interception. This is based on the fundamental law of quantum physics that observation itself disturbs the quantum state of a particle, i.e., if an eavesdropper tries to intercept the QKD signal, it will immediately change the signal's state. This makes the interception detectable and ensures any intercepted information is immediately discarded.



- QKD technology still faces some challenges, such as limited transmission distances (100-500 kms) due to decoherence and the high cost of specialised equipment.

QUANTUM TELEPORTATION-BASED COMMUNICATION

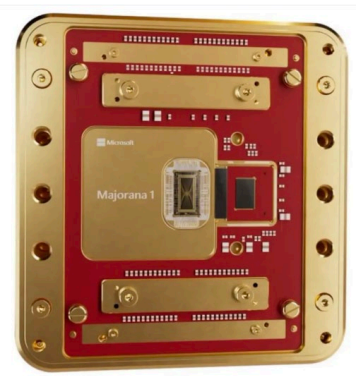
- Alternatively, a communication system can work by representing entire information using qubits of entangled photons instead of only the encryption key. The qubits may be instantaneously transferred owing to their entanglement. This instant transfer of "qubits" to send information is called "quantum teleportation".
- The entangled photons may also be used to send information through optical cables and received through a single-photon detector at the receiver end.

QUANTUM TELEPORTATION IN NEWS

NASA's quantum teleportation demonstration	NASA's jet propulsion laboratory has demonstrated 44 km teleportation of qubits of photons over a fibre-optic network and single-photon detectors.
Quantum Internet	• The future quantum internet is going to be built using communication links based on quantum teleportation. • Quantum internet would be able to transmit large volumes of data across immense distances at a rate that exceeds the speed of light. • It is said to be energy efficient because it does not involve transmission through electricity or light. • Quantum internet will be safe/tamper proof due to the interception-free nature of qubits.

4. QUANTUM CHIPS

MAJORANA 1



- Quantum chip developed by Microsoft.
- **Material:** Built using indium arsenide (a type of semiconductor) and part aluminum (a superconductor at low temperatures).
- When cooled to near absolute zero (−273.15 °C) and tuned with magnetic fields, this material exhibits topological superconductivity, leading to the emergence of Majorana zero modes. Two Majorana zero modes encode (act as) a single topological qubit in this chip.
- Majorana 1 chip contains eight qubits with the potential to scale to 1 million qubits on a single chip.

Superiority:

- Produces more reliable and scalable qubits (topological qubits) than its predecessors developed by Google (Willow) and IBM.
- Reduces error susceptibility, making it more reliable for practical applications.

MAJORANA ZERO MODES

- Exotic quasiparticles (not fundamental particles like electrons) that arise in certain types of topological superconductors.
- Exhibit unique behaviour and possess topological protection (inherent stability i.e, their quantum state is resistant to small disturbances). This makes them promising candidates for fault-tolerant qubits in quantum computing.
- Unlike traditional qubits which are prone to errors, Majorana zero modes offer a path to produce more stable and error-resistant qubits.

WILLOW

- Quantum computing chip developed by Google.
- Willow operates using superconducting **transmon qubits**—tiny electrical circuits exhibiting quantum behaviour at extremely low temperatures. These circuits are engineered to function like artificial atoms in a quantum state.

- **Benefits**: can reduce errors exponentially with increasing scalability (using more qubits).
- Willow performed a standard benchmark computation in under five minutes, that would have taken one of today's fastest supercomputers 10 septillion (10^25) years.

➔ NANOTECHNOLOGY

5. NANOTECHNOLOGY

- Nanotechnology involves the manipulation and control of matter at the nanoscale, typically in the range of 1 to 100 nanometers. (1 nm = 10-9 metres)
- At the nanoscale, materials exhibit novel properties and behaviours that are different from their bulk counterparts.
- These properties arise due to increased surface area, quantum effects, and the interactions between atoms on the surface and those at the interface (where two different materials meet).
- Nanotechnology aims to harness these properties to develop new materials, devices, and applications with enhanced performance and functionality.

APPLICATIONS OF NANO TECHNOLOGY

- **Medicine & Healthcare**: Drug delivery, bioimaging, diagnostics, antimicrobial coatings, cancer treatment.
- **Electronics**: Nanoelectronics, sensors, data storage devices, quantum dot displays.
- **Energy**: Solar cells, lithium-ion batteries, fuel additives, hydrogen generation and storage.
- **Automotive**: Fuel cells, lighter and stronger materials, advanced lubricants.
- **Aerospace**: Lighter and stronger composites, structural health monitoring.
- **Defence**: Lightweight soldier armour, surveillance, sensors, nano-coatings for equipment.
- **Environment:** Pollutant degradation, water treatment, nano-filtration, catalytic converters.
- **Agriculture**: Pesticides, fertilisers, pathogen detection, encapsulated agrochemicals.
- **Food & Packaging**: Antimicrobial films, nano-coatings with improved barrier properties, moisture control, and thermal stability.
- **Textiles**: Stain and wrinkle-resistant fabrics, lightweight protective clothing, moisture wicking.
- **Cosmetics**: UV protection in sunscreens, long-lasting colour pigments, anti-ageing creams.
- **Paints & Coatings**: Anti-corrosive coatings, thermal barrier coatings.

➔ IMPORTANT NANOMATERIALS

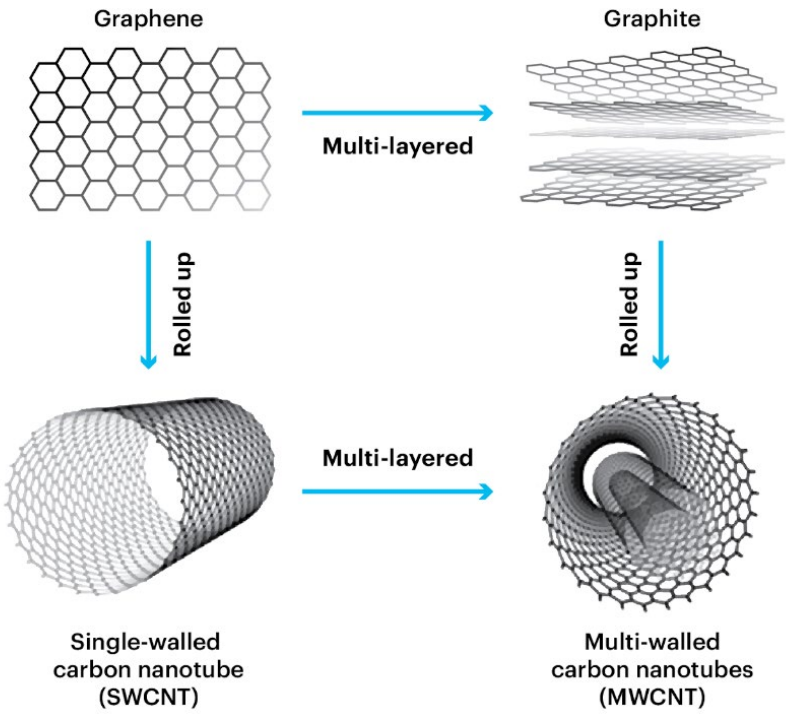
6. GRAPHENE

- An allotrope of carbon consisting of a single layer of Carbon atoms arranged in a two-dimensional hexagonal lattice.
- **Properties:**
  - Thinnest compound (one atom thick)
  - Lightest material known
  - Strongest material discovered (between 100-300 times stronger than steel) yet more flexible than rubber.
  - Best conductor of heat at room temperature and best conductor of electricity
  - Almost perfectly transparent as it absorbs only 2% of light.

- Impermeable to gases, even those as light as hydrogen and helium.

Applications of Graphene:

- **Flexible Electronics**: Can be used in wearable electronics, flexible displays, electronic textiles, and bendable sensors.
- **Energy Storage**: Enhance performance and capacity of energy storage systems like supercapacitors and high-performance batteries.
- **Water purification**: Suitable for water filtration and desalination systems.
- **Bio-sensors:** To sense chemical and biological agents, explosives, radiation, and other hazardous substances.



- **Aerospace and Defence:** Develop lightweight, high-strength armour and ballistic protection. Have potential to absorb and dissipate electromagnetic waves, making it valuable for developing stealth coatings.
- **Medical Applications:** Biosensing, bioimaging, targeted drug delivery, tissue engineering, phototherapy and cancer treatment. Has antibacterial properties.

7. CARBON NANOTUBES (CNTS)

- Graphene can be rolled up to form CNTs which are cylindrical in shape with diameter in nanoscale.
- **Properties:**
  - High tensile strength and light-weight
  - High electrical and thermal conductivity
  - Have large surface area and is chemically stable

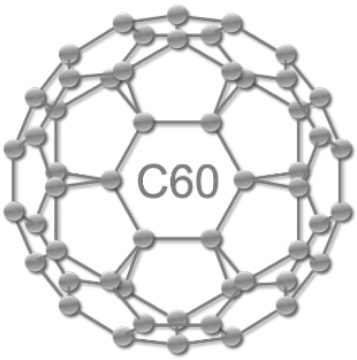
APPLICATIONS OF CNT

- Applications in aerospace and automotive engineering.
- Electronic devices, such as transistors, high sensitivity sensors, solar cells, flexible electronics and touchscreens, organic light emitting diodes.
- Used in energy storage (electrodes for Lithium-Ion batteries, capacitors), drug storage and drug delivery.

CHALLENGES

- Impurities such as residual metal particles in CNTs.
- Processing & manufacturing of CNTs is challenging.

8. FULLERENE



- Fullerene is a family of carbon allotropes that consists of cage-like or tubular structures.
- The most well-known fullerene is buckminsterfullerene (C60), which resembles a soccer ball with 60 carbon atoms forming pentagons and hexagons.
- Fullerenes (not typically abundant) are found naturally in soot, volcanic eruptions, interstellar dust clouds and lightning strikes.

APPLICATIONS OF FULLERENE

- **Electronics and Energy storage:** Can be used in organic photovoltaics (solar cells), organic light-emitting diodes and organic field-effect transistors.
- **Medicinal Chemistry:** Can act as antioxidants, scavenging free radicals, targeted drug delivery and imaging applications such as magnetic resonance imaging.
- **Catalysts:** Can act as catalyst supports in fuel cells and environmental remediation.
- **Water Purification**: Removal of pollutants and contaminants due to their adsorption and catalytic properties.

9. GOLDENE

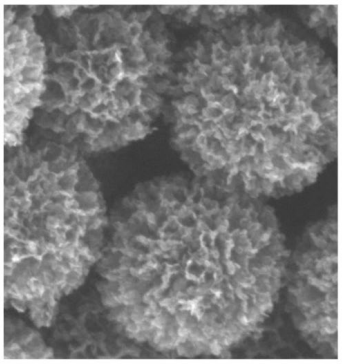
- Two-dimensional (monolayer) sheet of Gold. These sheets are roughly 100 nanometres thick and approximately 400 times thinner than the thinnest commercially available gold leaf.
- Each gold atom has only six neighbouring atoms compared to 12 in a three-dimensional crystal.
- Two-dimensional materials have unusual optical, electronic and catalytic properties, as they have extremely high surface area relative to their volume.

POTENTIAL APPLICATIONS

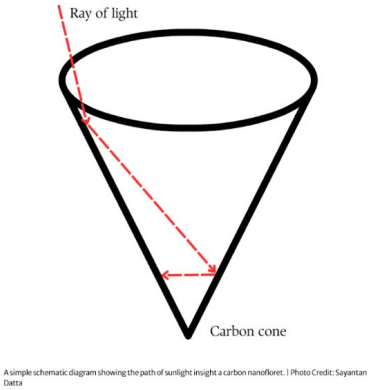
- Applications in electronics that use gold due to its electrical conductivity. A monolayer sheet would be more economically viable than thicker, three-dimensional gold.
- Promise as a great catalyst in various chemical processes. This could lead to advancements in areas like:
  - Hydrogen production
  - Carbon dioxide conversion
  - Selective production of valuable chemicals
- Water Purification technologies.

10. CARBON NANOFLORETS

- Unique nanostructure composed of carbon atoms arranged in a distinctive flower-like structure.
- They can absorb up to 87% of sunlight (infrared, visible, and ultraviolet light), and convert it into heat with exceptional efficiency, unlike traditional solar-thermal conversion materials that only absorb visible and ultraviolet light.
- Their cone-like shape minimises reflection allowing for maximum light absorption. Most of the light that falls on the material is reflected internally.
- **Applications:** Solar thermal applications such as water heating and desalination, heating homes, and sterilising surfaces in hospitals without relying on fossil fuels.



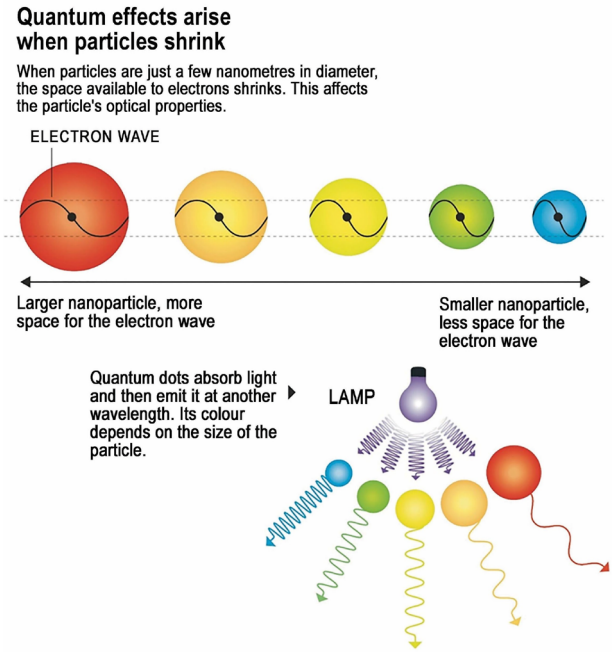
Carbon nanoflorets observed through a microscope. | Photo Credit: Ananya Sah



A simple schematic diagram showing the path of sunlight insight a carbon nanofloret. | Photo Credit: Sayantan Datta

11. QUANTUM DOTS (NOBEL IN CHEMISTRY 2023)

- Quantum dots are tiny particles or nanocrystals of semiconducting material made from a semiconductor such as Silicon with a diameter in the range of 2–10 nm (10–50 atoms).
- They behave like artificial atoms, as they can have a fixed number of electrons in a confined space, leading to unique properties that are size-dependent. QDs exhibit special properties when they interact with light.



APPLICATIONS OF QDS

- **Bioimaging**: QDs are 20 times brighter and 100 times more stable than traditional fluorescent dyes, are photostable and can be used in bioimaging.
- **Bio-sensors:** QD sensors can detect presence of pathogens in food or water, or monitor levels of pollutants in the environment.
- **Targeted Cancer treatment**: Because of their nano size, they possess large surface area ensuring higher drug loading capacity and can tag nanocarriers in biological systems.
- **Optical applications:** QLEDs are capable of emitting all colours depending on their size, thus, they provide high-definition, brighter and more colourful displays and efficient image sensors like CMOS sensors.
- **Flexible electronics:** Quantum-dot logic circuits layout building blocks for innovative devices, including printable electronics and flexible display screens.

12. LIQUID NANO UREA

- Developed by IFFCO. India will be the first country to start commercial production of Liquid Nano Urea.
- Nanotechnology-based fertiliser contains Nano-scale nitrogen particles which have more surface area and number of particles, which make it more impactful.

BENEFITS OF NANO UREA

- As compared to conventional urea, uptake of Nano Urea is more than 80%.
- Required in lesser amounts as compared to the conventional urea fertilisers. Other benefits:
  - Cheaper than conventional urea

- Easy to store
- Reduced input costs to farmers
- Easy to apply as Nano urea is directly sprayed on leaves (can be absorbed directly through the leaves' pores)
- Reduced import of conventional urea saving precious foreign exchange.

13. NANOPTA

- A platinum-containing synthetic nanozyme that mimics the function of oxidases (natural enzymes that remove hydrogen from substrates in the presence of oxygen to form water).
- Can degrade pollutants in industrial wastewater by oxidising them in the presence of sunlight, reducing wastewater toxicity.
- Exhibits robustness and can withstand pH and temperature variations.
- Exhibits impressive stability, lasting for up to 75 days at room temperature.
- Can oxidise neurotransmitters like dopamine and adrenaline, leading to a colour change in the solution, which can be used to measure their concentration.
- Holds potential for diagnosing neurodegenerative diseases like Parkinson's and Alzheimer's disease.

➔ MISCELLANEOUS

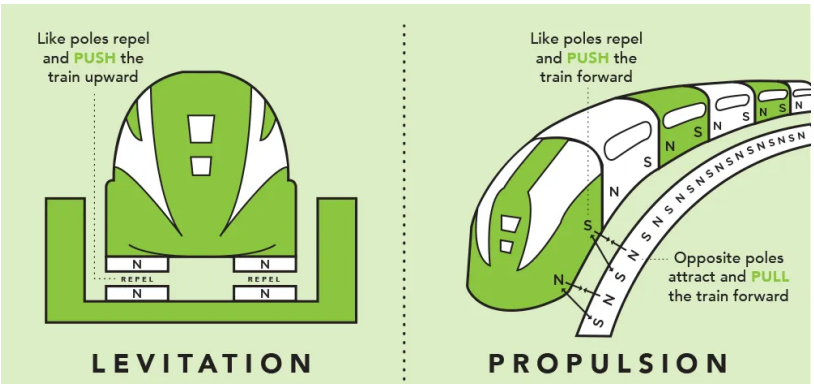
14. CRYOGENIC TECHNOLOGY

- Cryogenics is the study of the production and behaviour of materials at extremely low temperatures (below -150 degrees Celsius).
- Most common gases turned into liquid for cryogenics are Oxygen, Nitrogen, Hydrogen and Helium.

POTENTIAL APPLICATIONS OF CRYOGENIC TECHNOLOGY

- **Cryosurgery**: Kill unwanted or malignant tissues, such as cancer cells or moles.
- **Cryopreservation:** Preservation of organisms, tissue, and genetic material. **E.g.**, Freezing of eggs and embryos for fertility preservation, storage of vaccines, blood bank, food preservation.
- **Cryonics:** Cryopreservation of animals and humans with the goal of reviving them in the future.
- **Cryoelectronics:** Study of superconductivity and other electronic phenomena at low temperature.
- **Cryoengines**: Rocket engine using cryogenic fuels, including liquid hydrogen and liquid oxygen.
- **Research:** Used in Nuclear magnetic resonance (NMR) to determine the chemical structure of a molecule.

15. HYPERLOOP TECHNOLOGY



- Asia's first international Hyperloop summit was conducted at IIT-Madras. India's first hyperloop test track of 450 metres has been completed at the campus.
- Hyperloop (5th mode of transportation) is a high-speed train that travels in a **near-vacuum tube** (which eliminates friction and air drag) potentially allowing the pod to reach speeds of over 1000 kmph.
- The **basic principle** behind Hyperloop technology is magnetic levitation which allows the vehicle to be suspended and propelled on a guidance track made with magnets.

16. PIEZOELECTRIC EFFECT

- Phenomenon in which certain materials generate an electric charge in response to applied mechanical stress or pressure.
- Can be observed in materials like quartz, topaz, tourmaline, ceramics such as lead zirconate titanate (PZT), and even in certain biological materials like bone and tendons.
- Some materials also display an inverse piezoelectric effect, where the application of an electric current induces a mechanical strain or deformation.
- **Applications:** Used in pressure sensors, accelerometers, acoustic devices, high voltage generators, electronic frequency generators, microbalances, inkjet printers, electronic stethoscopes.

17. RHODAMINE-B

- Water soluble chemical compound or fluorescent xanthene dye which has various applications.
- Cause serious environmental and health problems- skin and eye irritation, stomach issues, breathing problems, organ damage, cancer.
- The chemical is considered substandard and unsafe under the Food Safety and Standards Act 2006.
- **Various applications include:**
  - Dyeing in textile, paper, paints, leathers etc., lends a vibrant pink hue.
  - Food dye (used in cotton candy to produce its iconic pink colour).
  - Used as a tracer dye, **E.g.,** To determine the rate and direction of water flow in rivers, lakes etc. To trace pollutants in water systems; detect leaks in pipelines due to its high visibility and water solubility.
  - Rhodamine dyes exhibit fluorescence, and can be used in fluorescence microscopy, spectroscopy and ELISA test (To detect the presence of specific molecules like proteins or antibodies).
  - Develop laser dyes employed in scientific and medical lasers.

18. THALLIUM

- Thallium is a highly toxic metal that is an abundant element on the Earth's crust.
- It does not exist in free elemental form, and exists in ores with other elements, mostly potassium.
- Tasteless, odourless, and colourless, which makes it dangerous for human consumption.
- Thallium poisoning can occur through ingestion, inhalation, or skin contact with thallium compounds.
- **Applications**:
  - Making electronics, such as, photoresistor, rectifier, detectors in infrared devices and in gamma radiation detection devices.
  - Used in the pharmaceutical industry to make various drugs and medicines.
  - Radioactive isotope Thallium-201 is used for nuclear medicine scan.

19. ACTIVATED CARBON

- Highly porous form of carbon with small, low-volume pores and large surface area, used for adsorption and chemical reactions.
- Produced by heating carbon-rich materials like wood, coconut shells, or coal in low oxygen conditions (anaerobic heating).

Properties:

- High adsorption capacity – can trap toxins, gases, and chemicals.
- Porous structure – large surface area for effective adsorption.
- Non-toxic & chemically stable – safe for various industrial and medical uses.

Potential Uses:

- **Water treatment plants:** Removes organic compounds, chlorine, and heavy metals from water.
- **Air Purification:** Used in air filters to capture volatile organic compounds (VOCs), odours, and pollutants.
- **Chemical Purification:** Activated carbon is used to purify chemicals and solvents in the pharmaceutical and chemical industries.
- **Food Industry:** Used for decolourisation and purification, such as sugar refining and wine purification.
- **Medical Uses:** Treating poisonings and overdoses by adsorbing toxins in the gastrointestinal tract.
- **Mining:** Extract gold from cyanide leach solutions.
- **Energy storage**: Used in supercapacitors. #Indian researchers have produced activated carbon from coconut husk suitable for supercapacitor fabrication.
- **Environment**: Carbon capture and adsorption; cleaning of oil spill; emissions control.

20. SLAG

- By-product generated during manufacturing of pig iron and steel. Primarily, slag consists of calcium, magnesium, manganese and aluminium silicates and oxides in various combinations.
- Among all the solid/liquid wastes, slag is generated at such large quantities that management of slag has become a critical component of steel production.
- **Potential Applications:**
  - Used mainly as road metal and bases, asphalt paving, track ballast, landfills and aggregate for concrete.
  - Used for soil conditioning as fertiliser and liming agent in agriculture due to presence of CaO content.
  - Used in making mineral wool for insulation purposes.
  - Barrier material remedy for waste sites where heavy metals tend to leach into the surrounding environment.
  - Used to filter and remove contaminants from water and treat acidic water discharges from abandoned mines.

Click here for INDEX

6 CHAPTER ENERGY TECHNOLOGY

➔ ALTERNATIVE FUELS IN TRANSPORTATION

1. GASEOUS ALTERNATIVES

- Looking for alternative fuels in transportation has multiple incentives for India as below.
  - Reducing import dependence for crude
  - Reducing carbon emissions
  - Reducing air pollution
- From solid to liquid to gaseous fuels, the fuels become cleaner in the increasing order. However, energy density decreases in that order.
- Characteristics:
  - Combustion efficiency is increased (Proportion of usable heat is more).
  - Cleaner combustion in the absence of residues and pollutants
  - Easy to transport. The challenge is to compress it in fuel pumps and tanks.
- However, the problem with gaseous fuels is higher NOx emission due to high temperature of combustion.

2. NATURAL GAS

- Natural gas is primarily a mixture of lightweight alkanes (aliphatic: straight-chain hydrocarbons).
- Before transportation, heavier hydrocarbons like propane, butane and pentane are typically removed.
- Raw natural gas often contains impurities: carbon dioxide (CO_2), hydrogen sulfide (H_2S), water vapour, and other trace gases.
- The CO_2 content in raw natural gas can be as high as 9%, but it is reduced to around 2% or lower before being transported. Carbon Capture Technology is used to remove CO_2 for more efficient combustion.
- Commercially available forms of natural gas are primarily a mixture of methane and ethane. Heavier alkanes (Propane and butanes) are extracted out, liquefied and sold as liquefied petroleum gases (LPG).

Natural Gas	Proportion
Methane	80-95
Ethane	5-10
Propane, Butane, Pentane etc.	<5

SOURCE OF NATURAL GAS: Natural Gas occurs in nature in the following forms:

GAS WELLS

- Bombay high, Assam's Digboi, KG basin, Rajasthan (Jaiselmer), Tripura.
- The largest gas fields are located in Iran, Central Asia and Russia. This is why the interest in TAPI (Turkmenistan, Afghanistan, Pakistan, India).

DILUTED IN OIL: PETROLEUM

- Petroleum wells are those in which liquid petroleum comes out and the top part is gas.

DILUTED IN WATER: GAS HYDRATES

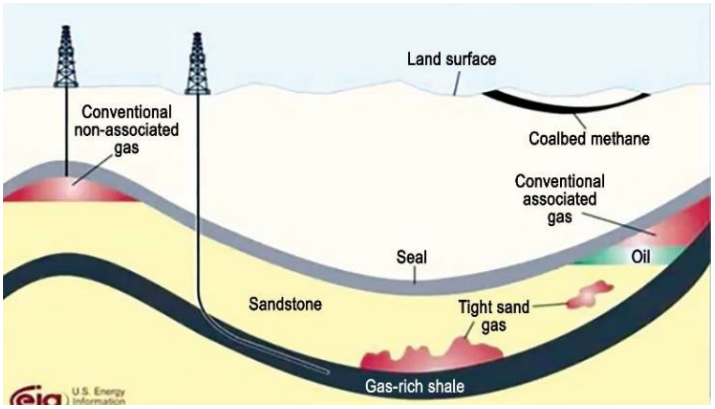
- Gas hydrates are natural gas diluted in water.
- Hydrates are water and gas molecules present in crystalline form.
- Normally they are in the form of ice in permafrost or sea floor which when heated releases methane.
- Gas hydrates are present all over the coast.

DILUTED IN ROCKS: COAL-BED METHANE

- Found in coal seams where hard coal deposits have methane absorbed in it under high pressure.
- Mining involves reducing the pressure of coal bed and releasing the trapped gases.

NATURAL GAS: STORAGE, TRANSPORT AND USE

- Since natural gas contains some amount of propane and butane, we remove them.
- This is because, being heavier and denser, propane and butane liquifies at low pressure of 5 bar and can be used as cooking fuel.
- The pressure in your LPG cylinders is around 12 bar to keep it liquid inside the cylinder. This is done to get high energy per volume.
- On the other hand, its major component methane is lighter and hence needs to be compressed or liquified in order to be used in vehicles.



- Accordingly, we make Compressed Natural Gas or Liquified Natural Gas. (more on this in a bit)

INDIA'S PUSH FOR GAS-RUN ECONOMY

- India has been taking a number of steps to leapfrog into a gas-run economy in the near future.
- Currently, the share of natural gas in India's energy basket is 6.7% compared to 23% worldwide. This India wants to increase to 15% by 2030.
- Accordingly, India has taken the following steps:
  1. Expansion of National Gas Grid to about 35,000 Km from current 17,000 Km.
  2. Expansion of city gas distribution network to cover 96% of India's population and 86% of its geographic area.
  3. Setting up of LNG Terminals.

3. COMPRESSED NATURAL GAS

- Methane, being lighter, is compressed at high pressure (around 225 bar) in the vehicle tanks to get a workable energy-density. This limits the size of the tank making it suitable for light-commercial vehicles like cars, auto-rickshaw etc.
- Thus, CNG vehicles have low range and are suitable only for intra-city travel.

NATURAL GAS ALTERNATIVE IN TRANSPORTATION

While CNG continues to grow rapidly in India, the next logical step is to step up LNG production to replace diesel in heavy-load trucks.

CNG IN TRANSPORT FLEET

- In 2022, annual sales of CNG vehicles in the passenger segment more than doubled. 49% of the light commercial vehicles sold in 2022 were CNG-powered.
- India plans to increase the number of CNG stations to 20,000 by 2030 from the current 6,000.

WORLD'S 1ST CNG TERMINAL: India is setting up the world's 1st CNG in Bhavnagar, Gujarat.

Advantage of CNG as alternative

- No sulphur emissions.
- No unregulated emissions like benzene.
- Low carbon emissions. (1 unit of natural gas produces half the emission as compared to 1 unit of coal)
- No PM pollution.

Challenge: NOX emissions will be higher due to higher temperature of combustion.

4. LIQUID ALTERNATIVES

- Liquid fuels can be alcohols or hydrocarbons. While hydrocarbon liquids are diesel and petrol, alcohol-based fuels are methanol and ethanol.
- Alternative liquid hydrocarbons include LNG, LPG, Dimethyl ether, liquid hydrogen (suitable only in rockets).

GAS-TO-LIQUIDS: HYDROCARBON-BASED ALTERNATIVES

- As you have already seen, hydrocarbons in the gaseous form are small chain molecules like C1, C2, C3 and C4 (methane, ethane, propane and butane)
- While gas is voluminous, its density is too small.
- Therefore, to transport and use gas either we need to pump it at high pressure, or we need to liquify it.
- Gas-to-liquid alternatives involve this conversion of gas to liquid fuels. (like LNG)
- When we are liquifying gas what we are really doing is only fusing small chain gas molecules into long-chain ones. This is fancifully called polymerising.
- This will increase the boiling point and eventually become liquid.
- Basically, **by polymerising hydrocarbon gas molecules to liquid, we are only making diesel.**

Advantage

- Higher energy density
- Ease of transport and storage
- Zero sulphur emissions
- Zero aromatics (no closed-rings remember?!)
- Low NOx as compared to gaseous fuels.

5. LIQUIFIED NATURAL GAS

- Alternatively, we can liquify methane in order to achieve higher energy density.
- However, at normal atmospheric pressure methane liquifies at a very low temperature of about -1700C because it has a low boiling point of about -1600C.
- Thus, in order to transport we need cryogenics which is not feasible for 2-wheelers and 4-wheelers.
- However, LNG, with its higher energy density, occupies three times less space than CNG.
- Thus, it is suitable for long-haul transport like trucks provided they are fitted with cryogenic tanks.

LNG in transport fleet:

- LNG is the best bet to replace heavy-duty diesel-based trucks.
- India has more than 10 million trucks, of which the government expects at least a million to run on LNG by 2035.
- India has planned to build 1,000 LNG stations in next 3 years along major highways, industrial corridors and mining areas.

6. LIQUIFIED PETROLEUM GAS

- As we have seen gaseous components of petroleum separation include methane, ethane, propane and butane. While methane and ethane are used in the form of gas (natural gas), the heavier gaseous hydrocarbons are liquified to form what is called liquified petroleum gas.
- This is because heavier gases like propane and butane liquify when compressed become liquid at lower pressures of around 5 bar.
- This is why your LPG cooking cylinders are maintained at a low pressure of 12 bar to hold enough fuel.
- LPG is a better alternative than LNG in transportation as it can be liquified at lower pressure and thus suitable even for 2, 3 and 4 wheeler segments.
- The only constraint has been lack of supply and thus LPG use is restricted as cooking fuel.

7. METHANOL

- Another approach to make liquid fuel is to compress the syn gas obtained from the gasification process. (remember syn gas is produced by burning any hydrocarbon, be it fossil or biofuel, with steam)
- The mixture of CO and H2 is adjusted in its pressure and temperature to form methanol. (CH3OH)
- $CO(g) + 2\ H_2(g) \rightarrow CH_3OH(l)$
- Methanol is also called wood alcohol.

Advantage

- Methanol can directly be used in IC engines or even converted to petrol using a zeolite catalyst.
- Note that methanol can be produced from any hydrocarbon not just coal including natural gas, biomass, and even captured carbon dioxide. The only thing you need to do is adjust the amount of hydrogen.

Significance of Methanol

- Methanol cars are becoming common, especially in China, where coal is in abundance.
- Methanol can also be used in fuel-cells and thus could drive the EV revolution.
- India by adopting Methanol can reduce its import dependence on oil and at the same time have a cheaper fuel (at least 30% cheaper than any available fuel)
- Niti-Aayog is set to come out with a roadmap for transition to the Methanol Economy.

8. DME: DI-METHYL ETHER

- The most significant thing about **DME is that it is liquid at room temperature.**
- To make DME all you need to do is to take 2 molecules of methanol and remove water from it.
- $CH_3OH + CH_3OH \rightarrow CH_3\text{-}O\text{-}CH_3 + H_2O$

Advantage

- Liquid at room temperature
- Liquifies at 5 bars and thus easily stored.
- Better for NOx reduction
- Low CO_2 emissions owing to high H-C ratio.

9. HYDROGEN

- A general trend towards development of better fuels is hydrogen-rich fuels.
- This means more of hydrogen in the fuel and less of carbon or more hydrogen to every carbon atom. Eg: Natural gas 4 hydrogen to every carbon as opposed to very little hydrogen in coal.

- This is because just like carbon, hydrogen is also combustible, i.e. it mixes with oxygen in the air and gives heat. In addition, moving from a solid to a liquid and then finally to a gaseous state energy carrier.

NATURE OF HYDROGEN

Hydrogen as an Energy Carrier

- It is important to understand that all fuels we have seen so far are energy carriers. Hydrogen is the best energy carrier as there are no harmful impact (read carbon emissions)
- This is because hydrogen is not freely available on earth, but it is in abundance in the form of water and hydrocarbons.
- However, to extract hydrogen from water or hydrocarbons you need to expend energy and the energy spent in extracting hydrogen is much more than the amount of energy the so-extracted hydrogen gives out. This is why hydrogen is an energy carrier and not an energy source. (in fact all fuels we have seen are energy carriers)

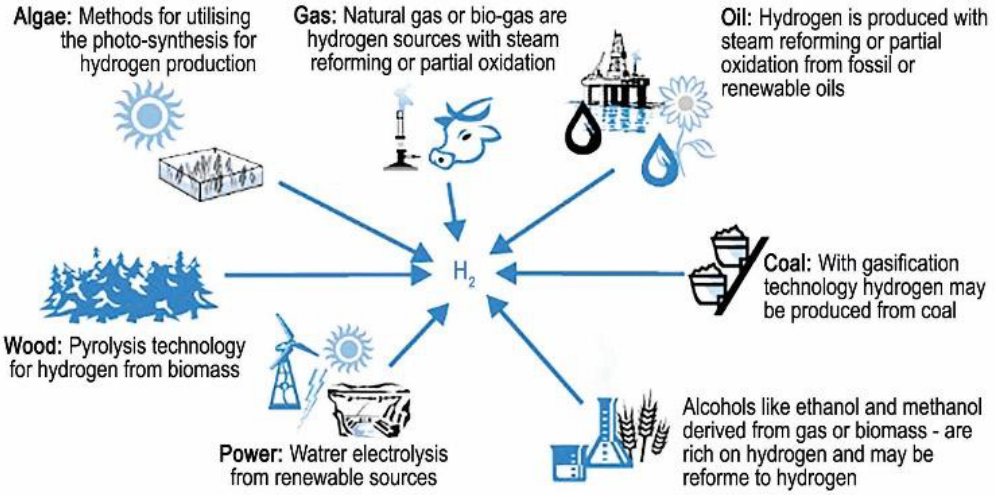
Combustible nature: Hydrogen is highly combustible i.e. it mixes readily with oxygen to produce heat.

High energy density per gram: Further the amount of energy out of this process is about 3 times higher than that you get when you burn petrol. (26 Kcal/gm for hydrogen compared to 10 Kcal/gm for gasoline).

Extremely low density per volume

- However, the problem with hydrogen is that it has low density, meaning the amount of hydrogen mass you can hold in 1 liter is about 71 grams. That means the tank size at normal temperature and pressure to hold hydrogen is very big.
- In other words, though the fuel itself is lightweight the tank size and therefore the weight of the tank goes higher. This puts a limitation on hydrogen being used as a fuel in private transport.
- Thus, hydrogen is suitable for large vehicles like buses which requires a limited range but can hold a large tank. This requires hydrogen to be compressed at high pressure. Alternatively you can liquify hydrogen by compressing and taking away heat.

HOW TO PRODUCE HYDROGEN?



- You can not mine hydrogen. There is virtually no hydrogen gas (or liquid) in the environment.
- But there is lots of hydrogen in water and in fossil fuels (hydrocarbons)- but not "free" hydrogen, the molecule H2. That is what we want for the hydrogen economy.
- Two major sources of hydrogen on earth are water and hydrocarbons.

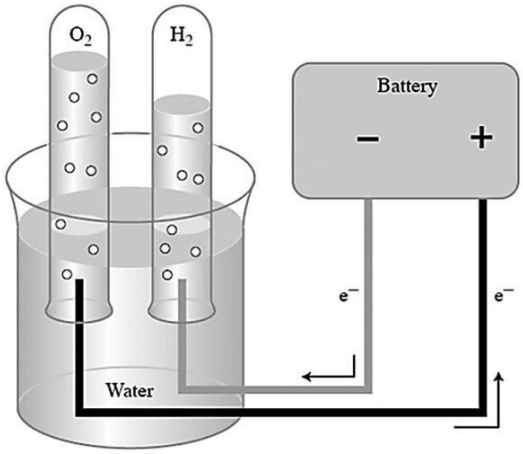
FROM HYDROCARBONS

- Take any hydrocarbon and treat it with steam we get syn gas which is a source of hydrogen.

- Any hydrocarbon + H_2O → CO + H_2
- Hydrocarbons could be either fossil or biofuel or even organic waste.
- However, the hydrocarbon source of hydrogen is again a problem because the leftover carbon has to go to the atmosphere.
- This makes it dirty. That is why hydrogen from these sources is colour coded with 'dirty' colours like grey, blue, black etc.

NATIONAL GREEN HYDROGEN MISSION

- Launched: 2023. Outlay: Rs. 19,744 crores from FY 2023-24 to FY 2029-30.
- **Aim:** To develop India into a global hub for production, usage and export of Green hydrogen and its derivatives.
- The scheme envisages generation of hydrogen from green power sources with a target of 5 MMT production capacity of Green Hydrogen per annum by 2030.
- **Initiative of**: Ministry of New and Renewable Energy.



ELECTROLYSIS OF WATER

- Pass electricity through water, it will split it into its constituent Hydrogen and Oxygen.
- Process where electricity is used to make a chemical change that would not happen otherwise.
- In a normal situation, oxygen pulls electrons and hydrogen pushes its electrons.
- In case of electrolysis, water is split into hydrogen and oxygen and for this to happen hydrogen has to gain electrons and oxygen has to lose electrons which is the opposite to what happens normally.
- This requires energy which is what electricity gives.
- So, take a battery, use the energy to pull the electrons out of oxygen and push it towards hydrogen.
- If the electricity you use to split water comes from renewable sources, it gives you green hydrogen, the cleanest source of hydrogen.

HOW TO USE HYDROGEN?

- Hydrogen being combustible can be used directly in IC (Internal Combustion) engines. However, the efficiency is very low, hardly 25-30%.
- Blend it with another gaseous fuel, or use it in a fuel cell. We can blend hydrogen with natural gas. This is what is called H-CNG.

ADVANTAGES OF HCNG

- The energy density increases.
- Carbon emissions are reduced.
- Low NOx emissions

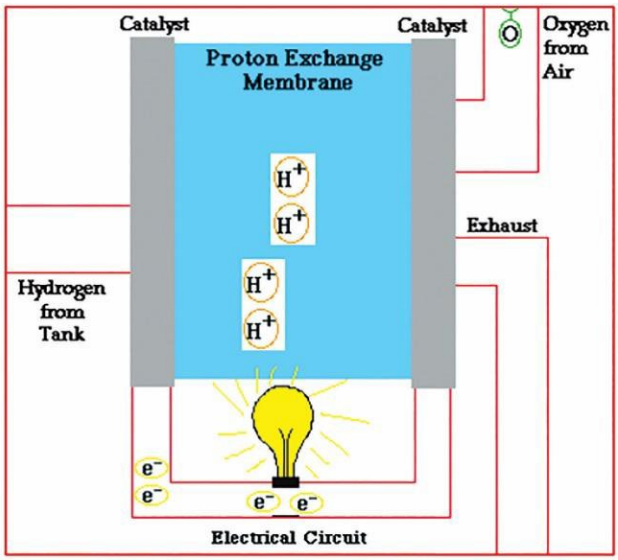
- No sulphur emissions

10. FUEL CELLS

- The best way to use the energy from hydrogen is fuel cells.
- The main advantage of using hydrogen in a fuel cell over using it as a fuel is the increase in efficiency which is around 60% for solid-oxide fuel cells and around 50% for PEM fuel cells. (heat energy is a poor form of energy compared to electric energy)
- Before you read about fuel cells do read the section on batteries below.

PRINCIPLE BEHIND FUEL CELLS

- A fuel cell is just electrolysis in reverse. Meaning in a fuel cell you pump hydrogen and oxygen which combines to form water, in the process you can derive electricity.



HOW DOES IT WORK?

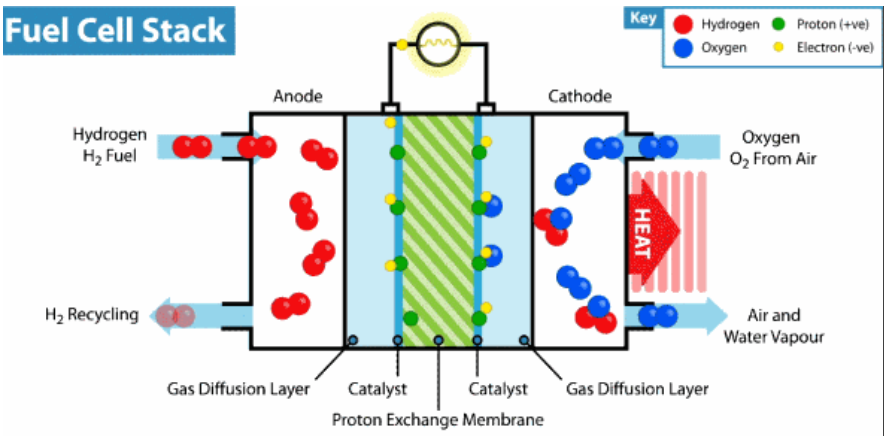
- You pump hydrogen on one side and separate it into its constituent ions and electrons.
- Make the electrons pass through a wire and you have electricity.
- In addition, we use an electrolyte which is simply a membrane that allows ions to pass through and not electrons to pass through.
- The separation of hydrogen into its constituent electron and hydrogen ion is brought about by a platinum catalyst. The challenge is its high cost.
- Finally, when ions and electrons recombine the resultant hydrogen is mixed with oxygen to produce water. This produces heat which is a deciding factor in choosing the fuel cell. **(operating temperature of fuel cell)**

TYPES OF FUEL CELLS

- Please note that the only fuel cells that are feasible for use in automobiles are the ones which have manageable operating temperatures.
- Following are the fuels cells that are relevant for use in electric vehicles.

PROTON-EXCHANGE MEMBRANE FUEL CELL

- The example we have taken to illustrate the working of fuel cells above is called proton-exchange fuel cell.



SOLID OXIDE OR METHANE FUEL CELLS

- Alternatively we can use methane or even ethanol out of which you can derive hydrogen inside the fuel cell itself.
- However operating temperatures are high upon formation of water (around 200 degree celsius).
- Due to high operating temperature, we can use solid electrolyte, typically a meta oxide. Thus, the name.
- Having a solid electrolyte is important for use in consumer electronics.
- However, one disadvantage of such fuel cells is that the solid electrolyte used are typically rare earth elements. (eg: cerium, gadolinium, yttrium called YSZ)
- These are expensive and also environmentally harmful)

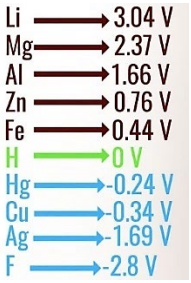
11. BATTERIES AND FUEL CELLS

PRINCIPLE

- Batteries or fuel cells are devices that derive electricity out of a chemical reaction that happens.
- The chemical reaction is simply pulling out electrons from something and making it travel along a metallic wire producing electricity. The thing about electrons is once you pull it out of something it needs a place to go.
- The terminals from where electrons are pulled out and sent to are called electrodes.
- The terminal from where electrons are pulled out are called anode and those where electrons are sent to are called cathode.
- The chemical reaction that leads to pulling out of electrons is called oxidation and that that leads to dumping of electrons is called reduction.
- So typically, oxidation happens at anode and reduction happens at cathode. In the process of electrons going from anode to cathode, the energy flow of electrons (electricity) is captured.

HOW DOES A BATTERY STORE ENERGY?

- A battery is a device where energy is stored for use later. So, in order to store energy, you need an energy source.
- The process of storing energy in a battery is what you know as charging the battery. Similarly, the process of using the energy is discharging the battery.



CHARGING AND DISCHARGING THE BATTERY

- Here you pump energy from an outside source to take electrons out of some chemical from where it does not want to come and send it to another chemical where it does not want to go. Once you do this the electrons want to run back to its original place.
- This is like sending electrons uphill(charging) and allowing the electrons to roll back downhill by itself (discharging).

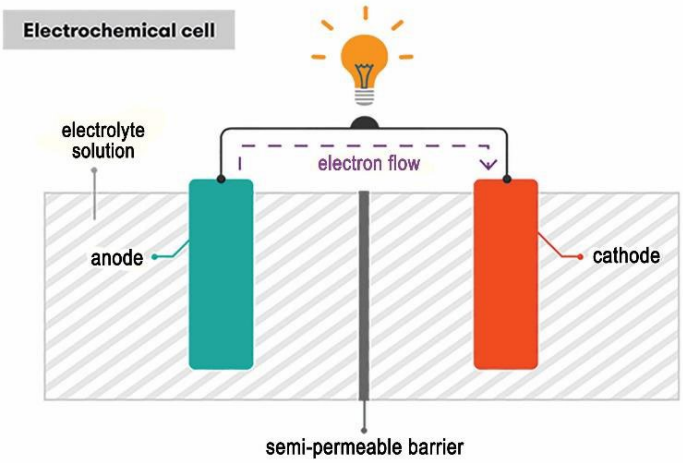
WHAT CHEMICALS TO USE?

- Everything depends on an element's affinity (likeability) for electrons. Simply elements which like electrons and those that do not like electrons.
- The likeability for electrons decides the electrochemical potential of that element.
- Electrochemical potential is simply how readily an element wants to lose or gain electrons.
- Higher the electrochemical potential the element wants to readily lose electrons, lower the electrochemical potential it wants to gain electrons.

WORKING OF A BATTERY

- The trick is to take two different metals with different electro-potential so that one wants to gain, and one wants to lose electrons. **E.g.,** Zinc (Zn) and Copper (Cu).

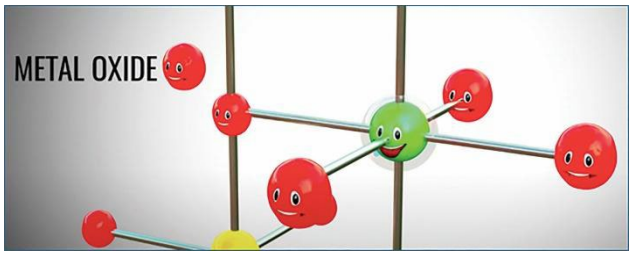
- As it is evident from the table, Zinc wants to give electrons and Copper wants to take it. (that is why Zinc is anode and Copper is cathode)
- So, if you somehow give a path for these electrons moving from Zn to Cu, you will have derived electricity.
- When all the electrons have come to Cu, you can no longer derive electricity.
- So, you have to forcefully pull electrons out of Cu (cathode) and put it back at Zn (anode). This is charging.
- Once you have forcefully pulled out all electrons from Cu and have put it back at Zn, you can again rely on the natural flow of electrons from Zn to Cu to derive electricity. This is discharging.
- One last thing is about the electrolyte. An electrolyte is like a semi-permeable membrane for electrons and ions. It allows the ions to pass through them but not electrons.
- **Putting all the pieces together**



- In a battery a chemical reaction takes place where you are separating electrons and ions from the anode material.
- Once separated you make 2 different paths for these electrons and ions to move. For electrons you connect the anode to a metallic wire and for ions you dip the anode in electrolyte.
- Electrons will move towards cathode along the wire and ions move towards cathode along the semi-permeable electrolyte.
- Electrons movement through the wire gives you electricity.

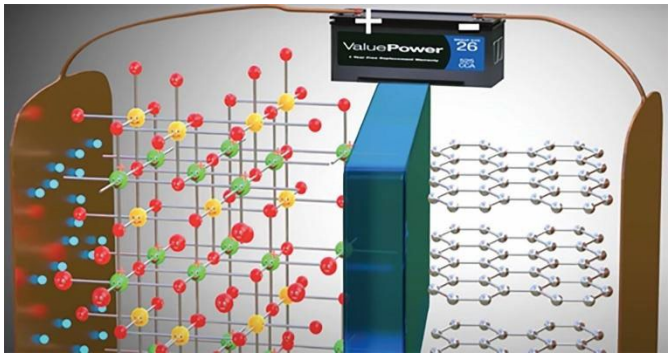
12. LITHIUM-ION BATTERY

PRINCIPLE

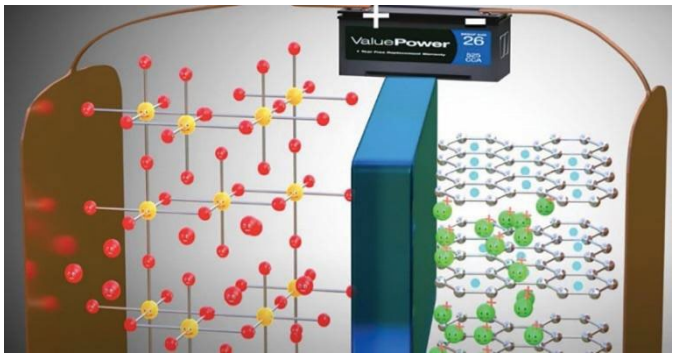


- Lithium, as you can see in the table above, has the highest electrochemical potential.
- It wants to lose electrons readily which makes it very reactive. That's why you don't get lithium in free form.
- However, when mixed with metal oxide lithium sits very stably.
- Thus, if we use this ability of lithium to be very unstable by itself to become very stable in metal oxide, we can derive electricity. This is what happens in a Li-ion battery.

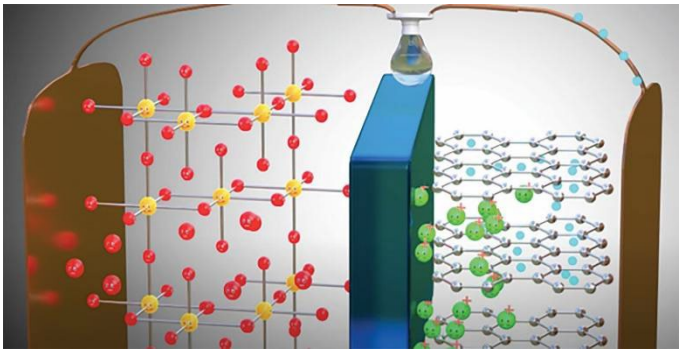
WORKING OF LI-ION BATTERY



- Lithium is mixed in metal oxide (typically cobalt, nickel, or manganese)) and is used as cathode.
- Graphite is used as a place to hold Li-ions which become an anode.
- As we have seen, Lithium in metal oxide is very stable.



- In the Li-ion battery we separate Lithium from metal oxide by pulling out its constituent electrons and ion forcefully by applying energy.
- The electrons and ions of Lithium are then given separate paths namely a metallic wire and an electrolyte.



- This is called charging as it requires external energy to separate electrons and Li-ions from Lithium metal oxide.
- The li-ions moving through electrolyte and electrons moving through the wire then recombine at anode which is graphite.
- Once all the electrons and ions are pulled out the battery is completely charged.
- The lithium ions and electrons that are sitting between graphite sheets are unstable and want to go back to metal oxide. If we again give separate paths to electrons and ions we can derive electricity.
- Thus a Li-ion battery is used to store energy by shuttling lithium ions back and forth between the anode(Li-ion in graphite) and cathode(Lithium in metal oxide).

ADVANTAGE OF LI-ION BATTERIES

- **Light weight:** Lithium being the lightest metal.

- **High Energy Density**
  - Lithium having highest electrochemical potential has very high energy density.
  - A typical automobile lead-acid battery weighs 6 kilograms more to store the same amount of energy than a lithium-ion battery.
  - In consumer electronics like mobile, laptops, camera etc.
  - Nickle-cadmium battery stores typically 60 to 70 watt-hours. A typical lithium-ion battery can store 150 watt-hours of electricity in 1 kilogram of battery.
- **Minimum losses:** A lithium-ion battery pack loses only about 5 percent of its charge per month, compared to a 20 percent loss per month for Ni-Cd batteries.
- **Low Maintenance:** Lithium-ion batteries can handle hundreds of charge/discharge cycles.

DISADVANTAGES

- **Faster discharge**
  - While quick discharge is an advantage in electric vehicle and consumer electronics applications, it is not suitable to store energy for longer than 4 hours.
  - Thus, it is **not suitable for grid-level storage** which is necessary for renewable energy like solar which suffer from intermittency problems.
- **Ageing:** Li-ion batteries suffer from ageing at room temperature. Therefore, in a consumer electronic, batteries need to be partially charged for longer life.
- **Transportation:** Another disadvantage of li-ion batteries is that there can be certain restrictions placed on their transportation, especially by air to protect against short circuits.
- **Cost:** Lithium-ion batteries are around 40% more costly to manufacture than Nickel cadmium cells owing to high cost of lithium refining, cobalt and nickel.

13. ALTERNATIVES TO LIB

SODIUM-SULPHUR BATTERY

Advantage

- Sodium-Sulphur batteries can be recharged 4500 times compared to 500 times of Lead-acid and Li-ion batteries.
- Furthermore, Lithium is way more costly than sodium (10 times more per kilo).

Limitation

- The problem is Sodium used in these batteries need to be in liquid state and that happens at high temperatures of 350 degree Celsius. So, Sodium-Sulphur cannot be used in laptops and mobile phones.
- Price per charge-discharge cycle is high.

FUEL CELL V/S BATTERY

FUEL CELL	BATTERY
Not an energy storage device in the strict sense. It produces electricity as long as the fuel is supplied	Energy storage device
Longer life	Shorter life as chemical is degraded during charge-discharge cycle t.

Lower efficiency as compared to battery: around 50-60%	Efficiency is high: around 80-90%
Requires less time to refuel fuel cell based electric vehicles as you directly pump hydrogen in the tanks.	Requires more time for recharging

MICROBIAL FUEL CELL

- It uses an organic electrolyte instead of inorganic ones in the above case.
- In other words, anaerobic oxidation of organic substances such as acetate, glucose, lactate, ethanol by microbes.

14. BIOFUELS

BIOFUEL: A NET-NEGATIVE PATHWAY

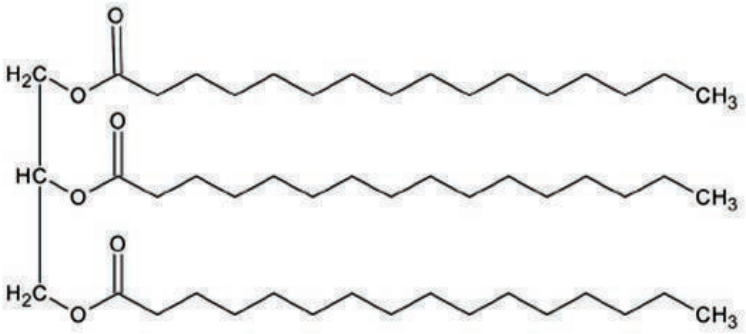
- Biofuels are a way to take out CO2 from the landmass above the surface of the earth and put it back in the atmosphere.
- This is unlike fossils where you are taking carbon stock from below the surface and putting it in the atmosphere increasing its concentration.
- Besides, the amount of CO2 you are putting by burning 1 kg of biomass is much lesser than the amount of CO2 the plants used to make 1 kg of plant biomass. (plants convert only 0.5% of photosynthesis to biomass)
- Thus, biofuels are net-negative in terms of carbon emissions.



(a) Petroleum diesel



(b) Biodiesel



(c) Vegetable oil or Triglyceride

As hydrocarbon: Biofuel v/s fossil

- Biofuel is small ring hydrocarbons with **lots of oxygen and benzene rings. (not suitable remember)**
- Fossil as we have seen is just biomass buried for millions of years under low oxygen conditions.

- Besides biomass are long polymers of sugar molecules. So, if we somehow remove oxygen and break the polymer, we can get biofuels.

BIOMASS AS ENERGY STORE

- Further biomass is basically solar energy trapped in the form of chemicals that make up the living beings.

BIOMASS COMPOSITION

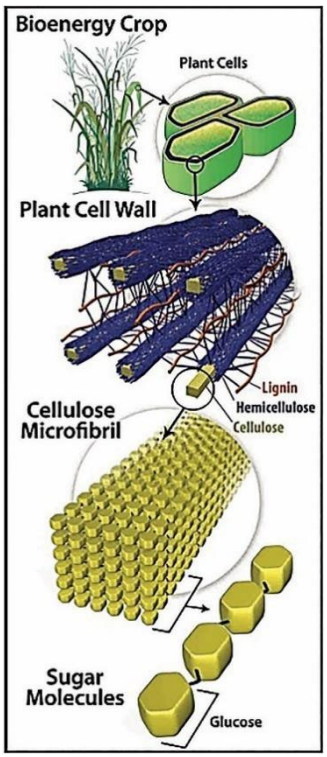
- C 50%
- O 40%
- H 5 %
- N 1%(From protein)
- P 0.5% (From protein)

BIOMASS CONSTITUENTS

- Primarily made of sugar polymers with lots O_2 and closed ring hydrocarbons. (Half the plant biomass is fundamentally sugar)
- Plant cells are made up of cell walls that give a protective layer to help plants stand tall.
- The cell wall is made of microfibrils of **cellulose** which is basically sugar polymer. (1/2 of cell wall)
- At later stages there is an additional protective layer in the form of hard cover in trees which is called **lignin**.
- Lignin is plastic-like. It is not degradable, and has lots of aromatics.

CLASSIFICATION OF BIOFUELS

- For the sake of simplicity, think of plant biomass in terms of edible and non-edible parts.
- The edible part of plants like fruits, vegetables, grains etc are usually made of simple sugar, starch.
- The other edible part is seeds that have edible oils.
- The non-edible part of biomass on the other hand is made of cellulose, hemi-cellulose and lignin.



ENERGY FROM BIOFUELS

- There are 3 pathways to derive energy out of biofuels:
  1. Burn them
  2. Make alcohols like ethanol and methanol
  3. Make biodiesel
  4. Make biogas

Burning of biofuels

- Burning solid biofuel will leave solid ash residue which is not desirable. Eg: Cow dung, wood etc
- So, we need to gasify or liquify biomass.

Liquifying biofuels

- Liquifying biofuels again include 2 pathways
  1. Make alcohols: ethanol and methanol: fermentation
  2. Make biodiesel: From vegetable oils: transesterification

Gasifying biofuels

- Biogas: Bio-methanation
- Biohydrogen: from syngas

ALCOHOLS: ETHANOL AND METHANOL

- Alcohol is the best alternative to petrol.
- You can directly replace petrol with ethanol or methanol in IC engines or alternately you may blend it with petrol. (Ethanol blending target 20% by 2025)

Fermentation

- Alcohols are made by fermentation of sugars by microorganisms.
- Depending on the microorganism used and sugar source you get different alcohols.

15. ETHANOL

- The most commonly formed alcohol through fermentation is ethanol.
- Fermentation is simply breaking down of sugar molecules in the absence of oxygen at the cellular level or in other words anaerobic cellular respiration to produce ethanol.

WHICH MICROORGANISMS PRODUCE ETHANOL?

- Microorganisms capable of producing ethanol include yeast, some species of bacteria, fungi and some species of micro algae.

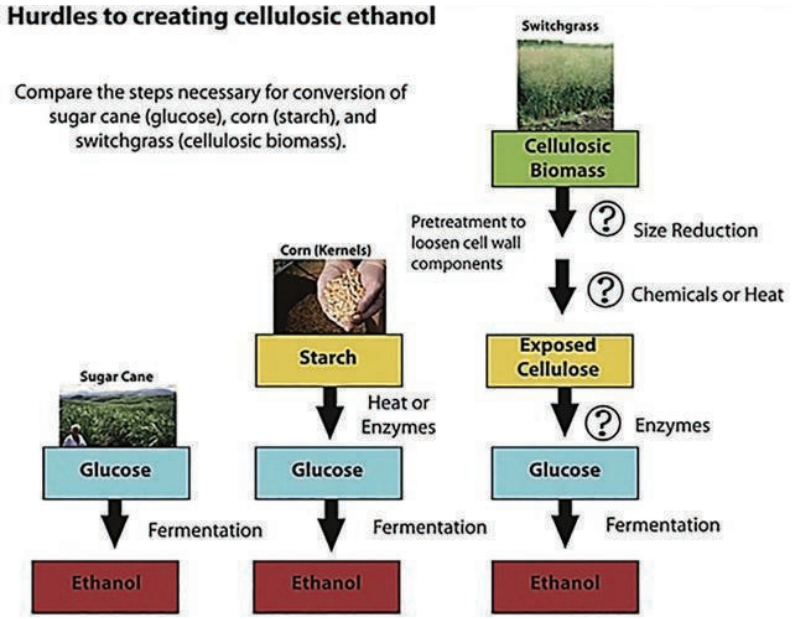
FEEDSTOCK

Edible part

- Ethanol can be made from both edible parts and non-edible cellulosic biomass.
- Edible sources include sugarcane, corn, and other starch-based crops (sugar beets, rice, wheat, potatoes)

Non-edible part

- The cellulosic matter from plant biomass are basically complex sugar molecules.
- Thus, they have to be broken down into simple sugar in order to be fermented into alcohols.
- In order to break them down hydrolysis is done in other words treatment with water. These simple sugars can then be used for fermentation by microbes to produce ethanol.
- However, lignin is made of phenols and not sugars. Thus, they cannot be converted into alcohol.



Advantage

- No SOX emissions
- Can be used as a substitute for petrol in SI engines
- Low carbon emissions (biofuels are net negative)
- Low NOx emissions

METHANOL

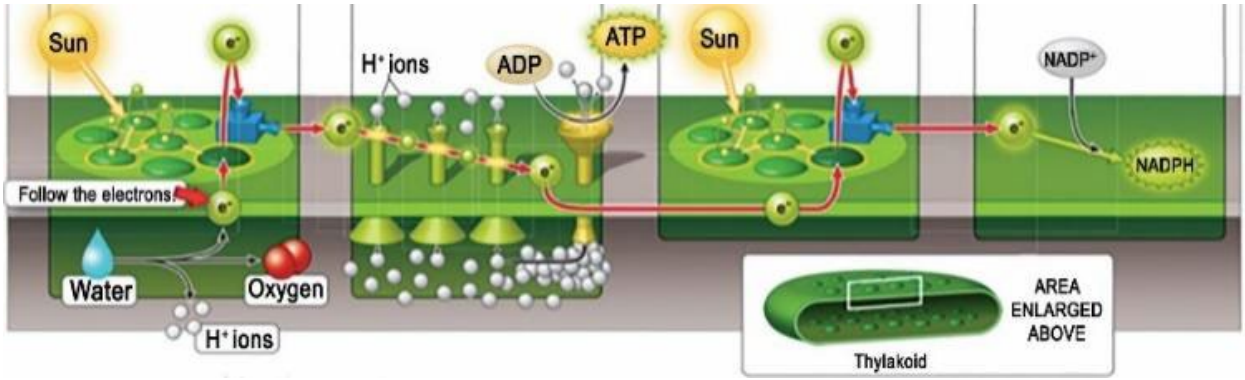
- Although methanol can also be made through fermentation, it is rare. This is because it is produced in small amounts as by-products during fermentation of certain bacteria and fungi species.

- Thus, the commonly used pathway to make methanol is producing syngas. (discussed in alternate fuels: Gas-to-liquid section)

16. PHOTOSYNTHESIS AND CELLULAR RESPIRATION: THE UNIVERSAL ENERGY PATHWAY

PHOTOSYNTHESIS: BUILDING THE ENERGY STORES

- Plants have simply collected energy from sunlight and stored it in sugar molecules through the process of photosynthesis.
- Under photosynthesis, photoreceptors in plants called chlorophyll absorb sunlight. When light falls on chlorophyll it excites electrons in the chlorophyll. This is how energy from sunlight is captured.



- The excited electron then carries this energy and transports it to different parts of the plant body and stores. (follow the path of electron in the figure)
- The energy carried by the electrons is stored in a kind of rechargeable batteries in plants called ATP molecules. (future civil servants need not know the details of this molecule)
- These rechargeable batteries called ATP are stored in the sugar molecules of plant biomass built by plants.
- To do this, plants use CO_2 and H_2O in the atmosphere.

CELLULAR RESPIRATION: USING THE ENERGY STORES

- Ultimately all living organisms access energy through the process of cellular respiration.
- The process of utilizing the energy contained in food is called cellular respiration.
- Cellular respiration is just a way of taking in energy stored in the sugar molecules to run life activities.
- The food we eat is first digested or broken into simple sugars and carried in the bloodstream which carries the energy-rich sugar to all the cells of the body.
- The cells in turn use the energy in the sugar to form the rechargeable batteries ATP in the cells which then acts as drivers of life activity. Once done CO_2 is released. Again, it is the electrons that carry energy.
- Once all the energy is delivered to ATP the electron needs a place to go. For this all living beings have electron acceptors.
- Oxygen is the most common electron acceptor. That is why we breathe oxygen.
- Finally, after oxygen receives electrons, it combines with hydrogen to form water completing the process of cellular respiration.
- Thus, for cellular respiration we need glucose and oxygen.
- $C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O$ + energy (needed for life activities)

CELLULAR RESPIRATION IN LOW OXYGEN CONDITION

- When we are doing strenuous jobs like running or swimming, we need more energy. This energy is obtained from breaking down glycogen (a storehouse of glucose in our body) that is stored in the liver.
- However, there is not enough oxygen to accept electrons. Thus, cellular respiration comes to a halt.
- Alternately humans use pyruvate molecules (broken sugar) as electron acceptor instead of oxygen.
- When pyruvate accepts electrons, it forms lactic acid instead of H2O.

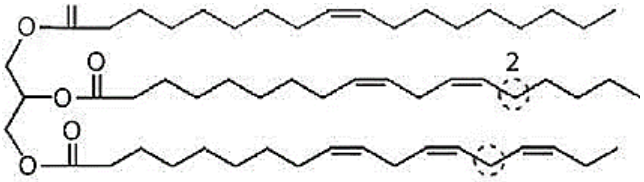
FERMENTATION: CELLULAR RESPIRATION IN THE ABSENCE OF OXYGEN

- Yeasts like humans also break down their food using oxygen.
- In the absence of oxygen, they produce alcohol that we drink.
- In yeasts after breaking down glucose into pyruvate. After this the pyruvate is converted into acetaldehyde which acts as electron acceptor in the absence of oxygen.
- When acetaldehyde accepts electrons, it forms ethanol.
- This anerobic respiration is called fermentation.

17. BIODIESEL: FROM TRANSESTERIFICATION

WHAT IS BIODIESEL?

- Oils from oilseeds are basically straight chain hydrocarbons but long ones.
- This is the only difference between biodiesel and diesel.
- If we take vegetable oil and break its long hydrocarbon straight chain into 1/3rd, 1/3rd, 1/3rd you get biodiesel.
- Biodiesel can directly replace diesel in diesel-IC engines.
- The resultant product, i.e., biodiesel is called ester.
- That's why the process of breaking long straight chain hydrocarbons into short chains is called transesterification.
- Breaking can be done by following ways:
  - Heat it: Pyrolysis
  - Apply pressure: Cracking
  - Replace double bonds with hydrogen: Hydrogenation



Feedstock

- Any oil seed can be used to extract oil. However, a better option is to use non-edible oil seeds.
- There are more than 200 varieties of oil seeds that can be used to produce biodiesel.
- Some common examples include rice bran, sal, neem, mahua, karanja, castor, linseed, jatropha, rubber seed etc.

Advantages

Fuel	Energy density] (in MJ/ kg)
Ethanol	24-25
Peirol	43-44
Biodiesel	40-41
Diesel	45.5

- High energy density
- Low energy input

- Nitrogen-fixation
- No Sulphur
- No aromatics

A CASE FOR BIODIESEL

- India uses 5 times more diesel than petrol, so an alternative for diesel is more important than that for petrol.
- De-sulphurisation of diesel is cost intensive.
- Rural development: growing oilseed-based crops for biodiesel will augment farmer's income.
- Converting degraded land
- Improves soil fertility as most oilseed-crops are leguminous crops which helps in nitrogen fixation.

18. GENERATIONS OF BIOFUELS

- Biofuels are classified into different generations based on the strategy adopted to derive energy from biomass. Accordingly, there are 4 generations of biofuels.

1ST GENERATION (MOSTLY EDIBLE SOURCE)

- Biofuels derived from edible sources.
- Includes both starch-based ethanol and biodiesel (made from edible vegetable oil)
- Sugarcane, corn, and other starch-based crops (sugar beets, rice, wheat, potatoes)
- Oilseeds include rape-seed, sunflower seed, soybeans, palm seeds, rice bran etc.
- Non-edible source Jatropha in India

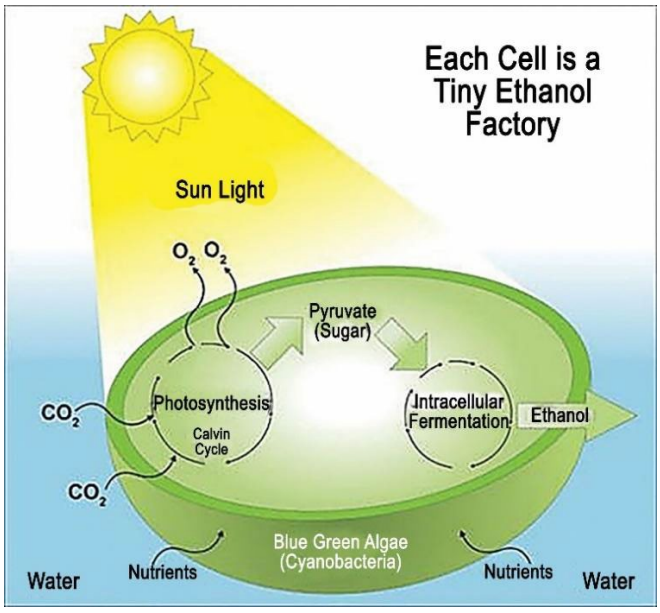
2ND GENERATION (CELLULOSIC BIOFUEL)

- Non-edible feedstocks like grasses, leaves etc.
- Ligno-cellulosic: Mostly wood
- Agricultural waste including rice husk, corn stalks etc

3RD GENERATION (MOSTLY ALGAL BIOMASS-BASED)

Biofuels from high-photosynthetic efficiency organisms like algae.

- Feed-stock include:
  - Macro-algae
  - Micro-algae
  - Aquatic plants (water hyacinth)
- While most plants are very inefficient in photosynthesis (only 0.5% of sunlight is captured), algae have very high photosynthetic efficiency capable of growing very fast, capturing the CO2 from the atmosphere.
- Algae extracts large amounts of the carbon dioxide to feed their growth and very little is left to emit to the open air.
- The so grown algae can be used as input for producing both ethanol and biodiesel.



Advantages

- Fastest photosynthesis

- Do not need arable land
- CO2 sequestration (CO2 source can be coal-based power plants too, see section on Direct capture under CCUS technologies)
- Lipids (can be used for biodiesel) and protein source (single-cell protein source)

LIQUID TREE: LIQUID 3.0

- Recently in Serbia scientists have created such a photo-bioreactor using algae to capture CO2 from the atmosphere.
- It contains 600 litres of water and works by using microalgae to bind carbon dioxide and produce pure oxygen through photosynthesis, in addition to producing biofuels.

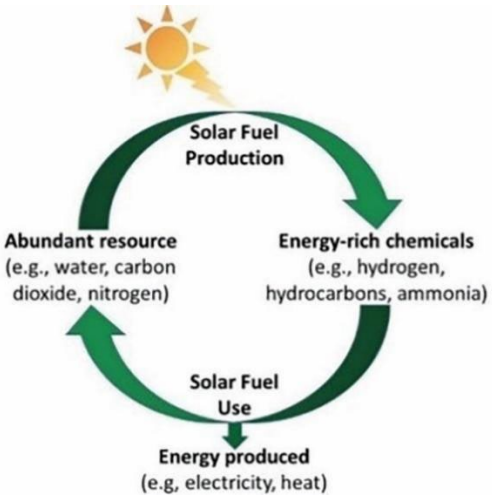
METHANOGENS: Additionally, there are organisms called Methanogens which capture sunlight to produce methane.

4TH GENERATION BIOFUELS: SYNTHETIC BIOFUELS

- Fuel sought to be extracted from GMOs; Carbon Capture and Utilisation (CCUS) technologies.

ARTIFICIAL LEAF AND PHOTOSHEETS

- This is application of synthetic biology to make biofuels.
- The approach mimics the biological process of photosynthesis to make energy-rich fuels. (Artificial photosynthesis)
- Currently there are two types of technologies performing artificial photosynthesis namely artificial leaf which produces syngas and photosheet technology which produces formic acid.
-
- An artificial leaf is a silicon-device powered by sunlight which produces synthesis gas (SynGas) by capturing CO2 from the atmosphere or flue gas from power plants.
- While artificial leaf is a cleaner way to produce syngas their main limitation is that it produces gas and takes an extra effort to store it in the form of liquid.
- Photosheet technology on the other hand produces formic acid which is stored as a liquid fuel.



19. GASIFYING BIOFUELS: BIOMETHANATION AND CBG

- Always remember the best way to biomass for energy production is to somehow extract only methane out of it and leave the rest for plant use.
- To gasify biomass is to copy what happens in a cow's stomach.
- Unlike humans, cows break down cellulosic biomass in its digestive process called anaerobic digestion.
- Cows have in its digestive system a large fermentation chamber filled with billions of microbes like bacteria and protozoa.
- The food that enters the fermentation chamber is broken down by these microbes, in the absence of oxygen, producing methane in its burps and fart. Mimic this process using bio-digesters you get biogas.
- Biogas is a mixture of methane, CO2 and hydrogen sulphide. So you need to separate CH4 before using it.
- Once separated, compress methane to store it and transport it. This process is called bio-methanation.

- Feedstock can be agriculture waste which is mostly cellulosic biomass, cattle dung, sugarcane press mud, municipal wet waste etc.

SATAT

- In 2018 Sustainable Alternative Towards Affordable Transportation (SATAT) initiative was introduced to promote CBG as an alternative to CNG.
- The target is to produce 15 million tonnes of gas aiming to reduce CNG use by 40%. (CNG is imported)
- Particularly useful in states like Punjab, Haryana and Uttar Pradesh which are infamous for stubble-burning.

➔ RENEWABLES

20. SOLAR ENERGY

NATURE OF SOLAR ENERGY

Ultimate source:

- Solar energy represents the ultimate source of all energy on earth.
- All energy sources we have seen so far are simply energy from the sun that was captured, be it in the form of ATP in plant biomass or energy in hydrocarbons in fossils.
- However, the amount of energy plant biomass or fossils have captured is a small proportion of energy from the sun.
- Thus, if we can find ways to access energy from the sun directly, we can increase the energy captured.

Intermittent nature:

- Besides since energy from the sun is accessible only during the day, solar energy, like other renewable sources, suffer from what is called an intermittency problem.
- As a result, an essential component of renewable energy including solar energy is energy storage.
- This is why India is mooting an Energy Storage Policy towards large-scale integration of renewable energy to the grid.

Decentralised source:

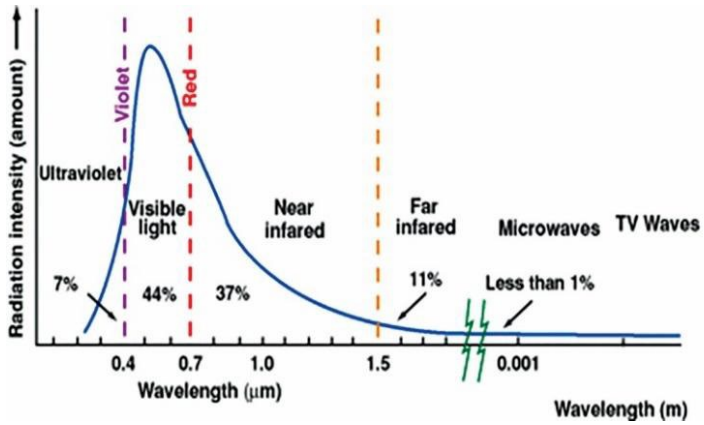
- Solar energy represents a decentralised energy source as sunlight represents a distributed energy resource.
- India's share of decentralised renewable energy is a meagre 6% of the total renewable energy capacity.
- The low share of decentralized solar energy, about 6.5 GW as against the target of 40GW, is the main reason for missing the target of 175 GW of RE by 2022.

India has set an ambitious target of achieving 500 GW of non-fossil fuel electricity capacity by 2030.

PATHWAYS OF CAPTURING ENERGY FROM THE SUN

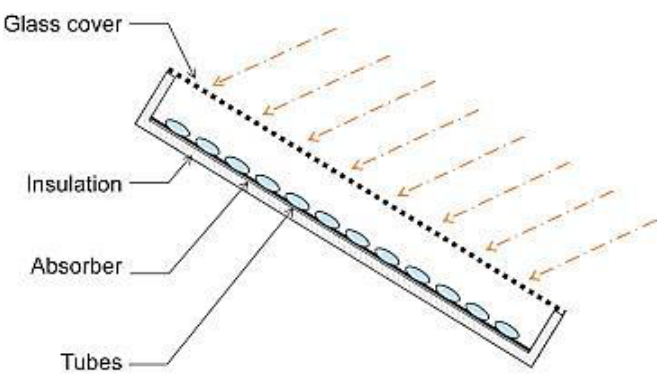
There are different ways to capture energy from the sun including:

- Solar thermal
- Solar concentrators
- Solar PV cells



21. SOLAR THERMAL

- The basic principle behind solar thermal is the green house effect. It is a way to capture the sun's energy in the form of heat.
- Since heat is low quality energy it has limited applications and thus is suitable only for domestic applications like solar water heaters, solar pumps etc.



PRINCIPLE: GREENHOUSE EFFECT

- Solar energy is an admixture of various electromagnetic radiation with visible light constituting the maximum proportion. See figure for sun's emission spectrum.
- Thus, in order to gain maximum from sunlight we need to capture visible light from the sun.
- In order to do this, we use the principle of greenhouse effect demonstrated by glass.
- Glass is transparent to visible light and opaque to infrared light.
- So, glass cover is used in the top layer to allow visible light and block the infrared from the sun.
- This visible light is made to fall on a metal plate coated black which absorbs most of the visible light. (absorber in the figure)
- As a result of absorption of energy, the black metal emits infrared radiation which cannot pass through the glass cover.
- Thus, energy from visible light is trapped between glass cover and metal plate.
- This energy is in the form of heat (convection current) which can be carried by water or air.
- Besides, the direction of the set up should be such that it should be able to maximize the sun's rays incident on it.



- In the Northern hemisphere, it should be inclined at an angle equal to the latitude and south-facing.

Advantage: The efficiency of solar thermal can reach up to 60%.

Disadvantage: It is suitable for domestic applications as it cannot be scalable (water is heated up to 60-70 degree celsius).

22. SOLAR CONCENTRATORS

- In applications where you need temperatures above the boiling point of water you use solar concentrators.
- Solar concentrators use a bunch of mirrors in such a way to focus all the incoming solar energy at one point from where it can be collected and carried.
- Solar towers are an example of solar concentrators.
- At the top of the tower is a sodium salt which can hold heat for a longer time thereby increasing the amount of heat that can be captured.



Advantage

- You can heat the salt and keep it for later use when there is no sunlight.
- Efficiency is as high as 50%.

Disadvantage: Capital cost is high as mirrors are expensive. High land requirement: 1.5 hectare per mega watt.

SOLAR TOWERS IN INDIA: Bhadla solar park in Jodhpur Rajasthan (biggest plant in the world)

23. SOLAR PV CELLS

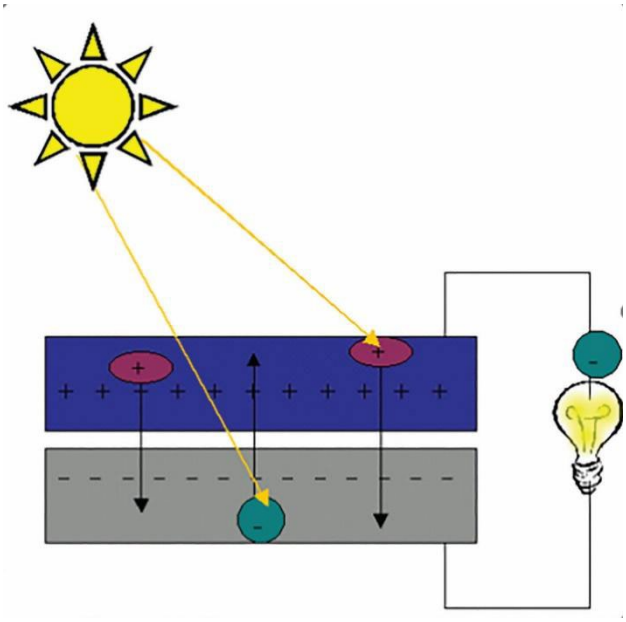
- Solar PV is the mainstay of solar energy particularly with decreasing cost of PV cells in recent times.
- A photo voltaic cell is an example of photodetector devices which work on the basis of photoelectric effect.
- It is a way of directly using sunlight to produce electricity.

PHOTOELECTRIC EFFECT

- The photoelectric effect, 1st demonstrated by Einstein, relies on the interaction of photons and electrons and their behaviour during interaction.
- Flash a beam of light onto some material (like metal). It will knock-off electrons, collect these electrons at an electrode. Connect the electrode to a wire and there you have electricity.

WORKING OF PV CELLS

- A photovoltaic panel creates electricity when a photon hits the silicon surface and pushes an electron out of the top layer of the silicon and across an electrical junction inside the panel.
- The movement of this electron creates a useful voltage. When wires are connected to produce a circuit, this voltage means that current will flow, eventually taking the displaced electron back to the top layer.
- Solar cells work best in strong sunlight but will also generate some power on an overcast day from the diffused light that gets through the clouds.
- Nature of electrical energy from PV cells: DC current.



TYPES OF PV CELLS

1. 1st gen: Single crystal PV cells

2. 2nd Gen: Thin-film PV cells
3. 3rd Gen: Perovskites

1st generation PV cells

- These are basically silicon-based.
- Single crystal silicon: have the highest efficiency, around 20-25%, high-cost manufacturing
- Amorphous silicon (powdered): efficiency is 5%, used in calculators

2nd generation PV cells: Polycrystalline PV cells

Cadmium Telluride

- CdTe can be deposited on thin sheets. So you can make flexible PV panels.
- CdTe absorbs sunlight readily to release electrons. So thin sheets can be made.
- Efficiency ~ 15%
- The disadvantage is that cadmium is highly toxic.
- Tellurium is not available as it is a rare earth metal. It is found in under-water ridges.

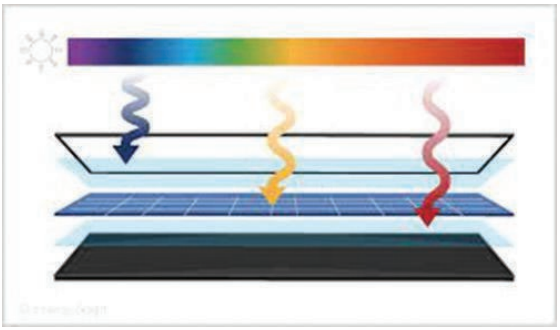
CIGS: Copper Indium Gallium Selenide

Advantage:

- Similar to CdTe, it can absorb sunlight readily. Besides, it is not toxic like CdTe.
- Disadvantage: Indium is in short supply

Multi-Junction PVs: Gallium-Arsenic, Indium, germanium

- Layers of semiconductor and metal
- The main advantage is high efficiency because different materials absorb different wavelengths from the sun.



3rd generation PV cells: Perovskites

- Perovskite is a crystalline material known for excellent light-absorbing properties and high photoconductivity.
- Perovskites can be used to make photovoltaic (PV) paints which can be applied on conformal surfaces like vehicles, buildings and even flexible electronics.
- Organic-perovskite solar cells are cheaper than conventional silicon-based PV cells and have higher efficiency (record-breaking efficiency of 24%).

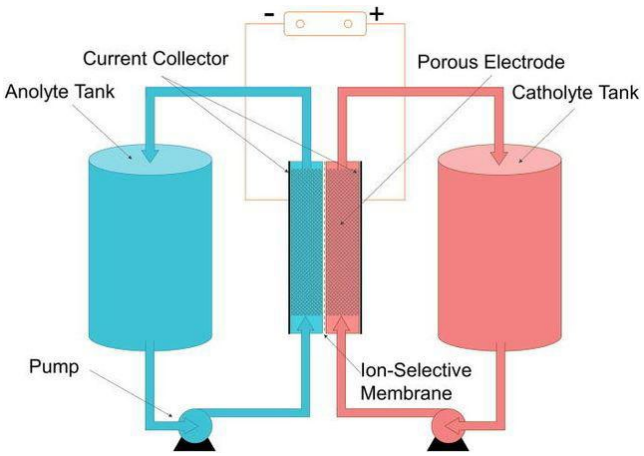
➔ ENERGY STORAGE

- Notwithstanding the advantages, Renewables have a peculiar problem called intermittency problem which means it is not available round the clock. Thus, energy storage via batteries becomes an important component of the renewable shift.
- The batteries including Li-ion batteries are not suitable to store energy for a longer time as the energy storage at the grid level requires us to store energy at least for 8 hours (night). The alternative suitable for grid-level storage is flow batteries.

24. FLOW BATTERIES

- The basic difference between conventional batteries and flow batteries is that while in convectional batteries we derived chemical energy from the electrodes, in flow batteries the energy is derived out of the flow of electrolytes.

- A flow battery uses two electrolyte solutions separated by a membrane to store electrical energy.
- A flow battery consists of two tanks - one for the positively charged electrolyte solution (called the catholyte) and one for the negatively charged electrolyte solution (called the anolyte).
- During charging the catholyte and anolyte flow on opposite sides of the membrane which is like going uphill.
- During discharge, the process is reversed. The catholyte and anolyte flow back into their respective tanks, and the electrical energy that was stored in the battery is released to power a device or a system.
- **Advantage:** Flow batteries are unique in their scalability which makes them suitable for grid-level storage.



25. SUPER CAPACITORS

PRINCIPLE

- A capacitor is a device that stores electric charges thereby electric energy.
- It is a set of two metal surfaces separated by an electric insulator.
- Put positive electric charge on one plate, negative on the other, and the combination can store energy for a long time, much longer than batteries can.
- Add more electric charge on the plate and you store more energy, but you also raise the voltage.
- The trick for storing energy in capacitors is making the insulator very thin, so that you can have lots of energy per unit volume while keeping the voltage low.
- These high energy-density capacitors are called supercapacitors or ultracapacitors.

ADVANTAGES

- Since capacitors don't depend on chemical reactions, they can release their energy extremely quickly thus suitable for EVs.
- Not subject to degradation with use and time.
- Because they can be charged so quickly, supercapacitors can be used to improve the efficiency of regenerative braking; they absorb the energy in charges and release it when needed.

DISADVANTAGE

- Supercapacitors can store about 1/3rd the energy of a same-weight lithium-ion battery.
- Costs three times more than Li-ion batteries.

26. BATTERY ENERGY STORAGE SYSTEM (BESS)

- BESS uses rechargeable batteries to store energy generated from different sources (generally renewable sources like sun and wind) and release it to the grid when needed.
- BESS are intelligent systems that use algorithms and software to coordinate with the electricity grid and decide whether to store or release energy.
- **Commonly used batteries in BESS**: Lithium-ion batteries due to their high energy density, long life span and declining costs.

- **Utility of BESS:** Store surplus energy; Reliable source of backup power; Enhance grid stability; Enhance integration of renewable energy into electricity grid.
- **Limitations**: High initial capital cost for installing BESS; Limited number of battery charge-discharge cycles; Environmental hazards (due to mining for critical minerals used in batteries & disposal).
- India aims to achieve 4 GWh of BESS capacity by 2026.

INDIA'S FIRST COMMERCIAL UTILITY-SCALE BESS

- Location: South Delhi's Kilokri village.
- Capacity: 20 MW (40 MWh) cluster of batteries.
- Capable of providing four hours of power every day.
- Electricity used to charge the system's batteries will come from renewable power purchase agreements.

Click here for INDEX

7 CHAPTER DEFENCE TECHNOLOGY

➔ MISSILES OF INDIA

1. CRUISE MISSILE AND BALLISTIC MISSILE

CRUISE MISSILE	BALLISTIC MISSILE
A cruise missile is guided towards a preset land-based target using a navigation system.	A ballistic missile is generally a projectile shot up in the atmosphere.
Cruise missiles are known for low altitude flight and high mobility and hence the name.	Ballistic missiles are launched outside the atmosphere where its warheads detach and hit the target. (uses gravity of earth and thus traces a parabolic path and hence the name)
Cruise missiles have short ranges of 300 to 1000 km. Brahmos – 290 to 450 KM Nirbhay – 700 to 1000	Ballistic missile have long ranges up to 1000 km (ICBMs) Agni V – 5000 KM
Easy to intercept because of high terminal speeds.	Difficult to intercept due to high terminal speeds of 5000 m/s
High precision due to navigation system	Low precision
Cheaper and thus suitable for firing conventional warheads.	Expensive and thus suitable for firing nuclear warheads.

2. SHORT-RANGE BALLISTIC MISSILES

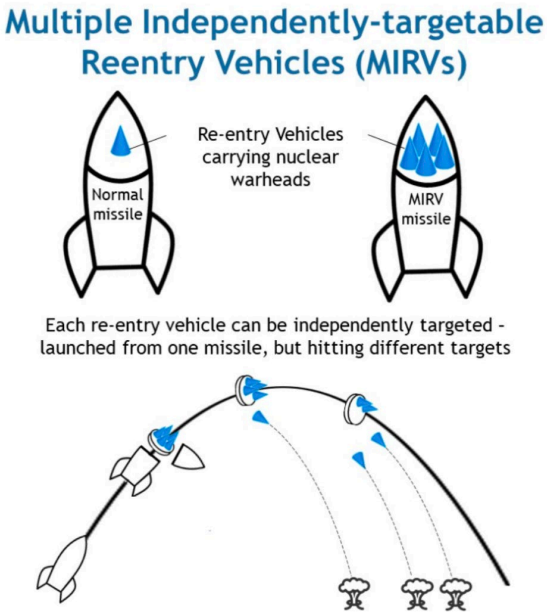
NAME	RANGE	WEAPON PAYLOAD	CHARACTERISTICS
PRITHVI-I	150 KM	1000 kg	Short range surface-to-surface tactical ballistic missile
PRITHVI-II	350 KM	500 kg	Air Force Version
DHANUSH (PRITHVI-III)	350 KM	1000 kg	Naval Version of Prithvi I and Prithvi II class missiles
SHAURYA	600-700 KM	1000 kg	Hypersonic Canister-launched Both conventional and nuclear warheads
PRAHAAR	150 KM	250 kg	To replace Prithvi I

			Weapons: High explosives, cluster munition, strategic nuclear weapon
AGNI-I	700–1200 KM	1000 Kg	Single stage solid fuelled Nuclear capable missile
PRALAY	150–500 KM	500- 1000 Kg	Surface-to-surface short-range missile Quasi-ballistic nature. Follows a low trajectory while retaining the ability to maneuver during the flight, making it harder to intercept. Tactical conventional missile developed by DRDO. Can be launched from a mobile launcher.

3. INTERMEDIATE-RANGE BALLISTIC MISSILES

NAME	RANGE	WEAPON PAYLOAD	CHARACTERISTICS
AGNI-II	2000-3000 KM	1000Kg	Two and a half stage solid fuelled missile
AGNI-III	3200 KM	2000-2500 Kg	2-stage solid propulsion system
AGNI-IV	3500 KM	1000 Kg	2-stage missile powered by solid propellant

4. INTER-CONTINENTAL BALLISTIC MISSILES

NAME	RANGE	WEAPON PAYLOAD	CHARACTERISTICS
AGNI-V #MISSION DIVYASTRA	>5000 KM	1500 Kg	3-10 Multiple Independently Targetable Re-entry Vehicle (MIRV) warheads MIRV-equipped missiles can render the missile defence system useless. 

AGNI-VI	11,000–12,000 KM (Under development)	3 tonnes nuclear payload	Four-stage intercontinental ballistic missile. MIRV warheads.

5. SUBMARINE-LAUNCHED BALLISTIC MISSILES

NAME	RANGE	WEAPON PAYLOAD	CHARACTERISTICS
K-15 Sagarika	750 KM	500 Kg	• Replica of land based Shaurya Missile. • K-15 Missile gets help from Indian Regional Navigation Satellite System. • K-series missiles are much faster than Agni Missiles
K-4	3500 KM	1000 Kg	• Nuclear-capable underwater missile • Meant for Arihant class submarines. • It is a solid-fuelled missile launched underwater capable of withstanding 50N of water pressure. • Powered by: Solid rocket propellant.
K5 (Under development)	5000 KM		
K6 (Under development)	6000 KM		

6. CRUISE MISSILES

NAME	RANGE	WEAPON PAYLOAD	CHARACTERISTICS
Nirbhay: Subsonic Cruise Missile	750-1000 Km	500 Kg	• Long range subsonic cruise missile. • 1st indigenously developed long range cruise missile flying at low altitudes. • Subsonic speeds of 0.7 Mach. • Can fly at tree-top altitudes as low as 10 m (now tested for 5 m) • Capable of delivering nuclear warheads of 200-300 kg. • 2-stage solid fuelled cruise missile. Go undetected by enemy radars.
Brahmos: Supersonic Cruise Missile	290 Km	300 kg	• Joint venture missile between India and Russia. • **Speed**: Top speed of 2.8 Mach (speed of sound) • **Range**: After India became a full member of the MTCR export control regime, the range of BrahMos has increased from 300 km to 450 km. • Fire and forget principle of operation • Capable of being launched from land, water and air. **BRAHMOS NG:** • Low weight Air-launched version capable of being carried by Light Combat Aircraft, Tejas LCA. • Top speed: 3.5 Mach

Brahmos II: Hypersonic	1000 Km	300 Kg	Over 7 Mach
RudraM-II	300-350 Km		• Solid-propelled air-to-surface anti-radiation supersonic missile. • Indigenously developed by DRDO. • Peak speed of Mach 5.5 (5.5 times the speed of sound). • Capable of carrying a payload of up to 200 kg i.e., explosives. • Can be launched from a height of 3 to 15 km. • The internal guidance system allows it to direct itself toward the target after the launch. • Capable of destroying aircraft hangars, bunkers, enemy ground radars (surveillance, tracking) and communication stations in SEAD missions.

7. SHORT-RANGE SURFACE-TO-AIR MISSILES

NAME	RANGE	WEAPON PAYLOAD	CHARACTERISTICS
TRISHUL	**9 – 12 Km**	**5 Kg**	• Short range low-level surface-to-air missile
AKASH	**30 Km**	**50 Kg**	• Supersonic speed: Mach 2.5. • Medium range surface-to-air missile • Part of Air-defence system • It can simultaneously engage multiple targets in Group Mode or Autonomous Mode. • It has built in Electronic Counter Counter Measures (ECCM) features. • Indigenous seeker technology.
MAITRI (QRSAM)	**25–30 km**	**10 Kg**	• Developed by DRDO in collaboration with BEL and Bharat Dynamics Ltd. • Most effective in combating low flying Aerial targets faced by forward tactical battlefield area formations like ○ Attack helicopters ○ UAVs ○ Armed drones ○ Subsonic Cruise missiles • QRSAM is part of India's air defence system. • Capable of striking targets on-the-move. • It can strike targets at various altitudes from 30m to 6km. • It has a speed of 700-800 metres per second • RF seeker (Radio Frequency) as a part of terminal guidance to hit the target. • It has a truck-mounted canister. • It is capable of multiple-target engagement. • It is light-weight, has high-mobility and shorter-reaction time compared to Akash Air Defence Surface-to-Air Missiles.

			• It is equipped with electronic counter measures against the aircraft jammers to deceive enemy radar.
REVATI	**25–30 km**	**10 Kg**	**(Naval-Version of Maitri)**
ROHINI	**25–30 km**	**10 Kg**	**(Air Force-Version of Maitri)**
BARAK-8	**70-100 Km**	**60 Kg**	• Indian-Israeli Medium Range Surface-to-Air-Missile • Part of naval air defence system To be used aboard INS Vikrant (under construction) • maximum speed of Mach 2
IGLA-S	Maximum range 6 km, altitude 3km		• Russia's Very Short Range Air Defence Systems (VSHORAD) • Man-Portable Shoulder launched Air Defence System • It can aim at enemy targets using Infrared signature, Laser targeting, Remote-controlled targeting

8. AIR-TO-AIR MISSILE

NAME	CHARACTERISTICS
ASTRA	• It is a BvRAAM (Beyond Visual Range Air to Air Missile). • 1st air to air missile developed by India. • Capable of engaging targets at varying range and altitudes. • Both short range targets at a distance of 20 km and long range targets up to a distance of 80-110 km.

9. ANTI-TANK GUIDED MISSILE

NAME	RANGE	WEAPON PAYLOAD	CHARACTERISTICS
NAG	3-7 Km	8 kg	• 3rd generation anti-tank missile, best in its class for Indian conditions. • Fire and forget capability. • Imaging Infrared (IIR) guidance with day and night capability • Weight- 48 Kg and fired from BMP-2 vehicle called **NAMICA (Nag Missile Carrier)** **DIFFERENT VARIANTS** 1. Mounted on an infantry combat vehicle: NAMICA. 2. Man Portable shoulder carried. 3. HeliNa: (Helicopter-launched Nag) Rudra Helicopter, Dhruv and LCH.
NAG MK-2	7-10 Km	8 kg	• Indigenous all-weather, fire-and-forget, lock-on after launch, third-generation anti-tank guided missile (ATGM).
HELINA	7 Km	8 kg	Helicopter-launched Nag
SPIKE			• Israeli 4th generation anti-tank guided missile. • Fire-and-forget capability

			• Available in man-portable, vehicle-launched, and helicopter-launched variants. • Change the target mid-flight as a result of dual-seeker. • Kill-probability of 95% • Better than Nag because Nag is facing seeker issues especially if the temperature of the target is high.
MILAN			• It is an anti tank guided missile for the Army. To be acquired from France.
SPICE 2000			• Israeli guided bombs for the Indian Air Force.

➔ INDIA'S MISSILE DEFENCE SYSTEMS

10. BALLISTIC MISSILE DEFENCE SYSTEM

- 2-tier missile defence system
- Aimed at intercepting aerial threats from ballistic missiles that have ranges up to 5000 km at altitudes both outside (exo) and inside (endo) the atmosphere.
- **1st layer: Endo**
  - The single stage solid rocket-propelled **Advanced Air Defence (AAD) low-altitude interceptor** missile. (**Ashwin**)
  - The AAD interceptor missile is primarily designed to intercept enemy missiles in the endo-atmosphere at altitudes of 20-40 kilometres.
- **2nd layer: Exo**
  - Prithvi Air Defense Vehicle known as Pradyumna Ballistic Missile Interceptor is designed to destroy missiles with ranges 300-2000 km at exo-atmosphere (about 80 km altitude).
  - For higher altitudes up to 150 km, Agni-V-based ballistic interceptors would be used. (because of 5000 km range)

11. MULTI-LAYERED AIR DEFENCE SYSTEM: S-400

- India has signed a deal with Russia to acquire five S-400 Triumf multi-layered air defence systems.
- S-400 is a surface-to-air missile system; one of the most advanced anti-aircraft systems developed by Russia.
- Capable of firing four types of missiles to create a layered defence.
- The layered defence system can intercept all types of aerial targets including aircraft, unmanned aerial vehicles (UAV), and ballistic and cruise missiles up to the range of 400 km, at an altitude of up to 30 km.
- **Note**: USA had offered **THAAD** and **Patriot** as alternatives to S-400 to India. Israel has Iron Dome: Israeli SHORT-RANGE AIR DEFENCE System.

12. BARAK-8 LONG AND MEDIUM RANGE SAM

- Medium-range surface-to-air missile system being developed jointly by India and Israel.
- It will have an interception range of 70-100 km.
- Part of naval air defence system To be used aboard INS Vikrant (under construction)
- Maximum speed of Mach 2.

13. AKASH MEDIUM-RANGE SURFACE TO AIR MISSILE SYSTEM

- India has 2 regiments of the indigenous Akash systems which are capable of multi-target engagement.
- It can strike targets up to a range of 25 km and altitude of 18,000m.

14. MULTI-LAYERED AIR DEFENCE SYSTEM FOR DELHI

- India is developing a multi-layered air defence system for its cities besides air defence systems for tactical battle areas.
- 1st layer: 2-tier Ballistic Missile Defence System
- 2nd Layer: S-400 layered defence system
- 3rd Layer: Barak-8 long and medium range SAM
- 4th Layer: Akash medium-range surface to air missile system

15. HYPERSONIC MISSILES

- Travels at Mach 5 or higher (more than one mile per second).
- They typically consist of a Supersonic Combustion Ramjet or Scramjet propulsion system to enable such high speeds.
- A **scramjet engine** collects oxygen from the atmosphere as it is travelling and mixes the oxygen with its hydrogen fuel, creating the combustion needed for hypersonic travel.
- DRDO is developing **Hypersonic Technology Demonstrator Vehicle (HSTDV)**. It is an unmanned scramjet (allowing supersonic combustion) demonstration vehicle that can cruise to a speed of Mach 6 (or six times the speed of sound) and rise up to an altitude of over 30 km in 20 seconds. HSTDV successfully demonstrated supersonic combustion in September 2020.
- There are 2 types of Hypersonic Weapon Delivery Systems
  1. Hypersonic Cruise Missiles (HCM)
  2. Hypersonic Glide Vehicle (HGV)

SIGNIFICANCE

- They are a mix of the speed of a ballistic missile and manoeuvring capabilities of a cruise missile.
- While cruise missiles achieve speeds of 550 miles per hour, the hypersonic missiles aircrafts can reach speeds more than 3500 miles per hour.
- Capable of penetrating any anti missile defence system currently available that are designed to intercept cruise and ballistic missiles.
- Specifically designed for increased survivability against modern ballistic missile defence systems.

IMPORTANT HYPERSONIC MISSILES

- DRDO conducted a successful flight-trial of **India's first long-range hypersonic missile** off the coast of Odisha late in November 2024.
- It can carry both conventional and nuclear warheads to a distance exceeding 1,500 km at a speed of nearly 3 km per second.
- Presently, the US, Russia and China have hypersonic missiles.
- **US:** Possesses hypersonic missiles, specific details are classified.
- **Russia:**
  - **Avangard**: Nuclear-capable hypersonic boost-glide vehicle (Mach 20, 6000+ km range).

- **Kinzhal:** Nuclear-capable air-launched ballistic missile (Mach 10, 2000+ km range).

- **China:**
  - **DF-17:** medium-range ballistic missile equipped with a hypersonic glide vehicle (Range around 1600 km)

➔ AIRCRAFT CARRIERS OF INDIAN NAVY

16. AIRCRAFT CARRIERS OF INDIAN NAVY

- **INS VIKRANT:** 1st aircraft carrier of India. Decommissioned in 1997 after serving for 37 years.
- **INS VIRAAT:** 2nd and the longest serving aircraft carrier of India. Decommissioned in 2017 after serving for 56 years. (26 years as HMS Hermes and 30 as INS Vikrant)
- **INS VIKRAMADITYA:** India and Russia signed $1.5 billion for the acquisition of the warship INS Vikramaditya inducted to the Indian Navy in 2013.
- **INS VIKRANT:** India's 1st indigenous Aircraft Carrier, inducted into the Indian Navy in 2022.
- **INS VISHAL:** A 65000 tonne Naval supercarrier on the lines of HMS Elizabeth is planned to be built by India.

➔ SUBMARINE ARM OF INDIA

CONVENTIONAL SUBMARINES OF INDIA

17. MAIN CLASSES OF SUBMARINES WITH INDIAN NAVY

1. **Sindhughosh-class:** Variant of the Russian Kilo-class submarines. E.g., INS Sindhurakshak, INS Sindhuvir, INS Sindhuratna.
2. **Shishumar-class:** Based on the German Type 209 design, these are diesel-electric submarines equipped with advanced systems. E.g., INS Shishumar, INS Shankush, INS Shalki, and INS Shankul.
3. **Kalvari-class (Scorpène-class):** Diesel attack submarines based on the Scorpène design developed by Naval Group (France). E.g., INS Kalvari, INS Khanderi, INS Karanj, INS Vela, INS Vagir, and INS Vagsheer (commissioned in Jan 2025).
4. **Arihant-class (SSBN):** India's indigenous nuclear-powered ballistic missile submarines (SSBNs). These submarines are equipped with nuclear propulsion and are capable of carrying ballistic missiles. Currently, INS Arihant (S2) is the operational submarine of this class. The second submarine INS Arighat (S3) was commissioned in August 2024. Two more submarines (S4 Aridhaman & S4* are launched but yet to be commissioned).
5. **Chakra-class (SSN):** Nuclear-powered attack submarines leased from Russia. Currently, INS Chakra-II (S1) is in service with the Indian Navy. Chakra-III is expected to be delivered by 2026 or later.

18. PROJECT 75

- 6 Scorpene-class submarines are constructed indigenously with Transfer of Technology from France.
- The 6 Scorpene class submarines are the core of India's conventional attack submarine arm. These include-
  - **INS Kalvari:** Inducted in 2017
  - **INS Kandheri:** Inducted in 2019
  - **INS Karanj:** Inducted in 2021
  - **INS Vela:** Inducted 2021
  - **INS Vagir:** Inducted 2023

- **INS Vagsheer:** Inducted 2025

- The submarines built under the project are capable of:
  - anti-surface warfare and anti-submarine warfare
  - Intelligence, Surveillance and Reconnaissance missions
  - underwater mining operations and naval mine laying.
- The submarines have Diesel Electric transmission systems.
- Can be upgraded with the Air Independent Propulsion (AIP) system in order to stay underwater for longer duration and thus increase their operational range and stealth capabilities.

Project 75-I (India)

- Project 75-I is a follow-up to Project 75. It improves upon the design and technology of its predecessor (P-75). The project is yet to begin.
- Indian Navy aims to acquire six conventional, diesel-electric attack advanced submarines equipped with fuel cells and an Air-Independent Propulsion System (AIP).

NUCLEAR-POWERED SUBMARINES

Nuclear submarines can be classified into two categories namely SSN and SSBN:

19. SSN (SUBMERSIBLE SHIP NUCLEAR)

- SSNs are fast attack submarines. They are propelled by nuclear power.
- Primarily designed for offensive operations (short-range assailant attacks), such as anti-submarine warfare and anti-surface warfare. They also carry out surveillance and intelligence missions.
- They are capable of launching **conventional weapons** like torpedoes and cruise missiles.

20. SSBN (SUBMERSIBLE SHIP BALLISTIC NUCLEAR)

- SSBNs are also propelled by nuclear power.
- They are usually equipped with nuclear weapons like ballistic missiles.
- Therefore they are usually used as deterrents and not as attack submarines.

21. NUCLEAR-POWERED SUBMARINE FLEET OF INDIA

SSN FLEET

CHAKRA-I

- 1st nuclear-powered submarine to be inducted into the Indian Navy. Served until 2021.

CHAKRA-II

- 2nd nuclear submarine to be inducted into Indian Navy. It was inducted in 2012.
- It is an advanced version of Chakra I with following features:
  - Higher operating depths (up to 600 metres).
  - Higher speed exceeding 30 knots. Powered by a 190 MW nuclear reactor.
  - Equipped with an advanced weapon system including tube-launched missiles.
  - Chakra II is deployed with the Eastern Naval Command.

CHAKRA-III

- In 2019, India and Russia signed an agreement for leasing of the Akula-class nuclear-powered submarine Chakra III for $3 billion for at least 10 years.
- Powered by a 190 MW nuclear reactor. Expected to be delivered in 2026 or later.

SSBN FLEET

ARIHANT

- Under the Advanced Technology Vehicle programme, India has indigenously-built Arihant, SSBN.
- India's 1st nuclear-powered ballistic missile submarine. Displacement capacity of 6,000 tons.
- Powered by an 83 MW pressurised light-water reactor with enriched uranium.
- Formally inducted in 2019 marking the Nuclear Triad capability of India.
- Capable of launching K-15 Missile with a range of 750 km and K-4 ICBM with a range of 3,500 km.

ARIGHAT

- 2nd Arihant-class submarine built under Advanced Technology Vessel Program.
- Powered by a pressurised water reactor.
- Maximum speed of 12–15 knots (22–28 km/h) when on surface and 24 knots (44 km/h) when submerged.
- 4 launch tubes can carry up to 12 K-15 Sagarika missiles or 4 four of the under-development K-4 missiles.

ARIDHAMAN

- Launched in 2021 (expected to be commissioned in 2025).
- Slightly larger than INS Arihant. Measures 125.4 metres and has a displacement of 7000 tons.
- Equipped with K-4 ICBM. Powered by a 83-MW pressurised light-water reactor with enriched uranium.

S4* (YET TO BE NAMED):

- Launched in October 2024 (yet to be commissioned). Has nearly 75% indigenous content.
- Measures 125.4 metres and has a displacement of 7000 tons.
- Equipped with K-4 ICBM. Powered by a 83-MW pressurised light-water reactor with enriched uranium.

After S4*, the next class of Indian SSBNs will be double the 6000 ton displacement of Arihant class and will be carrying nuclear missiles upwards of the range of 5000 kilometres and beyond.

22. P-81 POSEIDON AIRCRAFT

- Anti Submarine Warfare (ASW) aircraft which India has acquired from the US. It is a boost to maritime ISR capabilities (intelligence, surveillance and reconnaissance).
- Long-range maritime patrol aircraft capable of undersea surveillance from a height of up to 41,000 ft.
- Has an operational speed of 490 knots (564 mph), range over 2,222 kilometres with on-station time of four hours.
- Equipped with Active Electronically Scanned Array (AESA) radars capable of engaging multiple targets simultaneously. Magnetic Anomaly Detection radar will help locate submarines in deep seas.

23. IMPORTANT FIGHTER JETS OF INDIA

MIG 29	• Twin-engine, light-weight, air-superiority fighter aircraft developed by the Soviet Union. • Developed by the Soviet Union. Inducted into the Indian Air Force in 1987. • With about 110 Mig-29s operated by the Indian Air Force and Indian Navy combined, India is the second biggest operator of Mig-29s in the world after Russia. • Played an important role during the Kargil War of 1999.

	• Three squadrons of MiG-29s aircrafts are deployed in the strategically important Adampur Air Force Station, Punjab, which is around 100 km from Pakistan and 250 km away from China borders. • Capacity to launch Beyond Visual Range BVR missiles.
SUKHOI-30 MKI	• Twin-engine, multirole combat fighter aircraft jointly developed by Russia and India. • India's answer to Pakistan's F-16. • Top speed of 2120 km/h (Mach 2). • Capable of launching air-to-air, air-to-surface missiles, laser guided bombs and cluster bombs. • Provided significant strategic deterrence against China and Pakistan in multi-mission roles, including precision strikes on terror camps across the LoC. • Su-30MKI has been integrated with BrahMos supersonic cruise missile providing capability to engage high-value naval targets, including aircraft carriers.
TEJAS LCA	• 4th generation supersonic, single-seat, single-engine multirole light fighter aircraft. • Conceived in the early 1980s to replace the Russian-made MIG 21 of the Indian Air Force. • Indigenously designed and developed by Aeronautical Development Agency. • The combat-ready version comprises battle-time requirements such as mid-air refuelling, AESA radar, electronic warfare suites, bombs and weapons etc. • All 32 single-seat Tejas Mk 1 aircraft have been delivered to the IAF.
LCA-Mk2 (Tejas 2)	• 4.5 generation Advanced Medium Combat Aircraft. • IAF plans to have about six squadrons of Tejas Mk2. Estimated induction: 2028-29. • Powered by GE-414 engine (a high-performance afterburning turbofan engine). • Combat Range: 1500 kms. Maximum Speed: 1.8 Mach • With a weight of 17.5 tonnes, it can carry 6.5 tonnes of weapons. It has all indigenous weapons and about 11 weapon stations. • Designed to carry a range of air-to-air, air-to-surface and precision-guided weapons. • Capable of carrying 8 Beyond-Visual-Range (BVR) missiles at once.
RAFALE	• 4.5 generation twin-engine multirole combat aircraft manufactured by French Dassault Aviation. • France completed the delivery of all 36 Rafale jets to the IAF in December 2022. • Main roles include missions including Air-defence/air-superiority, reconnaissance, close air support, dynamic targeting, air-to-ground precision strike/interdiction, anti-ship attacks, and nuclear deterrence, buddy-buddy refuelling. • Capable of carrying over 9 tonnes of weapons, including air-to-air, air-to-ground, and air-to-ship missiles. • Maximum speed of 1.8 Mach. Combat range of more than 3700 km. • Weapon systems include SCALP and METEOR missiles. • **SCALP:** Precision long range ground attack missile that can take out targets with extreme accuracy. Has a range of 300 km, capped by the missile technology control regime. • **METEOR:** Beyond visual range air to air missile that is possibly the best in its class. Can take out enemy aircraft at a range of over 100 km.
ADVANCED MEDIUM	• 5th-generation fighter aircraft. Currently the only 5th generation fighter under development in India.

COMBAT AIRCRAFT (AMCA)	• Indigenously designed by: Aeronautical Development Agency under DRDO. • Full-stealth AMCA aircraft programme sanctioned in 2024. Estimated induction: 2034 • AMCA is a 25-tonne twin-engine aircraft. • Max speed: Nearly 2,600 kilometres per hour (Mach 2.15). Combat range: 1,620 km • Designed for multi-role missions, including air-to-air and air-to-ground operations, suppression of enemy air defenses (SEAD) and electronic warfare. • Stealth aircraft (capable of avoiding enemy radars and air defence mechanisms). • AMCA will be based on the 'first look, first kill' concept, where an AMCA pilot would see an enemy plane first, fire a missile and destroy it before the latter can react. • AMCA will carry: BrahMos-NG (next generation) air-to-ground missiles; Astra air-to-air missiles; Rudram air-to-ground anti-radiation missiles.

24. AIRBORNE EARLY WARNING & CONTROL SYSTEM (AWACS)

- Known as 'eyes in the sky', AWACS are Air-borne radar systems mounted on aircrafts for scanning and surveillance.
- With a 360-degree coverage, they are deployed to carry out surveillance on enemy Air Defence systems and to prepare an Air Situation Picture useful in aerial combats.
- **PHALCON:** AWACS procured by India from Israel. Detects aircraft at ranges exceeding 400 kilometres.

25. NETRA

- Netra Airborne Early Warning & Control System (AWACS) aircraft is a multi-sensor platform indigenously developed by DRDO.
- Netra has an indigenously developed Active Electronically Scanned Array (AESA) radar system mounted on the Embraer ERJ 145 twin-engine aircraft.
- The AESA radar is an airborne surveillance system with detection and tracking capabilities. It can track and find aircraft, missiles, ships and vehicles.
- Netra provides 240-degree coverage of airspace. Detects aircraft at ranges exceeding 400 kilometres.

26. AKASHTEER

- Akashteer (Sky Arrow) is an **Indian Air Defence Control & Reporting System** developed by Bharat Electronics Limited.
- It is designed to automate air defence control and reporting processes by digitising them.
- The Akastheer system is being integrated with the Integrated Air Command and Control System (IACCS) of the Indian Air Force to increase jointness of the Armed Forces for air defence.
- By integrating radar and communication systems at all levels into a unified network, 'Akashteer' aims to deliver an unprecedented level of situational awareness and control.
- This allows for effective monitoring of low-level airspace in battle zones and efficient management of Ground-Based Air Defence Weapon Systems.

27. INDRAJAAL

- Hyderabad-based Grene Robotics has launched Indrajaal, India's first AI-powered anti-drone system.

- Provides a 360-degree coverage/defence against all types and intensities of unmanned autonomous threats across regions up to 4000 square kilometres.
- Can quickly and accurately identify, categorise, track, and eliminate threats in real time.
- Offers protection from all kinds of autonomous drones, including low Radar Cross Section (RCS) threats, medium-altitude long-endurance (MALE) and high-altitude long-endurance (HALE) UAVs.
- Can also be used for loitering munitions, smart bombs, rocket showers, nano and micro drones, and even swarm drones.

28. MAN PORTABLE AIR DEFENCE SYSTEM (MANPADS)

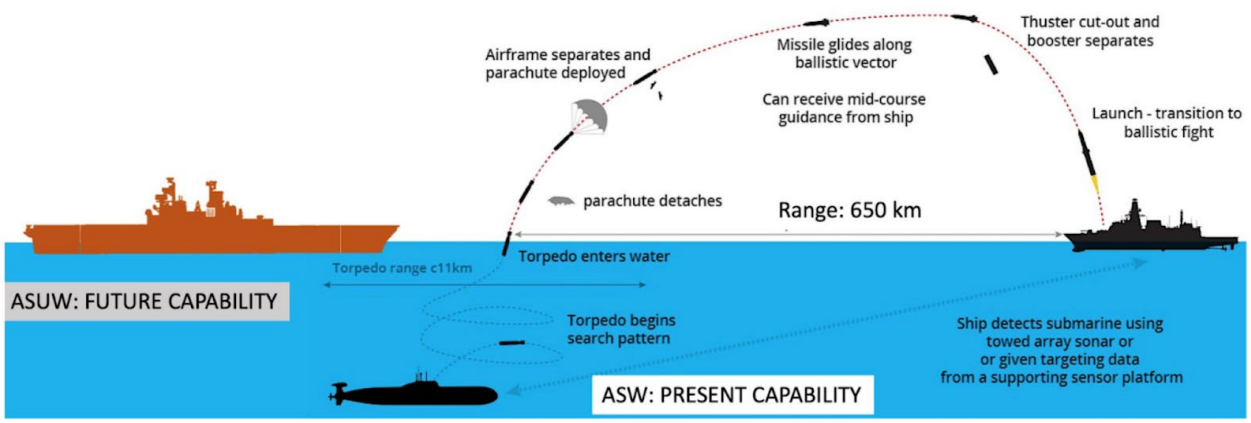
- MANPADS are short-range, lightweight and portable surface-to-air missiles.
- Can be fired by individuals or small groups to destroy aircraft or helicopters.
- Fitted with infrared (IR) seekers to identify and target the airborne vehicle through heat radiation being emitted by the latter.
- Have 'fire-and-forget' guidance systems. Target low-flying aircraft and shield troops from aerial attacks.
- **Note:** India has contracted 48 Igla-S launchers (MANPADS) from Russia. It will enhance Army's Very Short Range Air Defence (VSHORAD) capabilities.

29. VERY SHORT-RANGE AIR DEFENCE SYSTEM (VSHORADS)

- DRDO has successfully conducted three successive flight-trials of the VSHORAD system from Chandipur off the coast of Odisha.
- VSHORADS is a fourth generation technically-advanced miniaturised MANPAD.
- Indigenously developed by: DRDO's Hyderabad-based premier facility Research Centre Imarat.
- Interception Range: up to 6 kms. Limiting altitude of effective target destruction: 3.5 km
- Maximum speed: Mach 1.5
- Can neutralise low altitude aerial threats like drones and high-speed targets at short range.

30. SUPERSONIC MISSILE-ASSISTED RELEASE OF TORPEDO (SMART) SYSTEM

- DRDO tested a next-generation SMART system to boost the Navy's anti-submarine warfare capabilities.
- SMART is a missile-based system designed to significantly extend the range of lightweight torpedoes that can target submarines hundreds of kilometres away far beyond the conventional range of lightweight torpedoes (around 20-40 km).



- It carries an advanced lightweight torpedo missile as a payload along with a parachute-based release mechanism.
- It is mobile, and can be used both from coasts as well as from sea for immediate action when an enemy submarine is detected.

➔ MISCELLANEOUS

31. ANTI SATELLITE TECHNOLOGY

- Satellites form an integral part of a country's critical infrastructure. Intercepting satellites can halt various critical applications including navigation systems, communication networks, broadcasting, banking systems, surveillance etc.
- Anti-satellite technology is a counter-space capability of a country to neutralise space-based assets of an enemy country.

TYPES OF A-SAT

- **Missile-based A-SAT**: A missile is used to hit and destroy satellite using:
  - **Direct-ascent** kinetic kill vehicle (Chinese A-SAT in 2007, India now, USA and Russia)
  - **Co-orbiting** missile (Russia has this type)
- **Co-orbital drones**: Approach the target satellite and deviate it from its orbit. (China, UK, Russia are working on this technology)
- **High-energy lasers**: Blind the sensors of the satellites.
- **Interception and jamming** of signals from the satellites by sending more powerful radio signals.

HISTORY OF A-SAT MISSILE TESTS

- ASAT missile tests have been conducted by the USA and Russia in the cold-war era.
- The USA has had the anti-satellite weapon since 1959 followed by Russia in 1960.
- The cold-war witnessed the anti-satellite weaponry tests till early 1980s.
- China conducted the A-SAT weaponry test in 2007. (800 km orbit)

OUTER SPACE TREATY 1967

- According to this treaty, outer space shall be used only for peaceful purposes.
- It prohibits countries from placing into orbit around the Earth any objects carrying nuclear weapons or any other kinds of weapons of mass destruction.
- It requires that celestial bodies shall be used by all parties exclusively for peaceful purposes and no weapon can be stationed on them.
- However, the Outer Space Treaty by itself does not prevent an arms race in space.
- India is a party to the Outer Space Treaty.

32. MISSION SHAKTI

- Under Mission Shakti, India demonstrated the capability to destroy a satellite in low earth orbit using an anti-satellite missile in 2019.
- India became only the 4th country to conduct an Anti-Satellite missile test after the USA, Russia and China.

TARGET: Microsat R was a military imaging satellite placed in an orbit 274 km above the Earth surface with an orbital velocity of 7.8km/s.

TECHNOLOGY: 'HIT TO KILL'

- The anti- satellite test involved the 'hit to kill' missile technology.
- Under the 'hit to kill' technology, a missile is shot at the satellite in order to hit and kill the satellite.

KILL VEHICLE TECHNOLOGY

- The A-SAT missile was based on the exo-atmospheric kill vehicle technology. It includes:
  - an 18-tonne, 3-stage interceptor missile
  - with 2-stage solid propellants
  - with a long range tracking radar and
  - Infra-red and radar frequency seekers to reach the target satellite and hit it.
- Accordingly, the anti-satellite missile used was an advanced version of 'Prithvi Defence Vehicle' of India's Ballistic Missile Defence system. This is because the target satellite was in a 300 km orbit.

33. NAGASTRA-1 SUICIDE DRONE

- Nagastra-1 is the first indigenous Loiter Munition or suicide drone which can strike targets within a 15-30 km range using GPS-enabled precision.
- It can loiter over a target for a maximum of 60 minutes and has an accuracy of less than 2 metres.
- It can undertake shallow strikes with precision, without endangering the lives of soldiers and is capable of destroying various soft-skinned targets using its pre-fragmented high explosive warhead. The munition carries a day-and-night camera for surveillance as well.
- Nagastra-1 in its 'Kamikaze' mode can search and destroy any target by crashing into it. In case, a target is not detected or if the mission is aborted, it can be called back and made a soft landing with a parachute recovery mechanism, thus, enabling it to be reused multiple times.

34. HIGH ALTITUDE PSEUDO SATELLITES (HAPS)

- HAPS are unmanned air vehicles that can fly at an altitude of 18-20 km from the ground (stratosphere), almost double the heights attained by commercial aeroplanes.
- They have the ability to generate solar power, so they can remain in the air for months and even years, giving the advantage of a satellite.
- They move at just about 80-100 km per hour at a height of 20 km above the Earth's surface. These features help it to gauge an area for a long time. They can easily keep an eye over 200 km and can observe everything even over a 400 sq km area with 5 cm resolution.
- HAPS can work like a geostationary satellite and can be easily redeployed over another location, or can be reequipped with a different payload.
- **Primary utility** for surveillance and monitoring. Can be also useful in disaster management, emergency/public safety communications, maritime surveillance, land border control applications, etc.

35. BHAR OS SOFTWARE

- IIT Madras has recently developed 'BharOS' - an indigenous mobile operating system.
- The Android Open-Source Project comes with No Default Apps (NDA) which means that users are not forced to use apps that they may not be familiar with, giving more permission controls to users.
- It will be used in government and public systems and will reduce the reliance on foreign OS in smartphones.

36. MAYA OS

- Maya OS is an operating system developed by Indian Defence Research and Development Organisation, the Centre for Development of Advanced Computing (C-DAC), and the National Informatics Centre (NIC).
- The Indian Defence Ministry has decided to replace Microsoft's Windows with Maya OS on all its computers.
- It is an open-source Ubuntu-based operating system launched to prevent malware attacks by cybercriminals increasingly targeting critical infrastructure and government agencies.
- It is powered by an endpoint detection and protection system called **"Chakravyuh"** (anti-malware and antivirus software).